Computer Fundamentals

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COMPUTER FUNDAMENTALS

Publisher : **Mahbubur Rahman**
Director
Systech Publications Ltd.

1st print	:	July, 2002
2nd print	:	June, 2003
3rd pirint	:	January, 2005
4th print	:	May, 2008
5th print	:	March, 2010
6th print	:	February, 2011
7th print	:	January, 2012
8th print	:	September, 2012
9th print	:	March, 2013
10th print	:	January - 2015
11th print	:	August - 2018
Copy Right	:	Authors

Cover Design	:	Systech Publications Ltd.
Compose	:	M. A. Rahman & Co.

Price : Tk. 200

ISBN : 978-984-8980-53-8

PREFACE

Computer Fundamentals, intended for first level bachelor's course, covers introductory topics on computer, its organisation and applications. The important features of the book include complete coverage with unnecessary detail, easy and friendly style and practical orientation. The book offers just what the users need to know. The text presents information on microcomputers, which includes history, generations and organisation of microcomputers, computer hardware and software. It also covers topics on information and communication technology (ICT) and some of its modern applications, selection of computers and social impact of computers.

The primary goal, in presenting information, is to reach students by making presentations as clear, relevant and interesting as possible. The book is organised in a self learning style and the chapters are divided as lessons with objectives at the beginning and different categories of questions at the end for testing the outcome of reading the lesson.

The book is suitable for a 3-credit course on computer fundaments for bachelor's program in computer science/engineering, information and communication technology/engineering, and electronic and electrical engineering. The book can also be used for bachelor's program in all other branches of science and engineering. Presently business and commerce are the very important areas of applications of computers, and information and communication technology and the book is suitable for the 3-credit foundation course for bachelor's programs in business studies, e-commerce, management information systems, business information technology, financial engineering etc.

We consulted many books, articles and magazines in preparing the book and we are grateful to their authors and publishers. We are particularly grateful to our colleagues for their encouragement and Ms. Suzana Rahman for preparing the index of the book. We are grateful to Mr. Abdur Rahman for his help and cooperation in composing the manuscript for publication.

In spite of our sincere efforts, there are some inconsistencies and printing mistakes in the book and we are sorry for the inconvenience caused for the mistakes. We will appreciate helpful comments and suggestions by students, teachers and other readers of the book.

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and
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30 March 2002

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Chapter 1

Introduction to Computers

Lesson1: Basic Organization
Lesson2: Types of Computers
Lesson3: History and Generations

Lesson 1: Basic Organization

1.1 Learning Objectives

On completion of this lesson you will know:

- Basic concepts of a computer system
- Basic elements of a computer system
- Importance and limitation of computers

1.2 Computer Basics

A computer is an electronic machine that stores, retrieves, and processes data. It cannot think or reason; it can only carry out instructions given to it. A set of instructions that directs its actions is called a program. Different programs are used to solve different problems. Ability to accept, store and execute programs makes the computer an invaluable and all-purpose business tool.

The first step of solving a problem by a computer is to develop a computer program and then store it in the memory. The computer then executes the instructions in the program. The instructions generally direct the computer to perform three basic functions over and over again; these are input, processing, and output. Collectively, these functions constitute the data processing cycle.

Input: Input devices feed the computer facts or data to be processed.

Processing: The control and storing of data, numerical comparisons, and arithmetic operations are performed on the input data to produce the results.

Output: The computer feeds the processed data or information to the output devices.

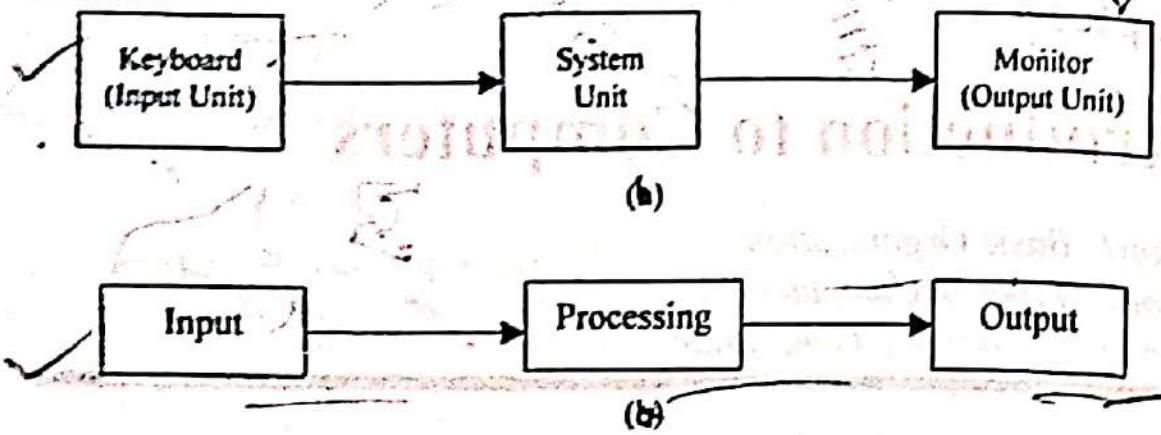


Figure 1.1: a) Solving a problem with a computer, b) equivalent block diagram.

A computer reads a program and stores it in the memory and executes instructions to:

- Input data from disk, keyboard, or other storage media,
- Process data and
- Output results to display screen, disk or other media.

1.3 Components of a Computer System

A computer is used to process data and a data processing system consists of more than just machines. A computer system must contain:

- Hardware
- Software
- Humanware, and
- Operational procedures.

Hardware

Hardware refers to machines or physical equipment that performs the basic functions of the data processing cycle. In addition to the computer itself, other hardware devices are also required. These devices may be off-line that is detached from the computer and operating independently or they may be on-line that is directly connected to and controlled by the computer. A printer is an off-line device and a keyboard is an on-line device.

Software

A program is a sequence of instructions, which directs a computer to perform certain functions. A computer must have access to stored programs and input devices for data, and it must make decisions, arithmetically manipulate data, and output results in the correct sequence. Programs are referred to as software. Software is generally categorized as:

- System software, and
- Application software.

System software consists of programs that help the use of a computer. These programs perform such standard tasks as organizing and maintaining data files, translating programs written in programming languages to a language acceptable to the computer, scheduling jobs through the computer, as well as aiding in other areas of general operations. Of all, the systems software the most important one is known as the operating system.

Application software includes programs to perform user applications. A computer program for preparing results of a public examination is an example of applications software. Application programs are either purchased or written by the computer users for specific applications.

Humanware

Humanware refers to the persons who design, program, and operate a computer. There are numerous categories of jobs, but the three principal positions for a large computer installation are: systems analyst, programmer, and computer operator. People in each of these areas generally perform special-purpose tasks under the supervision of a director or manager.

A systems analyst requires a broad background and extensive understanding of the above three job categories. The main task of a systems analyst is to study information systems and their processing requirements. A systems analyst defines the applications problem, determines systems specifications, recommends hardware and software and designs information-processing procedures.

A programmer requires a comprehensive knowledge of programming languages and standard coding procedures. This position does not require the broader understanding of the structure and inner workings of an application. A programmer's principal job is to code or prepare programs based on the specifications made by the systems analyst.

A computer operator requires the least extensive background of the three categories. A computer operator generally performs a series of well-defined tasks to keep the computer operating at maximum efficiency. The efficiency of a computer installation depends on the quality and ability of the operational staff.

Operational Procedures

Operations of a data processing center require an extensive and clearly defined set of procedures for performing the essential functions. These functions generally include obtaining, preparing, and entering data into the computer, processing jobs, initiating new programs and changing or deleting old ones etc. Such procedures require actions to be taken in the event of hardware or software malfunctions.

1.4 Importance and limitations of Computers

Storage, processing and dissemination of information play a very important role in modern information society. Computers, telecommunications, data and computer communications contribute tremendously to these activities. Computers and electronic communications are getting closer and closer every day. Computers, electronic communications and other technologies associated with automation come under the general term of information technology. Information technology influences our lives and its impact on society is increasing day by day.

A computer can perform calculations at a speed beyond human ability and with accuracy and consistency which people can not match. Some of the knowledge we process today would be unknown without the fast computational power of computers. The rapid social, economic cultural and technological changes cannot be imagined without the fast computational speed of present day computers.

Present world is so much dependent on computers that if they could be taken away overnight, the airlines, electronic communications, financial operations and official activities would collapse. There are many more examples like these. It is not possible for an average person to appreciate the direct and indirect impact of computers in society. Computers have penetrated human society far more deeply than average persons can really appreciate.

The following are some important criteria, which may be used to justify the use of computers:

Volume of data: The computer is suited to handling large volume of data.

Accuracy: Computers ensure high degree of accuracy and their consistency is reliable.

Repetitiveness: Processing cycles that repeat time and time again are ideally suited to computers. Once a computer is programmed, it goes on automatically performing the task as many times as required.

Complexity: Computers perform complex calculations. A computer provides the required answers after running the program for complex calculations.

Speed: Computers work at very high speeds and this enables them to respond quickly to a given situation.

Common data: One item of data can be used for different computer procedures.

A combination of two or more of the above criteria usually justifies the use of computers. These criteria are used to carry out the preliminary survey to judge the suitability of applications for computerization. Technological innovations in the recent past have increased the range of technically feasible applications and they also provide cost-effective solutions to a range of problems.

If the general criteria justify that a particular application may be suitable for computerization, then the following questions should be satisfactorily answered before taking any decision for computerization:

- Is the use of a computer technically feasible?
- Would it be cost effective?
- Would it be socially acceptable?

Besides the advantages of using a computer, there are a few limitations of it. These are:

- Computers can not think
- Computers can not do anything without human instructions.
- Computers can not make any adjustment as human being can do.

1.4 Exercise

Multiple choice questions

1. The three basic functions of a computer system are:

a. Input, addition and output
b. Input, multiplication and output.
c. Input, processing and output.
d. None of the above.

2. In general the categories of software are:

   a. Package programs and application software.
   b. System software and application software.
   c. Programming language and operating system.
   d. Application programs and package programs.

Questions for short answers

3. What is a computer?
4. Name the elements of a computer system.
5. What is the difference between system software and application software?
6. What is meant by operational procedures?
7. What are the limitations of a computer?

Analytical questions

8. What is data processing cycle? Discuss the elements of the cycle.
9. Describe the components of a computer system.
10. Write the importance and limitations of computer.
11. Name and briefly explain important criteria, which may be used to justify the use of computers.

Lesson 2: Types of Computers

2.1 Learning Objectives

On completion of this lesson you will know classification of computers:

- On the basis of purposes
- On the basis of signals
- On the basis of Capacity.

2.2 Classification based on purposes

There are either special-purpose or general-purpose computers. A special-purpose computer is designed for a specific application. It is also known as dedicated computer. Many such computers have instructions permanently programmed into them that are designed to perform only one major function. Special-purpose computers are used to control traffic lights, to control the collection of tolls on highways, and in automobiles, weapons, appliances and games etc.

General-purpose computers are used to handle a variety of tasks. This is possible by the stored-program concept. By this concept, a program containing a series of instructions is prepared for an application and temporarily stored in memory. Once stored in the computer's memory, the program can be executed to perform the specific function. After the completion of the execution of one program, another program can be used for some other task. That is, the same hardware can be used to execute many different programs.

General-purpose computers are more versatile than special-purpose computers. But typically general-purpose computers are less efficient and slower than special-purpose computers when applied to the same task.

2.3 Classification based on signals

Two types of computers based on internal signaling are analog computers and digital computers. An analog computer represents quantities by physical analogies. It represents physical quantities, such as distance, velocity, acceleration, temperature, pressure, or angular position, force etc. by mechanical or electrical parameters.

An automobile speedometer is an example of an analog-computing device. It converts the rotational rate of the drive shaft of an automobile into the numerical value of the speed of the vehicle. Similarly, a thermometer functions as an analog device by converting the movement of a column of mercury into a temperature reading.

Analog computers are ideal in situations where data can be accepted directly from measuring instruments The ability to collect data at high speeds and to process data at equally high speeds, makes analog computers uniquely suited to controlling processes of oil refineries, steel mills, weapon systems and similar other operations. An analog computer does not require any storage capability. The out from an analog computer is generally in the form of readings on a dial (as in the speedometer of a car) or a graph plotted on paper.

Analog computers were in use before the invention of the digital computers. There are far more digital computers in use today than analog computers. This book concentrates on digital computers.

2.4 Classification based on capacity

Capacity of a computer refers to the volume of data that a computer system can process. Previously a computer's size was an indication of its capacity. With the current state of miniaturization, measurement of capacity is based on throughput of the computer. Throughput is the amount of processing that can be performed in a given amount of time. Based on throughput computer systems can be divided into four major categories:

- Microcomputers
- Minicomputers
- Mainframe computers and
- Supercomputers.

Microcomputers

Microcomputers (also known as personal computers - PCs) are microprocessor based small laptop or desktop or notebook systems with varying capability. Personal digital assistants (PDAs) are very small portable computers. PDAs are also known as palmtop computers. The brain of a microcomputer is the microprocessor; it is a silicon chip containing necessary circuits to perform arithmetic/logic operations and to control input/output operations. A microprocessor is an integrated circuit (IC) which generally contain millions of transistors squeezed onto a tiny chip of silicon. A microcomputer system is formed by adding input/output capability and memory to the microprocessor.

Early microcomputers had limited processing power and limited choice of input/output devices. Present day microcomputers have wider processing capabilities and they support a wide range of input/output devices. Today microcomputers are available with a selection of input/output devices varying from a cassette recorder to a voice synthesizer. In addition to general-purpose computations, microcomputers are also used for special purpose applications in automobiles, airplanes, toys, clocks, appliances etc.

Workstations: High-end microcomputers are known as workstations. The workstation represents the bridge between the microcomputers and minicomputers. It is a microcomputer with many of the capabilities of larger minicomputers but costing much less. Initially it was designed for use by engineers and designers who need exceptionally powerful processing and output capabilities.

Servers: Servers are not designed to be used directly. They make programs and data available for users having access to a computer network. A computer network is a collection of computers connected together.

Clients: To use servers, users run desktop programs called clients, which know how to contact the server and obtain information from the server. Use of desktop clients and centralized servers is called client/server computing.

Terminals: Although terminals look like the personal computers, they have some limitations when compared with personal computers. Terminals have only a screen and a keyboard and the electronics that allow them to communicate with the computer to which they are connected. Because they lack the ability to process data on their own, they are called dumb terminals. There is a variety of dumb terminals that can perform limited processing. These are called smart (or intelligent) terminals. A personal computer is an example of a smart terminal.

Minicomputers

A minicomputer system performs the basic arithmetic and logic functions and supports some of the programming languages used with large computer systems. They are physically smaller, less expensive, and have small storage capacity compared to mainframes. Minicomputers are ideally suited for processing tasks that do not require access to huge volumes of stored data. As a result of low cost, ease of operation, and versatility, minicomputers have gained rapid acceptance since their introduction in the mid-sixties. Some of the larger and expensive minicomputers are capable of supporting a number of terminals in a time-shared mode. Use of minicomputers are gradually being diminished with the rapid development of microcomputers.

Mainframe Computers

A larger computer generally consists of modules mounted on a chassis (or mainframe) and is known as a mainframe computer. They vary in size, from those slightly larger than a minicomputer to supercomputers. Mainframe computer systems offer substantial advantages over minicomputers or microcomputers. Some of these are: great processing speed, greater storage capacity, a larger variety of input/output d support for a number of high-speed storage devices, multiprogramming, sharing.

Owing to tremendous expense, a mainframe computer system must be operated efficiently. Operating a mainframe at the required level of efficiency requires a very large and highly trained staff. Mainframe Computers are generally used by large businesses, universities, governmental agencies, and the military. These systems are often coupled with other computer systems in a large network to provide enormous computing power. This is referred to as a distributed data processing system.

Supercomputers

A very large and very powerful mainframe computer is called a supercomputer. The astronomical cost of super-computers has limited their development to only a few hundred worldwide. The Cray X-MP is an example of a supercomputer. Such supercomputers are applied to the solution of very complex and sophisticated scientific problems and for national security purposes of some advanced nations.

2.5 Exercise

Multiple choice questions

1. A PC is a:

   a. General purpose computer b. Microprocessor
   c. Special purpose computer c. None of the above.

2. Based on capacity, types of computer systems are:

   a. Microcomputers, minicomputers, mainframe computers and supercomputers.
   b. Microcomputers, personal computers and IBM computers.
   c. Minicomputers, supercomputers and digital computers.
   d. Analog computers and digital computers.

Questions for short answers

3. Distinguish between an analog computer and a digital computer.
4. Classify computer systems on the basis of capacity.
5. What is a PC? Differentiate between a client and a server?

Analytical questions

Lesson 3: History and Generations

3.1 Learning Objectives

On completion of this lesson you will know:

- History of computers
- Generations of computers.

3.2 Beginning

Different devices and tools have been used in calculation and processing of data. An ancient calculating device is the abacus, a mechanical calculating device first used around 2500 BC to add and subtract. Scientists and mathematicians later sought other means for calculations. John Napier, a Scottish mathematician, developed (about 1610) a series of rods made of bones (commonly called Napier's bones) that could be arranged to generate the products of selected numbers. He used these rods to produce the first table of logarithms. In 1865 French mathematician Blasé Pascal improved on this concept and produced a mechanical calculator called Pascaline. It was more compact and easier to use than Napier's bones. The Pascaline was capable of performing addition and subtraction. All attempts to produce a calculator capable of performing all the four arithmetic operations and producing mathematical tables quickly and accurately were not successful until 1820. Thomas de Colmar of France produced the arithmometer, the first four-function practical mechanical calculator in 1920.

A young English mathematician named Charles Babbage of Cambridge University contributed substantially towards the development of computers. Babbage gave much thought to the design of a device to produce mathematical and navigational tables and came upon a principle that used the "differences" between previous values in a table to produce new values. Babbage was able to construct a working model to illustrate the principle of the difference engine. Babbage thought of a steam-driven version of the difference engine capable of calculating and printing results at a rate of two twenty-digit numbers per minute.

Babbage built part of the difference engine but abandoned it in favor of a more powerful and versatile machine, the analytical engine. The analytical engine was designed to use punched cards to provide a constant flow of information through the machine's elaborate series of columns, gears, wheels, and levers. The analytical engine included all the functional units of a modern computer: input of data, arithmetic unit for computation,

memory for data and instructions, and display for output. This was an ambitious project during a time when electronics, transistors, and chips did not exist. The engine was a puzzle to all but a few mathematicians. This machine, however, was never built. Nearly a century later a new generation of scientists and engineers equipped with new developments brought Babbage's vision back into focus.

Insignificant progress took place over the next decades. In the United States the 1890 census was approaching, and there was no way that it could be completed by 1900, as required by the constitution. Herman Hollerith, an employee of the Census Office in Washington, started to develop an automated device to complete this task in ten years. The result was Hollerith's tabulator. A manual card puncher, a card reader, and an electromechanical card sorter were the main components of this machine.

With this tabulator, Hollerith was able to complete the census calculations in only two years. Hollerith's success paved the way for further research and development. Analog computers, a new class of computing devices, emerged. These devices used electrical voltages to represent physical quantities. They functioned by establishing an analogy between a physical quantity and a voltage level. They were very fast but not sufficiently accurate.

The first electronic computers, the ABC (Atanasoff Berry Computer) and the ENIAC (Electronic Numerical Integrator and Calculator) were built in USA in the early 1940s. The ABC, built by Atanasoff and Berry, was the first electronic computer using vacuum tubes. The ENIAC, built by Eckert and Mauchly, was an special-purpose computer. In 1949, at Cambridge, the EDSAC (Electronic Delay Storage Automatic Computer), the first general-purpose electronic computer operating under the control of a stored program, was completed. A stored program is a set of instructions stored in memory that guides the computer, step by step, through a process.

John Von Neumann, originator of the stored-program concept, developed the IAS (Institute for Advanced Study) computer at Princeton University. This machine was the realization of John Von Neumann ideas on computer design. Most computers built after the IAS computer have "Von Neumann" characteristics.

A group of scientists at MIT in USA headed by Ken Olsen developed the Whirlwind computer, which was more than twenty times faster than the ENIAC. Both the IAS and Whirlwind computers introduced computational innovations of astronomical proportions. Computers were not available commercially until early 1951. In 1951, the Sperry Rand Corporation of USA built the UNIVAC I (Universal Automatic Computer). The

UNIVAC I built for the Bureau of Census of the United States was the first commercially available computer. The first computer installation to handle business applications was set up in 1954 at Louisville, Kentucky, USA. The UNIVAC I is now on display in the Smithsonian Institution in Washington D.C. in USA.

Day by day computers were becoming smaller, faster, and more powerful. In 1956, MIT in the USA introduced its compact TX-0 transistorized computer system. A few years later, in 1960, the first integrated circuit was produced by Jack Kilby of Texas Instruments in USA and the DEC PDP-1 minicomputer was developed. In the Mid-1960s, minicomputers began appearing as a forerunner of mainframes and supercomputers. The invention of microprocessor in 1971 began a new ara and microcomputers based on microprocessors began to appear from the Mid-1970's. Microcomputers provide the driving force of the present information age.

3.3 Computer Generations

Developments over the years have resulted in machines with greatly increased speeds, storage, memory, and computing power. The developments were so far-reaching and numerous that they are generally categorized by generations. Each generation is initiated by significant advances in computer hardware or computer software.

First Generation (1942-1959)

First-generation computers utilized vacuum tubes in their circuitry and for storage of data and instructions. The vacuum tube was bulky, caused tremendous heat problems, and was never a reliable device, it caused a great number of breakdowns and inefficient operations. Magnetic cores began to replace vacuum tubes as the principal memory device in the early machines. Small doughnut-shaped cores were strung on wires within the computer. Programs were written in machine language employing combinations of 0 and 1. Examples of first generation computers are IBM 650, IBM 704, IBM 705, IBM 709, Mark II, Mark III etc.

Second Generation (1960-1965)

The second generation computers saw the replacement of the vacuum tubes by transistors. A transistor can be thought of as a switch, but with no moving parts. Because of high speed operation and small size, computers of this generation could perform a single operation in microseconds and were capable of storing tens of thousands of characters. Manufacturers began producing business-oriented computers with more efficient storage and faster input and output capabilities. Second generation computers

were reliable, compact in size, and virtually free of heat problems. Programming was done in both machine and symbolic languages. Symbolic languages utilize symbolic names for computer commands and allow the use of symbolic names for items of data. This language is also known as assembly language. Examples of second generation computers are: IBM 1400, CDC 1604, RCA 501, NCR 300, GE 200, IBM 1600 etc.

Third Generation (1965-1971)

These computers were characterized by integrated circuits with components so small that in many cases they were hardly visible to the naked eye. Third generation computers were characterized by increased input/output, storage, and processing capabilities. Input/output devices could communicate with computers over distances via ordinary telephone lines or could scan a page and input the information directly into the computer, could display pictures on a television-like screen, could make musical sounds, and could even accept voice input.

Storage capabilities were increased and millions of characters could be stored and randomly accessed in fractions of a second. Third-generation computers could process instructions in nanoseconds. In addition, computers were able to process several programs or sets of instructions simultaneously. Programmers were able to make use of high-level problem oriented and procedure oriented languages that closely resembles the commonly used form of expressions. Examples of third generation computers are IBM 360, IBM 370, PDP-8, PDP-11, GE 600 etc.

Fourth Generation (1972 - Present)

The fourth generation computers possess still greater input/output, storage, and processing capabilities. In the fourth generation computers, semiconductor storage devices were introduced. In the early 1970s IBM introduced the concept of virtual storage into their 5000 and 370 series of computers. Machines previously limited to a maximum internal storage capability of approximately one million characters now possessed a virtual storage capability in billions and trillions of characters. With this capability a machine could execute a program many times the size of the machine's actual memory. Microcomputers using microprocessors as the CPU proliferated in the fourth generation.

Now a days, the compact disk (CD) promises to become the data storage medium of choice. A compact disk read-only memory (CD-ROM) is encoded with on and off bits. Bits are stored on the disk's (3.5-inch diameter) aluminum surface as tiny pits at varying depths. The average CD can store about 4,800 million bits or 600 million characters of

data. This is approximately a quarter of a million pages of text.

The most impressive advancement has occurred in software. As a result of these changes, access to substantial computer power, previously only affordable by very large business concerns, is now economically feasible for small business and personal applications. Personal computers are examples of fourth generation microcomputers.

Fifth Generation

Fifth generation of computers is yet to come. They will be capable of reasoning, learning, making inferences and behaving in ways usually considered exclusive of humans. These computers will be equipped with massive primary-storage capabilities and extremely fast processing speeds. Software will proliferate and get much bigger. Hardware will continue to shrink in size but internal memory will increase dramatically. "Talking machines" will be common place. Voice-recognition, the ability for a machine to understand and obey spoken words will be developed. Industrial and personal robots will roll and walk. Expert systems software will place the knowledge of experts and consultants (such as doctors, lawyers and teachers) at the disposal of general users. Huge number of computers will be linked in parallel offering computing power of an inconceivable magnitude. Artificial intelligence will be used extensively to enhance the system behavior in the future.

3.4 Exercise

Multiple choice questions

1. Who developed the stored program concept?

   a. Blaise Pascal. b. John Von Neumann.
   c. Herman Hollerith. d. Charles Babbage.

2. Influence of artificial intelligence is considered in:

   a. First generation b. Third generation
   c. Fourth generation d. Fifth generation

3. Who developed the concept of the difference engine?

   a. Blaise Pascal. b. John Von Neumann.
   c. Herman Hollerith. d. Charles Babbage.

Questions for short answers

4. Who was Charles Babbage? Mention his contributions in the history of computers?
5. What is meant by computer generations?
6. What is the name of first commercial computer?
7. Who was John Von Neuman? What is his contribution in the history of computer?

Analytical questions

8. Write an essay on the history of computers.
9. Briefly describe the computer generations.
10. Describe the trend and features of fifth generation computers.

Chapter 2

Number Systems and Codes

Lesson 1: Number Systems
Lesson 2: Conversion of Numbers
Lesson 3: Binary Arithmetic
Lesson 4: Data Representation and Codes

Lesson 1: Number Systems

1.1 Objective

On completion of this lesson you will know:

- Different number systems
- Decimal, binary and hexadecimal numbers
- Explanation of numbers with examples.

1.2 Number Systems

Popular number systems in connection with digital computers are: decimal, binary, octal and hexadecimal. Basic characteristics of these number systems are given below.

Decimal System

It is the most commonly used number system in real life. It has the following features:

Digits : 1, 2, 3, 4, 5, 6, 7, 8, 9 and 0. (total 10 digits)
Base : 10 (since total number of digits is 10)

Binary System

Digital computers use binary numbers for internal operations. Features of the binary system are:

Digits : 0 and 1 (total 2 bits, a bit is a short of binary digit)
Base : 2 (since total number of digits is 2)

Octal System

This is not a commonly used number system. Basic features of this system are:

Digits : 0, 1, 2, 3, 4, 5, 6 and 7 (total 8 digits)
Base : 8 (since the total number of digits is 8).

Hexadecimal System

It is a popular system for computer scientists, engineers and information technologists. Basic features of this system are:

Digits : 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F (total 16 digits)
Base : 16 (since the total number of digits is 16).

1.3 Decimal System

A positive integer, represented as a string of decimal digits, may be expressed as a sum of powers of 10.

For example, 8253 can be expressed as:

$$\begin{aligned} 8253 &= 8\times10^{3}+2\times10^{2}+5\times10^{1}+3\times10^{0} \\ &= 8\times1000+2\times100+5\times10+3\times1 \quad (\text{as } 10^{\circ}=1) \\ &= 8000+200+50+3 \end{aligned}$$

Here the powers of ten, are called the place values of the digits.

Any fractional value, represented by a string of decimal digits together with an embedded decimal point, may also be expressed by negative powers of 10.

For example, 837.526 is expressed in expanded notation as:

$$\begin{aligned} 837.526 &= 8\times10^{2}+3\times10+7\times10^{0}+5\times10^{-1}+2\times10^{-2}+6\times10^{-3} \\ &= 8\times100+3\times10+7\times1+5\times0.1+2\times0.01+6\times0.001 \\ &= 800+30+7+0.5+0.02+0.006 \end{aligned}$$

Here, $10^{-1}=0.1$ $10^{-2}=0.01$ $10^{-3}=0.001$

This decimal number has three decimal places, that is the number of digits to the right of the decimal point.

1.4 Binary System

A binary number is a sequence of bits. The place values in the binary system are the powers of the base 2, just as the place values in the decimal system are the powers of ten. The place values of the integral part of a binary number are the positive powers of two.

Table 1.1 shows counting in decimal and binary numbers.

Table 1.1: Decimal and binary counting.

Decimal counting	Binary counting
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010
11	1011
12	1100
•	•
•	•
•	•

The place values of the fractional part of a binary number are the negative powers of two (Table 1.2).

Table 1.2: Binary place values and corresponding decimal values

Binary place values	Decimal values
2^{-4}	0.0625
2^{-3}	0.125
2^{-2}	0.25
2^{-1}	0.5
2^{0}	1
2^{1}	2
2^{2}	4
2^{3}	8

Examples of binary number are: 1011, 101010, 1011.0010, 1111.1101 etc.

1.5 Octal System

Each octal digit has a unique 3-bit binary representation. Table 1.3 shows the octal digits and their decimal and binary equivalents.

Table 1.3: Octal digits and their decimal and binary equivents.

Octal digit	Decimal values	Binary equivalent
0	0	000
1	1	001
2	2	010
3	3	011
4	4	100
5	5	101
6	6	110
7	7	111

The place values in the octal system are powers of 8; some of these powers are shown in Table 1.4.

Table 1.4: Octal place values in decimal.

Octal place values	Decimal values
8^{-3}	1/512 = 0.001953125
8^{-2}	1/64 = 0.015625
8^{-1}	1/8 = 0.125
8^{0}	1
8^{1}	8
8^{2}	64
8^{3}	512
8^{4}	4096
8^{5}	32768

Examples of octal numbers: 147, 56, 747, 71, 15.632 etc.

1.6 Hexadecimal System

Each hexadecimal digit has a unique 4-bit representation which is shown in Table 1.5.

Table 1.5: Hexadecimal digits and their decimal and binary equivalents.

Hexadecimal Digits	Decimal Values	Binary Equivalents
0	0	0 0 0 0
1	1	0 0 0 1
2	2	0 0 1 0
3	3	0 0 1 1
4	4	0 1 0 0
5	5	0 1 0 1
6	6	0 1 1 0
7	7	0 1 1 1
8	8	1 0 0 0
9	9	1 0 0 1
A	10	1 0 1 0
B	11	1 0 1 1
C	12	1 1 0 0
D	13	1 1 0 1
E	14	1 1 1 0
F	15	1 1 1 1

Examples of hexadecimal numbers: 16, 1A2, FFFF, 25C.A2 etc.

The place values in the hexadecimal system are the powers of 16, some of which are listed, along with their decimal values, in Table 1.6.

Table 1.6: Hexadecimal place values in decimal.

Octal place values	Decimal values
16^{-3}	1/4096 = 0.000244140625
16^{-2}	1/256 = 0.00390625
16^{-1}	1/16 = 0.0625
16^{0}	1
16^{1}	16
16^{2}	256
16^{3}	4096
16^{4}	65536
16^{5}	1048576

1.7 Exercise

Multiple choice questions

1. The base of octal number system is

   a. 2 c. 10

   b. 8 d. 16

2. The place values in the decimal systems are the powers of

   a. 2 c. 10

   b. 8 d. 16

3. The place values of the fractional part of a binary number are the

   a. nonnegative powers of 2 c. negative powers 10

   b. negative powers of 2 d. negative powers of 8

4. Each hexadecimal digit has a unique

   a. 2-bit binary representation c. 4-bit binary representation

   b. 3-bit binary representation d. 5-bit binary representation

Questions for short answers

5. What is a bit?
6. What are the bases of binary, decimal, octal and hexadecimal numbers?
7. Give binary values for hexadecimal digits: A, F and 3.
8. Give two examples each for binary, octal and hexadecimal positive integer numbers.

Analytical questions

9. How can decimal numbers 625.536, 0.326 and 1278 be expressed in expanded notation?
10. Write down the digits for decimal, binary, octal and hexadecimal numbers.
11. Briefly describe four popular number systems.

Lesson 2: Conversion of Numbers

2.1 Objective

On completion of this lesson you will know:

- Conversion between binary and decimal numbers
- Conversion between hexadecimal and decimal numbers
- Conversion between hexadecimal and binary numbers.

2.2 Binary-to-Decimal Conversion

Binary number can be expressed as the sum of the product of each of its digit and the digit's place value.

For example,

$$110101 = 1\times 2^5 + 1\times 2^4 + 0\times 2^3 + 1\times 2^2 + 0\times 2^1 + 1\times 2^0$$
$$110.101 = 1\times 2^2 + 0\times 2^1 + 1\times 2^0 + 1\times 2^{-1} + 1\times 2^{-2} + 0\times 2^{-3}$$

Since each power of two is weighted by either 0 or 1, and the binary number is the sum of the place values in which the bit 1 appears. This sum gives the decimal equivalent of the binary number.

Table 1.7 lists the binary representations of the integers from 0 to 20, with the place value of the bits shown at the top.

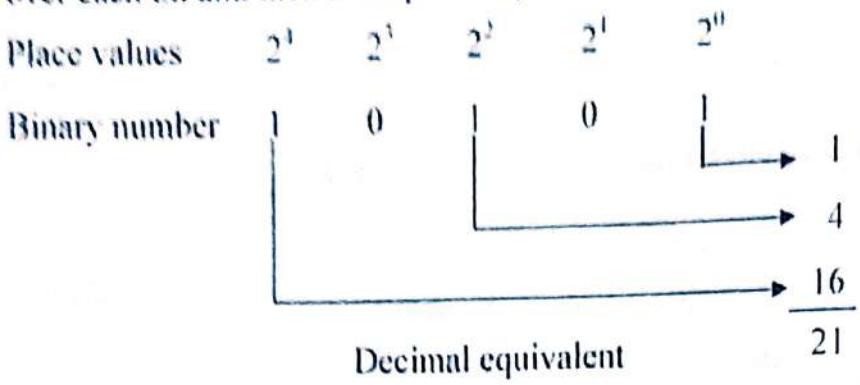
Table 1.7: Binary representation of decimal number

Decimal	Binary number					Decimal	Binary number				
Number	16	8	4	2	1	number	16	8	4	2	1
0					0						
1					1	11		1	0	1	1
2				1	0	12		1	1	0	0
3				1	1	13		1	1	0	1
4			1	0	0	14		1	1	1	0
5			1	0	1	15		1	1	1	1
6			1	1	0	16	1	0	0	0	0
7			0	1	1	17	1	0	0	0	1
8		1	0	0	0	18	1	0	0	1	0
9		1	0	0	1	19	1	0	0	1	1
10		1	0	1	0	20	1	0	1	0	0

Example 2.1: Convert a) 10101 and b) 101.110 to decimal.

Ans : a) To convert 10101_2 to its decimal equivalent, write the appropriate place value

over each bit and then add up those powers of two which are weighted by 1 :

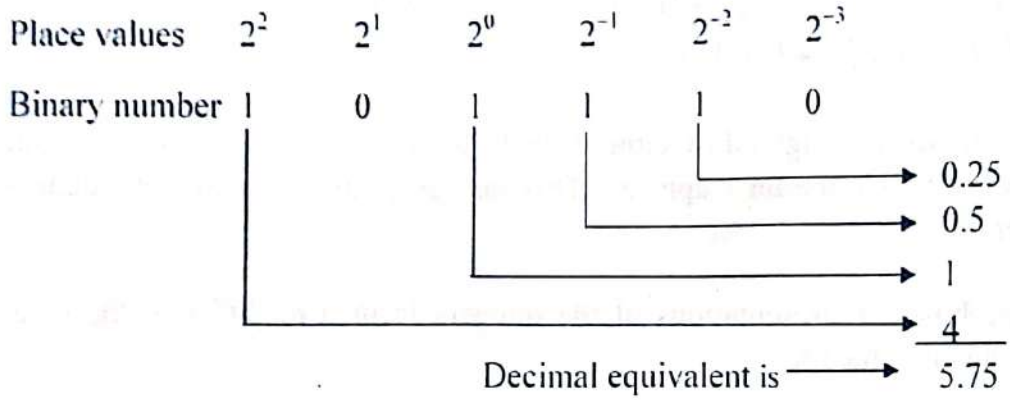


Alternatively, $1 \times 2^4 + 0 \times 2^3 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$= 16 + 0 + 4 + 0 + 1$

$= 21$

b) To convert 101.110_2 to its decimal equivalent, use the decimal values of the powers of two



Alternatively, $1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3}$

$= 4 + 0 + 1 + 0.5 + 0.25 + 0$

$= 5.75$

2.3 Decimal-to-Binary Conversion

It is possible to find binary representation of a decimal number N by converting its integral part (N_I), and its fractional part (N_F) separately.

Example 2.2: Convert 109.78125 to binary number

Ans : a) To convert the integral part N_I, that is 109, to binary, divide N_I and each successive quotient by 2, noting the remainders, as follows :

Divisions	Quotients	Remainders
109 ÷ 2	54	1 ↑
54 ÷ 2	27	0 |
27 ÷ 2	13	1 |
13 ÷ 2	6	1 |
6 ÷ 2	3	0 |
3 ÷ 2	1	1 |
1 ÷ 2	0	1 |

The zero quotient indicates the end of the divisions. The sequence of remainders from the bottom to up, as indicated by the arrow, yields the required binary equivalent. That is, $N_I = 109 = 1101101_2$.

In practice, the above divisions may be shown as :

	Remainders
2)109	
2)54	1
2)27	0
2)13	1
2)6	1
2)3	0
1	1

Here the division processes is stopped when the quotient, 1, is less than the divisor 2. Again the arrow indicates the sequence of bits that give the binary equivalent of the number.

b) To convert, the fractional part N_F, that is 0.78125, to its binary equivalent, multiply N_f and each successive fractional part by 2, noting the integral part of the product, as follows:

Multiplications			Integral parts
0.78125 × 2	=	1.56250	1
0.5625 × 2	=	1.1250	1
0.125 × 2	=	0.250	0
0.25 × 2	=	0.50	0
0. 50 × 2	=	1.00	1 ↓

The zero fractional part indicates the end of the multiplications. It is observed that the integral part of any product can only be 0 or 1. The sequence of integral parts as

indicated by the arrow, yields the required binary number.

That is, $N_F = 0.78125 = 0.11001_2$.

In practice, the above multiplications may be shown as:

$$
\begin{array}{r}
0.781\,25 \\
\times 2 \\
\hline
\underline{1}.562\,50 \\
\times 2 \\
\hline
\underline{1}.125\,00 \\
\times 2 \\
\hline
\underline{0}.250\,00 \\
\times 2 \\
\hline
\underline{0}.500\,00 \\
\times 2 \\
\hline
\underline{1}.000\,00
\end{array}
$$

It is observed that the integral part of each product is underlined and is not used in the next multiplication. Again the arrow indicates the sequence of digits that give the required binary representation.

The binary equivalent of N is simply the sum of the binary equivalents of integral and fractional parts,

$$N = N_I + N_F = 1101101.11001$$

Example 2.3: Convert 13.6875 to decimal

	Remainders
2)13	
2)6	1
2)3	0
1	1

Integral parts

$$
\begin{array}{r}
0.6875 \\
\times 2 \\
\hline
\underline{1}.3750 \\
\times 2 \\
\hline
\underline{0}.7500 \\
\times 2 \\
\hline
\underline{1}.5000 \\
\times 2 \\
\hline
\underline{1}.0000
\end{array}
$$

Thus, $13.6875 = 1101.1011_2$.

Remark: The binary equivalent of a decimal fraction does not always terminate. For example, for the decimal number 0.6.

Multiplications	Integral parts
$0.6 \times 2 = \underline{1}.2$	1
$0.2 \times 2 = \underline{0}.4$	0
$0.4 \times 2 = \underline{0}.8$	0
$0.8 \times 2 = \underline{1}.6$	1

This shows that the above four steps will be repeated again and again. That is,

Decimal 0.6 = 0.1001 1001 1001 ... ...

2.4 Hexadecimal-Decimal Conversion

Conversion between the hexadecimal and decimal systems is accomplished using the algorithms of Section 2.3 with b = 16. Additionally one has to know how to handle the hexadecimal digits A, B, C, D, E and F. One can also convert from hexadecimal to decimal by decimal evaluation of the expanded hexadecimal form.

Example 2.4: Convert $73D5_{16}$ to its decimal equivalent.

$$73D5_{16} = 7 \times 16^3 + 3 \times 16^2 + 13 \times 16^1 + 5 \times 16^0, \text{ (as D = 13)}$$

$$= 7 \times 4096 + 3 \times 256 + 13 \times 16 + 5 \times 1$$

$$= 28672 + 768 + 208$$

$$= 29653$$

Example 2.5: Convert $39.B8_{16}$ to its decimal equivalent.

$$39.B8_{16} = 3 \times 16^1 + 9 \times 16^0 + 11 \times 16^{-1} + 8 \times 16^{-2}, \text{ (as B = 11)}$$

$$= 3 \times 16 + 9 \times 1 + 11 \times 0.0625 + 8 \times 0.00390625$$

$$= 48 + 9 + 0.6875 + 0.03125$$

$$= 57.71875$$

Example 2.6: Convert the decimal number 9719 to its hexadecimal equivalent.

The base of hexadecimal number is 16. Divide the number and each successive quotient by 16, noting the remainders, as follows:

Divisions	Quotients	Remainders
9719 ÷ 16	607	7
607 ÷ 16	37	15 (= F)
37 ÷ 16	2	5
2 ÷ 16	0	2

Thus $9719_{10} = 25F7_{16}$

Here the sequence of remainders, with F for remainder 15, in reverse order gives the hexadecimal form the decimal number

Example 2.7: Convert the decimal fraction 0.78125 to its hexadecimal equivalent.

For this conversion apply the integral-part algorithm, with base 16, as follows:

Multiplications	Integral parts
$0.78125 \times 16 = \underline{12}.50000$	12(=c)
$0.50000 \times 16 = \underline{8}.00000$	8

Thus $0.78125_{10} = 0.C8_{16}$

In this case a zero fractional part is reached. The sequence of integral parts, with C for 12 gives the required hexadecimal form.

Example 2.8: Convert the decimal number 9719.78125 to its hexadecimal form.

From the prevision example we know, $9719_{10} = 2517_{10}$ and. $78125_{10} = C8_{16}$
Combining the two parts, $9719.78125_{10} = 25F7.C8_{16}$

2.5 Hexadecimal-Binary Conversion

This is accomplished exactly as octal-binary conversion, except that 4-bit equivalents are now involved.

Example 2.9: Convert to binary form a) $3D59_{16}$. b) $27.A3C_{16}$.

Replacing each hexadecimal digit by its 4-bit representation

a) $3D59_{16} = 0011\ \ 1101\ \ 0101\ \ 1001$
That is, $3D59_{16} = 11110101011001_2$.

b) Similarly, $27.A3C_{16} = 0010\ \ 0111 \cdot 1010\ \ 0011\ \ 1100$

That is, $27.A3C_{16} = 100111 \cdot 1010001111_2$.

Example 2.10: Convert the binary numbers a) 10110100101110_2, b) 11100.1011011011_2 to hexadecimal forms.

Partitioning each binary number into 4-bit blocks to the left and right of the fractional point and then replacing each 4-bit block by its equivalent hexadecimal digit,

a) Binary 0010 1101 0010 1110 Hexadecimal 2 D 2 E

That is, $2D2E_{16}$ is the required hexadecimal form.

b) Similarly, $0001\ \ 1100 \cdot 1011\ \ 0110\ \ 1100_2 = 1C.B6C_{16}$
Hence, $1C.B6C_{16}$ is the required hexadecimal form.

2.6 Exercise

Multiple choice questions

1. The decimal equivalent of 1110_2 is

   a. 8 c. 12
   b. 10 d. 14

2. The decimal equivalent of 10110_2 is

   a. 15 c. 22
   b. 18 d. 24

3. The binary equivalent of 109 is

   a. 100110_2 c. 1101101_2
   b. 1111001_2 d. 1110101_2

4. The decimal equivalent of 25F716 is

   a. 1719 c. 9719
   b. 9610 d. 09919

Analytical questions

5. Convert the following binary numbers to decimal.

   a. 10101_2 c. 1011.101_2
   b. 100101_2 d. 101.1101_2

6. Convert the following decimal numbers to binary

   a. 653.625 c. 367
   b. 13.6875 d. 235.07

7. Convert the following hexadecimal numbers to decimal

   a. $129A.B86_{16}$ c. 0.7825_{16}
   b. $73D5_{16}$ d. $39.C8_{16}$

8. Convert the following hexadecimal numbers to binary

   a. $129A.B86_{16}$ b. $3D59_{16}$

9. Convert the following binary numbers to hexadecimal

   a. $101101001011 10_2$
   b. $1011011011 10.1000110_2$

Lesson 3: Binary Arithmetic

3.1 Objective

On completion of this lesson you will know:

- Addition of two binary numbers
- Multiplication of two or more binary numbers
- Subtraction of one binary number from another
- Division of binary numbers.

3.2 Binary Addition

The execution of numerical calculations is essentially the same in all positional number systems. The addition of two binary numbers is accomplished according to the following three-step algorithm:

Step 1 : Add the first (rightmost) column.
Step 2 : Record the unit digit of the column sum. If the sum exceeds one, carry '1' goes to the next column.
Step 3 : If there are additional columns or if there is a carry from Step 2, add the next column and repeat Step 2. Otherwise stop.

The facts needed for binary addition are shown below.

0 + 0 = 1	
0 + 1 = 1	
1 + 0 = 1	
1 + 1 = 0.	with a carry of 1
1 + 1 + 1 = 11.	with a carry of 1

Example 3.1: Evaluate the binary sum

```
  111   Addend
  101   Augend
```

According to the three-step algorithm.

```
  1 1    Carry
  1 1 1  Addend
+ 1 0 1  Augend
-------
1 1 0 0
```

3.3 Binary Multiplication

Binary multiplication is similar to decimal multiplication but is simpler then decimal multiplication.

Example 3.4: Obtain the value of the binary product 101011 × 1011.

multiplying 1101011 by the digits, 1, 1, 0 and 1 and then adding the four bottom rows.

```
  1 0 1 0 1 1
  × 1 0 1 1
  1 0 1 0 1 1
 1 0 1 0 1 1
0 0 0 0 0 0
1 0 1 0 1 1
1 1 1 0 1 1 0 0 1
```

The final sum is the required product. Here it is extremely important to line up the numbers in the correct columns.

3.4 Binary Subtraction

Subtraction in the binary system can be performed using the following two-step algorithm:

Step 1 : If the lower (subtrahend) digit is greater than the upper (minuend) digit, borrow from the next column to the left.

Step 2 : Subtract the lower value from the upper value.

The facts needed for binary subtraction are listed below:

0 – 0 = 0
1 – 0 = 1
1 – 1 = 0
0 – 1 = 1, with a borrow of 1 from the next column

The last entry comes from:

$$10 - 1 = 1$$

That is, the difference 0 – 1 requires borrowing, which then yields 10 – 1 = 1.

Example 3.5: Calculate the value of 11101–1011.

Applying the subtraction facts stated above,

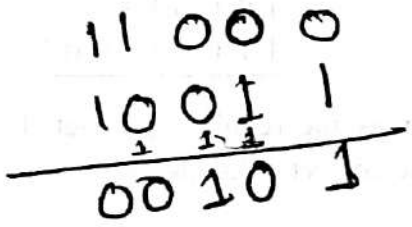
$$\begin{array}{r} 11101 \\ -1011 \\ \hline 10010 \end{array}$$

It is observed that 1 is borrowed from the third column because of the difference 0-1 in the second column.

As with decimal subtraction, binary subtraction becomes more complex when a borrow is needed from a digit which is 0. In such a situation, a borrow is taken from the first nonzero digit to the left, but each intervening 0 becomes 1 (as in 10-1=1).

Example 3.6: Consider the difference

$$\begin{array}{r} 11000 \\ -10011 \\ \hline \end{array}$$



Here,

$$\begin{array}{r} 011 \\ 11000 \\ -10011 \\ \hline 00101 \end{array}$$

In this case, a difference 0–1 occurs in the first column; hence it is required to borrow from the fourth column, where the first nonzero digit to the left appears, and the two intervening 0s become 1s.

Example 3.7: Calculate the difference 1100101001–110110110.

Here,

$$\begin{array}{r} 01111011 \\ 1100101001 \\ -110110110 \\ \hline 101110011 \end{array}$$

3.5 Binary Division

Division of decimal numbers can be reduced to multiplying the divisor by individual digits of the dividend and subtraction.

Example 3.8: Calculate 42558 ÷ 123.

Here 123 is the divisor. The algorithm for division yields:

```
        346
123)42558
    369
    565
    492
    738
    738
      0
```

That is, multiply 123 by 3 and subtract the product from 425; then multiply 123 by 4 and subtract the product 492, from 565; lastly multiply 123 by 6 and subtract the product 738 from 738 to obtain a 0 remainder.

The above algorithm also works for binary division. In fact, multiplying the divisor by the nonzero digit, 1, does not change the number; hence the algorithm for division reduces to repeated subtraction of the divisor.

Example 3.9: Evaluate 1010001 ÷ 11.

This gives

```
     11011
11)1010001
   11
   100
    11
    100
     11
     11
     11
      0
```

Thus the quotient is 11011.

As in decimal division of integers, a remainder is possible.

Example 3.10: Evaluate 1110111 ÷ 1001.

Applying the usual division algorithm,

```
      1101
1001)1110111
   1001
    1011
    1001
     1011
     1001
       10
```

The quotient is 1101, with a remainder 10.

Also, the division of binary fractions is handled in the same way as the division of decimal fractions.

3.6 Complements

Complements come up in storing numbers in the computer. While human beings use the signs, + and −, to denote positive and negative numbers, the computer can process data only in terms of bits. Although it is possible to reserve a bit to denote the sign of a number (say, 0 for + and 1 for −), most computers store negative numbers in the form of their arithmetic complements.

Complements also arise in the operation of subtraction. In fact, complements can be used to reduce subtraction to addition. This is especially useful as it avoids the possibility of repeated borrowing from one column to another.

There are two types of complements, the one's complement and two's complement.

For instance, if A is a binary number, then one's complement of A is obtained by inverting each digit of A, and the two's complement of A is its one's complement plus 1.

Example 3.11: Binary number and one's and two's complement

Binary number	: 111100001111
One's complement	: 000011110000
Two's complement	: 000011110001

Observe that the one's complement simply inverts each digit, i.e. 0 is replaced by 1 and 1 is replaced by 0.

Binary subtraction is performed by adding two's complement.

Example 3.12: Evaluate the difference Y = B − A.

Where,

A = 10001110 and B = 11110000.

a) First by ordinary binary subtraction:

```
   011
 11110000  B
-10001110  A
 01100010  Y
```

Observe that the borrowing was propagated to the third digit to the left.

b) The two's complement of A is 01110010. Add this to B:

11110000	B
+01110010	Two's complement of A
①01100010	

Deleting the 1 (which would be an overflow in a 8-bit register) gives the difference Y.

Example 3.13: Using 5 bits for storing numbers.

Calculate the values of a) 9–4, b) – 9 + 4 by 2's complement addition.

a) Positive and negative number

Binary number for + 9	:	01001	(1st bit is sign bit)
Two's complement of 4	:	11100	(1st bit is sign bit)
By addition		① 00101	(Difference)

Deleting the 1 (which would be an overflow in a 5-bit register) gives the difference. Since the sign bit 0, the result is a positive number.

b) Negative and positive number

Two's complement of 9	:	10111	(1st bit is sign bit)
Binary number for 4	:	00100	(1st bit is sign bit)
By addition		11011	(Difference)

Since the sign bit is 1, the result is a negative number. To find the binary form of the result, the result needs 2's complement form. Thus, the 2's complement of 11011 is 00101, that is in decimal the results is 5.

It is concluded that the subtraction operation in 2's complement system can be carried out by addition as shown above.

3.7 Exercise

Multiple choice questions

1. Addition in the binary system can be performed using

   a. 2 step algorithm
   b. 3 step algorithm
   c. 4 step algorithm
   d. 5 step algorithm.

2. Complements arise in the operation of

a. addition
b. subtraction
c. multiplication
d. conversion.

Analytical questions

5. Add the following binary numbers

a. 10110_2
b. 100101_2
c. 1100_2
d. 10100_2

4. Multiply the following numbers

a. 100011_2
c. 1101_2
c. 101_2
d. 1011_2

3. Evaluate the following

a. 10010 – 11011
b. 10101 – 00110
c. 11101 – 1011
d. 1010 – 1011

6. Evaluate the following

a. 11010011 ÷ 11
b. 11110111 ÷ 1001
c. 110100 ÷ 101
d. 111101 ÷ 1101

7. What do you understand by 1's complement and 2's complement?

8. Find the 1's complement and 2's complement of the following numbers

a. 101101_2
b. 1110101_2
c. 111100001111_2
d. 111100111_2

9. Using 5-bits for representing numbers perform following subtraction using 2's complement method

a. 5 – 4
b. 6 – 8
c. 12 – 14
d. 10 – 12

Lesson 4: Data Representation and Codes

4.1 Objectives

On completion of this lesson you will know:

- Different methods of coding
- Definitions of data, information and code
- Different methods of data representation in computers

4.2 Data, Information and Codes

Data are the names given to basic facts such as names and numbers. Examples of date are unit price, quantity sold, times, dates, product, names, addresses, tax codes etc.

Information is processed data that is, information is data which have been converted into a more useful form. For example: total price = unit price × quantity sold. Here total price is information and unit price, quantity sold are data.

Codes are used to reduce the volume of data. By coding, recording of data can be made less laborious, less prone to error and the data become more manageable and easier to manipulate.

4.3 Numeric Data Representation

Numeric data are represented in the computer using straight binary coding, which encodes an entire number as a whole. Straight binary coding requires that numbers be stored in computer memories as a fixed number of bits. A group of bits so treated as a unit is called a word, and the number of bits is called the length of the word.

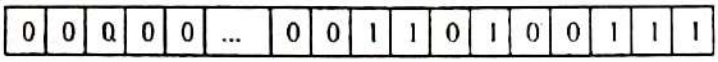
Integer Representation

Integers or fixed-point numbers have no decimal points. An integer I is represented in the memory of the computer by its binary form if I is positive, and by its 2's complement if I is negative.

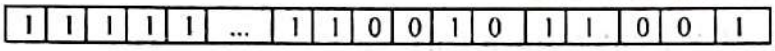
Example 4.1: Storing $1101001\overline{11}_2$ that is 423_{10}, and -423_{10} in 32-bit memory location

For simplicity we assume that the word length is 32. The computer stores 423 that is 110100111_2 in a 32-bit memory location by introducing 0s at the beginning of the binary

form. Thus:

423 

The computer stores – 423 in a memory location in 2's complement form for 423, that is

– 423 

In the first display the dots represent omitted 0s; in the second omitted 1's.

The computer can tell whether an integer I in memory is positive or negative by looking at the left most bit. If the left most bit is 0, then I is positive; if the left most bit is 1, then I is negative. Accordingly, the largest (positive) integer that can be stored in a 32-bit memory location is

$$\underbrace{0\,1\,1\,1\,1\,\ldots\,1\,1\,1\,1\,1}_{31\text{ ones}}$$

or $2^{31} - 1$, which is approximately 2 billion. Similarly, the smallest (that is negative) integer that can be stored in a 32-bit memory location is -2^{31}, or approximately –2 billion.

4.4 BCD Code

There are many ways of representing numerical data in binary form. One way is simply to write the numbers to the base 2. This is called straight binary coding. Another way is to encode decimal digits. These code, which require 4 bits for each decimal digit, is called BCD (binary-coded decimal) code. The BCD code words for decimal digits 0 to 9 are shown in Table 1.8.

Table 1.8: The BCD code words for decimal digits.

Decimal digits	BCD code (8-4-2-1)
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1

The BCD code is a weighted code, in which from left to right the bits are given the weights 8, 4, 2, and 1, respectively. Since these weights are just the place valves in the binary system, a decimal digit is encoded as its binary representation.

Example 4.3: The BCD representation of 469 is

4	6	9
0100	0110	1001

On the other hand, the straight binary representation of 469 is 111010101_2 which involves 3 fewer bits.

4.5 Alpha numeric codes

Various 8-bit codes have been developed for representing characters and digits. They are called alphanumeric codes. Each coded character, or byte, is normally divided into four zone bits and four 8-4-2-1 numeric bits, as shown

zone bits				Numeric bits			
Z	Z	Z	Z	8	4	2	1

More generally, the word 'byte' is used to denote any group of eight bits. It is seen that a byte may be represented by two hexadecimal digits, the first corresponding to the zone bits and the second to the numeric bits.

There are two 8-bit alphanumeric codes predominant in the computer industry today. They are ASCII and EBCDIC. EBCDIC is pronounced 'ebb-see-dick' and is short for Extended Binary-Coded Decimal Interchange Code. This code, developed by IBM, is used mainly by IBM-compatible computer systems.

ASCII pronounced 'ass-key' and is short for American Standard Code for Information Interchange. This code was developed as a 7-bit standardization of various special codes. It is used mainly by personal computer systems. Table 1.9 shows the ASCII code for 0 to 9 and A to Z only.

In this system a digit has its binary representation as the numeric portion of its code. For the zone portion of the digits, ASCII uses 0011 as shown in the table. The zone bits for the characters A to Z can also be observed from the table.

Table 1.9: Alphanumeric code for 0 to 9 and A to Z only.

Char.	Zone Numeric	Hex	Char.	Zone Numeric	Hex	Char.	Zone Numeric	Hex
0	0011 0000	30	A	0100 0001	41	P	0101 0000	50
1	0001	31	B	0010	42	Q	0001	51
2	0010	32	C	0011	43	R	0010	52
3	0011	33	D	0100	44	S	0011	53
4	0100	34	E	0101	45	T	0100	54
5	0101	35	F	0110	46	U	0101	55
6	0110	36	G	0111	47	V	0110	56
7	0111	37	H	1000	48	W	0111	57
8	1000	38	I	1001	49	X	1000	58
9	0011 1001	39	J	1010	4A	Y	1001	59
			K	1011	4B	Z	0101 1011	5A
			L	1100	4C			
			M	1101	4D			
			N	1110	4E			
			O	0100 1111	4F			

Unicode: Unicode is a sixteen-bit code. It is designed to support international languages (like Bangla, Hindi, Chinese, Japanese, Arabic etc). Many of the international languages have too many characters to be represented by eight-bit ASCII and EBCDIC codes. Unicode, a 16-bit code system, is developed by Unicode Inc. with support from Apple, IBM and Microsoft and is standardised by ISO.

4.6 Exercise

Multiple choice questions

1. Codes are used to reduce the volume of
   a. information
   b. data
   c. members
   d. files

2. Integers numbers have
   a. decimal points
   b. no decimal points
   c. embedded decimal points
   d. no embedded decimal points

3. BCD codes require
   a. 4 bits for each decimal digit
   b. 7 bits for each decimal digit
   c. 8 bits for each decimal digit
   d. 10 bits for each decimal digit

Analytical questions

4. What do you understand by data, information and code?
5. Describe positive and negative representations of numbers in computers.
6. Write the BCD representation of the decimal numbers 469 and 4793.
7. What do you understand by BCD, ASCII and EBCDIC codes?

Chapter 3

Digital circuits

Lesson 1: Logic Functions and Logic Gates
Lesson 2: Boolean Algebra and Logic Simplification
Lesson 3: Karnaugh Map and Logic Simplification
Lesson 4: Latches and Flipflops
Lesson 5: Shift Registers and Counters

Lesson 1: Logic Functions and Logic Gates

1.1 Learning Objectives

On completion of this lesson you will know:

- Elementary concepts of Boolean algebra
- Primary logic gates
- Truth table for logic expressions
- Operation of secondary logic gates.

1.2 Introduction

Data and control instructions move inside a computer by means of pulses of electricity. Certain components of computers combine these pulses following a set of rules. These components are the logic elements.

Pulses of electricity are called digital signals. A digital signal has two discrete levels or values. These two discrete signal levels HIGH and LOW can also be represented by binary digits 1 and 0 respectively. A binary digit (0 or 1) is referred to as a bit. A digital signal can have only one of the two possible levels 1 or 0, and the binary number system can be used for the analysis and design of digital systems. The two levels (or states) can

also be designated as on and off (or TRUE and FALSE). George Boole introduced the concept of binary system in 1854 and developed its algebra known as Boolean Algebra.

1.3 Primary Logic Gates

The common use of logic elements is to act as switches, although they have no moving parts. They open to pass on a pulse of electricity or close to shut it off. This is why they are known as gates. The primary gates are OR, AND and NOT.

OR Gate

An OR gate has an output 1 if any of its inputs are 1. The symbol and truth table for a two-input OR gate are shown in Figure 3.1. Here output, Y = A + B, where + denotes OR operation.

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(a)

(b)

Figure 3.1: Two-input OR gate, (a) Truth Table, (b) Symbol

Figure 3.2 Illustrates, the close relationship between the 2-input OR gate and an electrical switching circuit. Such a circuit normally contains some source of energy (a battery), an

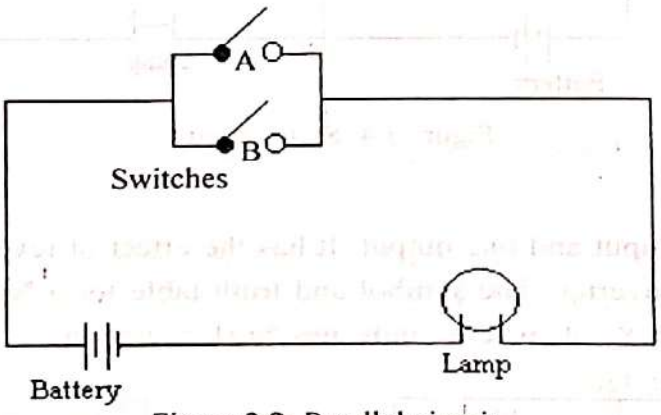


Figure 3.2: Parallel circuit

output device (a lamp), and one or more switches — all connected by wires. A switch is a two-state device that is either closed (on) or open (off). In Figure 3.2 switch A and B, are connected in parallel. The lamp will light if switch A is closed, or if switch B is closed, or if both switches are closed. This is the property described by the truth table for the OR

gate, where 1 denotes that the switch or lamp is on and 0 indicates that it is off.

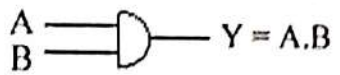
AND Gate

An AND gate has an output 1 if all of its inputs are 1. The symbol and truth table for a two-input AND gate are shown in Figure 3.3. Here output, Y= A.B, where . denotes AND operation.

Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(a)



(b)

Figure 3.3: Two-input AND gate, (a) Truth Table, (b) Symbol

Figure 3.4 is a circuit showing two switches, A and B, connected in series. The lamp will light only when both A and B are closed. This is exactly the property described by the truth table for the AND gate, here 1 denotes that the circuit element is on and 0 denotes that it is off.

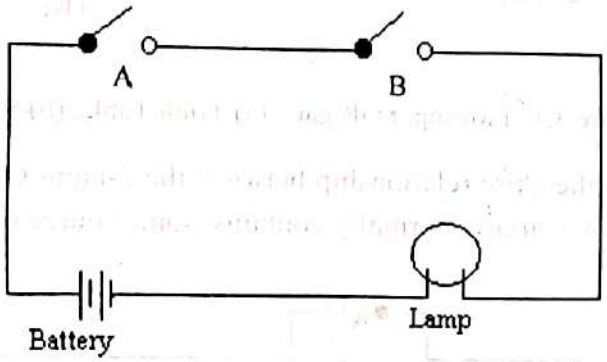


Figure 3.4: Series circuit

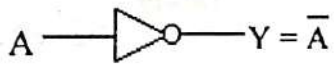
NOT Gate

A NOT gate has one input and one output. It has the effect of reversing the input signal and is also called an inverter. The symbol and truth table for a NOT gate are shown in Figure 3.5. Here output Y= $\bar{A}$, where - indicates NOT operation.

Truth Table

A	Y
0	1
1	0

(a)



(b)

Figure 3.5: NOT gate, (a) Truth Table, (b) Symbol

1.4 Secondary Logic Gates

Some secondary gates are NAND, NOR, EXOR etc. NAND and NOR gates are called universal gates because any one them can be used to realize any logic expression.

NAND Gate

A NAND gate has the same effect as an AND gate followed by a NOT gate. Hence the output will be opposite of the AND gate. The symbol and truth table for a two-input NAND gate are shown in Figure 3.6.

Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(a)

(b)

Figure 3.6: NAND gate, (a) Truth Table, (b) Symbol

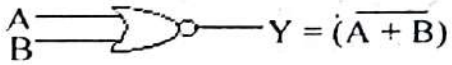
NOR Gate

A NOR gate has the same effect as OR gate followed by a NOT gate. Hence the output will be the opposite of OR gate. The symbol and truth table for a two-input NOR gate are shown in Figure 3.7.

Truth Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(a)



(b)

Figure 3.7: NOR gate, (a) Truth Table, (b) Symbol

EXOR Gate

It is widely used in digital circuits. Symbol and truth table for a two-input EXOR gate are shown in Figure 3.8. Here output Y= A ⊕ B, where ⊕ denotes EXOR operation.

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(a)

(b)

Figure 3.8: EXOR gate, (a) Truth Table (b) Symbol

Basic gates (AND, OR, NOT) and universal gates (NAND, NOR) can be used in combination to make up digital computer circuits.

1.5 Exercise

Multiple choice questions

1. The primary gates are

   a. OR, AND, NAND
   b. NOR, AND, NOT
   c. OR, AND, NOT
   d. NOR, NAND, EXOR.

2. Pulses of electricity are called

   a. analog signals
   b. digital signals
   c. voltage
   d. current.

3. An AND gate has an output 1 if

   a. all of its inputs are 1
   c. any of its inputs is 1
   b. any of its inputs is 0
   d. all of its inputs are 0.

Analytical questions

4. What do you know about Boolean algebra?
5. Draw the symbol and truth table of a two-input OR gate.
6. Explain the operation of the following gates:
   AND, NOT, NOR, XOR.
7. What do you understand by basic gate and universal gate?

Lesson 2: Boolean Algebra and Logic Simplification

2.1 Learning Objectives

On completion of this lesson you will know:

- Basic and other Boolean theorems
- Simplification of logic functions using Boolean algebra
- DeMorgan's theorems and their proof by truth tables.
- Universality of NAND gate and NOR gate
- Realization of logic circuits by NAND gates only.

2.2 Boolean Theorems

Logic circuits can be simplified by Boolean algebra and simplification of logic functions and logic circuits is an important application of Boolean algebra. A simplified circuit costs less and works efficiently.

Boolean theorems are used to simplify or manipulate logic functions. Important Boolean theorems are given below. The theorems can easily be verified by assigning binary values (0 or 1) to the variables.

Basic Theorems

OR	AND	NOT
$A + 0 = A$	$A \cdot 0 = 0$	$A + \overline{A} = 1$
$A + 1 = 1$	$A \cdot 1 = A$	$A \cdot \overline{A} = 0$
$A + A = A$	$A \cdot A = A$	$\overline{\overline{A}} = A$
$A + \overline{A} = 1$	$A \cdot \overline{A} = 0$	

Other Theorems

$A + B = B + A$

$A \cdot B = B \cdot A$

$A + (B + C) = (A + B) + C$

$A + B.C = (A+B)\,(A+C)$

2.3 Examples of logic Simplification

Example 1 : $X = ABC + AB\overline{C}$

$X = AB(C + \overline{C}) = AB,$ as $C + \overline{C} = 1$

Example 2 : $X=(A+\overline{B})(\overline{A}+B)$

$X = A\overline{A} + AB + \overline{A}\,\overline{B} + B\overline{B}$

$= 0 + AB + \overline{A}\overline{B} + 0 = AB + \overline{A}\overline{B}, \quad$ as $\quad A\overline{A} = 0$

Example 3 : $X = B\overline{C} + \overline{B}\overline{C} + BC$

$X = B(\overline{C} + C) + \overline{B}\overline{C} = B + \overline{B}\overline{C}, \quad$ as $\quad C + \overline{C} = 1$

$= (B + \overline{B})(B + \overline{C}) = B + \overline{C}$

2.4 DeMorgan's Theorem

French Mathematician DeMorgan formulated two theorems in 1953 and they are known as DeMorgan's theorems. Demorgan's theorems for two variables are:

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$
$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

The proof of the DeMorgan's theorem for two variables using truth table follows.

A	B	A + B	A. B	$\overline{A+B}$	$\overline{A}$	$\overline{B}$	$\overline{A}.\overline{B}$	$\overline{A}+\overline{B}$	$\overline{A.B}$
0	0	0	0	1	1	1	1	1	1
0	1	1	0	0	1	0	0	1	1
1	0	1	0	0	0	1	0	1	1
1	1	1	1	0	0	0	0	0	0

The validity of the theorems can be observed from the table.

The theorems are valid for any number of variables. For three variables, the theorems are:

$$\overline{A + B + C} = \overline{A}.\overline{B}.\overline{C}$$
$$\overline{A.B.C} = \overline{A} + \overline{B} + \overline{C}$$

The DeMorgan's theorem are used for simplification of logic functions associated with NOT gates.

2.5 Universality of NAND and NOR gates

Any logic can be realised by OR, AND and NOT gates. But it is possible to realise any logic circuit by NAND gates only. This is possible because all the basic logic gates can be relised by NAND gates only. Similarly any logic circuit can be realised by NOR gates only. This is known as the universality of NAND and NOR gates.

Realisation of Basic logic gates by NAND gates only

NOT gate: Figure 3.9 (a) shows this realisation. In this realisation $X = \overline{A.A} = \overline{A}$ and the NAND gate functions as a NOT gate.

AND gate: Figure 3.9 (b) shows the realization of AND function with the help of two

NAND gates.

Here, $X = \overline{\overline{A.B}} = A.B = AB$ and the circuit acts as an AND gate.

OR gate : Figure 3.9(c) shows the realization of an OR gate by NAND gates only. In this case and the circuit acts as an OR gate.

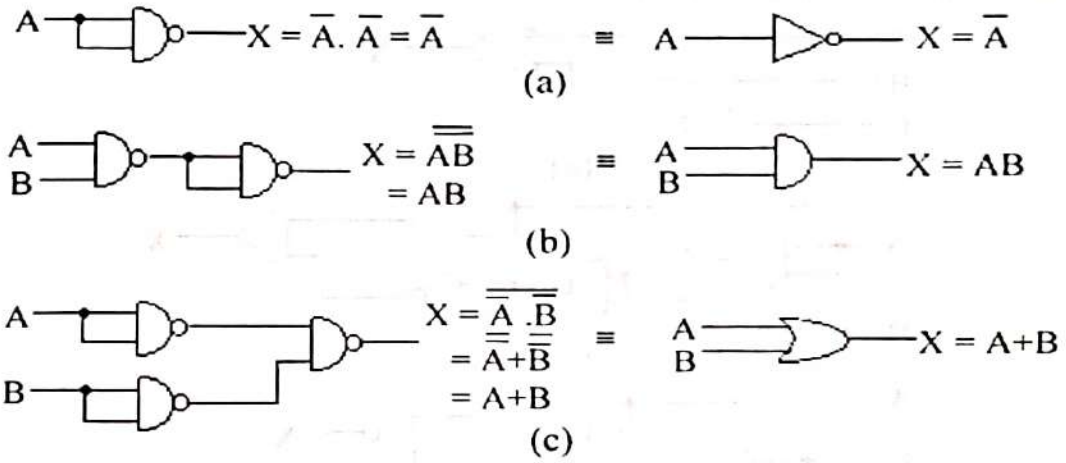


Figure 3. 9: Realization of basic logic gates by NAND gates only.

Realization of Basic Logic Gates by NOR Gates Only.

NOT gate: Figure 3.10 (a) shows this realisation. In this case

$X = \overline{A + A} = \overline{A}$ and the circuit functions as a NOT gate.

OR gate: Figure 3.10 (b) shows the realization of a 2-input OR gate with the help of two NOR gates only.

Here, $X = \overline{\overline{A + B}} = \overline{A + B}$ and the circuit acts as an OR gate.

AND gate: Figure 3.10 (c) shows the realization of a 2-input NAND gate by NOR gates only. In this case

$X = \overline{\overline{A} + \overline{B}} = \overline{\overline{A}}.\overline{\overline{B}} = A.B$

and the circuit acts as an AND gate.

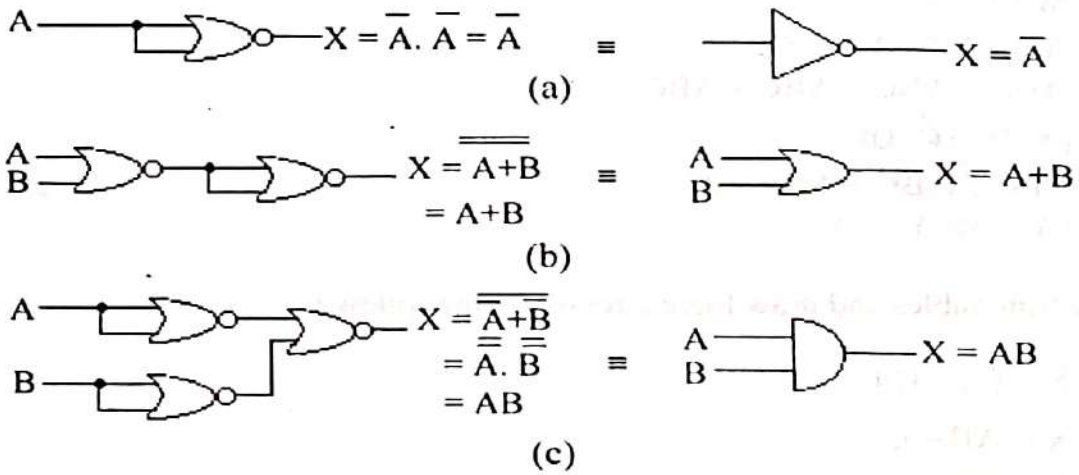


Figure 3.10: Realization of basic logic gates by NOR gates only.

Example: Realise the logic circuit of Figure 3.11(a) by NAND gates only.

Solution: The basic logic gates are first replaced by their NAND equivalents as shown in Figure 3.11 (b). Two serial NAND operations do not perform any logic operations at all and can be eliminated as shown in Figure 3.11(c).

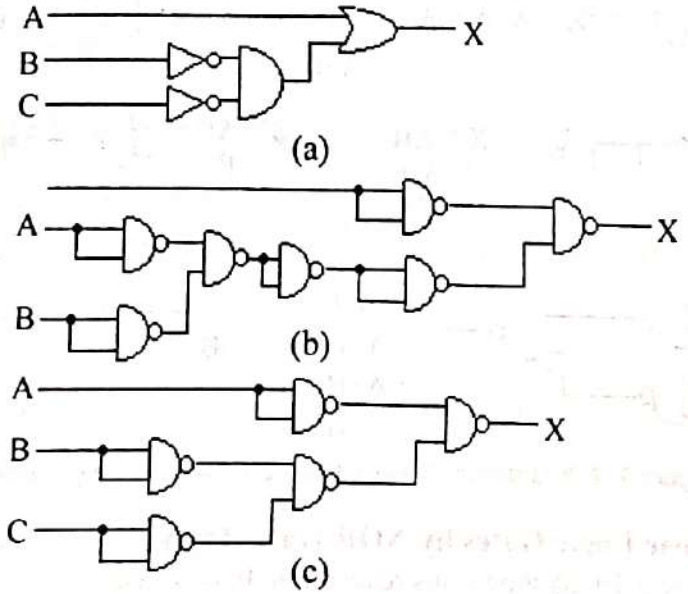


Figure 3.11: A circuit realized by NAND gates only.

Similarly any logic circuit can be relised by NOR gates only.

2.6 Exercise

1. Simplify the following expressions using Boolean algebra

   a. $X + XY$

   b. $\overline{\overline{WXY} + Z}$

   c. $XY + XZ + \overline{X}Z + ZX$

   d. $\overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$

   e. $\overline{(\overline{A} \cdot \overline{B}) \cdot (\overline{C} \cdot \overline{D})}$

   f. $\overline{A}BC + A\overline{B}C + ABC + \overline{A}C$

   g. $(A + B)(A + C)$

2. Give truth tables and draw logic circuits for the following:

   a. $X = \overline{(A + B)}C$

   b. $X = \overline{\overline{AB} + \overline{C}}$

   c. $X = \overline{A}B + A\overline{B}$

3. a. Realise logic circuits for the following using basic logic gates.

$$X = AB + CD$$

$$X = \overline{A}B\overline{C} + BC + \overline{A}$$

   b. Replace the basic gates by their NAND equivalents and convert the circuit for NAND gates only.
   c. Repeat 2(b) for NOR gates only.

4. Write DeMorgan's theorems for four five variables.

LESSON 3: Karnaugh Map and Logic Simplification

3. 1 Learning Objectives

On completion of this lesson you will know:

- Characteristics of 2-, 3- and 4-variable K-maps.
- Simplification of logic functions using K-maps.
- Making groups of 1's for 2-, 3- and 4-variable K-maps.

3.2 Introduction

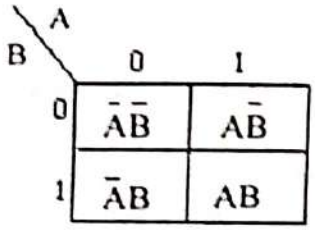
Logic simplification using Boolean Algebra was discussed in Lesson 2. There is a graphical technique for simplifying and manipulating Boolean expressions. In this technique information from a truth table or from a logic expression is represented on a map called Karnaugh map (or K-map). This is a popular tool for simplifying logic functions. The technique can be used for any number of variables, but the technique becomes cumbersome for six or more variables.

3.3 Two-variable K-map

Figure 3.1(a) shows K-maps for two variables A and B. In this map A= 1 for the two right cells and A = 0 for the two left cells. Similarly B = 0 for the upper two cells and B = 1 for the lower two cells. The values for the cells are shown inside the cell itself.
For example:

$\overline{A}\overline{B}$ stands for A = 0 and B = 0
$\overline{A}B$ stands for A = 0 and B = 1
$A\overline{B}$ stands for A = 1 and B = 1, and
AB stands for A = 1 and B = 0

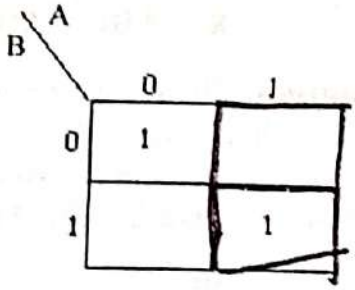
Figure 3.1(b) shows the truth table and K-map for a the function $X = A\overline{B} + AB$. In this figure the A = 0 and B = 0 condition of the truth table corresponds to the $\overline{A}\overline{B}$ cell in the K-map. Since X = 1 for this case 1 is placed in the $\overline{A}\overline{B}$ cell. Similarly 1 is placed in the AB cell as A = 1 and B = 1 for this cell.



(a)

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

$X = \bar{A}\bar{B} + AB$



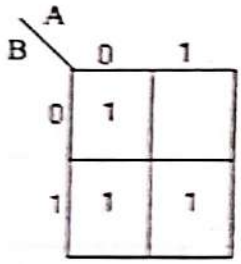
(b)

Figure 3.12: K-maps for $\bar{A}\bar{B} + AB$

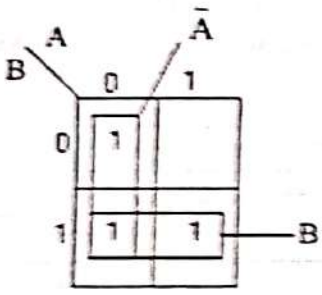
Example: Simply the following using K-map

$$X = \bar{A}\bar{B} + \bar{A}B + AB$$

Solution: The two-variable K-map for this function is shown in Figure 3.13(a). In a two-variable K-map, the adjacent cells are grouped for simplification (Figure 3.13(b)). It is to be noted that a cell may be used more than once for making groups of two cells. For the left vertical group B = 0 and B = 1 and B should not be considened. The value for this group is therefore $\bar{A}$ only. Similarly for the lower horizontal group, A = 0 and A = 1 and A is not considened and the value for the lower group is B. Thus for the two groups, from the map, $X = \bar{A} + B$.



(a)



(b)

Figure 3.13: Simplification using K-map.

3.4 Three-variable K-map

Figure 3.14 shows three-valuable K-maps with the values of each cell. There are eight cells in this map for the three variables. In this map, groups of two or four adjacent cells are considered for simplification. For this map,

- A single cell is identified by three variables
- A group of two adjacent cells is identified by two variables
- A group of four adjacent cells is identified by one variable only.

Example: Simplify the following using the K-map method.

$$X = \overline{A}BC + ABC + A\overline{B}C + \overline{A}B\overline{C}$$

Solution: The K-map for the function is shown in Figure 3.14(a). For simplification the groups should be as large as possible and the number of groups should be minimum and all the 1's should be considered. The two possible groups each consisting of two 1's are shown in Figure 3.14(b). From figure 3.14(b) we can write: $X = \overline{A}B + AC$

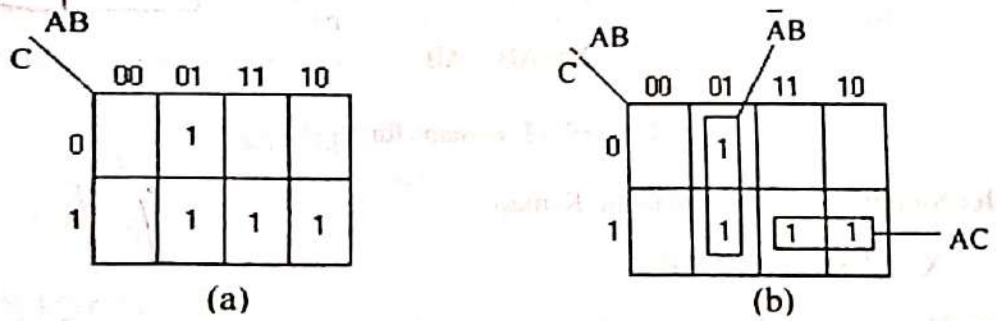


Figure 3.14: Simplification using K-map.

Example: Obtain the simplified expressions from the K-maps shown in Figure 3.15

Solution: The largest possible groups for the map is shown in Figure 3.15 The simplified expressions from the maps are given below:

From Figure 3.15(a) : $B + \overline{C}$
From Figure 3.15(b) : $\overline{B} + AC$
From Figure 3.15(c) : B
From Figure 3.15(d): $B + AC + \overline{A}\overline{C}$

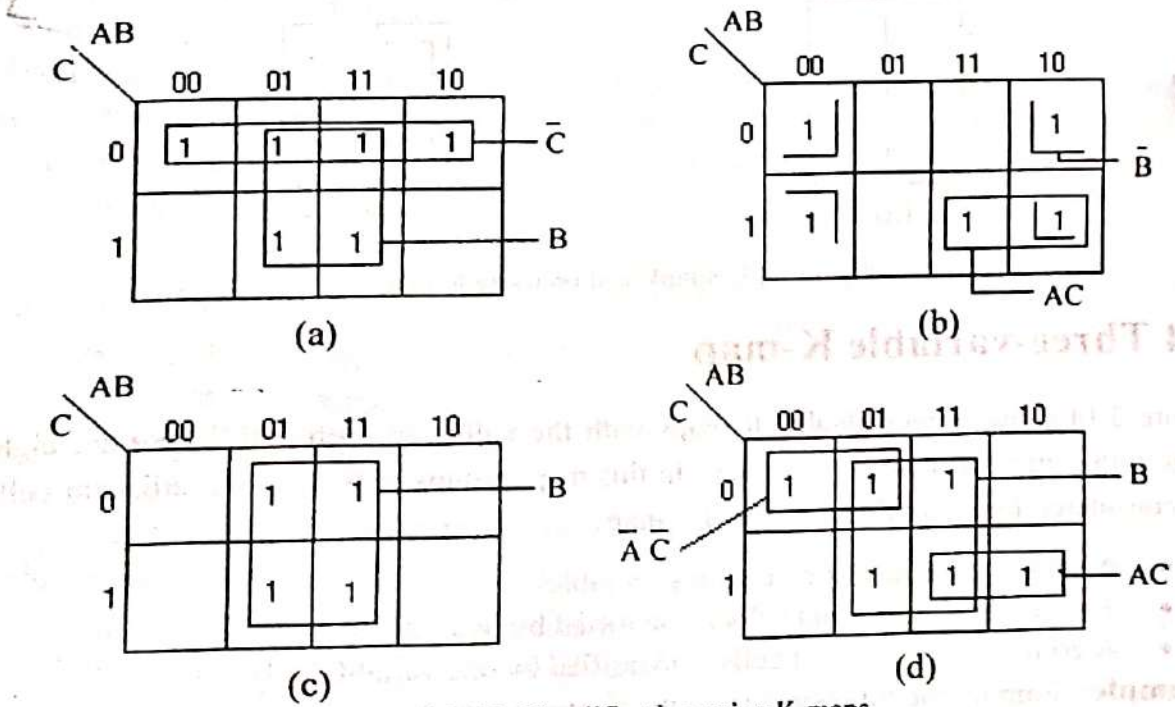


Figure 3.15: Simplification using K-maps.

3.5 Four-variable K-map

Figure 3.16 shows the K-map and the values for each cell for a four-variable K-map. There are sixteen cells in this map. The rules for simplification using four variable K-maps are given below.

- A single cell is identified by four variables.
- A group of two adjacent cells are identified by three variables.
- A group of four adjacent cells are identifes by two variables.
- A group of eight adjacent cells are identified by one variable only.

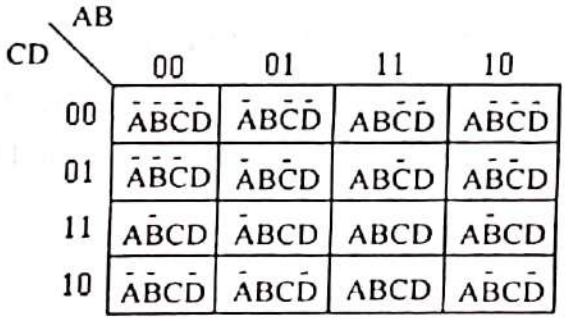


Figure 3.16: A four-variable K-map.

Example: Obtain simplified expressions from the maps of Figure 3.17.

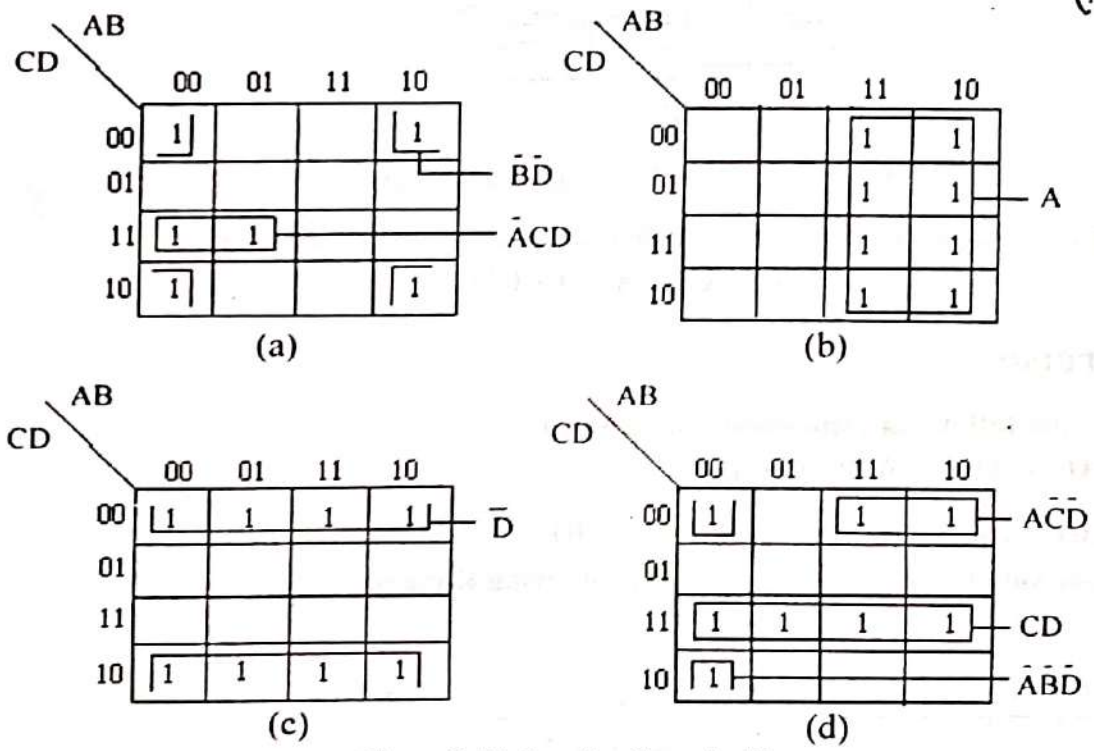


Figure 3.17: Reading from the K-map

Solution: In Figure 3.17 (a) there are three groups and the simplified expension is:

$\overline{A}CD + \overline{B}\overline{D}$

From Figure 3.17(b), the simplified expression for the group of eight 1's is given by A.
From Fig 3.17(c), the single group of eight 1's can be read by $\overline{D}$ only.
From Fig 3.17(d), the simplified expression is given by: $CD + \overline{A}\overline{B}\overline{D} + A\overline{C}\overline{D}$

Example: Simplify the following Boolean equation

$$X = \overline{A}\overline{B}CD + ABCD + \overline{A}C\overline{D} + \overline{A}CD + A\overline{B}D$$

Solution: Figure 3.18 shows the K-map for the equation. In this case the three-variable terms are expanded and plotted in the Map. The expansion techniques are given below:

$$\text{For } \overline{A}C\overline{D} : \quad \overline{A}C\overline{D}(B + \overline{B}) = \overline{A}BC\overline{D} + \overline{A}\overline{B}C\overline{D}$$
$$\text{For } \overline{A}CD : \quad \overline{A}CD(B + \overline{B}) = \overline{A}BCD + \overline{A}\overline{B}CD$$
$$\text{For } A\overline{B}D : \quad A\overline{B}D(C + \overline{C}) = A\overline{B}CD + A\overline{B}\overline{C}D$$

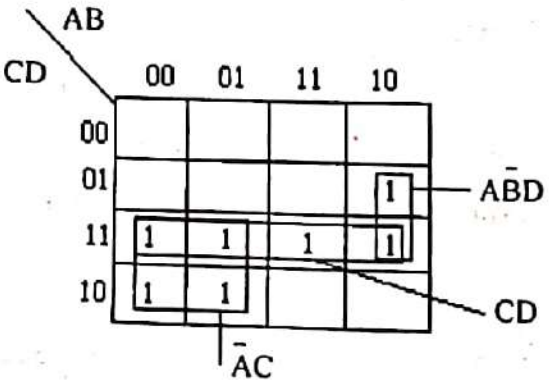


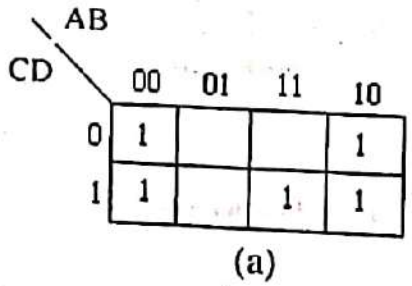
Figure 3. 18: Simplification using K-map.

Three groups can be identified from the map and the simplified expression is,

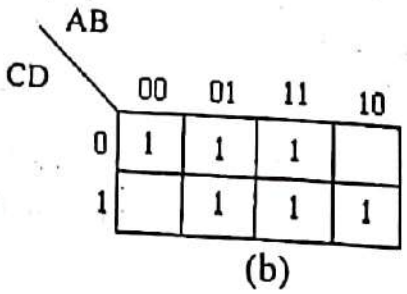
$$X = \overline{A}C + A\overline{B}D + CD$$

3.6 Exercise

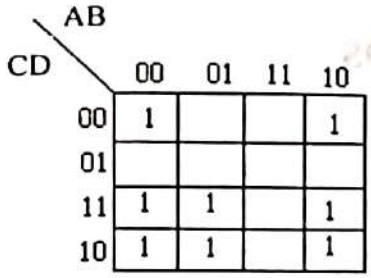
1. Simplify the following expressions using K-map:
   a. $\overline{A}\overline{B}\overline{C} + ABC + \overline{A}\overline{B}C + A\overline{B}C$
   b $\overline{A}BD + AB\overline{C}\overline{D} + \overline{B}\overline{C}D + \overline{A}B\overline{D} + (\overline{A+B})$
2. Obtain the simplest expressions from the following K-maps.



(a)

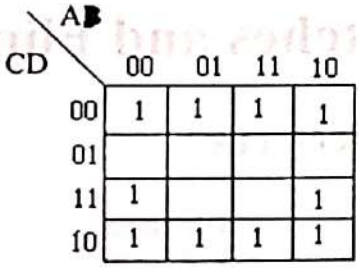


(b)



CD \ AB	00	01	11	10
00	1			1
01				
11	1	1		1
10	1	1		1

(a)



CD \ AB	00	01	11	10
00	1	1	1	1
01				
11	1			1
10	1	1	1	1

(b)

Lesson 4: Latches and Flipflops

4.1 Learning objectives

On completion of this lesson you will know:

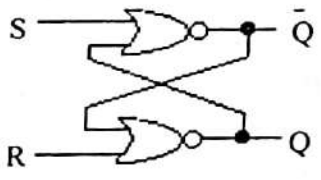
- Latches and their functions
- NOR latch, NAND latch and Clocked Latches
- Characteristics of a clock signal
- Positive- and negative-edge triggering of flipflops
- Symbols and truth tables of SR, JK, T and D flipflops

4.2 Latch

The basic memory circuit is known as a latch. A latch is used for storing bits (0 or 1). A latch has two states: state 1 and state 0. A latch can be realised by NAND gates or NOR gates.

NOR Latch: The logic diagram and the truth table for a NOR latch is shown in Figure 3.19. The circuit has the following characterishes:

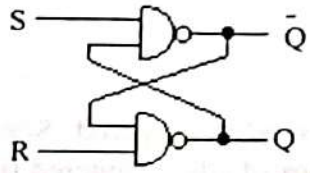
- The outputs Q and $\overline{Q}$ are always complentary
- In the set state $Q = 1$ and $\overline{Q} = 0$
- In the reset state $Q = 0$ and $\overline{Q} = 1$
- In the set state the circuit stores 1
- In the reset state it stones 0.



S	R	Q	$\bar{Q}$
0	0	No change	
1	0	1	0 (set)
0	1	0	1 (reset)
1	1	Not allowed	

Figure 3. 19: NOR latch, (a) circuit, (b) truth table

NAND Latch: Figure 1.2 shows the logic circuit and the truth table for a NAND latch. Like a NOR latch it is also used to store 0 or 1.

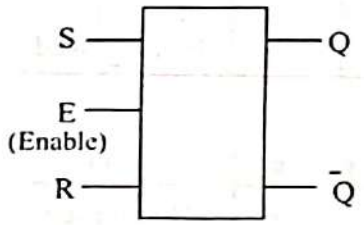


S	R	Q	$\bar{Q}$
0	0	Not allowed	
1	0	1	0 (set)
0	1	0	1 (reset)
1	1	No change	

Figure 3. 20: NAND latch, (a) circuit, (b) truth table.

Clocked Latch

It is often required to set or reset the memory cell in synchronism with an enable pulse. Such a circuit is called a clocked latch (Figure 3.20) and it changes its output state only if it is enabled, that is when E = 1



E	S	R	Q	$\bar{Q}$
0	X	X	No change	
1	0	0	No change	
1	1	0	1	0
1	0	1	0	1
1	1	1	Not allowed	

Figure 3.20: Clocked latch.

4.3 Clock Signal

Digital systems can operate either in asynchronous mode or synchronous mode. In asynchronous systems, the outputs of logic circuits can change state at any time when one (or more) input changes. On the other hand, in a synchronous system the exact times at which an output can change state is determined by a signal called the clock. The clock signal is generally a square wave (Figure 3.21). Generally, in synchronous systems, the outputs can change state only when the clock makes a transition. The transitions are also called edges. When a clock changes from 0 to 1, it is called positive-edge transition and when the clock goes from 1 to 0, it is called negative-edge transition.

There are several types of clocked flipflops. In most clocked flipflops the clock input is edge-triggered, that is the flipflop is triggered by a transition of the clock pulse. The clocked flipflops can thus be positive-edge triggered or negative-edge triggered.

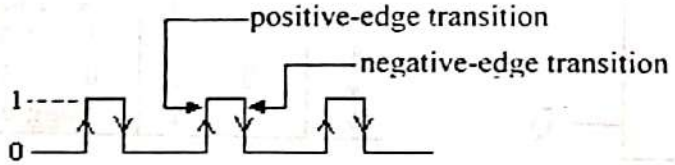


Figure 3.21: A clock signal.

4.4 Flipflops

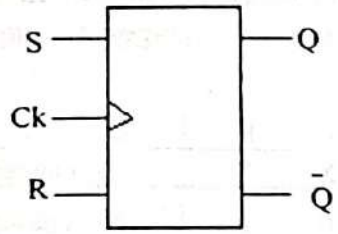
The popular edge-triggard flip-flops are listed below:

- SR flipflop
- JK flipflop
- D flipflop, and
- T flipflop

They can be either positive edge-triggered or negative edge triggered. Special symbols are used with the clock connection in the representations of edge-triggered flipflops.

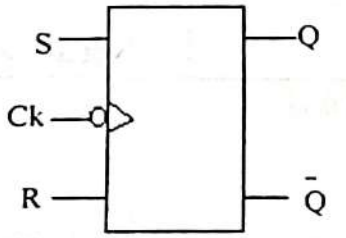
SR flipflop

The symbol and truth table for positive-edge triggered SR flipflop and negative-edge triggered SR flip-flop are shown Figure 3.22(a) and (b) respectively. It is to be noted that the changes in the output states take place at the point of clock (Ck) transition only.



Ck	S	R	Q	$\bar{Q}$
↑	0	0	No change	
↑	1	0	1	0
↑	0	1	0	1
↑	1	1	Not allowed	

(a)



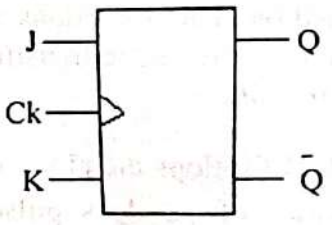
Ck	S	R	Q	$\bar{Q}$
↓	0	0	No change	
↓	1	0	1	0
↓	0	1	0	1
↓	1	1	Not allowed	

(b)

Figure 3.22: a) Positive-edge triggered SR flip-flop, b) Negative-edge triggered SR flip-flop.

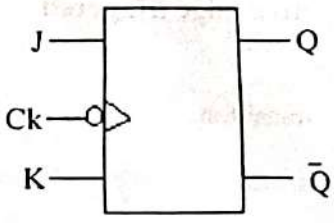
JK Flipflop

The symbols and truth table for JK flipflops are shown in Figure 3.23. It is to be noted



Ck	J	K	Q	$\bar{Q}$
↑	0	0	No change	
↑	1	0	1	0
↑	0	1	0	1
↑	1	1	Toggles	

(a)



Ck	J	K	Q	$\bar{Q}$
↓	0	0	No change	
↓	1	0	1	0
↓	0	1	0	1
↓	1	1	Toggles	

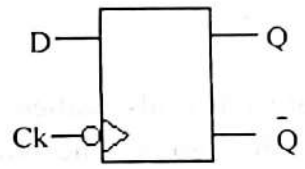
(b)

Figure 3.23: a) Positive-edge triggered JK flipflop, b) Negative-edge triggered JK flipflop.

that unlike SR flipflops, the output of JK flipflops toggles, that is, goes to opposite state when both the inputs are high.

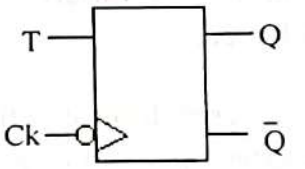
T and D flipflop

T and D flopflips can be positive-edge triggered or negative-edge triggered. They can be realised by the JK flipflops or SR flipflops. The block repretation and truth table for negative-edge trigged T and D flipflops are shown in Figure 3.24. The block representations and truth tables for positve-edge triggerd T and D flip-flops can be obtained by using the appropriate symbols and transitions of the clock signal.



Ck	D	Q	$\bar{Q}$
↓	1	1	0
↓	0	0	1

(a)



Ck	T	Q $\bar{Q}$
↓	1	Toggles
↓	0	No change

(b)

Figure3.24: Symbols and truth tables for negative-edge triggered a) T flipflop and b) D flipflop.

4.4 Exercise

Short questions

1. When a flipflop is set, what are the values of its outputs?
2. Draw a clock signal and show its positive and negative edge transitions.
3. Distinguish between (a) T and D flipflops, and (b) SR and JK flipflops

Analytical questions

4. Draw the logic circuits for NAND and NOR latches and give their truth tables.
5. Give the symbols and truth tables of positive-edge triggered and negative-edge triggered flipflops.

Lesson 5: Shift Registers and Counters

5.1 Learning objectives

On completion of this lesson you will know:

- Basic principles of shift registers and counters
- Types of shift registers
- Modulo-numbers and basic working principle of counters.

5.2 Shift register

A flipflop can store only 1-bit of information (1 or 0). It is also called a 1-bit register. A number of flipflops is required to store binary information. The number of flipflops required is equal to the number of bits in the binary word. A register is formed by combining a number of flipflops. Registers are used in a variety of digital systems including microcomputers, calculators, digital watches etc.

Data can be entered in shift registers bit by bit in serial form or all bits together in parallel form. Data can also be read from shift registers in serial or parallel forms. For the serial input, only one line is required for data input and for the serial output only one line is required for data output. The number of lines required for parallel input and parallel output are equal to the number of bits for parallel inputs or parallel inputs.

Registers are classified on the basis of the way in which data are entered and retrieved. The four possible modes (Figure 3.25) of operations are:

- S1S0: Serial-in Serial-out
- S1P0: Serial-in parallel-out
- P1S0: Parallel-in Serial-out, and
- P1P0: Parallel-in Parallel-out.

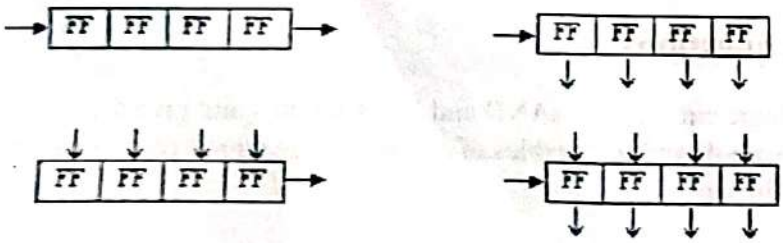


Figure 3.25: Shift register configurations, (a) S1S0, (b) S1P0, (c) P1S0 and (d) P1P0.

Besides these, there are other configurations based on the above classification. In a

bidirectional register data can be shifted serially from left to right or from right to left using a mode control input. A register is referred to as a universal register if it can be operated in all the four possible modes as well as a bidirectional register.

5.3 Counter

Counters are used for counting pulses. A counter is constructed by connecting a number of flipflops (Figure 3.26). The number of states that a counter can count is called its modulus. For example, a decimal (or decade) counter has modulus 10 and it can count from 0 to 9. When the pulses to be counted are applied to a counter, it goes from one state to the next state and the counter comes back to its starting state after counting N pulses in the case of modulo-N counter.

A flipflop has two stable states and a group of n flipflops has 2^n states in total. Thus it is possible to make a modulo 2^n counter using n flipflops. Basically there are two types of counters, they are:

- Asynchronous counter (or ripple counter),
- Synchronous counter

In an asynchronous counter, the flipflops are not clocked simultaneously, whereas in a synchronous counter all the flipflops are clocked simultaneously, that is, at the same point of time. Synchroms counters are faster than the asynchroms counters.

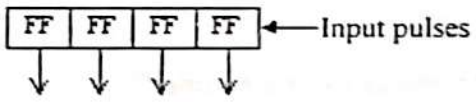


Figure 3.26: Block representation of a 4-bit counter.

5.4 Counters and Counting

Figure 3.27: shows the block representation of a decade counter together with its one input and four outputs: A, B, C, D. The input pulses are applied in the input and the outputs, indicate the count states. The count table for the counter is shown in Table-3.1.

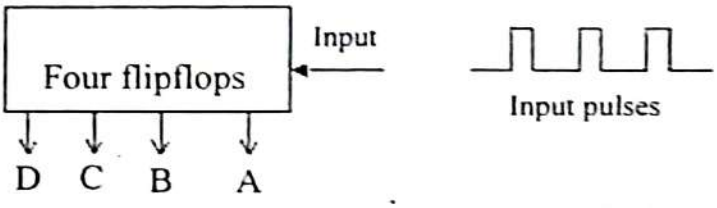


Figure 3.27: Block repreatation of a decade connected.

Table 3.1 Count table for the decade counter.

Count states				Number of
D	C	B	A	Input pulses
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

5.5 Exercise

Short Questions

1. Give the main functions of a shift register and a counter.
2. What is a universal register?
3. What do you understand by modulus of a counter?
4. Distinguish between synchronous and asynchronous counter.

Analytical Questions

5. Name the four possible modes of shift registers and show their operation with the help of illustrations.
6. Draw the block representation of a decade (or decimal counter) and give its count table

Chapter 4

Microcomputer System

Lesson 1: Microcomputer basics
Lesson 2: Organization of a microcomputer
Lesson 3: Bus organized architecture
Lesson 4: Working principle of a microcomputer
Lesson 5: Motherboard and adapter

Lesson 1: Microcomputer Basics

1.1 Objectives

On completion of this lesson you will know:

- Main components of a microcomputer
- Functions and characteristics of processor, memory and I/O devices
- Types of programs and Basic Input / Output System (BIOS)
- Personal computers, clones, servers and workstations
- Chronological events for the development of microcomputers.

1.2 Microcomputer Components

The five key parts of a computer are:

- Processor (also called the central processing unit, or CPU),
- Memory,
- Input/output devices,
- Disk storage, and
- Programs.

Figure 4.1 shows the basic components of a microcomputer and how they fit together.

Before the advent of integrated circuits, individual portions of computers were housed on separate boards or even in separate units made up of many boards. Today, the majority of the circuits of the microcomputer are housed on a single printed circuit called the system board or motherboard.

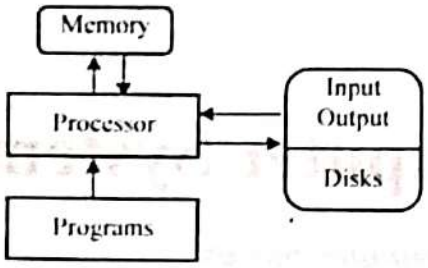


Figure 4.1: Major components of a microcomputer.

The usual components on the motherboard include the main processor and its support circuits, memory, I/O interface and a bus system which enables the CPU to talk to other components that are not integrated with the motherboard.

The processor is the brain of the microcomputer. It executes instructions and programs. The processor performs arithmetic and simple logical operations. In a microcomputer, like a PC, the main processor is a microprocessor. Memory is the microcomputer's workplace.

The size of a microcomputer's memory sets a practical limit on the kinds of work that can be undertaken. The variety of tools that can be used at any time is largely determined by the size of the memory of the microcomputer. Microcomputers are rated by the amount of memory they have.

The computer takes in and sends out date through input/output devices. They include what we type on the keyboard and what the computer shows on the display screen or prints on the printer. The I/O devices are also called peripheral devices.

Disk storage is a very important I/O device. It is the computer's reference library, filing cabinet, and toolbox all in one. The computer keeps data in the disk when they are not in use in the computer's memory. The disks are the most practical and important medium for storing data.

Programs bring a computer to life, make it go, and turn it into a powerful working tool. Programs tell the computer what to do, when to do and how to do.

1.3 Processor

The processor executes or runs a program. The processor must have the capability to

read and write information in the computer's memory. Both the program instructions that the processor carries out and the data on which the processor works are stored in the memory. The processor must recognize and execute a series of commands or instructions provided by programs. The processor must tell other parts of the computer what to do. so that, the processor can carry out the operations of the microcomputer.

1.4 Memory

The processor finds programs and data from memory when it does its assigned task. The memory is the place where every thing is kept. The microcomputer's memory is just a temporary space like a chalkboard where the computer scribbles while work is being done. Unlike human memories, the computer's memory is not a permanent repository. The computer's memory simply provides a place where computing can happen.

The processor makes a vital distinction between programs and data. But to the computer's memory there is no difference between programs and data. It is like a piece of paper that neither knows nor cares what one writes on it. It is the same with the microcomputer's memory. Only the processor recognizes the difference between program and data.

Anything can be written on the memory, and the writing can be changed by writing over it. Unlike a chalkboard, the computer's memory doesn't have to be erased before something new can be written to it; the act of writing automatically erases what was in the memory before. Reading information from the memory is simple and straightforward as reading any thing written on paper. Both the processor and the I/O devices have the capability to read and write data from and to the main memory.

1.5 I/O Devices

In general, the I/O devices are a bridge between a user and a microcomputer. A microcomputer gets what the user types on the keyboard, the user sees what the computer writes on the printer or displays on the screen.

The disk storage is a special category of I/O that is intended for a computer's own use. Information on a disk can be read and written only by the computer and it can not be read or written by users.

The microprocessor and the memory make up a closed world and the I/O devices open the world to the users. The computer communicates through I/O devices. These devices

include keyboard, display screen, mouse, printer, telephone line connected to the computer, and any other channels of communication into or out of the computer.

1.6 Programs

Programs tell a computer what to do. There are two different kinds of programs: systems programs and applications programs. All programs accomplish some kind of work. Systems programs help operate the computer itself; the inner systems of a computer are controlled by systems programs. An applications program carries out a task for the user. In fact, applications programs get users work done, and systems programs help a computer manage itself.

A few of the systems programs, that a PC needs to manage its operating, are permanently built into it. These are called firmware, because they are permanently stored in read only memory. These programs do the most fundamental supervisory and support work, such as providing essential services that all the application programs use. These service programs are called the Basic Input/Output System (BIOS).

Other systems programs built on the foundation created by the BIOS provide a higher level of support services. Operating systems, such as DOS and Windows are examples of these higher-level systems programs.

1.7 PC and PC Clones

Microcomputers come in a wide variety of designs. They are known as "personal computers" and are relatively small, relatively inexpensive, and are designed primarily for use by one person at a time. However there are significant exceptions to this; powerful personal computers are used as data or application servers in a networking environment or for huge data processing applications.

The term "PC" generally contrasts computers based on the Intel family of micro-processors and compatibles with Apple computer's Macintosh, and other non-Intel systems. This started with IBM's original Personal Computer, which was based on the Intel 8088 processor and labeled "the PC" when it reached the market in the early 1980s.

Most of the personal computers sold today do not come from IBM at all. Most are designed, built, and sold by a number of manufacturers who offer a wide range of PCs for business, education and home use. These machines are called "clones" because they were made to look, feel and work like an IBM PC. The name clone seems to stay with these machines, though manufacturers have moved in their own directions.

1.8 Servers and Disk Arrays

Servers and disk arrays are the network configurations of personal computers. Networks enable multiple computers and their users to share processing power, disk storage, printers and other peripheral devices.

Most of the computers mentioned before can function as a server or as the controller for a disk array. Personal computers are evolving in such a way that differentiation between clients (or workstations) and servers are diminishing. A server is usually a high end PC having a lot of memory and several large disks for storage and is dedicated to managing network traffic, storing common programs or data for users across the network, or directing communications via telephone lines (or dedicated links) outside the local network.

A disk array is a special kind of server that uses a large portion of its processor power simply to manage I/O for a collection of disk drives. A regular PC may contain one or two hard drives with several GBs of storage, whereas a disk array may contain many times more storage space for programs and data.

1.9 Chronological Events

1971	First Microprocessor (Intel 4004)
1972	First 8-bit Microprocessor (Intel 8008)
1975	First Microcomputer (Altair 8800)
1977	Apple II Computer, Microsoft Corporation Founded by Bill Gates and Paul Allen.
1978	Intel 16-bit Microprocessor (Intel 8086)
1979	Motorola 16-bit Microprocessor (M 68000)
1981	IBM PC Introduced
1983	IBM PC XT
1984	Apple Macintosh
1988	First Computer Virus
1990	Microsoft Introduced Windows 3.0
1992	First 64-bit RISC Microprocessor (Alpha)
1993	Intel Pentium Microprocessor
1994	300 MHz, 64-bit RISC Processor (Alpha)
1996	Intel PentiumPro Microprocessor
1997	500 MHz Alpha 2114 RISC Processor
1998	Intel Pentium II
1999	Intel Pentium III
2000	Intel Pentium IV

1.10 Exercise

Multiple choice questions

1. Which is not an I/O choice?

   a. disk storage
   b. keyboard
   c. mouse
   d. microprocessor

2. The permanently stored program in a read only memory is called

   a. firmware
   b. application software
   c. customized software
   d. permanent program

3. I/O devices are also called

   a. peripheral devices
   b. electronic devices
   c. electro-mechanical devices
   d. memory devices

Questions for short answers

4. List the basic parts of a microcomputer and show how they fit together.
5. What is a program? Distinguish between system program and application program.
6. What are I/O devices? List common I/O devices.
7. Distinguish between PC and PC clone.

Analytical questions

8. Briefly describe the functions of processor, memory and I/O devices of a microcomputer.
9. What is a server? Mention its main functions.

Lesson 2: Organization of a Microcomputer

2.1 Objectives

On completion of this lesson you will know:

- Simplified block representation of a PC
- General characteristics of major components of a PC
- Microcomputer system unit and their packaging styles.

2.2 Hardware Organization

A basic PC system has a number of major hardware elements. Figure 4.2 shows the simplified block diagram of a microcomputer. These elements are contained in the system unit of the microcomputer.

Processor: In most PCs this is an Intel Pentium microprocessor, or one of the Intel-compatible processors available from several companies.

Input and output devices: Input devices include keyboard, mouse, scanners, CD-ROM drives, bar-code readers etc. The most common output devices are printers and display monitors. These devices are attached to the microprocessor through ports that attach to the system unit's microprocessor buses.

Memory devices: They are used to save programs and data.

Buses: They connect the microprocessor to its memory and to the adapters that enable connection to other devices through their ports or expansion slots. The buses inter-connect the hardware components of the microcomputer. The system's buses define system performance and adapter compatibility.

Adapters: They enable the microprocessor to communicate with and control I/O or storage devices. The adapters are a set of hardware circuitry that attach to the system's buses and convert each bus to an interface port supporting the attachment of specific I/O device. For example, a serial port adapter attaches to a bus and it has a COM port where a modem can be attached. The adapter is thus the bridge between the microcomputer bus and the device that needs to be attached to the microcomputer. Adapters may be "integrated" on the motherboard or adapter boards that plug into expansion slots.

Ports: The ports of the adapters support the attachment of I/O devices.

Expansion slots: They are physical connectors where adapters can attach to the buses so that the microcomputer can be configured with a variety of adapters. Expansion slots permit easy expansion or upgrading of a system with new I/O devices and memory.

Low-speed storage devices: They store data and programs for long-term. This includes nonvolatile magnetic media and optical media storage such as hard drives, CD-ROMs, and floppy disks

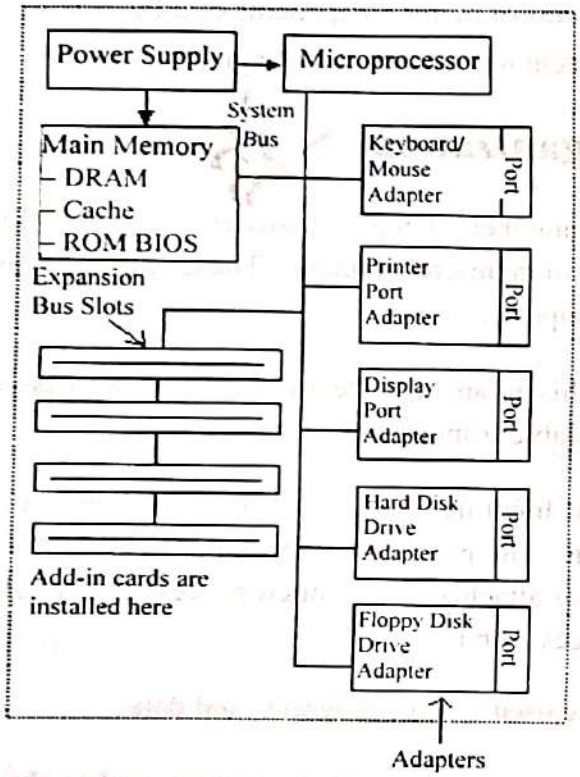


Figure 4.2: Organisation of a PC system.

Display

The main output device for a microcomputer is the display device. In most microcomputers the display device is a Cathode Ray Tube (CRT). In notebooks, it is often a flat panel device known as Liquid Crystal Display (LCD). The term monitor is used to describe the CRT display. This display monitor is very similar to a TV, but is designed to be viewed at much higher rates, typically more than 72 frames per second. The TV displays images at only 30 frames per second.

The PC systems today use 14-, 15-, and 17-inch displays. Larger displays are available but are very costly. High resolution and high refresh rates are needed to display computer generated text and images that do not flicker when viewed at close range. At preset PC

displays are able to display images with pixel resolutions in the 1024 × 768 range. This means 768 horizontal lines where each line contains 1,024 picture elements or pixels. A pixel is the smallest picture element that can be created and displayed by the display monitor.

The monitor is attached to the computer through a short cable attached to the system unit's display adapter port. The display adapter attaches to the system bus. The display adapter is either integrated on the motherboard with the processor or installed as a separate adapter board in one slots of the expansion of the motherboard.

Keyboard

Keyboards are sold with adapter plugs that can connect the keyboard to the keyboard port. An 8-bit microprocessor inside the keyboard is programmed to continuously scan the keyboard and report key presses and releases to the keyboard adapter. A keyboard adapter actually contains another 8-bit microprocessor that communicates with the keyboard's microprocessor and sends the keystroke to the main processor through the system bus.

Printer

A printer is attached to the system unit through a parallel port, also called the printer port. In a parallel port, data are transferred 8 bits at a time in parallel, as compared to the keyboard port where data are transferred in a serial mode, one bit at a time. The printer or parallel port is sometimes referred to as Centronics printer port. Early printers manufactured by a company called Centronics defined a parallel port interface for their printers and this interface was selected by IBM on the original PC.

Most microcomputers come with parallel or printer port as standard; sometimes the parallel port adapter is integrated on the systems unit's motherboard, and sometimes it is on an adapter board installed in one of the expansion bus slots. To save expansion slots, the printer port function may be included on a multi-I/O port adapter board supporting several commonly used ports and adapters.

Three basic types of printers are popular for microcomputers. They are wire matrix impact printers (often called dot-matrix printers), laser printers, and ink jet printers.

Mouse

The mouse is now the preferred input device for navigation through an application. The mouse is attached to the system unit in different ways. On most microcomputers, the

mouse is attached using one of the two asynchronous communications ports, or COM ports. The mouse can also be installed in one of the expansion slots using a mouse adapter board. The mouse detects motion on a surface and sends tracking information to the mouse port.

Other Ports and Adapters

Most microcomputers today come with two serial communication (COM) ports and a game port. Often one of the serial communications ports is used to attach the mouse. These serial ports support data rates up to 115 kilobits/second using an asynchronous start/stop serial protocol. This protocol frames the serial data that are transmitted to the interfaces. The electrical interface is called an RS-232-C standard interface. The two COM ports are accessible at the rear of the system unit as either 9-pin or 25-pin D connectors. The COM ports are typically created using a UART (Universal Asynchronous Receiver Transmitter) chip attached to the processor's bus. Sometimes the chip is integrated on the system unit's main board, but most often it is installed on an adapter board in one of the bus expansion slots. Often it is combined with other adapters on a single board called a multi-I/O adapter board. A game port connector is often on the same multi-I/O adapter board.

Power Supply

The PC power supply converts the standard wall outlet 220 volt AC to DC voltages to run the electronics and peripheral devices in the microcomputer. This power supply unit also supplies AC voltage to the systems display monitor through an outlet on the power supply. The power supply powers the keyboard and mouse through the interface ports and cables. The power supply converts the AC voltage to four DC voltages: +5 volts, +12 volts, -5 volts and -12 volts. Today most desktop units come with power supplies rated at between 150 and 220 watts. Tower designs are typically rated at 200 to 400 watts. If a large number of hard drives and adapter boards are to be installed in a tower system a minimum of 275 watts should be sufficient.

Cooling

In most desktop and mini-tower systems cooling is provided by the power supply fans. This fan is used to cool both the power supply and the system unit. In larger tower systems, separate cooling fans may be necessary. One fan cools the power supply and the second fan cools the motherboard and the adapter boards. With the newer high-performance CPUs often a fan is attached directly to the CPU.

Floppy Disk Drives

All microcomputers with the exception of some notebooks and palmtop devices, support

one (or more) floppy disk drive (FDD). The floppy disk is a data distribution medium because it is removable after it is read or written. Data and programs can be moved from system to system by storing them on floppy disks. Also, software is sold or distributed on floppy disks. Early floppy disks were 5.25 inches in diameter and used to store from 160KB to 1.2MB of data. A smaller 3.5-inch disk introduced later stores up to 2.88MB of data (1.44MB is the common one). Most PCs are equipped with 3.5-inch floppy disk dives. Most PCs come with floppy disk drive adapters that support a minimum of two drives.

Hard Disk Drives

The hard disk drive (HDD) is used for high-speed access and storage of programs and data. It is a nonvolatile rotating medium similar to the floppy disk. However, it is not flexible, hence the name hard disk. It is capable of storing much more data at significantly higher speeds than the floppy disk drives. Today's typical hard drive stores from 8GB to more than 40GB of data with access times below 10 milliseconds. Hard drives come in several package sizes with disk media or platter sizes from 8 inches to under 1.8 inches in diameter. Most microcomputer hard drives are 3.5 inches in diameter, enabling them to fit in the same drive bays as 3.5-inch floppy disk drives. Hard drives attach to a microcomputer through an IDE (Intelligent Drive Electronics) or SCSI (Small Computer Systems Interface) adapter port.

CD-ROM Drives

A CD-ROM drive is capable of storing over 600MB of data on a single 5-inch disk. CD-ROMs are suitable for distribution of large software packages and can contain all types of multimedia data. The CD-ROM technology is low cost and easy to produce. In future it will be possible to save data or programs on a CD-ROM in real-time, similar to a floppy disk. CD-ROM drives can be installed in the same size drive bays as a 5.25-inch floppy disk drive. The CD-ROM drive interfaces to the microcomputer through one of several port adapters. Often, a SCSI adapter comes with the CD-ROM drive, or it can use one that is often included on a sound board. However, many CD-ROM drives are now capable of being attached to the IDE adapter port. The IDE Port adapter standard has recently been extended to support additional IDE ports with CD-ROM attachment capability.

Microcomputer System Unit

Most of the intelligence and functioning of a PC system resides in the system unit. All external input/output devices are attached to the system unit and upgrade and expansion features are installed in the system unit. The system unit is the heart of the

microcomputer Figure 4.3 is a view of the inside of a desktop microcomputer system unit. The drive bays accept hard disk drives, floppy disk drives, and CD-ROM drive of various physical sizes. The motherboard is installed on the base of the system unit. The processor, cache memory, system DRAM, core logic chip set, integrated adapters and ports, BIOS ROM, and expansion slots are mounted on the motherboard. What is actually in a microcomputer depends on the PC's packaging style and its bus architecture.

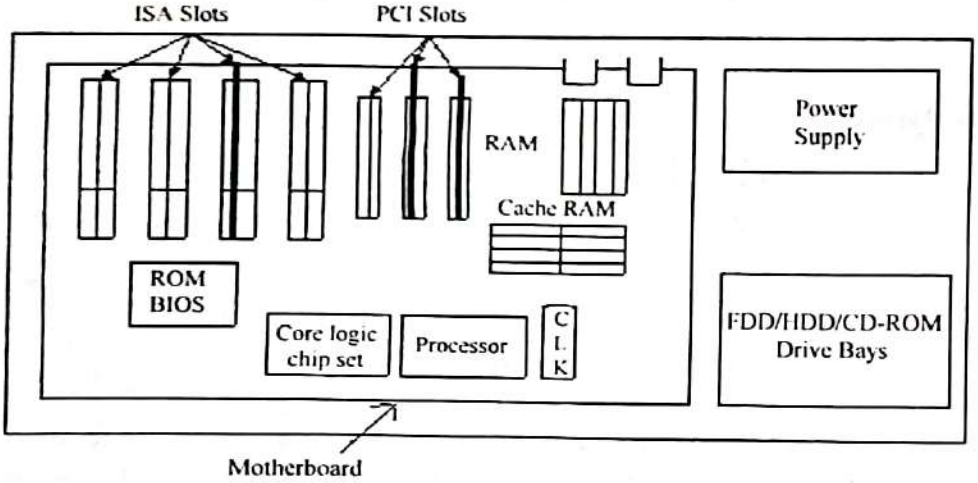


Figure 4.3: Inside a typical desktop microcomputer.

2.3 Packaging Styles

Microcomputer system units have evolved to a number of packaging styles, primarily to support different levels of expansion capability and ergonomic needs of the users.

For desktop designs the packaging changed very little over the years. Developers have maintained the basic design and simply dress up the systems with unique front designs and custom colours and logos. The basic layout and partitioning of function usually remains intact. Other mechanical packages have evolved to meet specific needs not addressed by the basic desktop design. The tower package added significant expansion capacity. This includes space for additional disk drives and hard drives and in some designs, additional bus expansion slots. Tower designs often had larger power supplies and additional cooling fans to support these expansion features.

There are a number of packages supporting portable microcomputers. These range from the large, barely transportable systems to the very small palmtop devices. The basic concept of a portable system is to integrate all the functions into a single device. Some compromises have to be made to achieve the size and weight desired. Displays are

usually smaller flat panel devices, expansion slots are limited, and keyboard size and layout are reduced. In the smallest designs, floppy and hard disk drives are replaced with other more compact nonvolatile memory technologies.

Notebooks are partitioned into two basic categories: notebooks and sub-notebooks. The primary difference is the presence or absence of a floppy disk drive and the smaller screen in the sub-notebooks. Sub-notebooks seem to be losing some of their popularity as notebooks become lighter, approaching the weight of a sub-notebook. Also contributing to their decline is the difficulty of typing on small keyboards and smaller screen sizes.

2.4 Exercise

Multiple choice questions

1. Most desktop units come with power supplies rated at between

   a. 100–200 watts
   b. 150–220 watts
   c. 200–300 watts
   d. 400–500 watts

2. The Hard drive attaches to a microcomputer through

   a. IDE port
   b. COM port
   c. Printer port
   d. Centronics port

3. Most microcomputers today come with

   a. Two serial ports and a game port
   b. One serial ports and two game ports
   c. One serial port and one game port
   d. None of these

4. SCSI means

   a. Small communication system interface
   b. Small computer support interface
   c. Small computer systems interface
   d. Small communication support interface

5. IDE means

   a. Intelligent drive electronics
   b. Intelligent data electronics
   c. Intelligent delay electronics
   d. None of these

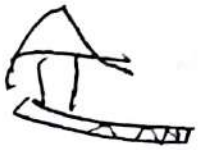
Questions for short answers

6. Explain the terms: computer bus, adapter, port, and expansion slot.

7. What is the system unit of a PC and what major parts does it contain?
8. List the popular packaging styles of PCs.
9. Explain the term 1024 × 768 indicating the resolution of a PC monitor.
10. Write short notes on: CD-ROM drive and tower case PC?

Analytical questions

11. Draw a simplified block organization of a PC showing the system bus, adapter ports, main memory, microprocessor and other items.
12. What is the function of the power supply unit of a PC? Describe its main characteristics.
13. Distinguish between HDD and FDD and describe their main characteristics.



Lesson 3: Bus Organized Architecture

3.1 Objectives

On completion of this lesson you will know:

- Basic elements of a microcomputer and their interconnection
- Address, data and control buses and their functions
- Microcomputer system buses: XT, AT, ISA, EISA, Micro channel, VL and PCI
- Typical organization of PCI/ISA bus system.

3.2 Basic Microcomputer Elements

A microcomputer contains many elements including the microprocessor. The microprocessor is typically a single large scale IC chip that contains the control and arithmetic circuits of a microcomputer. The RAM and ROM devices constitute the main memory. The RAM section consists of one or more IC chips. This section of memory is used to store programs and data that change during the course of operation. It also stores intermediate and final results of operation during execution of programs.

The ROM section, consisting of one or more IC chips, is used to store data and instructions that do not change. The input and output interface circuits, are required to allow the I/O devices to communicate with the rest of the computer. In some cases, the interfaces are the IC chips which interface the microprocessor to a variety of I/O devices. In other cases, the interface circuits are contained in a small circuit board popularly known as interface card or adapter.

3.3 The Buses

The microcomputer has three local buses. These are:

- Address bus
- Data bus
- Control bus

The buses connect the CPU (microprocessor) to each of the memory and I/O devices. The CPU is involved in sending or receiving information to or from memory location, input or output device and secondary memory device (FDD or HDD). When the CPU sends data to a device or memory, it is called WRITE operation and when the CPU receives data it is called READ operation. The function of buses are described now.

Address Bus

The address bus is unidirectional, information flows over it only in one direction, from the CPU to the memory or I/O devices. The CPU generates addresses on the lines of the address bus. Each of the addresses corresponds to one memory location or one I/O device. When the CPU wants to communicate with (READ or WRITE), certain memory location or I/O device, it places the appropriate address on its address output. These address is then decoded to select the desired memory location or I/O device. This decoding process usually requires address decoder circuits.

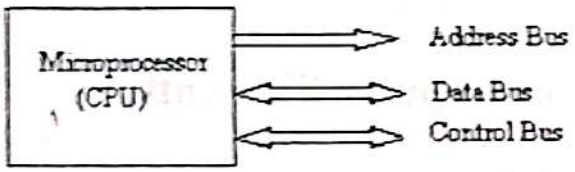


Figure 4.4: Microprocessor local buses

Data Bus

The data bus is bidirectional and data can flow to and from the CPU through it. The data bus can be either input or output depending on whether the CPU performs a READ or a WRITE operation. During READ operation the data bus receives data that have been placed on the data bus by the memory or I/O device selected by the address. During WRITE operation the data bus acts as the output and places data on the data bus which are sent to selected memory location or I/O device.

Control Bus

The control bus consists of a set of signals that is used to synchronize the activities of separate microcomputer elements. Some of these control signals are sent by the CPU to the other components to tell them the type of operation in progress. The I/O devices can send control signals to the CPU. Read/Write, Reset, interrupt are examples of control signals used in a microcomputer.

3.4 The System Bus

The control lines, the address bus and the data bus constitute the system bus and are guided in parallel to the bus slots. Several bus architectures used in microcomputers are: AT, ISA, EISA, IBM Micro channel, VESA Local Bus, PCI etc. Figure 4.5 illustrates different system bus architectures and the branches represent processor types that have developed over the years.

The term XT stands for eXtended Technology. It is the successor of the IBM PC with an 8086 processor and an internal 16-bit data bus. The term AT stands for advanced technology. It is the successor of the PC XT with 80286 microprocessor and 16-bit slots. The AT bus developed by IBM is commonly known as ISA which stands for Industrial Standard Architecture. The ISA bus has a speed of 8MHz, and a maximum throughput of 8MB per second. EISA is the abbreviation of Extended ISA. EISA defines a 32-bit extension for the ISA or AT bus to integrate the 80386/80486 32-bit processors. EISA is downward compatible with XT and AT buses. XT and ISA adapters can be used with EISA bus. At a bus speed of 8MHz, EISA can achieve a maximum throughput of 33MB per second.

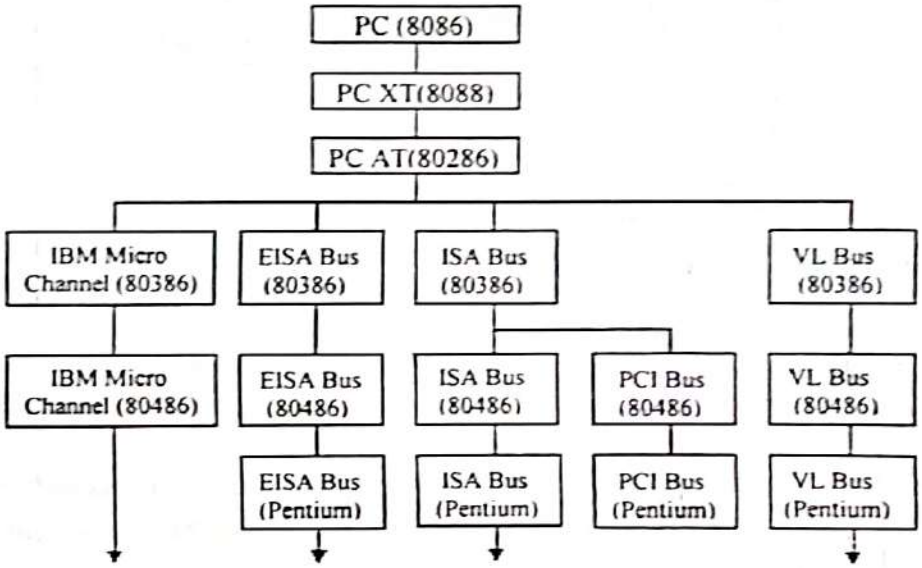


Figure 4.5: PC system buses.

Micro channel is a bus system of IBM PS/2 personal computer. This bus is designed for an 8-bit, to 32-bit data bus and it supports multitasking operating systems on a hardware level. Unlike EISA, the micro channel is incompatible with PC XT and ISA bus.

VL bus is also known as VL local bus. It is an abbreviation of VESA local bus, a bus architecture introduced by Video Electronics Standard Association (VESA). The VL bus is a 32-bit bus, running at either 33 or 40 MHz. The maximum throughput is 133MB per second at 33MHz or 148MB per second at 40MHz: The most common VL bus adapters are video adapters, hard-disk controllers and network interface cards.

3.5 The PCI Bus

The term PCI stands for peripheral component interconnect. The PCI specification

introduced by Intel defines a local bus that allows up to ten PCI complaint expansion cards to be plugged into the computer. One of these ten cards must be the PCI controller card, and others are video card, network interface card, SCSI interface or any other basic I/O function.

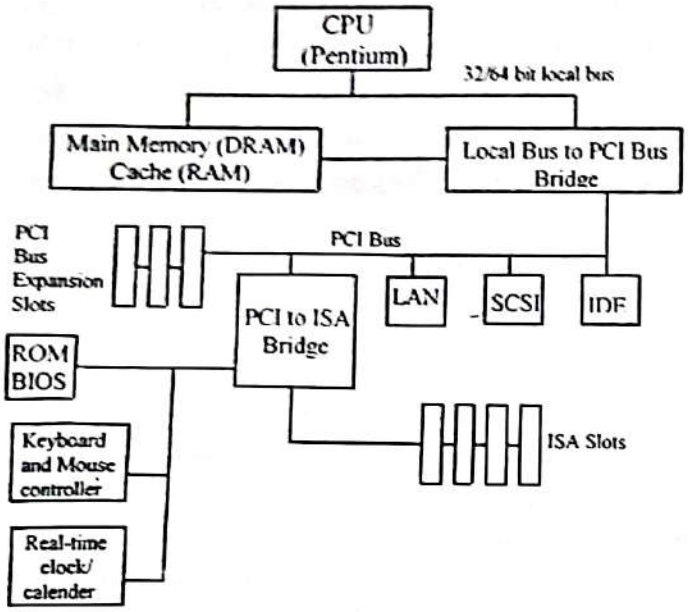


Figure 4.6: Typical PCI / ISA bus system.

The PCI controller exchanges information with the computer's processor as 32-bit or 64-bit words and allows intelligent PCI adapters to perform certain tasks concurrently with the main processor.

PCI can operate at a bus speed of 32MHz, and can manage a maximum throughput of 132MB per-second with 32-bit data or 264MB per-second with 64-bit data path. Unlike the VL bus, PCI is processor independent. Figure 4.6 shows the typical PCI/ISA bus system.

With the exception of IBM micro channel bus, all PCI and VESA local bus systems support ISA slots. EISA system also support ISA slots as part of their basic design.

3.6 Exercise

Multiple choice questions

1. ISA stands for

   a. Institute standard architecture
   b. Industrial standard adapter
   c. Industrial standard architecture
   d. Industrial standard association

2. Which one is the fastest bus?

   a. ISA
   b. EISA
   c. PCI
   d. VL

Questions for short answers

3. What do you understand by system bus?
4. List the popular system buses for microcomputers.
5. Give the main characteristics of the PCI bus system.
6. What are the local buses?
7. What is Micro channel architecture

Analytical questions

8. Draw a bus organized architecture of a microcomputer and explain its operation.
9. Give the main characteristics of address bus, data bus and control bus of the CPU of a microcomputer.
10. Draw the block diagram of a typical PCI/ISA bus system showing the major components which can be connected to them.

Lesson 4: Working Principle of a Microcomputer

4.1 Objectives

On completion of this lesson you will know:

- How the CPU (i.e. microprocessor) executes an instruction
- Execution of a sample program
- Instruction execution and fetch-execute cycle.

4.2 CPU and Instruction Execution

The CPU controls the operation of a microcomputer. It fetches binary coded instructions from main memory, decodes the instruction into a number of simple actions and executes them. The arithmetic logic unit (ALU) of the CPU performs add, subtract, OR, AND, invert or exclusive-OR operations when instructed. The CPU contains an address counter, called instruction pointer (IP) which is used to hold the address of the next instruction or data to be fetched from memory. The CPU also contains general purpose registers, which are used for temporary storage of binary data, and circuits for generating control signals.

4.3 Execution of a Simple Program

Let us suppose that the only three instructions of a sample program are:

1. Input a value from a keyboard connected to a port at addresses 07H.
2. Add 6 to the value read in
3. Output the result to a display connected to a port at address 05H.

Let us assume that the CPU fetches instructions and data from memory one byte at a time. We further assume that the binary codes for the instructions are stored in memory locations starting at address 00100H. The binary codes that would be required to execute this program on Intel 8086 processor would be as follows.

Memory Address	Contents (Binary)	Contents (Hex)	Operation
00100H	11100100	E4	Input from
00101H	00000111	07	Port 07H
00102H	00000100	04	ADD
00103H	00000110	06	06H
00104H	11100110	E6	Output to
00105H	00000101	05	Port 05H

The CPU of a microcomputer fetches an instruction from memory decodes the instruction to determine the actions to be taken and execute the instruction by carrying out the actions. The sequence of actions for the execution of the three instructions of the programs is given below.

1. a) CPU sends address of the first instruction to memory
   b) CPU sends memory read signal to read from memory
   c) Instruction byte is sent from memory to the CPU through the data bus.

2. a) CPU addresses next memory location to get rest of the instruction
   b) CPU sends memory read control signal to read from memory
   c) Port address byte is sent from memory to CPU through data bus.
   d) CPU sends port address on address bus
   e) CPU sends read signal to enable part 07H
   f) Data from port is sent to CPU through data bus.

3. a) CPU sends address of next instruction to memory
   b) CPU sends memory read signal to read from memory
   c) Instruction byte from memory is sent to CPU through data bus.

4. a) CPU sends next address to memory to get rest of the instruction.
   b) CPU sends read control signal to read from memory
   c) Number 06H from memory is sent to CPU through data bus.

5. a) CPU sends address of next instruction to memory.
   b) CPU sends memory read signal to enable memory.
   c) Instruction byte from memory is sent to CPU through data bus.

6. a) CPU sends next address to get rest of instruction
   b) CPU sends memory control signal to enable memory.
   c) The port address byte 05H is sent to CPU from memory through the data bus.
   d) CPU sends port address (05H) to the address bus.
   e) CPU sends result to the port (05H) through the data bus.
   f) CPU sends write signal to enable the port 05H.

All the steps described above may seem like a great deal of work. But even a simple computer can execute such steps in a few microseconds. In the process of execution of program a microcomputer fetches each instruction in sequence decodes the instruction and executes it. The Fetch-execute cycle is shown in Figure 4.7.

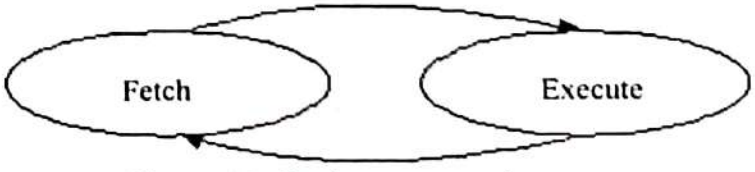


Figure 4.7: Fetch execute cycle.

4.4 Exercise

Questions for short answers

1. How the CPU executes an instruction?
2. Explain how the processor reads an instruction from the memory.

Analytical questions

3. What do you understand by fetch-execute cycle? Explain fetch-execute cycle with the help of an instruction execution.

Lesson 5: Motherboard and Adapter

5.1 Objectives

On completion of this lesson you will know:

- The motherboard and its main components
- Adapter boards, their functions and uses
- Display adapters with characteristics and standards
- Types and characteristics of SCSI adapters
- Other adapters used in microcomputers.

5.2 Motherboard and its Components

The motherboard is the main printed circuit board in a computer. It contains the CPU, appropriate support chips, device controllers, memory and also expansion slots to give access to the computer's internal bus.

The motherboard is the heart of the system. It is called motherboard because all other features and peripherals plug into it to operate. Figure 4.6 (lesson 3) presents the contents of a typical PCI/ISA motherboard. Some of its major components are described below.

Microprocessor and Upgrade Socket: Most systems do not solder the microprocessor permanently to the motherboard. The processor is in a socket that permits the processor's easy removal from the motherboard. The socket is generally designed to support several types of microprocessors. This enables a system to be upgraded in the future with more powerful processor by simply replacing the older with a better one.

Cache Memory and DRAM Controller: The main memory DRAM (dynamic RAM) is physically packaged on small circuit boards called SIMM or DIMM. SIMM stands for Single In-line Memory Module and DIMM stands for Dual In-line Memory Module. The DRAM SIMMs are installed in SIMM sockets on motherboards. For timing, interface and refresh of DRAM, a DRAM controller device is needed between the local bus and the DRAM chips.

The main system memory (DRAM) can not keep up with the performance of the processor and becomes a performance bottleneck. To overcome this limitation, a second smaller memory, called cache memory, is placed between the microprocessor and the

system memory. This memory (SRAM) is much faster and can keep up with the processor's performance. The circuits that manage the cache memory is called the cache controller. The DRAM controller and cache controller functions are typically in a single chip on the motherboard. The high-speed cache memory is also on the motherboard and can be expanded in size by adding chips into empty sockets on the motherboard.

PCI Bus Bridge Chip: This chip converts the microprocessor's local bus to the PCI bus. This chip is different for different microprocessors. In modern microcomputers the "core logic" chip set includes the microprocessor local bus to PCI bridge chip, the system memory controller and cache controller chip and the PCI to ISA bridge chip.

Keyboard/Mouse Controller and Ports: A motherboard generally contains a keyboard and a mouse controller device and connector ports. The keyboard is attached to the motherboard through a serial interface cable. The keyboard clock and data interface signals attach to a micro-controller chip on the motherboard. This chip in turn attaches to the main microprocessor bus. Figure 4.8 illustrates how microcomputer's keyboard attaches to the system bus. The micro controller does not actually send keyboard character codes, it actually sends special 8-bit scan codes, that are translated by the system unit's software to character codes.

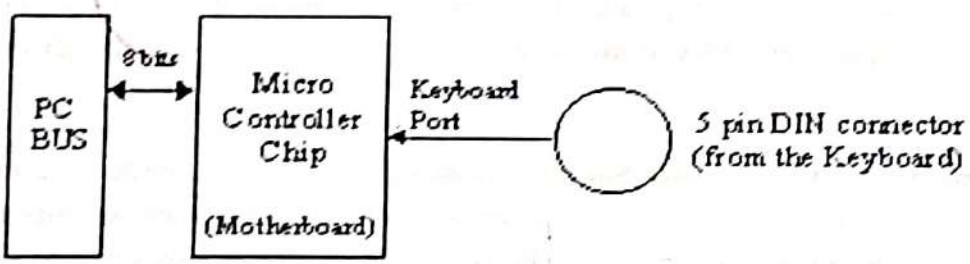


Figure 4.8: Keyboard /mouse adapter.

ROM BIOS: BIOS stands for Basic Input/Output System. The ROM BIOS is a read only memory that contains software. Its primary purpose is to load the basic operating system software from the system's boot or start up device and perform power on system tests. The operating system boot device is usually the system's hard disk or the LAN port. The read only memory is about 128KB in size. Most of the times its contents are moved to faster DRAM memory for execution. Some BIOS also incorporates features to support system configuration, detection and resource allocation as defined by the new plug and play specifications.

Real-time Clock and Calendar: Every motherboard contains a chip called a real-time clock/calendar. This device also contains a small nonvolatile memory (normally 64 bytes). This device always runs even when the system is powered off. The power is supplied by a

small battery installed on the motherboard. This device supplies date and time to the operating system and stores basic system configuration data in the nonvolatile memory. Loss of this data by disconnection of the battery or a low battery condition temporarily cripples the system by destroying key information for configuration.

PCI to ISA Bridge: The PCI motherboard contains a bridge chip. This chip performs transformation of the PCI bus to the older ISA bus. This chip contains the Intel core logic including DMA chips or their equivalents.

5.3 Adapter Boards

An adapter board is a printed circuit board that plugs into an expansion bus slot to provide added capabilities. The common adapters for the microcomputer include display adapters, memory expansion adapters, input/output adapters and adapters for other devices such as internal modems, CD-ROMs or network interface cards. One adapter can often support different devices. For example, an input/output adapter may support one parallel port, a game or joystick port, and several serial ports. Some microcomputers incorporate many of the functions previously performed by individual adapters on the motherboard.

Multi-I/O Port Adapter Board: This board supports a number of peripheral interface adapters. Typically all the device adapters and ports on the multi-I/O port boards are integrated into a single chip called a super I/O chip. The adapter functions which may be included in the multi-I/O board are: floppy disk drive adapter and port, the IDE adapter

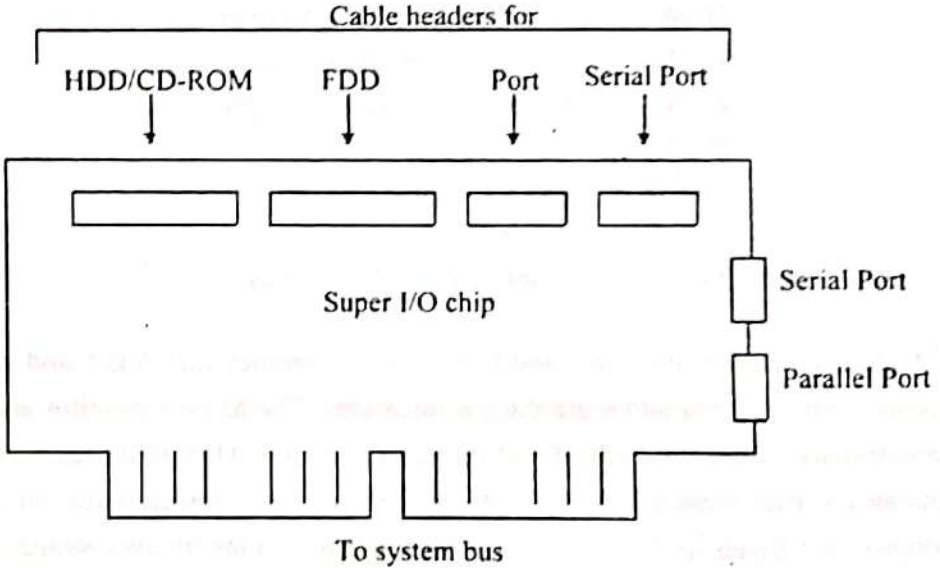


Figure 4.9: Typical PC multi-I/O adapter board.

and port for hard disk drive, two serial communication (COM) ports for RS 232C or standard asynchronous transfer protocols, game controller adapter and port, parallel printer adapter and port etc. (Figure 4.9).

There have been several enhancements in the IDE interface that supports more than two drives and a significant increase in interface data transfer performance. These enhancements are part of an improved IDE specification called EIDE (Enhanced IDE). It is possible to attach CD-ROM drives to the EIDE port.

5.4 Display Adapters

The display adapter and the video monitor work together. The video display adapter is a small card that fits into a spare slot of the microcomputer. In some microcomputers, the electronics for this card are integrated in the motherboard. In conjunction with the monitor, the display adapter interprets the data stream of the system bus and formats it into a data stream that the monitor can understand and display.

The basic type of video adapter used now are VGA (Video Graphics Array) and SVGA (Super VGA). They can display information in text-form or in graphical form. The graphical standards for display adapter developed over the years are: MDA, HGA, MGA, CGA, EGA, VGA, MCGA, SVGA, XGA, 8514/A etc. The full description for these display standards are:

MDA : Monochrome Display Adapter
HGA : Hercules Graphics Adapter
CGA : Colour Graphics Adapter
EGA : Enhanced Graphics Adapter
VGA : Video Graphics Array
MCGA : Multi-Colour Graphics Array
SVGA : Super VGA
XGA : Extended Graphics Array

The SVGA includes higher resolution and better colour vibrancy than VGA and is the best display standard for high resolution graphics applications. The 8514/A monitor and display adapters are primarily used for medical and seismic imaging, CAD/CAM presentations and other applications that require a very high resolution. The characteristics of common display adapters are given in Table 4.1. The present day display adapter supports all the modes of the preceding display adapters.

Table 4.1: Characteristics of display adapters

Adapter	Maximum Resolution/Colours
CGA	640 × 200/2
MDA	720 × 350/3
EGA	640 × 350/64
HGA	720 × 350/3
MCGA	640 × 480/2
VGA	640 × 480/262,144
SVGA	up to 1024 × 768/256
8514/A	1024 × 768/262,144
XGA	1024 × 786/262,144

Figure 4.10(a) shows a display system without using a graphics coprocessor and Figure 4.10(b) shows the scheme using a graphics coprocessor. The speed of the graphics coprocessor limits the speed at which monitor can be refreshed. Video controller cards also include video RAM(or VRAM). VRAM is dual-ported, that is can send screenful of data to the monitor while at the same time receiving next screenful of data from the CPU.

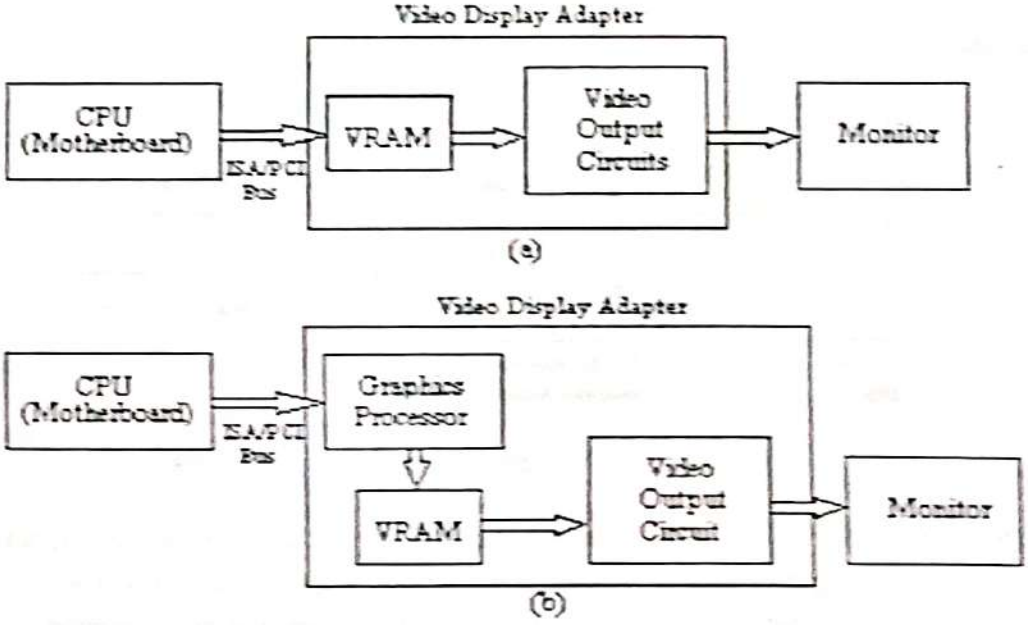


Figure 4.10: PC display systems.

The CPU sends information of every pixel on the screen to the video display adapter. At the resolution of 1024 × 768, there are 786432 pixels to control. For 256 colours, each pixel requires one byte of information. But for maximum amount of colour each pixel require 24 bits(millions of colours possible) and the CPU must send 1024×768×24 bits, that is 1024×768×24 bytes which is actually 2,359,295 bytes, to the monitor for each screen.

5.5 SCSI Adapter

SCSI stands for small computer system interface. It is used to connect a microcomputer to several peripheral devices using just one port. Devices connected in this way are said to be "daisy chained" together, and each device must have a unique identifier or priority numbers.

An SCSI adapter can be installed in an IBM compatible computer as a single expansion board. SCSI is often used to connect hard disks, tape drives, CD-ROM drives, mass storage media, scanners and printers. Figure 4.11 shows an arrangement for SCSI ports on a IBM compatible computer.

There are several different SCSI interface definitions: SCSI-1, SCSI-2, wide SCSI and SCSI-3. The SCSI-1 is a 8-bit interface with a maximum data transfer rate of 5 MB per second. The SCSI-2 is 16-bit or 32-bit wide (called wide SCSI) and its data transfer rate is 10 to 20 MB per second (also called fast SCSI). Wide-Fast SCSI has 16-bit data bus and maximum data transfer rate of 20 MB per second. SCSI -2 is backward compatible with SCSI-1.

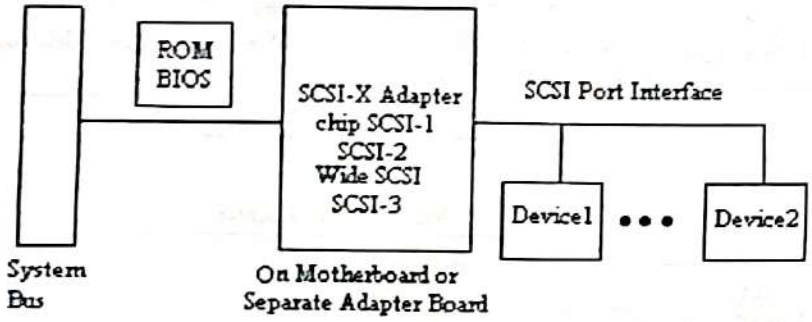


Figure 4.11: SCSI ports on personal computers.

The SCSI-3 definition increased the number of connected peripherals from 7 to 127, increased cable lengths, added support for a serial interface and for a fiber optic interface. For SCSI-3 data rats in excess of 100 MB per second are possible.

5.6 Other Adapters

The other adapters used in microcomputers are: PCMCIA (Personal Computer Memory Card Interface Association) cards for notebook computers, MPEG (Motion Picture Expert Group) cards for full motion video playback, sound cards for audio applications, network adapters for networking applications, modem cards for telephone lines etc. Any of these cards can be plugged into the bus slot for a peripherals device.

5.7 Exercise

Multiple choice questions

1. Cache memory is placed between
   a. microprocessor and main memory
   b. main memory and secondary memory
   c. HDD and FDD
   d. none of these

2. EIDE means
   a. extra IDE
   b. extended IDE
   c. enhanced IDE
   d. increased IDE

3. Parallel printer adapter and port is also known as
   a. entromics printer port
   b. adapter port
   c. printer control adapter
   d. none of these

Questions for short answers

4. What is the SIMM or DIMM?
5. Describe the functions of PCI bus bridge chips.
6. Describe the functions of ROM BIOS.
7. What is the PCI to ISA bridge?
8. Describe the use of display RAM and RAMDAC.
9. Describe the function of a typical multi-I/O adapter.
10. Describe the use of MPEG adapter and modem.

Analytical questions

11. What is the motherboard? List the main components of a motherboard and briefly describe their uses.
12. What is the display adapter? List the standard display adapters and mention their resolutions.
13. Give the block representation of a display adapter and briefly explain its operation.
14. What is the SCSI adapter? Describe its characteristics.

Chapter 5

Input and Output Devices

Lesson 1: Input/Output Operations
Lesson 2: Input Devices
Lesson 3: Output Devices
Lesson 4: Other Peripheral Devices

Lesson 1:Input/Output Operations

1.1 Objectives

On completion of this lesson you will know:

- Input/output interfaces
- Input/output methods and communications
- Speed and efficiency of input/output operations
- Media for I/O devices.

1.2 I/O Interface

Communications between an input/output device and the microcomputer take place through an interface. The interface converts the data from a form used by a device to a form acceptable by the other. It must also adjust for speed differences between the processor and the input/output device. The interface circuits used in microcomputers correspond to the I/O control units of larger systems. Two general types of interface devices in use are:

- Serial interface, and
- Parallel interface.

Serial Port and Serial Interface

A serial port is associated with the serial interface and a serial input/output device is connected to the PC through the serial port. The serial port is simple in concept, it uses one line to send data, another line to receive data, and a few lines to regulate the flow of data. For its simplicity, the serial port is used to make a PC communicate with just about any device imaginable (modem, printer, plotter etc.). Common uses for a serial port are with mouse or modem. A serial port is not a very efficient way to transfer data. It can only send data in series, one bit of data at a time. This inefficient data transfer, however, is acceptable for a mouse, which transmits so little data that speed is not crucial. The serial port is often referred to as an RS-232 port. Both 9-pin and 25-pin connectors are used as serial ports.

Parallel Port and parallel Interface

A parallel I/O device is connected to the PC through the parallel port associated with the parallel interface. The parallel port has been synonymous with printer port since early days of microcomputers. A serial port sends data one bit at a time, a parallel port can send several bits of data across parallel wires simultaneously. A serial connection sends a single bit, a parallel port can send an entire byte. Some early printers and plotters used serial ports to communicate with printers. But today, graphics and scaleable fonts are common in printed documents and they require that vast amounts of data be sent to the printer, making a parallel port the only practical choice.

1.3 Input/Output Methods

There are three basic methods by which data can be read from or written to a peripheral device and RAM. These are:

- Programmed I/O
- Interrupt I/O, and
- Direct memory access.

Programmed I/O: In this technique, the microprocessor is in direct control of all transfers and other I/O operations. This is accomplished with specific input or output instructions. When an input operation is desired, the microprocessor simply issues an input command and awaits the arrival of the data at the bus. From the bus the data are moved to memory. Similarly, if an output operation is desired, the microprocessor transmits data to the bus and issues a command to the output device through the appropriate interface. Once the data transfer is initiated, the microprocessor must wait for its completion and the bus to be freed before beginning a new transfer. This method is commonly used in personal computers because it is effective and easily implemented.

Interrupt I/O: This is a device initiated transfer. An interrupt transfer involves the I/O device sending a request to the processor through interrupt input (INT, IRQ etc.) to inform the CPU that the I/O device is ready for data transfer. In response, the CPU interrupts the execution of its current program and jumps to a new program called interrupt service routine which contains instructions to transfer data to or from the interrupting device.

Direct Memory Address: This method is the fastest of the three methods. It requires a multi-bus architecture and allows the microprocessor to be bypassed completely. A special direct memory access controller is connected between RAM and an input or output device. This method is the most expensive of all and therefore is not used except with very high-speed input/output devices and only when absolutely necessary.

1.4 Speed and Efficiency

Applications of computers in business are characterized by limited computations and by the large volume of data that are input to, and output from computers. Handling volumes of data is one of the functions of the I/O devices, supporting a computer. These devices are responsible for sensing or reading data from the input media and converting the data into a form that the computers can understand and process. They are also responsible for converting computer stored data into information that can be read and understood by users.

I/O devices perform significant functions and form an indispensable part of a computer system. The faster one can input data to the computer, the more efficiently the computer will function and the faster output can be obtained. Thus it is essential to be familiar with the various I/O devices and know what devices are available for different applications and when to use each. Different media are used to record data associated with different I/O devices. Specific details of I/O devices are presented for purposes of comparison and should be examined with the purpose in mind. Such specific information is included in the specifications available from vendors.

1.5 Input Media

An input device of a computer-based information system is used to capture raw data and convert them into computer-usable form as quickly and efficiently as possible. The approaches devised to accomplish this can be categorized according to the type of material on which the data are recorded. These materials are called media. Although in few sophisticated systems data can be captured in very special forms, the following media are used most often:

- Paper medium
- Magnetic medium.

For many years, the most widely used media to record input data were paper-based products such as punched cards and punched paper tape. The operator of a keypunch machine transcribes data from a source document by punching holes in the cards via a keyboard according to a special code.

Paper-based input media are also used with scanners, connected to computers, that can "read" typed or handwritten data. The banking industry has long relied on paper-based input media. Cheques are printed using a special magnetic ink that can be scanned for magnetic retrieval of data. Optically oriented input devices are available for use with microcomputer systems. For example, even a small independently owned store may have a microcomputer at the check stand and a hand-held optical scanning device to read the product numbers on purchased items. The most widely used input media today are magnetic media in tape or disk form. The major advantages of magnetic media are their reusability, low cost, great flexibility for accessing the data during the input process, and high speed compared to punched cards and manual keyboard entry.

1.6 Exercise

Multiple choice questions

1. The efficient data transfer port is:

a. serial port	b. parallel port
c. RS-232	d. none of these.

2. Printer port is a:

a. serial port	b. parallel port
c. Rs-232	d. none of these.

Questions for short answers

3. What is an I/O interface?
4. Distinguish between serial port and parallel port.
5. Name the different I/O methods.
6. What are the most common input media?

Analytical questions

7. Describe the serial and parallel ports of computer systems.
8. What are the different I/O methods? Describe in detail.

Lesson 2: Input Devices

2.1 Objectives

On completion of this lesson you will know:

- QWERTY keyboard and Dvorak keyboard
- Reading, pointing and scanning devices
- Some other input devices.

2.2 Keyboard

The most common of all input devices is the keyboard. Microcomputers offer enhanced keyboards for easy entry of numbers. This is accomplished with a smaller group of keys known as a numeric keypad at the right of the keyboard. These keys generally consist of the digits, a decimal point, a negative sign, and an ENTER key. This type of keyboard is ideal for accounting operations which require a large volume of numbers to be entered.

A keyboard generally utilize integrated circuits to perform essential functions, such as determining binary code, to send to the CPU corresponding to each depressed key, switching between shifted and non-shifted keys, repeating a key code etc.

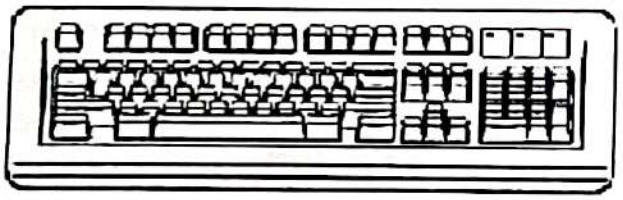


Figure 5.1: Standard keyboard

The keyboard arrangement provided as standard on most keyboards is the QWERTY arrangement, named for the six letters beginning the row at the top left of the keyboard. This arrangement was chosen intentionally to slow expert typists, since they type too fast to cause the keys on a mechanical typewriter to jam. Slowing down the typist was accomplished by scattering the mostly used keys around the keyboard, making frequently used combinations of letters slower to type. The QWERTY keyboard has been in use for nearly a century.

The Dvorak simplified keyboard arrangement, designed in 1932 by August Dvorak, is the result of extensive ergonomic studies. Dvorak noted that with the QWERTY keyboard arrangement, typists use the weakest fourth and fifth fingers of their left hand a large proportion of the time. Dvorak rearranged the keyboard so that the five more

frequently used vowels (a, o, e, u, and i) and the five most frequently used consonants (d, h, t, n, and s) were positioned on the home row where the fingers of the left and right hands rest, respectively. Thus, 70 percent of the typing is done on the home row. He then placed the next most frequently used characters in the row above the home row and the least frequently used characters in the row below the home row. This resulted in a reduction of finger movement of approximately 80 percent, and an increase in productivity of nearly 40 percent.

2.3 Reading Devices

A character reader is capable of accepting printed or typed characters from source documents and converting these data into a computer-acceptable code. Currently available high-speed character readers are capable of reading source documents at rates of up to several thousand documents per minute and are costly. The basic types of character readers are:

- Magnetic-ink reader
- Optical mark reader
- Optical character reader
- Hand written character reader
- Smart cards etc.

Magnetic-ink Character Reader

Magnetic-ink Character Recognition (MICR) was developed by the Stanford Research Institute of USA for use by the Bank of America. This system can read data prerecorded on checks and deposit slips with a special ferrite-impregnated ink. The magnetized characters can be read and interpreted by MICR equipment.

The American Bankers Association (ABA), in the mid-1950s, adopted the magnetic-ink character recognition (MICR) technology, as its primary means for processing checks. This technology involves reading of numeric characters and a few special symbols printed on cheques with magnetic ink. Figure 5.2 shows the layout of the MICR encoding on a personal cheque. The data are divided into five groups:

- Cheque routing symbol
- ABA transit number
- Customer account number
- Cheque number, and
- Cheque account.

The cheque amount in the bottom-right corner is keyed in manually at the branch where the cheque is received for processing. The advantages of the MICR system are:

- Human involvement is minimal, thus the potential for errors is small.
- It is fast, automatic, and reliable.

However, some human involvement is still required to encode the cheque amount in MICR characters thus, some room for error does remain.

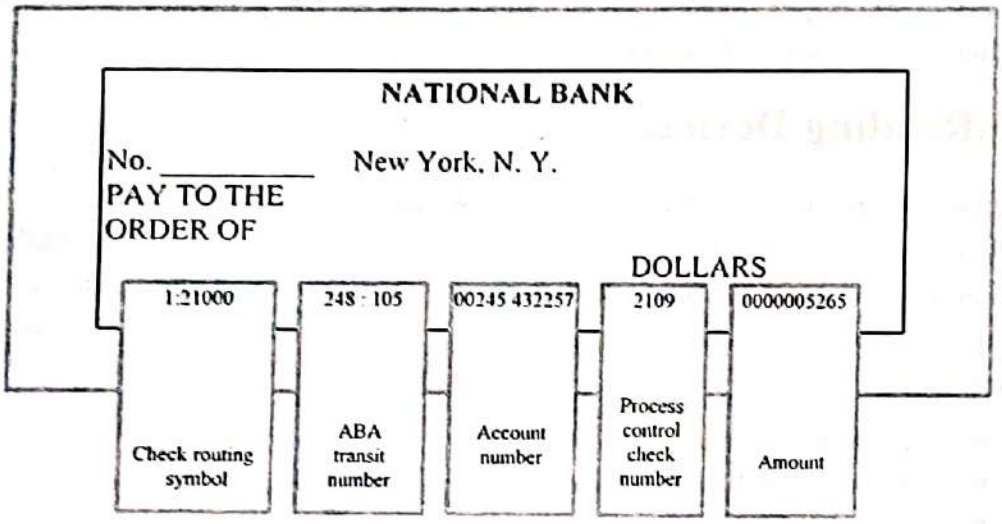


Figure 5.2: The layout of the MICR encoding on a personal cheque.

Optical Mark Reader

An optical mark reader (OMR) optically reads marks on carefully printed forms. Optical mark forms are relatively expensive, as they must be printed with exact tolerances. The most popular use of such devices is optical readers for scoring examinations in educational institutions.

Optical Character Reader

Optical character recognition (OCR) devices can convert data from source documents to a machine-recognizable form. Applications of optical scanning include billing, insurance premium notices, invoices etc. An OCR device can reliably read and interpret script or handwriting. However, to read handwriting certain general guidelines are observed when the data are written. Generally, optical character readers are limited with respect to hand-written characters and can only read handwritten digits and some symbols. Many OCR devices are available for reading typed characters, including digits, letters and some special characters. Not all printed characters can be read reliably on OCR readers. Generally, each reader is capable of reading only selected character styles.

Even if the character style and spacing are acceptable, errors can result from reading a character that is not written perfectly. To reduce errors, OCR devices generally compare the pattern read with the patterns of acceptable characters. The read character is assumed

to be the character whose stored pattern most closely matches the read pattern.

Handwritten-Character Reader

Data recorded by hands have dropped substantially over the last decades, however quite a bit of data are still recorded this way. As OCR technology advanced, designers felt that it would be possible to convert handwritten data into computer-usable form in much the same way that typewritten data are.

Devices with this capability are the most sophisticated and versatile of the OCR devices. Because handwriting varies widely, specific guidelines must be followed so that the OCR reader will interpret the character accurately.

Smart Cards

A smart card is small in size and to use it, the card is inserted into a special card-reading terminal and then a password is entered. The cards have microchips that can keep permanent records, which are updated each time the card is used. The transaction data stored on the card can later be read into the computer to update the user's bank records. Now a days smart cards are used to pay bills, buy merchandise, make phone calls, buy postal money orders, get examination results, store emergency medical information, and to perform some other activities. Electronic card readers are installed in stores, restaurants, post offices, phone booths, banks, and so on. Some manufacturers are already talking about a "super smart card" that will incorporate a keypad and a display unit along with the memory and processing capabilities.

2.4 Pointing Devices

Computer users frequently find it easier to point to something on a screen or at an item of text or graphical material they are entering into the computer. A number of devices are available to assist in fulfilling this need.

Mouse

A mouse is a hand-movable device that controls the position of the cursor on a screen. It has a box with buttons on the top and a ball on the bottom. The box is placed on a flat surface (or mouse mat), with the user's hand over it. The ball's movement on the surface causes the cursor to move.

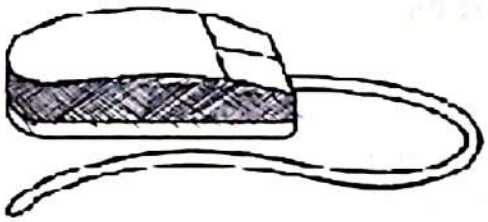


Figure 5.3: Mouse

The mouse has become most popular pointing device for graphic interfaces by windows. With the mouse, one can control the PC by pointing to images instead of typing in commands. However, mouse can never replace the keyboard, but it can supplement the keyboard by doing tasks such as moving and pointing to on-screen objects, task for which the cursor keys are ill-suited.

Light Pen

The earliest pointing device is the light pen. This device is placed close to a screen or monitor and turned on. A photo sensor inside the light pen detects the scanning beam sweeping back and forth across the screen. Accompanying circuitry converts the pen's reading into the position of the pen on the screen. Light pens are used to select items from a list or menu displayed on the screen and to draw graphic displays on the screen.

Digitizer Pad

A digitizer pad looks like a graph pad with a pointer. It functions like a light pen on a display screen except that the pad is mounted horizontally. As the pointer is moved on the pad, the corresponding point on the screen is illuminated. The digitizer pad is useful in converting graphic input, such as charts, graphs, and blueprints into patterns that can be stored by the computer.

Joystick and Trackball

Joysticks are used with video games for user input. These devices may also be used to move the cursor around a screen to facilitate input to a graphical display. A trackball is similar in operation to the joystick. It uses a billiard-sized ball to position the cursor. Several keyboard manufacturers have integrated them directly into their keyboards.

Touchscreen

Touchscreen detects the touch of a human finger. One popular technique used to detect the touch of a finger utilizes infrared light beams. In this technique, infrared light beams shine horizontally and vertically on the face of the screen. A pointing finger interrupts both horizontal and vertical beams, pointing its exact location.

2.5 Scanning Devices

Scanners

Scanners allow a PC to convert a drawing or photograph into code that a graphics or desktop publishing program can use to display the image on the screen, to reproduce the image with a printer. A scanner can also convert printed text into editable text with the

help of optical character-recognition software. The three basic types of scanners are:

- Sheet-fed scanner
- Flatbed scanner
- Hand-hold scanner.

These scanners differ from each other primarily in the way that the page containing an image and the scan head that reads the image move past each other. In a sheet-fed scanner, mechanical rollers move the paper past the scan head. In a flatbed scanner, the page is stationary behind a glass window while the head moves past the page, similar to the way a photo-copying machine works. Hand-held scanners rely on the human hand to move the scan head.

Each method has its advantages and disadvantages. The flatbed scanner requires a series of mirrors to keep the image that is picked up by the moving scan head focused on the lens that feeds the image to a bank of sensors. As no mirror is perfect, the image undergoes some degradation. But the advantage of a flatbed scanner is that it can scan oversized or thick documents, such as a book. With a sheet-fed scanner, the image is captured more accurately, but it is limited to scanning single, ordinary-sized sheets. A hand-held scanner is a compromise. It can scan pages in books, but often the scanning head is not as wide as that in either a flatbed or a sheet-fed scanner.

Bar Code Reader

A bar code reader is a device for scanning or reading a bar code. Bar code is printed code that consists of parallel bars of different width and spacing. The application most commonly observed is the coding on food and other goods that is read at the checkout counter and translated into a line of print on the bill showing product and cost. The information is also used to update stock records and provide sales statistics. In the United States the code used for this purpose is the Universal Product Code (UPC) and in Europe it is the European Article Numbering (EAN) code. The UPC decodes initially into two five-digit numbers. The first five identify the supplier and the next five are the item number within that supplier's range of goods. From this information the checkout terminal can access the details to be printed on the bill. The EAN code has a two-digit number to indicate country of origin, then the two five-digit numbers, followed by a check digit. The EAN arrangement simplifies the allocation of codes to suppliers. Only the two-digit code and the format need to be agreed internationally.

Other codes are used for shop-floor data collection, library systems, and monitoring the circulation of confidential document. The advantage of bar codes is that they can be

produced and read by relatively simple equipment.

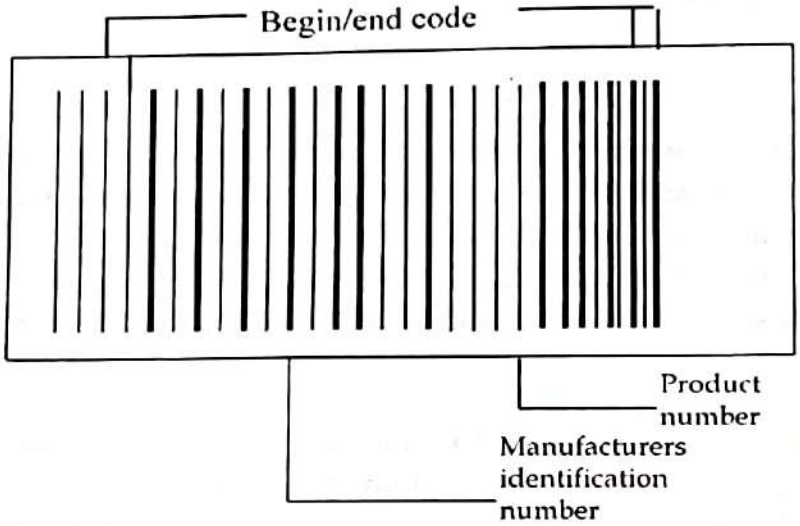


Figure 5.4: UPC type bar codes.

Digital Camera

A digital camera uses light-sensitive electronic technique instead of film to capture images. Instead of a photo album, one can store pictures in PC's hard drive or in a writable CD-ROM. Digital photography is not new, although most digitized photos have made their way to disk via conventional photography and scanners. Digital cameras that eliminate the scanners already exist. The technology is no mystery, and any of the cameras need to replace film cameras is to improve resolution and come down in price.

The film and electronic photographs have a lot in common. For example, a good photograph appears to be a seamless image of widely varying color, darkness, and contrast. It consists of the same elements found in electronic photography. Color film is made up of thin layers of chemicals that react to three different colors, just as PC displays are made of three colors. Chemical film has its own equivalent of a screen's pixels. The film uses the grains of silver compounds that react to light by clumping together to create tiny dots of black and white or different colors. The more and smaller the clumps, the finer is the film's resolution. On fine-grained film the naked eye can not see the clumps.

However, there is a great deal of variety in how digital cameras store the visual data once the shutter is snapped. The image can be stored in RAM, to the camera's own hard or floppy drives, to a PC Card, or through a direct connection to a desktop PC. Eventually, of course, all digital photos are stored up on a PC's hard disk. There, the images can be

manipulated, which is the digital equivalent of darkroom retouching. The images are then displayed or printed. The manipulation of digital photographs is a distinct jump from chemical photography, where retouching is a difficult job. Image editing software, such as Adobe Photoshop or Painter, can be used to improve the quality of photos.

2.6 Other Input Devices

Punched Card

Punched cards served as an input medium to automated computational devices. It has undergone little or no change, and most users have phased out and replaced it with the more efficient data entry media. Among the punched card devices still in use is the punched card reader. The reading of punched cards takes place at speeds, ranging from hundred fifty to more than two thousand five hundred cards per minute.

Key-to-Tape and Key-to-Disk Systems

In a key-to-tape system, data entered at a keyboard are recorded directly on magnetic tape. The magnetic tape used is similar to the tape cartridge or cassette used with home recorders. Accuracy is verified by placing the recording tape into a magnetic tape verifier and having the original data retyped. Magnetic tape encoders and verifiers are generally housed in the same physical unit. Errors detected are corrected simply by erasing the mistakes and substituting correct characters.

Voice Input Device

A voice input device, or a voice recognition system, converts spoken words into electrical signals by comparing the electrical patterns produced by a speaker's voice with a set of prerecorded patterns. If a matching pattern is found, the computer accepts this pattern as a part of its standard "vocabulary". This technology is also used by people who are not able to use traditional input devices. The biggest problems with this technology are limitations on the size of the computer's vocabulary, pronunciation differences among individuals, and the computer's inability to accept continuous speech. Several voice input units are currently available for use with microcomputers. Speaker-independent voice recognition systems are being designed so that the system need not be tuned to the voice of any one individual.

Touch-Tone Device

Touch-tone devices have wide spread use in many developed nations.Touch-tone devices can be used to send data to a computer. Touch-tone devices are also used to run credit card checks in retail stores. After appropriate keys are touched, the device sends data

over a phone line to a central computer, which then checks the data against files and reports credit information back to the store.

2.7 Exercise

Multiple choice questions

1. Increase in productivity using Dvorak simplified keyboard is about:

   a. 60 percent b. 30 percent
   c. 40 percent d. 50 percent

2. Which one is used for scoring examinations?

   a. MICR b. OMR
   c. OCR d. none of them

3. Which one is used with video games for user input?

   a. Touchscreen b. Mouse
   c. Digitize pad d. Joystick.

4. Touchscreen is usually used to detect the touch of a

   a. Human finger b. Pen
   c. Wooden stick d. none of them.

5. Which one is not a pointing device?

   a. OMR b. Mouse
   c. Touch screen d. Light pen

Questions for short answers

6. Briefly describe the advantages of Dvorak simplified keyboard.
7. What is the basic difference between OMR and OCR?
8. What is a mouse in a computer system?
9. Write down the application of digitized pad and touch screen.
10. What are the different types of scanners?

Analytical questions

11. Describe the keyboard as input device.
12. Describe basic types of character readers.
13. Name different pointing systems and describe any one of them.
14. Name the different type of scanners and mention their advantages and disadvantages.

Lesson 3: Output Devices

3.1 Objectives

On completion of this lesson you will know:

- Functions and characteristics of monitors
- Flat-panel display devices
- Characteristics of Printers and Plotters
- Some other output devices

3.2 Monitor

The monitor is the commonly used display device. The monitor utilizes a cathode ray tube (CRT). CRT monitors generally produce images by the raster-scan method. In this method, an electron beam with varying intensity, is moved back and forth horizontally across the face of the monitor. As the beam is directed to each spot on the phosphor-coated screen, it illuminates the spot in proportion to the voltage applied to the beam. Each spot represents a picture element or pixel. When the electron beam has scanned the entire screen and illuminated each pixel, one can see a complete image. The image that can be seen is the one traced on the retinas of eyes by the light beam. However, this image will fade unless it is refreshed. Thus, the electron beam must scan the screen very rapidly (at least 60 times per second), so that the intensity of the image remains approximately the same and the screen does not appear to flicker.

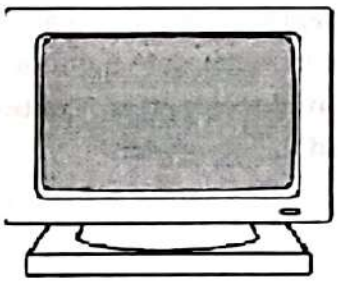


Figure 5.5: CRT monitor

The screen resolution of a monitor is determined by the number of pixels making up the screen. Monitors are currently available with 64,000 to more than 2 million pixels per screen. The greater the resolution of a monitor the greater is the storage demand on the computer. This is because the image must be stored in memory before it can be displayed. Two techniques used to store computer images are bit-mapped and character-

addressable. Bit mapped techniques are popular with present day microcomputers.

Monochrome or Colour: Some monitors display images in only one colour while others are capable of producing images in colours. Monochrome monitors use a single electron beam and display one colour, generally green, amber, or white, on a black background. The phosphor composition of the screen determines the colour. Colour monitors produce multi-colour images by combining the red, blue, and green colours in varying intensities. Each pixel is made up of three colour dots: red, blue, and green. It will appear to glow in different colours depending on the intensity of individual dots in the pixel. A critical specification of a colour monitor is its dot pitch. Dot pitch is the distance between the phosphor dots that make up a single pixel. A dot pitch no greater than 0.28 mm is very good for a 15" monitor.

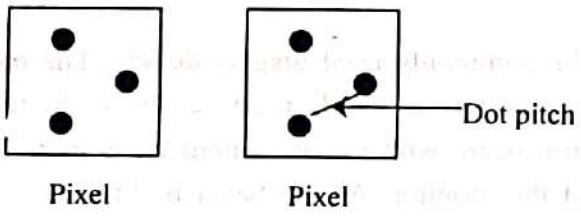


Figure 5.6: Dot pitch.

Colour monitors are commonly referred to as RGB monitors since they employ three beams, one for each colour. Colour monitors are categorized as CGA, EGA, VGA and SVGA depending on the resolution. CGA monitors provide the least resolution and SVGA monitors provide the greatest resolution (1024 × 768 pixels and more).

Monitor Interface: A monitor requires an appropriate interface to communicate with a computer. For example, a colour graphics interface board is needed for a colour monitor. This interface may not work with a monochrome monitor. A variety of monitor interface boards are available for use with microcomputers. Caution must be exercised to match the interface to both the monitor and the computer.

3.3 Flat-Panel Display

The CRT monitor has many disadvantages, some of these are:

- Large size
- High power consumption
- Fragility, and
- Occasional flickering images.

These have led to the development of flat-panel display. The flat-panel display is particularly useful for laptop computers, which can be used in the office and then taken

home or on trips. Producing a truly lap-size, or laptop computer, that is fully functional and light, has not been easy, and designing the video has been the most difficult problem. The most common flat-penal displays are:

- Liquid crystal
- Electro-luminescent and
- Gas plasma display.

Liquid Crystal Display: The liquid crystal display (LCD) uses a clear liquid chemical trapped in tiny pockets between two pieces of glass. Each pocket of liquid is covered both front and back by very thin wires. When a small amount of current is applied to both wires, a chemical reaction turns the liquid dark, thereby blocking light. The point of blocked light is the pixel. The principal advantages of LCDs are:

- Low power consumption
- Low cost
- Small size.

The major disadvantages of LCDs are:

- LCDs do not emit light, as a result, the image has very little contrast.
- Screen is susceptible to glare, so the optimum viewing angle is narrow.
- LCDs have less colour capability.
- Resolution is not as good as that of a CRT.

Research and development work is in progress to overcome the limitations of LCD screens, thereby making the screen easier to look at. It is anticipated that LCD will continue to be a popular technology for some time. Currently popular laptop computers, using flat-screen LCDs are manufactured by a number of companies.

3.4 Printer

The printer is a common output device. It produces permanent visual record of the output from a computer. It is capable of producing business reports and documents. Currently available printers are capable of printing from 150 to over 20,000 lines per minute, with each line having up to 150 characters. Thus, a maximum printing speeds of approximately 50,000 characters per second is possible.

Printers print on plain paper or on specially prepared single-or multiple copy forms, such as invoices, labels, checks, bills and other special-purpose forms used in business and industry. They can print both text and graphics in black and white or in colour. Printers can be subdivided into two broad categories,

- Impact printers and
- Non-impact printers.

Impact Printers

In an impact printer, printing occurs as a result of a hammer striking a character form and the character form in turn striking an inked ribbon, causing the ribbon to press an image of the character on paper.

Character printer devices print one character at a time at speeds of about 10 to 500 characters per second. The fastest of these printers is the wire or dot-matrix printer. It prints characters made up of a pattern of dots formed by the ends of small wires. By extending certain wires beyond the others, a dot pattern can be created that gives the appearance of numbers, letters or special characters.

These extended wires are pressed against an inked ribbon to print the characters on the paper. Some slower and less expensive matrix printers print a character as a series of columns, each one dot wide. It can be used to print special character shapes that can be used with graphics.

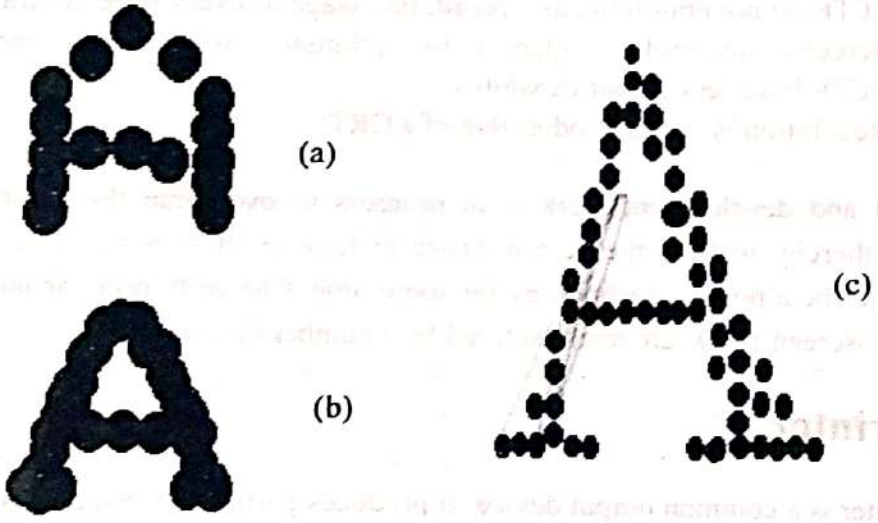


Figure 5.7: The letter A is shown printed by (a) 9-pin printer, (b) 24-pin printer. (c) 9-pin letter-quality dot-matrix printer capable of overlapped dot printing.

For a typewriter-quality output, a special dot-matrix or daisy metal print element, similar in appearance to the arrangement of petals on a daisy flower is used. This element is rotated until the correct character is in position, and then pressed against an inked ribbon. The process is repeated for each character to be printed on a line. Printing speed for such printers range from 25 to 100 characters per second. Impact character printers are the

common output devices used with personal and small business microcomputer systems. They are significantly cheaper than the line printers.

Nonimpact Printers

Nonimpact line printers, using laser, xerographic, electrostatic, or ink jet methods are the fastest printers. Before the development of the ink jet, thermal and laser printers, nonimpact printers were not heavily used for the following reasons:

- Special and more expensive paper was required.
- Printed output was not as sharp or as clear as with impact printers.
- Output could not be easily or satisfactorily copied on office copiers.

Ink-Jet Printer: Many types of ink-jet printers are available. The simplest of these contains a series of ink-jet nozzles in the form of a matrix. Vibrating crystals force tiny ink droplets, from selected nozzles to form an image in the same manner as an image is formed by a matrix printer. Coloured inks are combined to form additional colors for color printing.

Several hundred nozzles are employed in the sophisticated ink jet printers to direct a continuous stream of droplets across the page to form an image. These charged ink droplets travel at speeds of up to 40 miles per hour as they move between a set of plates that deflect the droplets. Droplets not needed are electrostatically attracted away from the paper for reuse. A stream of more than 100,000 droplets can form approximately 200 characters per second.

Table 2.2: Comparison of nonimpact printers.

Printer Type	Technology	Advantages	Disadvantages	Typical Speed
Thermal	Temperature sensitive, paper changes colour when treated, characters are formed by selectively heating print head	Quiet, high quality colour output	Special paper Required, expensive, slow	Several dozens toseveral hundred Characters per Second.
Ink-jet	Electrostatically charged ink drops hit paper	Quiet,prints colour, Lessexpensive, fast	Relatively slow, clogged jets, lower dpi	Several hundred Characters per Second
Laser	Laser beam directed to a drum, "etching" spots attract toner, which is then transferred to paper	Quiet,excellent quality, very high speed, colour or black and white.	High cost,	Up to several Hundred pages per minute.

Thermal Printers: Thermal printers use heat to produce an image on special paper. The

print mechanism, like a dot-matrix print head, is designed to heat the surface of chemically treated paper and a dot is produced based on the reaction of the chemical to the heat. No ribbon or ink is involved. For the highest-quality desktop colour printing, thermal printers are the answer. However, they are also expensive, and they require special and expensive paper.

Laser Printers: Laser printer technology is much less mechanical than impact printing, resulting in much higher speeds and quieter operation. The process resembles the operation of a photocopy machine. A laser beam is directed across the surface of a light-sensitive drum and fired as needed to record an image in the form of a pattern of tiny dots. The image is then transferred to the paper, a page at a time, in the same fashion as a photocopy machine, using a special toner.

The major advantages of laser printers are:

- Very high speed
- Low noise level
- Low maintenance requirement
- Very high image quality
- Excellent graphics capabilities
- A variety of type sizes and styles

3.5 Plotters

An inexpensive portable plotter is capable of generating multicolor plots from data stored on magnetic tape or disk. Plotters with multicolor capabilities generally use a writing mechanism containing several pens, each capable of producing a different color. Some devices for automated drafting are equipped with plotting surfaces larger than 10 square feet and is very costly.

Plotters are available for general (such as designing, mapping, or plotting schematics) or more specialized applications (such as three-dimensional data presentation, structural analysis, contouring, or business charts).

3.6 Microfilm Devices

Computer output microfilm (COM) devices convert computer output to a human-readable form stored on rolls of microfilm or as microfilm frames stored on cards called microfiche. At speeds of 10,000 to over 30,000 lines per minute, COM is one of the fastest computer output techniques more faster than the fastest impact printer. A single roll of microfilm can store approximately 2000 frames and costs less than half the cost to

print the same amount of data on paper.

Because of high cost of COM equipment, it is generally practical for larger businesses or industries generating several thousand documents per day. COM devices are commonly used in libraries, mail-order concerns, defense installations, government agencies, and for similar large operations.

3.7 Voice Output System

Voice output systems are relatively new and can be used in situations where traditional display screen output is inappropriate. It is more helpful to hear a warning than to take eyes off the road to check a display. The difficult part for voice output technology is that every individual perceives speech differently; that is, the voice patterns, pitches, and inflections are different for all of us. It is not always easy to understand an unfamiliar voice pattern. Two different approaches to voice output are: speech coding and speech synthesis. Speech coding relies on human speech as a reservoir of sounds in building the words and phrases to be output. Sounds are coded and stored on disk to be retrieved and translated back as sounds. Speech coding has been used in applications such as automobiles, toys, and games. Many cars having voice output devices pronounce cautionary words like: "please fasten your seat belt" or "you are exceeding speed limit" etc.

Speech synthesis relies on the use of a set of basic speech sounds that are created electronically without the use of a human voice. The largest application for the speech synthesis approach for converting text into "spoken" words, has many potential uses, which includes providing reading machines for the blind. And, of course, sound output does not have to be in voice form, it can be music or special-effect sounds such as the sound accompaniment for computer animation.

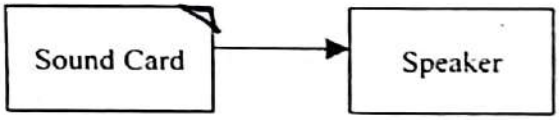


Figure 5.8: A sound card driving a speaker

3.8 Exercise

Multiple choice questions

1. Multi-colour images are produced in combination of:

   a. red, green and blue
   b. yellow, red and blue
   c. black, blue and white
   d. none of the above

2. An LCD as compared to a CRT monitor:

a. is more costly
b. has little contrast
c. is larger in size
d. none of these.

3. Dot matrix printer is:

a. an impact printer
b. a non-impact printer
c. a laser printer
d. none of these.

4. In general, which one of the following is the best quality printer?

a. dot matrix
b. ink-jet
c. desk-jet
d. laser.

Questions for short answers

5. What is the pixel of a monitor?
6. What are the following acronym stands for: COM and LCD?
7. Describe flat-panel display.
8. Give the difference between ink-jet and laser printers?
9. Give the difference between printer and plotter?
10. What are the applications of voice output system?

Analytical questions

11. Describe monitor display of computer systems.
12. Describe details of different impact and non-impact printers.
13. Draw the diagram of an ink-jet printing process and explain it briefly.
14. Compare thermal, inkjet and laser printer.

Lesson 4: Other Peripheral Devices

4.1 Objectives

On completion of this lesson you will know:

- Some special peripheral devices
- Characteristics and mechanism of such devices.

4.2 Terminal

A terminal is a popular input/output device. Terminals are used for two-way communications with a CPU or with other terminals a few feet or thousands of miles away. With the aid of a terminal, a user can access computers around the world. Terminals, also called workstations, allow a user to interact with a computer. It uses a keyboard to enter data and a cathode ray tube (CRT) screen, or monitor, for displaying data. Because data must be keyed into these devices one character at a time, the possibility of error is high and the data transmission rate is very low; thus limiting the use of terminals to small-volume input and inquiries only. Some of the functions that can be performed using terminals are described below.

Messaging: The communication of information from one terminal to one or more remote terminals.

Data collection: Data are received by terminals and recorded on secondary storage media for subsequent processing. This eliminates needs to record the information on a source document and then to key the information from the source document into a computer.

Inquiry or transaction processing: Data stored in central data files can be accessed from remote terminals for updating or to determine answers to inquiries about information stored in these files. The system employed by most airlines to maintain and update flight information is an example of such a function.

Remote job processing: Programs can be received from remote terminals directly to a CPU for processing. After execution, the results can be transmitted back to the terminal or to other terminals for output.

Graphic display and design: Data can be displayed in graphic form, and can also be manipulated and modified. Interactive graphic displays, from simple video games displayed on a television set to sophist'cated computerized systems, provide complex

designs and three-dimensional displays.

Terminals are available with features to suite the multitude of applications to which they are applied. In general three broad types of terminals are: dumb, smart and intelligent.

4.3 Speech Recognition and Voice Response Devices

Speech recognition devices were introduced in the early 1970s. Typically, these systems contain a database of stored voice patterns. The database of voice patterns is generally stored in a recognition unit or in secondary storage. A microphone, attached to the keyboard or recognition unit, records the spoken word patterns. A built-in microprocessor then compares, words with the stored patterns and transmits the result of the comparison to a computer for processing. A sentence must be spoken as a series of disjoined words and numbers spoken as a series of digits and not as a single number. Speech recognition devices are generally used in situations where access to a switch is not possible or where a user's hands are otherwise occupied.

Because voice patterns vary greatly from person to person, most speech recognition services are speaker-dependent and must be fine-tuned to each operation. This is generally accomplished by having the operator speaks each of the words or digits to be stored in the recognition unit dictionary several times. An average of the spoken voice patterns is taken and stored as the standard for future voice communications

Speaker-independent systems are less common and have a very restricted vocabulary: generally ten digits and "yes" or "no" response. Despite their restricted vocabulary, speaker-independent systems are widely usable since they do not have to be fine-tuned but can be understood by anyone. Clearly, speaker-independent systems are more desirable than speaker-dependent systems. Expense, large database requirements and the limitations of current technology have made the development of voice recognition systems slow. Manufactures are beginning to offer sophisticated speech recognition devices for the popular microcomputers.

One of the strongest impacts made on the use of voice response has come from the manufacturers of microcomputers. The pricing and availability of voice response units are economically feasible for even the smallest concern. Voice response is no longer an isolated discipline but another among the multitude of computer output techniques.

4.4 Vision Systems

A vision system utilizes a camera, digitizer, computer, and a technique known as image

processing. Image processing is concerned with digitizing and storing of computer-processed images and with pattern recognition.

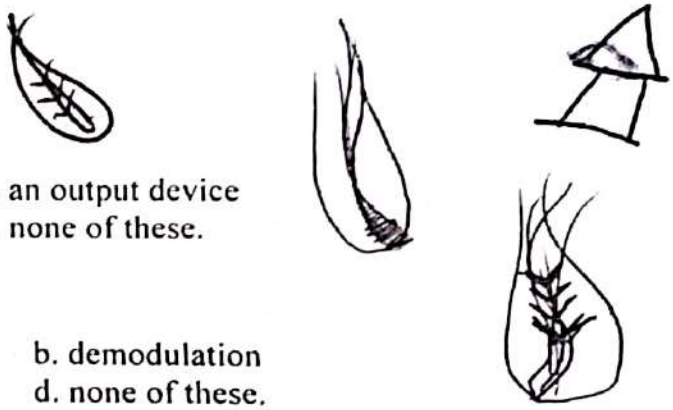
Familiar examples of computer-processed images are: computer generated digitized portraits at amusement parks, computer-produced special effects in movies, digitized images of Jupiter and Saturn beamed from image processors of spacecraft to earth etc. All of these examples have one thing in common, that is to digitize an image. In a visual system, all images that must be recognized or interpreted are digitized and stored in a database. Only after the database has been established, the visual system can be applied to pattern recognition. Pattern recognition, the process of interpreting images, begins when the system digitizes the image of the object to be interpreted. The digitized image is then compared to those in the database to determine a probable match. As it is unlikely that a perfect match will be achieved, there is always a small possibility of error.

4.5 Modem

A modem is the bridge between digital and analog signals. It converts digital data into analog signals by varying or modulating an electrical wave, a process known as modulation. Modulation is carried out for sending digital data through analog telephone system. On the receiving end of a phone connection, a modem does just the opposite. It demodulates the analog signals back into digital code. The two terms MOdulate and DEModulate give the modem its name. Modem communications involve three elements of personal computing: serial ports, modem commands, and communications software.

4.6 Exercise

Multiple choice questions



1. The terminal is:

   a. an input device
   b. an output device
   c. an input/output device
   d. none of these.

2. A modem does:

   a. modulation
   b. demodulation
   c. modulation and demodulation
   d. none of these.

Questions for short answers

3. What is a terminal?
4. Briefly describe the functions of a terminal.

5. What is the purpose of the vision system?
6. What do you understand of speech recognition?
7. What are the different types of terminals?
8. What are the functions of a modem?

Analytical question

9. Explain in detail the I/O devices that can be used as both input and output.
10. Describe speech recognition and voice response devices.

Chapter 6

Microprocessors

Lesson 1: Introduction to Microprocessors
Lesson 2: ALU and Control
Lesson 3: Microprocessors: Past, Present and Future
Lesson4: Parallel Processing

Lesson 1: Introduction to Microprocessors

1.1 Objectives

On completion of this lesson you will know:

- Functions of a microprocessor
- Organization of a microprocessor
- Components of a microprocessor.

1.2 Microprocessor

Every computer system has a unit whose primary purpose is to process data. This unit is the control centre of the entire computer system. It accepts data from input devices, processes data, and sends results to the printer or other output devices under control of a stored program. This unit is referred to as the microprocessor in a microcomputer and the central processing unit (CPU) in large computer systems. Both units perform basically the same functions.

Functions of Microprocessors

The microprocessor is an electronic device. It is the heart and brain inside every microcomputer. This tiny chip of silicon determines the speed and power of the entire computer by handling most, if not all, of the processing of a microcomputer. This can be

explored by using a flow diagram as shown in Figure 6.1. Functions of a microprocessor can be summarized as follows:

- The processor first fetchs an instruction from the main memory.
- The instruction is then decoded to determine what action is required to be done.
- Based on instruction the processor fetchs, if required, data from main memory or I/O module.
- The instruction is then executed which may require performing arithmetic or logical operations on data.
- In addition to execution, CPU also supervises and controls I/O devices. If there is any request from I/O devices, called interrupt, the CPU suspends execution of the current program and transfers control to an interrupt handling program.
- Finally, the results of an execution may require transfer of data to memory or an I/O Module.

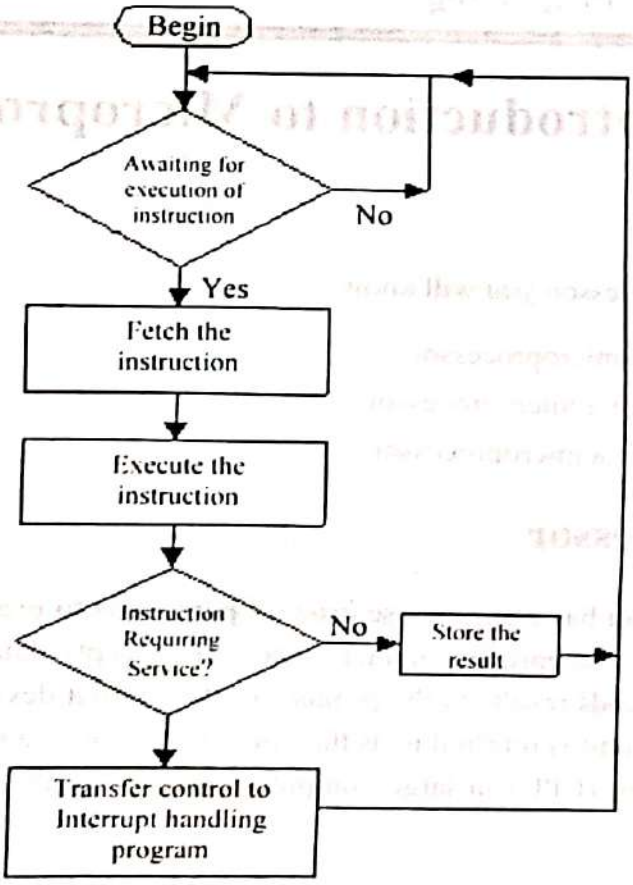


Figure 6.1: Functions of a typical CPU.

1.3 Organization of a Microprocessor

Figure 6.2 shows a block diagram of a typical microprocessor with its components. The

major components are:

- An arithmetic/logic unit (ALU) and
- Control unit (CU).

The ALU performs the actual computation or processing of data. The control unit controls the movement of data and instructions into and out of the CPU. It also controls the operation of the ALU. In addition, the figure shows a minimal internal memory, called registers. The data transfer and logic control paths are indicated. The internal CPU bus is needed to transfer data between the various registers and the ALU. The ALU operates only on data in the internal CPU memory. The figure also shows basic elements of the ALU.

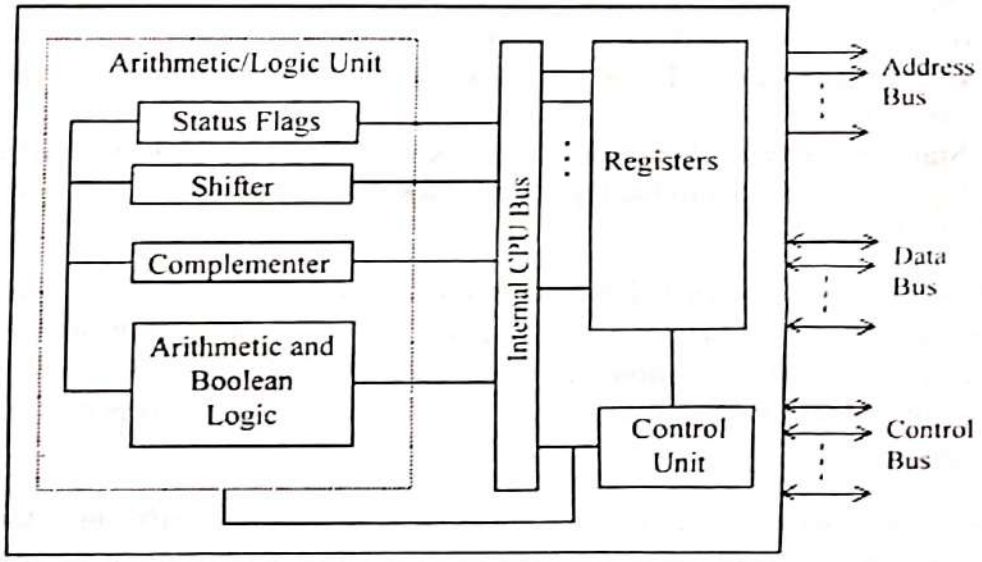


Figure 6.2: Internal organization of a microprocessor.

Within the CPU there is a set of registers that function as a level of memory above main memory and cache in the hierarchy. The registers in the CPU serve two functions:

- **User-Visible Registers:** These registers enable the programs to minimize main-memory references by optimizing use of registers.
- **Control and Status Registers:** They are used to control the operations of the CPU and to control the execution of programs.

User-Visible Registers

User-visible registers can be characterized in the following categories:

- General Purpose Registers
- Data Registers
- Address Registers
- Condition Code Registers.

General-purpose registers can be assigned to a variety of functions by the programmer. A general-purpose register can contain the operand for any opcode. This provides true general purpose register use. However, there are restrictions. There may be dedicated registers for floating-point operations.

General-purpose registers can be used for addressing functions. In other cases, there is a partial or clean separation between data registers and address registers. Data registers may be used only to hold data and cannot be employed in the calculation of an operand address. Address registers may be somewhat general-purpose, or they may be devoted to a particular addressing mode, some examples are:

- **Segment Pointers:** A segment register holds the address of the base of a memory segment. There may be multiple registers; for example, one for the operating system and one for the current process.
- **Index Registers:** These are used for indexed addressing and may be autoindexed.
- **Stack Registers:** Typically the stack is in memory and there is a dedicated register that points to the top of the stack. This allows push, pop, and other stack instructions.

A final category of registers which is at least partially visible to the user, holds condition codes or flags. Condition codes are bits set by the CPU as the result of operations. For example, an arithmetic operation may produce a positive, negative or zero result or an overflow. In addition to the result, it may subsequently be tested as part of a conditional branch operation.

Condition code bits are collected into one or more registers. Usually, they form part of a control register. Generally, machine instructions allow these bits to be read, but they cannot be altered by the programmer.

Control and Status Registers

A variety of CPU registers control the operation of the CPU. Most of these, are not visible to the user. A reasonably complete list of register types, with brief description are presented here. The following four registers are essential for instruction execution:

- **Program Counter (PC):** Contains the address of an instruction to be fetched.
- **Instruction Register (IR):** Contains the instruction most recently fetched.
- **Memory Address Register (MAR):** Contains the address of a location in memory.
- **Memory Buffer Register (MBR):** Contains a word of data to be written to memory or the word most recently read.

The program counter contains an instruction address. Typically, the program counter is updated by the CPU after each instruction fetch and thus, it always points to the next

instruction to be executed. A branch or skip instruction modify the contents of the PC. The fetched instruction is loaded into an instruction register, where the opcode is analyzed. Data are exchanged with memory using the MAR and MBR. In a bus-organized system, the MAR connects directly to the address bus, and the MBR connects directly to the data bus. User-visible registers, in turn, exchange data with the MBR.

The four registers mentioned above are used for the movement of data between the CPU and memory. Within the CPU, data must be presented to the ALU for processing. The ALU may have direct access to the MBR and user-visible registers. Alternatively, there may be additional buffer registers at the boundary to the ALU; these registers serve as input and output registers for the ALU and exchange data with the MBR and user-visible registers.

All CPUs include a register or set of registers, often known as the program status word (PSW), that contains status information. The PSW typically contains condition codes plus other status information. Common fields or flags include the following:

- **Sign:** Contains the sign bit of the result of the last arithmetic operation.
- **Zero:** Set when the result is 0.
- **Carry:** Set if an operation resulted in carry (addition) into or borrow (subtraction) out of a high-order bit.
- **Equal:** Set if a logical compare result is equality.
- **Overflow:** Used to indicate arithmetic overflow.
- **Interrupt Enable/Disable:** Used to enable or disable interrupts.
- **Supervisor:** Indicates whether the CPU is executing in supervisor or user mode. Certain privileged instructions can be executed only in supervisor mode, and certain areas of memory can be accessed only in supervisor mode.

In addition to the PSW, there may be a pointer to a block of memory containing additional status information. In machines using vectored interrupts, an interrupt vector register may be provided. If a stack is used to implement certain functions (e.g., subroutine call), then a system stack pointer is needed. A page table pointer is used with a virtual memory system. Finally, registers may be used in the control of I/O operations.

In addition to the above components modern microprocessors may also have the following components:

- **Integer unit (iu):** To handle the integer operation efficiently.
- **Floating point unit (fpu):** To perform floating point operations.
- **Memory management unit (mmu):** This unit handles run-time memory management problems to enhance performance.
- **Data cache and instruction cache:** These are also called built-in cache. The built-in cache is a fast primary memory.

It is also noted that currently some microprocessors may have a special component for a

specific purpose, for instance multimedia. Some processors have more than one ALU unit to enhance performance. Finally, it is essential to say that the components of a microprocessor may change or new components may be incorporated in the future.

1.4 Exercises

Multiple Choice Questions

1. Microprocessor is the CPU of a
   a. microcomputer
   b. main frame computer
   c. supercomputer
   d. none of the these

2. Major components of a microprocessor are
   a. ALU, CU and I/O devices
   b. ALU, and I/O devices
   c. ALU, and CU
   d. none of these

3. Data register is a
   a. user-visible-register
   b. status register
   c. control register
   d. none of these

4. IR stands for
   a. internal register
   b. instruction register
   c. individual register
   d. none of these

Questions for Short Answers

5. What is a microprocessor?
6. Distinguish between CPU and microprocessor.
7. What are the functions of registers?
8. What are the types of user-visible-registers?
9. What are the control and status registers?
10. What is the function of program counter?

Analytical Questions

11. Describe the functions of a microprocessor with the help of a flowchart.
12. Draw the structure of a microprocessor and explain briefly its components.
13. Describe the register sets of a microprocessor.

Lesson 2: ALU and Control

2.1. Objectives

On completion of this lesson you will know:

- Arithmetic logic unit
- Operation of the control unit
- Difference between hardwired control and micro-programmed control.

2.2 Arithmetic/Logic Unit (ALU)

The arithmetic/logic unit, or ALU, is the data processing unit of the microprocessor (Figure 6.3). Functions of the ALU are:

- Arithmetic operation (addition, subtraction, multiplication and division)
- Logical operations (OR, AND, NOT etc.)
- Decision making.

ALU is the computer's calculator. A few current processors use multiple ALUs to attain high processing speeds. However, most microprocessors have a single ALU.

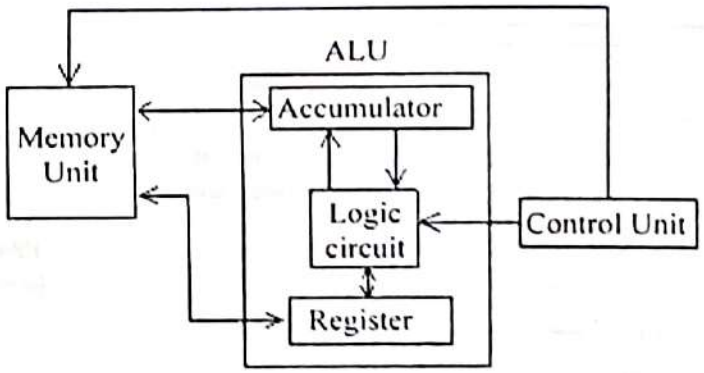


Figure 6.3: Block diagram of an ALU and its relation to control and memory units

Arithmetic operations include addition, subtraction, multiplication, and division. The data operated on can be stored in various forms: the binary, BCD, EBCDIC and ASCII representations.

The ALU makes use of temporary storage areas referred to as registers. Data to be arithmetically manipulated are copied from memory and placed in registers for processing. Upon completion of the arithmetic operation, the result can be transferred from the Accumulator to memory. The ALU uses one or more adders (shown as logic

circuit) that actually add, subtract, multiply, or divide the binary digits.

Decision making is the ability to compare two numbers to determine, if the first number is smaller than, equal to, or greater than the second number and to take appropriate action based on the result of the comparison. For example, if the question is $x > 100$: the ALU would determine answer as being either true or false depending on the value of x. It is also possible to test a condition during the processing of an application and to alter the sequence of instructions accordingly.

The various circuits used in the execution of data processing instructions by a processor are usually merged into the ALU. The complexity of an ALU is determined by the manner in which its arithmetic instructions are realized. Simple ALUs for fixed-point operations can be constructed around the circuits developed for multiplication and division. More expensive data processing and control logic is necessary to implement floating point arithmetic in hardware. Some microprocessor families having fixed-point ALUs employ special-purpose units called arithmetic coprocessors to perform floating-point and other complex numerical functions.

2.3 Control Unit

The control unit performs the computer's traffic control. It coordinates and controls

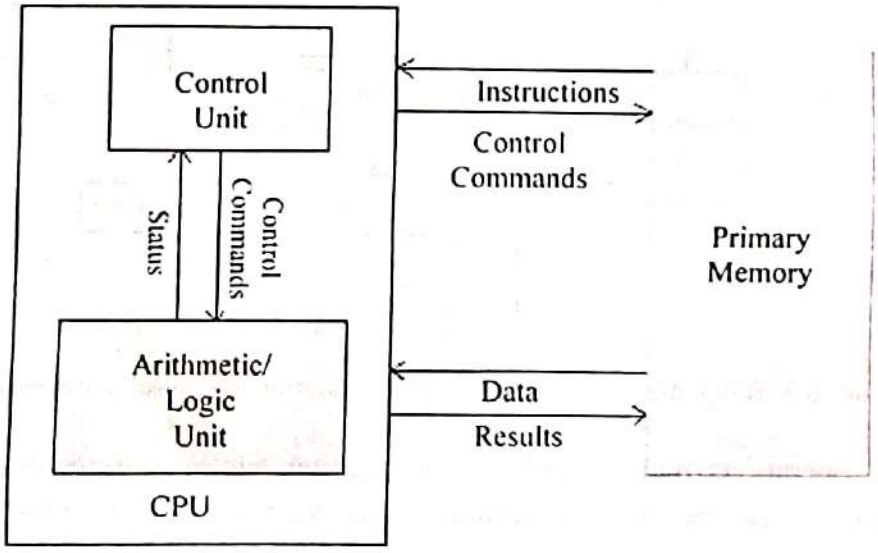


Figure 6.4: Communication of Control Unit with ALU and Primary memory.

operations of the central processing unit. It does this like the human brain coordinates and controls the activities of the human body. The control unit does not input, output, process, or store data; rather, it initiates and controls these operations. The control unit

also communicates with input devices to begin the transfer of data or instructions into memory and with output devices to begin the transfer of results from memory.

Data transfer involves the moving of data or instructions from one location to another. When an item of data is stored in a given location, it replaces the previous contents of that location. But when an item of data is moved from one location to another the data item is not physically removed from its initial location; actually the data item is copied to the new location. When the computer executes a program contained in primary memory, the control unit obtains the instructions in the sequence in which they are executed, interprets the instructions, and issues signals to execute them. To accomplish this, the control unit must communicate with ALU and primary storage.

The control unit generally performs all or most of the following functions for execution of instructions:

- Determines the instruction to be executed
- Determines the operation to be performed
- Determines what data are needed and where they are stored
- Determines where results, if any, are to be stored
- Determines where the next instruction is located
- Causes the instruction to be executed
- Transfers control to the next instruction.

Machine Cycle

The activities of the control unit, are composed of thousands of individual steps, each of which takes place in a fixed interval of time. These intervals are controlled by an internal electronic clock that emits millions of regular electronic pulses every second. Clock speeds are generally measured in megahertz (MHz) or million pulses per second. In general, operations within the CPU of the computer take place in terms of a fixed number of clock pulses. This number determines the machine cycle for the computer. During a machine cycle, the computer can perform one machine operation. The number of machine operations required to execute a single instruction vary from instruction to instruction. Execution of instructions takes place under the direct supervision of the control unit.

Instruction Format

Program instructions stored in memory must be in a machine-readable form. In general, these instructions consist of two distinct parts.

- An operation code, and
- One (or more) operand

The operation code (Opcode) tells the machine what task is to be performed; the operands specify what are to be used to perform the task. Examples of operands are:

- Address of a data item or an instruction in primary memory.
- Address of a data item or a program outside the CPU on a secondary memory.
- Address of an input or an output unit.
- Address of a register or special-purpose temporary memory area.

For example, the instruction

MOV 8000 9000

contains the operation code MOV and the operands 8000 and 9000. This could mean that the machine is to move the contents of storage location 8000 to storage location 9000. In addition to the address of an item of data in primary storage, an operand can be used to indicate the address of a register, the address of data stored in secondary storage, or even the address of an input or output device.

2.4 Instruction Execution

The actual execution of an instruction, such as the move instruction, generally requires a number of machine cycles. The instruction is retrieved from storage, interpreted, and executed. These functions generally require at least two machine cycles called

- Instruction cycle, and
- Execution cycle.

Instruction Cycle

The instruction cycle is the first machine cycle in the execution of an instruction. Four distinct steps are performed during the instruction cycle (Figure. 6.5). These are:

- The instruction is fetched from primary memory and placed in a register called a storage register.
- The opcode is moved to the instruction register and decoded. This is accomplished with the aid of operation decoders.
- The operands are placed in one or more address registers, depending on the number of operands in the instruction.
- The address of the next instruction is determined.

Instructions are executed one after the other. Normally next instruction will be located immediately after the current instruction. Its address is simply the address of the current instruction plus the number of locations occupied by the current instruction. Some times it may be necessary in a program to branch, or transfer control, to an instruction other

than the next instruction. In such cases, the current instruction indicates the address of the next instruction. This address is then placed in the instruction counter, and the sequential execution of instructions is resumed until another branch instruction is encountered.

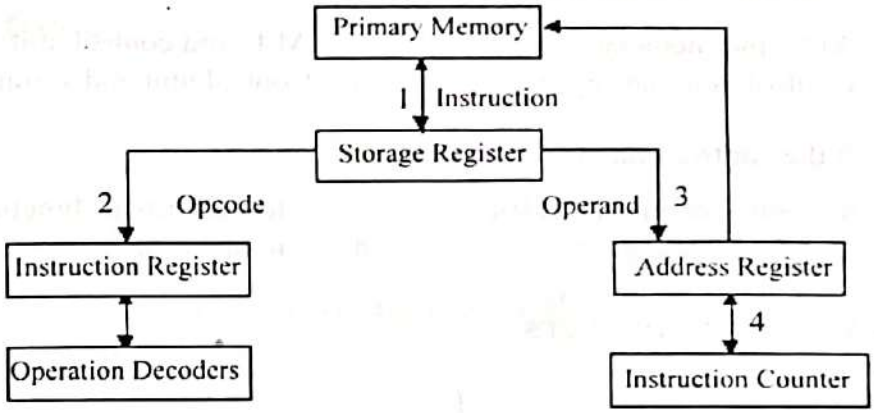


Figure 6.5: The instruction cycle.

Execution Cycle

The execution cycle represents the remaining machine cycles in the execution of an instruction. The execution cycle may consist of more than one machine cycle. The time required to complete the execution cycle is referred to as execution time. Throughout this cycle, operations are controlled by the instruction register.

The steps performed in an execution cycle depend on the instruction being executed. Let us assume that a program requires that two numbers be added. During the instruction cycle the control unit would decode the instruction and direct the ALU to perform the ADD operation. It would also have placed the addresses of the operands in the address register. The execution cycle would cause the following:

- Operands would be obtained from the primary locations indicated in the address register. One value would be placed in a storage register and the other would be placed in the accumulator register.
- ALU would then add these numbers.
- The sum would be stored in the accumulator register.

The instruction cycle of the next instruction would begin next. This instruction would most likely involve moving the content of the accumulator register (the sum of the two numbers) to a location in the primary memory.

2.5 Exercise

Multiple choice questions

1. The functional units of a microprocessor are:

   a. ALU and memory
   b. Control unit and register
   c. ALU and control unit
   d. Control unit and storage

2. Function of the control unit is:

   a. to control computer traffic
   b. to store information
   c. to perform arithmetic operation
   d. none of the above

Questions for short answers

3. What is an ALU?
4. What are the functional units of an ALU?
5. Briefly write the functions of ALU.
6. What is the purpose of an accumulator?
7. What is meant by an operand?

Analytical questions

8. Write an essay on functional units of an ALU.
9. Describe the functions of the control unit.
10. Describe the steps performed during the instruction cycle.

Lesson 3: Microprocessors: Past, Present and Future

3.1 Objectives

On completion of this lesson you will know:

- Classification of microprocessors
- Future microprocessors.

3.2 Microprocessor Technology and Performance

To satisfy high performance demands, microprocessor technology has developed at a rapid pace in recent years. This is based on:

- Processing speed,
- Processing ability,
- Communication ability, and
- Control ability.

Every year brings new devices, new functions, and new possibilities. Over the past three decades, the microprocessor has achieved a total performance growth of about 150 times. Microprocessor can be classified on the basis of generations or their characteristics.

3.3 Classification Based on Generations

Microprocessors have passed several generations. Fundamentally each generation has been driven by advances in hardware technologies and attempt to extract higher degree of parallelism.

First Generation

The first generation includes the serial processors, which began with the first electronic digital computers. Serial processors fetch and execute each instruction before going to the next.

Examples: Intel 4004, Intel 8085.

Second Generation

The second generation is distinguished by pipelining and similar methods for overlapping instruction execution. Pipelining is a technique of decomposing a sequential

process into suboperations, with each subprocess being executed in a special dedicated segment that operates concurrently with all other segments.

Examples: Intel 80486, 80i860.

Third Generation

The third and current generation is characterized by superscalar processors. A superscalar processor is one in which multiple piplines are provided to allow instructions to execute in parallel.

Examples: Pentium processors; RISC processors- Power PC 601, 602, etc.

Fourth Generation

The high performance processors of the next fourth generation will be composed of multiple superscalar pipelines, with some higher level control that dispatches groups of instructions to the individual superscalar pipes. Major design goals of a fourth generation processor are to:

- Make parallel instructions more visible
- Dynamically partition hierarchical parallelism

An example of fourth generation processor is trace processor.

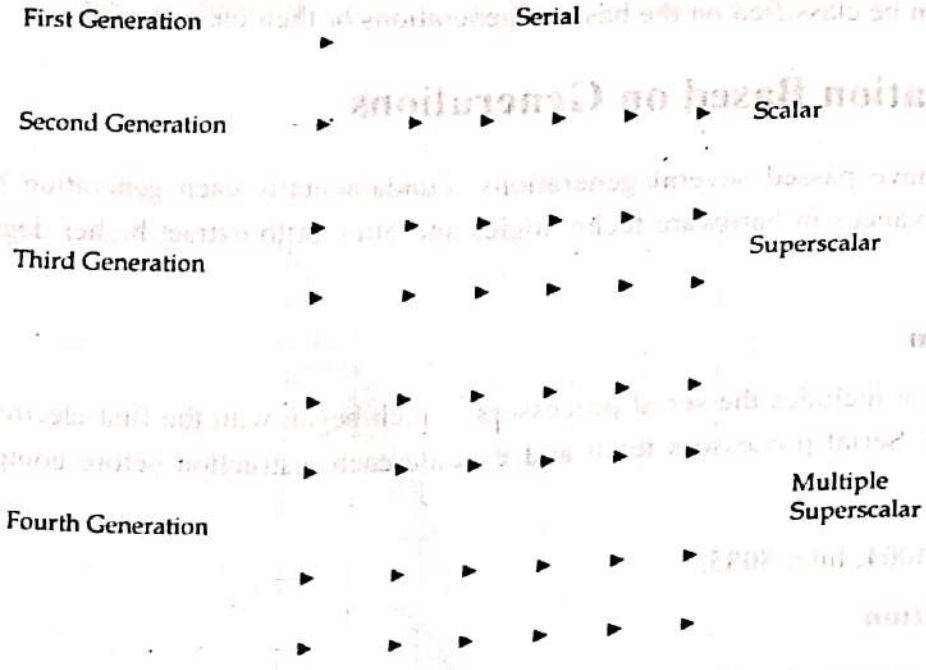


Figure 6.6: Four-processor generations.

3.4 Classification Based on Characteristics

Based on characteristics, the microprocessors are classified into three categories. These are as follows:

- RISC processors
- CISC processors
- Special processors.

RISC Processors

RISC is an acronym for reduced instruction set computer. The concept of RISC processor involves an attempt to reduce execution time by simplifying the instruction set of the computer. The major characteristics of a RISC processors are:

- Relatively few instructions
- Relatively few addressing modes
- Memory access limited to load and store instructions
- All operations done within the registers of the CPU
- Fixed-length, easily decoded instruction format
- Single-cycle instruction execution
- Hardwired rather than microprogrammed control.

Some architectural features of RISCs are:

- Relatively large number of registers in the processing unit
- Use of overlapped register windows to speedup procedure, call and return
- Efficient instruction pipeline
- Compiler support for efficient translation of high level language programs into machine language programs.

Some popular RISC processors are:

- Power PC : 601, 604, 615, 620
- DEC Alpha : 21064, 21066, 21068, 21164
- MIPS : TS (R10000) RISC processor
- PA-RISC : HP 7100LC

CISC Processor

CISC is an acronym for complex instruction set computer. Major characteristics of a CISC processors are:

- Large number of instructions
- Some instructions that perform specialized tasks
- Variety of addressing modes

- Variable length instruction formats
- Instructions that manipulate control
- Several cycles may be required to execute one instruction.

Examples: Intel's x86 family: Motorola's 68000, 68020, 68030, 68040 etc.

For many years, Intel is the only major supplier of x86 processor with a few authorized and controlled alternate sources. Today a large number of companies offer compatible processors with no assistance from Intel. They are:

- AMD (Advanced Micro Device) processors
- Cyrix processors
- Texas Instrument (TI) processors
- NexGen processors.

All CISC microprocessors that run PC software were derived from the early Intel 8086 or x86 architecture. Today Intel controls the development of CISC microprocessor and other industries produce the Intel compatible processors.

Intel presented a new processor in 1993 called Pentium. With this processor, Intel opened a new era of technology. Day by day, new features are added to the processor and as a result new highly developed processors come out. Some of them are:

PentiumPro: This is an optimized processor for high-performance network server implementation.

Pentium III: This is the recent Intel processor in the market, which has both the technology of PentiumPro and also MMX (multimedia extension) technology. The new PIII system gives very good combination of price and performance and graphic applications.

3.5 RISC Versus CISC

In early 50's and 60's IBM 360 mainframe machine used simple fixed length instructions. The IMB 360's instruction set was a classic CISC design, but the microcode engine that actually executed the instruction set was a simple RISC processor. So RISC concepts are not new, they were just waiting for the cost of technology to justify the RISC approach.

RISC got a significant boost as the result of software work being done at IBM. As a result of IBM's work on compiler technology, it was recovered that most of the complex instructions in the CISC design were rarely used. By inspecting the compiler generated machine language output, it is found that often as little of 10% of the CISC machine

language instruction set accounted for over 90% of the code generated by the compiler. IBM and several universities began to do research into the development of simple computer designs that had very few instructions, large register arrays, simple load-and-store access to main memory and with instructions executed in one clock cycle. As a result, RISC processors arrived.

RISC processors were extensively used in high performance workstations for scientific and engineering applications. As the RISC processors evolved, their true advantage

Table 6.1 The major differences between RISC and CISC microprocessors

RISC	CISC
• Reduced instruction set computer.	• Complex instruction set computer.
• The microprocessor is designed using hardwired control.	• The microprocessor is designed using code control.
• It executes at least one instruction in a cycle.	• Several cycles may be required to execute one instruction.
• The instructions have simple fixed formats with few addressing modes.	• The instructions have variable formats with several complex addressing modes.
• It has several general purpose resisters and large cache memory.	• It has a small number of general purpose registers
• The instruction set of RISC micro-processors typically includes only register to register load and store.	• The instruction set of CISC microprocessor includes several instructions to access memory and CPU registers.
• Pipelining and superscalar architectures are the base methods to design a RISC processor.	• Pipelining and superscalar features are not the bases to design such processors, although, many CISC microprocessors use several features if RISCs such as: pipelining, to increase its performance.

began to emerge. In the final analysis, RISC designs appear superior. At similar internal clock speeds, RISC processors consistently outperform CISC processors. Since RISC processors do not require lots of silicon, it adds superscalar execution units, more powerful floating point units and large on-chip caches. Or more simply RISC designs take less space and allow more functions on a chip than CISC design.

Even with all RISC's advantages, the major point favouring continued use of CISC processors is that of software compatibility. If one is to buy a new RISC based PC, the full set of PC industry software would not be available to run on the machine. Most PC software is designed for and compiled to run exclusively on PC platforms.

RISC ideas have infiltrated to every high performance design. Intel, the main designer of CISC microprocessors, also uses basic RISC features to improve the performance of the processor. For example:

- Intel 80486 uses features like scalar, pipeline
- Intel Pentium uses features like superscalar and superpipeline.

The upcoming new Pentium PC processors also contain more or less many characteristics of RISC processors to increase their speed and performance. So all the next generation CISCs will evolve with the features of RISC processors.

3.6 Special Processors

There are several other processors, which are used for special purposes. Some of these processors are briefly discussed below.

Coprocessor

A coprocessor is a microprocessor but unlike a general purpose microprocessor, it is dedicated to its specific function as a special purpose device. Because, the coprocessor is designed for a specific purpose, it can handle its particular function many times faster than the ordinary general purpose microprocessors. The most well know coprocessor is the math-coprocessor.

Until the introduction of the 486DX microprocessor, the math-coprocessor was a separate chip. But 486DX and Pentium microprocessors have the math-coprocessor integrated on the same chip with the microprocessor. Intel math-coprocessors are:

8087 – used with 8086
80287 – used with 80286
80387 – used with 80386.

Cyrix and Weitek also market math-coprocessors.

Input/Output Processor

One of the most important type of processors is I/O processors. An I/O processor has a local memory of its own right. With this architecture, a large set of I/O devices can be controlled, with minimal CPU involvement. A common use for such architecture has been to control communication with interactive terminals. The I/O processors take care of most of the tasks involved in controlling the terminals. I/O processors are used in the following fashion:

- The CPU executes a series of data-transfer instructions that send a set of input operands and command information to registers in the I/O processor.
- The I/O processor decodes and executes the command received from the CPU, generating a result that is placed in registers accessible to the CPU.
- The CPU determines that the I/O processor has completed its task either by checking its status or else by receiving an interrupt signal from the I/O processor.
- The CPU then obtains the results from the I/O processor by executing more data transfer instructions.

Common examples of I/O processors are:

- DMA (Direct Memory Access) controller
- Keyboard/mouse controller
- Graphic display controller
- SCSI port controller etc.

Transputer

The transputer (Transister Computer) is a high performance microprocessor designed to facilitate inter-process and inter-processor communications and is targeted at the efficient exploitation of very large scale integration (VLSI) technology. The most important feature of the transputer is its external links, which enable it to be used as a building block in the construction of low cost, high performance multiprocessing systems. Communication takes place (via these links) only between pairs of devices and is distributed throughout the system. The transputer family consists of several types of VLSI devices including the 16-bit T212, the 32-bit T425 and the floating point T800, T805 and T9000 processors.

DSP (Digital Signal Processor)

This processor is specially designed to handle real-world analog signals that have been converted to digital representations. Some of the applications that DSP can support are:

- Sound and music synthesis
- Modem function
- Audio and video compression
- Speech synthesis and decompression
- Speech recognition
- Video signal processing
- 2D and 3D graphics acceleration.

Examples: Texas Instrument's TMS 320 series, e.g., TMS 320C40, TMS320C50.

3.7 Future Processors

Future trend of processor architecture is explored in this section. Some future architectures are:

- Advanced Superscalar Processor
- Superspeculative Processor
- Simultaneous Multithreaded (SMT) Processor
- Trace Processor
- Vector IRAM Processor
- Raw Processor.

The circumstance in which processor designs will find themselves in the next years are truly daunting. In future, microprocessors having more than a billion transistors on a single chip will present opportunities for new level of computing capability.

At the same time, it is important to mention that over the next few years media processing will become the dominant force in computer architecture and microprocessor design. Dramatically different nature and demands of media processing will force profound changes and it is difficult to look ahead to the next generation.

3.8 Exercise

Multiple choice questions

1. Current generation of microprocessor is

   a. first generation
   b. third generation
   c. second generation
   d. fourth generation

2. Compared to CISC, RISC processors (at the same clock) are

   a. faster
   b. slower
   c. similar
   d. none of these

Question for short answers

3. What is the basis for the rapid developments of microprocessor technology?
4. What are the different categories of processors based on generations?
5. Classify the microprocessors based on characteristics.
6. Distinguish between CISC and RISC processors.
7. What is the transputer?
8. Give examples of current CISC processors that are used in personal computers.

Analytical questions

9. Describe the microprocessors of different generations.
10. What is the RISC and CISC controversy? Distinguish between RISC and CISC processors.
11. Briefly describe the classification of microprocessors based on characteristics.
12. Write an essay on future microprocessors.

Lesson 4: Parallel Processing

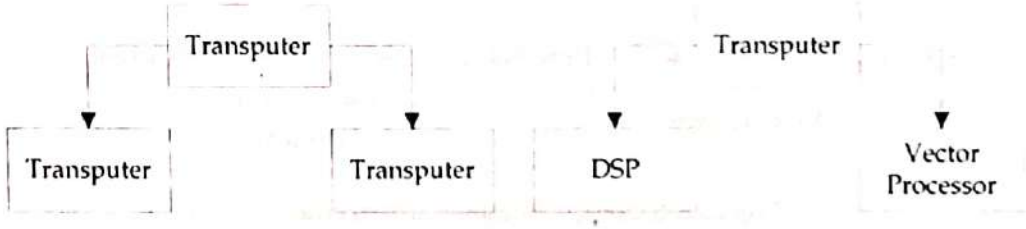
4.1 Objectives

On completion of this lesson you will know:

- Concept of parallel processing
- Classification of parallel architectures
- Performance measuring parameters and issues
- New technologies for future computing.

4.2 Introduction

In general, the term parallel processing (PP) can be defined as processing of an algorithm simultaneously by several processors in a computing platform. On the other hand, if a number of algorithms are processed simultaneously by several processors in a computing platform, it is known as distributed processing. Parallel processing or distributed processing or both can be called multiprocessing. The hardware that can be used for parallel processing is know as parallel architecture. In a conventional parallel system all the processors or processing elements (PEs) are identical. This architecture can be described as homogeneous (Figure 6.7a). However, for variety of computational demands of many algorithms, mixed PEs based architectures are becoming popular. A parallel architecture comprising of a variety of PEs is described as heterogeneous system (Figure 6.7b).



(a) Homogeneous architecture. (b) Heterogeneous architecture.

Figure 6.7: Homogeneous and heterogeneous parallel architectures.

Parallel processing has emerged as a key technology in modern computing to meet the increasing demand for higher performance, lower costs and sustained productivity in real-life applications. The concept of PP on different problems or different parts of the same problem is not new. Discussions of parallel computing machines are found in the literature at least as far back as the 1920s. There have been continuing research efforts to

understand parallel computation. Such efforts have intensified dramatically in the past few years, with thousands of projects around the world involving scores of different parallel architectures for all kinds of applications, including signal processing, control, artificial intelligence, pattern recognition, robotics, computer vision, computer aided design, discrete event simulation etc.

4.3 Classification of Parallel Architectures

Parallelism appears in various forms, including lookahead, pipelining, vectorization, concurrency, simultaneity, data parallelism, partitioning, interleaving, overlapping, multiplicity, replication, time sharing, multitasking, multiprogramming, multithreading, and distributed computing at different processing levels. The most commonly used classification of parallel architectures is that given by Flynn in 1966. Flynn considered the presence or absence of potential multiplicity in the instruction and data streams of computer systems. The instruction stream is the programmed instructions of opertions. Thus, an architecture is categorized by the multiplicity of the hardware used to manipulate the instruction and data streams. In this manner, it is possible to obtain four classes of computers as described below.

Single Instruction Single Data Stream

Conventional computers belong to the single instruction single data stream (SISD) category where there is one central processing unit in the computer which can perform one instruction at a time. This represents the serial architecture illustrated in Figure 6.8. This, until recently, has formed the basis for nearly all computers.

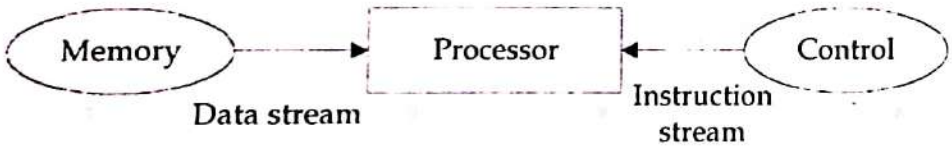


Figure 6.8: SISD computer architecture.

Single Instruction Multiple Data Stream

The single instruction multiple data stream (SIMD) systems are those in which the same instruction is executed, at the same time, by several processors, as illustrated in Figure 6.9. Clearly such a system comprising of N processors can have a throughput of N times more than an SISD machine. Examples include pipelined vector processors and array processors.

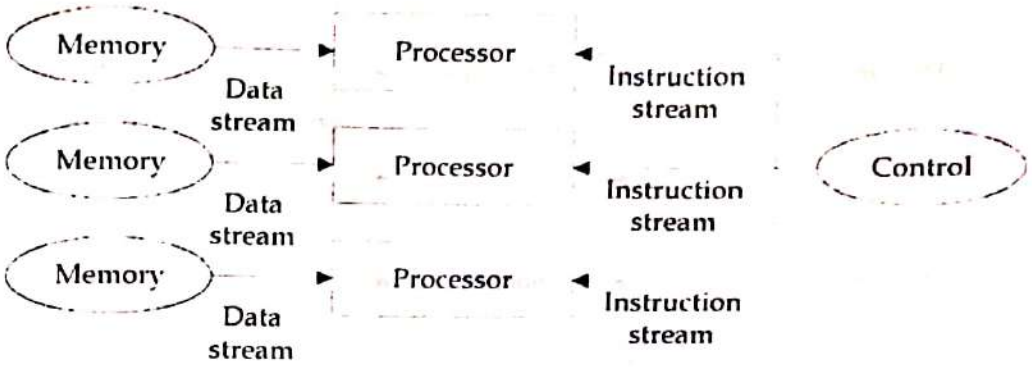


Figure. 6.9: SIMD computer architecture.

Multiple Instruction Single Data Stream

In a multiple instruction single data stream (MISD) architecture, the processors can share common memory and each processor operates according to the instructions it receives via its own instruction stream (Fig. 6.10). This class of computers is difficult to conceptualize since it implies that several instructions operate on a data item simultaneously. For most applications, MISD computers are rather awkward to use, but can be useful in applications of rather specialized nature. A typical example of one of such specialized application is robot vision. For example, a robot inspecting a conveyor belt by sight may have to recognize objects that belong to different classes.

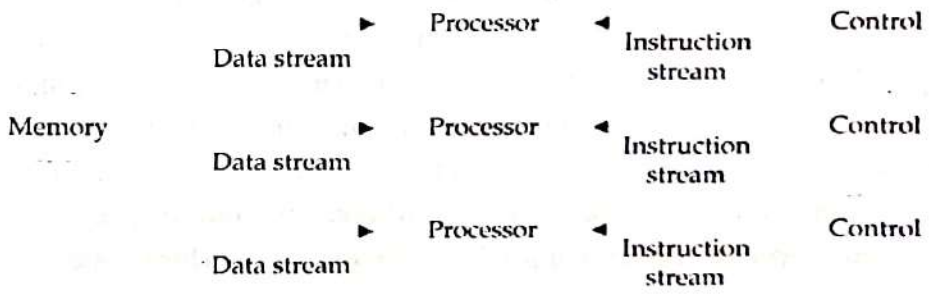


Figure 6.10: MISD computer architecture.

Multiple Instruction Multiple Data Stream

The multiple instruction multiple data stream (MIMD) is Flynn's definition of the most versatile PP machine where there are several instruction streams and several data streams (Figure 6.11). This class includes all forms of multiprocessor configurations from linked main-frame computers to large arrays of microprocessors. Multiple processors capable of independent operations is the main characteristic of MIMD computers.

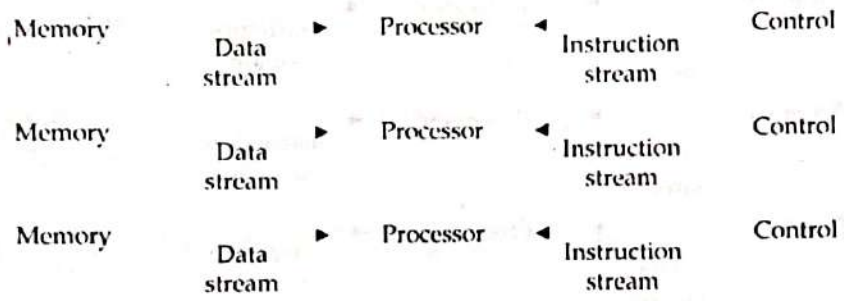


Figure. 6.11: MIMD computer architecture.

4.4 Performance Measuring Parameters

The most widely accepted measure to evaluate the performance of a parallel system is speedup. Speedup (S_N) is defined as the ratio of the execution time (T_1) on a single processor, to the execution time (T_N) on N processors:

$$S_N = \frac{T_1}{T_N}$$

The theoretical maximum speed that can be achieved with a parallel architecture of N identical processors working concurrently on a problem, is N. This is known as the ideal speedup. In practice, the speedup is much less, since some architectures do not perform to the ideal level owing to conflicts over memory access, communication delays, inefficient algorithms and mapping for exploiting the natural concurrency in a computing problem. But in some cases, the speedup can be obtained above the ideal speedup, due to anomalies in programming, compilation, architecture usage, etc. For example, a single processor system may store all its data off-chip, whereas the multiprocessor system may store all its data on-chip providing an unpredicted increase in performance.

Another useful measure in evaluating the performance of a parallel system is efficiency (E_N) which can be defined as

$$E_N = \frac{S_N}{N} \times 100\% = \frac{T_1}{NT_N} \times 100\%$$

Efficiency can be interpreted as providing an indication of the average utilization of "N" processors, expressed as a percentage. Furthermore, this measure allows a uniform comparison of the various speedups obtained from a system containing different number of processors.

4. 5 New Technologies for Parallel Processing

Two promising technologies that will have a far-reaching impact on parallel processing are:

- Neural Networks and
- Optical Computing.

Neural Networks

Neural Networks consist of many artificial neurons, or processors, that have densely parallel interconnections. The processors communicate across the connections in terms of "activation" and "inhibition" signals that excite or inhibit responses by connected processors. Neural networks differ from one another in the way the nodes are interconnected and in the rules they use to produce a given output signal for different sets of input signals. Some networks are fully interconnected gride, with each processor directly connected to every other processor; others embody tree-like or layered architectures.

Neural networks represent a new approach to computing and artificial intelligence that shows great promise for solving problems that are clearly not suited for conventional computers and current established techniques.

Optical Computing

The main features of optical computers are parallelism and global connections. Optically interconnecting a large number of computing elements in a structure creates many independent channels that can be exploited to increase the computational power by decreasing the communications overhead. Optical signals can propagate through separate channels with essentially no interaction and through parallel channels without interference.

A neural network computer built from optical elements consists of two main components. One is a two-dimensional array of optical switching elements (neurons) which switch states to which they are connected. Each element of the array can be interconnected to all other elements by light beams. The second component consists of interconnections among the elements. Such interconnections constitute the memory of the neural computer and must be modifiable in a general purpose optical neural computer. A hologram whose volume is one cubic centimeter can specify more than a trillion connections. This is enough for all possible interconnections of one million processing elements. It is not surprising that optical neural computers combining the technologies of neural networks and optical computing form an exciting new area of parallel processing.

4.6 Exercise

Multiple choice questions

1. The most established classification of parallel processing architecture was made by
   a. Babbage
   b. Neuman
   c. Flynn
   d. None of them

2. Heterogeneous architecture is constituted by
   a. Identical processors
   b. Vector processors
   c. Mixed processors
   d. None of these

3. Speedup is the ratio of
   a. Execution time of one processor to execution time of multiple processors
   b. One processor to the number of processors
   c. Efficiency of one processor to efficiency of several processors
   d. None of these

4. The two promising technologies on parallel processing are:
   a. Networks and computing
   b. Neural networks and optical computing
   c. Optical computing
   d. None of the above

Questions for short answers

5. What is parallel processing?
6. Name the different architectures based on Flynn's classification.
7. Define the efficiency of a parallel architecture.
8. What is the speedup of an architecture possessing five processors, if the speed of a single processor is 5 sec and for five processors it is 3 sec?
9. What is the interprocessor communication?
10. What is optical computing?

Analytical questions

11. What is meant by parallel processing? Describe the measuring parameters of parallel processing.
12. Write an essay on performance evaluation issues on parallel processing.
13. Distinguish between parallel processing and multiprocessing architectures.
14. Describe the Flynn's classification of parallel processing with examples.
15. Describe the new technologies of parallel processing.

Chapter 7

Memory Organization

Lesson 1: Memory Basics
Lesson 2: Main Memory
Lesson 3: Secondary Memory

Lesson 1: Memory Basics

1.1 Objectives

On completion of this lesson you will know:

- Classification of memory
- Characteristics of memory devices
- Memory hierarchy.

1.2 Classification of Memory

A computer system contains a variety of memory devices to store instructions and data for its operation. In general, memory components of a computer system can be divided into three categories:

- Internal processor memory (or built-in memory)
- Main memory or primary memory
- Secondary memory

Internal Processor Memory: Basically it comprises of a small set of high-speed register used as a working memory for temporary storage of instructions and data. Some microprocessors also employ another type of built-in memory called cache.

Main Memory: It is relatively a large memory. This memory is used for program and

data storage during computer operation. Locations in main memory can be accessed directly and rapidly by the CPU. The technology used for main memory is based on semiconductor technology. Common examples of main memory are RAM (Random Access Memory) and ROM (Read Only Memory).

Secondary Memory: This is generally much larger in capacity but is slower than main memory. It is used for storing system programs and large data files, which are not continually required by the CPU. Information in secondary storage is accessed indirectly via input-output programs. Common types of secondary memory are magnetic disk, magnetic tape and optical disks.

1.3 General Properties of Memory Devices

Memories are based on a number of physical phenomena and principles. Knowledge of these properties is essential in evaluating memory technologies. The characteristics and principles of some specific technologies are discussed below.

Access Time: Performance of a memory device is primarily determined by the rate at which information can be read from or written into the memory. A convenient performance measure is the average time required to read a fixed amount of information (e.g. one word), from the memory. This is termed as the access time. Access time depends on the physical characteristics of the storage medium, and its access mechanism. Clearly, low cost and low access time are desirable memory characteristics. Memory units with low access time are generally expensive, while low cost memories are relatively slow. Figure 7.1 shows the average access time and cost for different memory technologies.

Cost: The cost of a memory unit is most meaningfully measured by the purchase price to the user. The price should include not only the cost of the information storage but also the cost of the peripheral equipment or access circuitry essential to the operation of the memory.

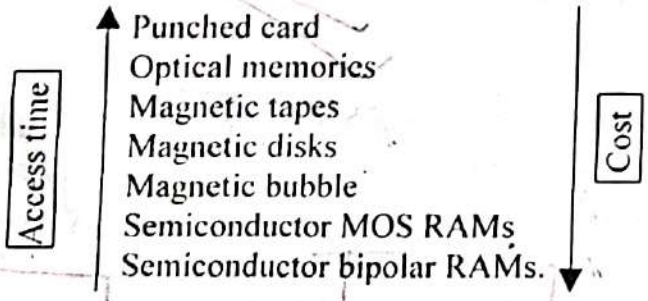


Figure 7.1: Access time and cost hierarchy for memory technology.

Access Modes: One of the important properties of a memory device is the order or sequence in which information can be accessed. Two main types of memory access techniques are:

- Random-access, and
- Serial-access

If memory locations may be accessed in any order and access is time independent of the location being accessed, the memory is termed a random-access memory (RAM). Ferrite-core and semiconductor memories are of this type. If storage locations can be accessed only in certain predetermined sequence then the memory is called serial-access memory. Magnetic-tape, magnetic-bubble, and optical memories employ serial access method.

Alterability: Two major types of main memories are ROM and RAM. Memories whose contents cannot be altered are called read-only memories (ROM). A ROM is a nonerasable storage device. ROMs are widely used for storing permanent programs. A ROM whose contents can be changed with the help of special instruments is called programmable read-only memory (PROM). RAM (random access memory) is basically a read/write memory. Information can be written into or read from a RAM while it is in use. However, it loses its content if the power supply is switched off.

Cycle Time and Data Transfer Rate: The minimum time between two consecutive memory access operations is called the cycle time. It is generally convenient to assume that cycle time is the time needed to complete any read or write operation in the memory. The maximum amount of information that can be transferred to or from the memory every second is called the data transfer rate.

Data-Transfer Rate and Bandwidth: The data-transfer rate is measured in bits or words per second. A factor limiting memory bandwidth is the memory bus width, which is the number of bits that can be transferred simultaneously over the memory bus.

Physical Characteristics: A variety of physical properties of matter are used for information storage. The important properties used for the purpose are electronic, magnetic, mechanical and optical. Table 7.1 shows the physical characteristics of some major memory technologies.

Table 7.1: Physical properties of some memory technologies.

Technology	Access mode	Alterability	Medium
Bipolar Semiconductor	Random	Read/write	Electronic
MOS Semiconductor	Random	Read/write	Electronic
Magnetic disk	Semi-random	Read/write	Magnetic
Magnetic tape	Serial	Read/write	Magnetic
CD ROM	Semi-random	Read/write	Optical
Punched card	Serial	Read/write	Mechanical

Memory Capacity

Memory capacity is the number of 0s or 1s that a memory can hold. The following terms are used to denote memory capacity.

Bit: In binary system 0 or 1 is called a bit, which is short for binary digit

Byte: A group of eight bits is called a byte. A byte is used to represent one character (A, a etc), digit (0, 4 etc) or some other special characters. The capacity of a computer's memory is expressed in number of bytes.

Kilobyte: A kilobyte (K or KB) is about 1,000 bytes (precisely 1,024 bytes).

Megabyte: A megabyte (M or MB) is about one million (precisely 1,048,579) bytes. Memory capacity of microcomputers is expressed in megabytes.

Gigabyte: A Gigabyte (G or GB) is about one billion (precisely 1,048,876) bytes. Capacity of hard disks is expressed in gigabytes.

Terabyte: A terabyte represents about one trillion (precisely 1,073,741,824) bytes.

1.4 Memory Hierarchies

A variety of factors should be considered to explore the hierarchy of memories. The major components are: nature of connection between memory and processor, cost per bit, access time and storage capacity. Figure 7.2 shows these factors in hierarchical form. It can be observed from the figure that the cost per bit increases with fall of access time and increasing closeness with the CPU (microprocessor).

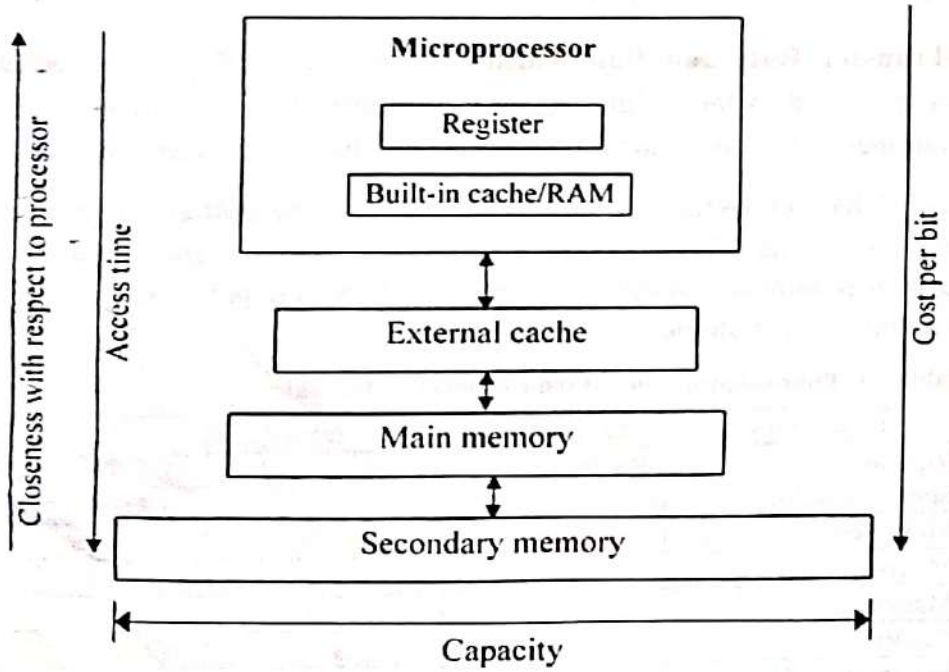


Figure7.2: Memory hierarchy.

1.5 Exercise

Multiple Choice Questions

1. Which one is fastest?

   a. MOS RAM
   b. Bipolar RAM
   c. Optical memory
   d. Magnetic disk.

2. Cost/bit of a primary memory as compared to secondary memory is

   a. low
   b. high
   c. similar
   d. none of the them.

Questions for short answers

3. What is meant by access time?
4. What do you understand by bandwidth?
5. Show the memory hierarchy by a diagram.
6. What are the categories of memory components of computer system?

Analytical questions

7. Describe the categories of computer memories.
8. Describe the general physical properties of memory devices.
9. Describe the memory hierarchy with a diagram

Lesson 2: Main Memory

2.1 Objectives

On completion of this lesson you will know:

- Permanent and programmable memories
- ROM and RAM characteristics.

2.2 Read only Memory

ROM

ROM (Read Only Memory) is used for storing programs and data permanently. The data and programs required for some applications are permanently stored in the ROM chip. Thus a user may read a program or data from ROM but may not write on ROM. The content of ROM is not altered when power is removed from the computer. It is therefore referred to as nonvolatile memory. ROM is generally used to store programs, routines, and language interpreters for immediate access by the user of the system. Such a program, usually supplied by the manufacturer, is referred to as firmware. Many microcomputers are supplied with ROM chips that contain the BIOS (basic input/output system).

PROM

PROMs (Programmable Read Only Memory) are blank chips which have nothing recorded on them. Once instructions or data are recorded into the chip by a special programming device, the PROM chip permanently stores the information like ROM. The programming of the PROM chip is generally done by the manufacturer of computer systems. PROM chips are used primarily to provide special purpose programs, which can be plugged into the main computer board.

EPROM

EPROM (Erasable Programmable Read Only Memory) chips can be programmed, as the PROM chips. But they can be erased and reprogrammed by a special programming device. The contents of an EPROM chip can be erased by exposing it to ultraviolet light. The light passing through a quartz window in the plastic container exposes the silicon chip that contains memory cells storing information.

EEPROM

EEPROM (Electrically Erasable and Programmable Read Only Memory) is similar to EPROM except that it is erased by applying electrical pulses to the chip. It is possible to reprogram an EEPROM chip via keyboard commands without removing the chip from the computer.

2.3 Random Access Memory

RAM (or Random Access Memory) is temporary storage that can be read from or written into by the user. RAM is used primarily to store user programs and data. The content of RAM is generally volatile, and is lost when power is removed. When the system is powered up, programs and data to be used must be read into RAM through an input device. Access time for data stored in RAM generally ranges from a microsecond to a few nanoseconds.

Some laptop microcomputers utilize special MOS chips that draw almost no power and preserve their contents indefinitely using an internal battery. However, these low-power chips are more expensive than their high-power counterparts.

RAMs are characterized by the fact that every location can be accessed independently. The access and cycle times for every location are constant and independent of its position. Single-chip RAMs are manufactured in a wide range of sizes from a few hundred bits to a megabit (2^{20} bits) or more. Both bipolar and MOS transistor circuits are used for RAM chips. The present limitations of IC manufacturing makes it impossible to manufacture, say, a gigabit (2^{30}-bit) RAM on a single chip. Consequently, very large semiconductor RAMs must be constructed by interconnecting a number of RAM chips. Semiconductor memories fall two main categories,

- Static RAM and
- Dynamic RAM.

Difference between SRAM and DRAM follows:

SRAM	DRAM
• SRAM stands for Static Random Access Memory	• DRAM stands for Dynamic Random Access Memory
• High speed RAM	• Relatively slower
• SRAM is Costly	• Relatively low cost
• Does not require refreshing	• Requires refreshing periodically
• Generally used for cache	• Used for main memory
• Control complexity is less	• Relatively high control complexity

Differences between RAM and ROM are as follows:

RAM	ROM
RAM stands for Random Access Memory	ROM stands for Read Only Memory
User can read or write into it	User can only read from it
For any type of processing, user must store program and data into RAM	Manufacturer stores instruction into the ROM permanently
Temporary or volatile memory	Permanent or non-volatile memory

2.4 Cache Memory

As mentioned earlier the memory is slower than the CPU. System performance suffers when a fast device must wait to access data from a memory system. This may be an I/O device or the CPU, and the memory system may be main memory or an external device. Because computer users always seem to want larger memories, computer designers must carefully decide memory cost and performance. One way to improve the performance is to incorporate a cache memory between the CPU and main memory. Cache memories are high-speed buffers for holding recently accessed data in main memory. By adding a cache memory between the fast device and the slower memory system, an apparently fast memory system can be provided.

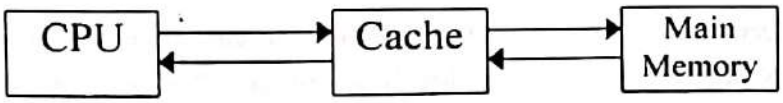


Figure 7.3: Cache and main memory.

2.5 Exercise

Multiple choice questions

1. Which one is the permanent primary storage?

   a. RAM b. EPROM
   c. ROM d. None of them.

2. Cache is a high speed

   a. DRAM b. EEPROM
   c. SRAM d. Floppy disk

3. The EPROM stands for

   a. Electrically and programmable read-only memory.
   b. Erasable and programmable read-only memory.
   c. Extended programmable read-only memory
   d. none of the above.

4 The two types of RAMs are

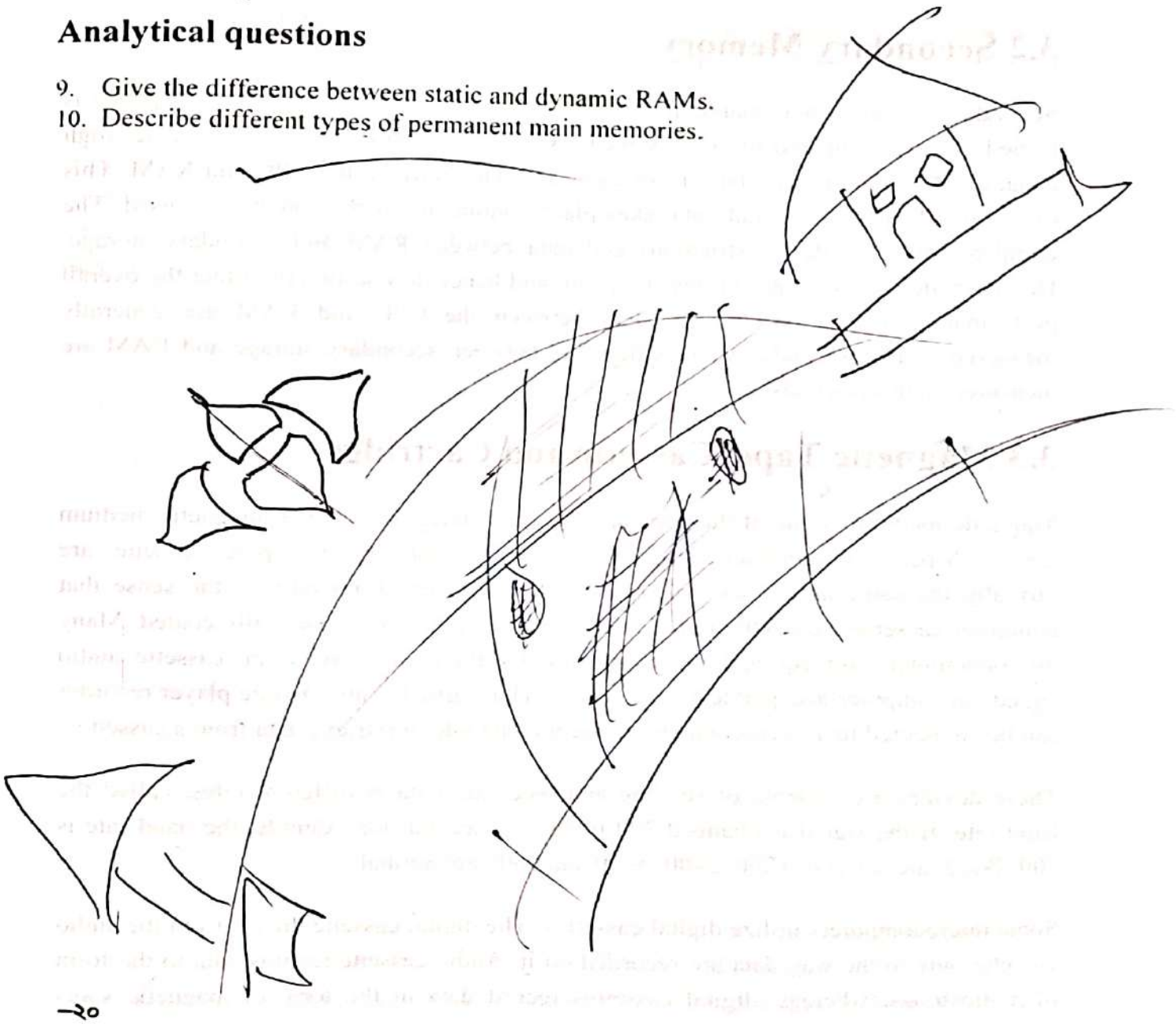
a. dynamic and ROM
b. static and dynamic
c. static and magnetic core
d. none of them.

Questions for short answers

5. Differentiate between static and dynamic RAMs.
6. Differentiate between RAM and ROM
7. Differentiate between EPROM and EEPROM
8. What is the cache memory?

Analytical questions

9. Give the difference between static and dynamic RAMs.
10. Describe different types of permanent main memories.



Lesson 3: Secondary Memory

3.1 Objectives

On completion of this lesson you will know:

- Importance of secondary storage
- Characteristics of secondary storage
- Main difference between primary and secondary storage.

3.2 Secondary Memory

Secondary storage is nonvolatile, that is information stored in it is not lost when power is turned off. As compared to primary memory it is slow, less costly and possesses high capacity. A computer transfers instructions and data between the CPU and RAM. This exchange of instructions and data takes place continually as the computer is used. The computer also transfers instructions and data between RAM and secondary storage. However, this transfers are far less frequent, and hence do not directly affect the overall performance of the system. Transfers between the CPU and RAM are generally measured in nanoseconds, whereas transfers between secondary storage and RAM are measured in milliseconds.

3.3 Mágnetic Tape, Cassette and Cartridge

Magnetic medium is one of the popular secondary storages. Common magnetic medium used with personal computers is the tape, cassette or cartridge. Computer cassettes are virtually the same as the standard audio cassettes. They differ only in the sense that computer cassettes generally do not have a leader that is nonmagnetically coated. Many microcomputers are equipped with standard interfaces to convert the cassette audio signals to computer-acceptable binary signals. Thus virtually any cassette player/recorder can be connected to a microcomputer to record data into or retrieve data from a cassette.

These devices are capable of sending and receiving data at different rates, called the baud rate. If the signal is changed 300 times in a second, for example, the baud rate is 300. Baud rates of 500, 1200, 2400, 5600 and 9600 are normal.

Some microcomputers utilize digital cassettes. The digital cassette differs from the audio cassette only in the way data are recorded on it. Audio cassette records data in the form of audio tones; whereas, digital cassettes record data in the form of magnetic spots

representing binary 1s and 0s. The storage capacity and cost of a digital cassette can be several times that of an audio cassette. The principal use of digital cassettes is a backup medium for high-capacity storage units.

The main advantages of the tape medium is simplicity and low cost. Its disadvantages include low speed. and that it can only read and write sequentially.

3.4 Floppy Disk

The advantage of the floppy disk. or diskette. over the magnetic tape. cassette or cartridge is that the disk can be accessed directly. Any item of data or program stored on it can be accessed without accessing information stored previously. A Floppy disk consists of a piece of magnetically coated mylar enclosed in a plastic or cardboard jacket (Figure 7.4). Floppy disks are generally available in either a 5.25 inch diameter or a 3.5 inch floppy diskette. The 5.25 inch diameter floppy disk. is sometimes referred to as a mini-floppy and the 3.5 inch disk is called a micro-floppy. Other than their outward appearance they are very similar.

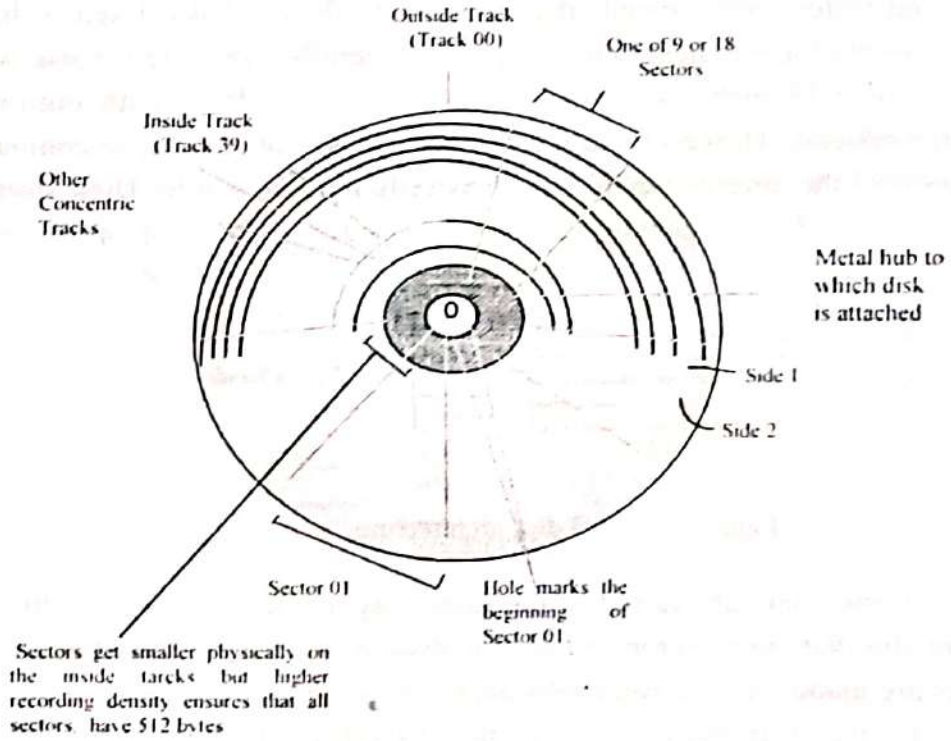


Figure 7.4: Sectors and tracks on a floppy disk

The floppy disk is held in the center of a disk drive by a motor-driven shaft that spins the disk within its protective jacket at 300 to 360 rotations per minute. Data are read from or

written onto the disk by a read/write head. This head moves in or out over the head-access slot in the jacket to access different tracks on the surface of the disk.

Floppy disks are organized into sides, tracks, and sectors. The surface of some micro diskettes are subdivided into 80 concentric circular areas or tracks and each track is divided into 9 or 18 pie-shaped segments or sectors (Figure 7.4).

Capacity: A disk has two sides. The total number of sectors in a 3.5-inch commonly used floppy disk with 80 tracks per side and 18 sectors per track is

$$2 \text{ sides} \times 80 \frac{\text{tracks}}{\text{side}} \times 18 \frac{\text{sectors}}{\text{track}}$$

that is, 2880 sectors. Generally a sector stores 512 bytes (approximately 0.5 KB) and the capacity of the two-sided floppy disk is approximately 2880×.5 KB that is 1.44 MB.

3.5 Hard Disk

For more capacity and faster access speeds than those with floppy disks, fixed or hard disk storage is used in microcomputers. The hard disk generally available for use with microcomputers (3.5 or 5.25 inch) is similar to those disks available with mini-and mainframe computer systems. However, unlike the hard disks for the larger computer systems, the hard disks of the microcomputers are generally not removable. Their storage capacity can range from hundred megabytes to many gigabytes.

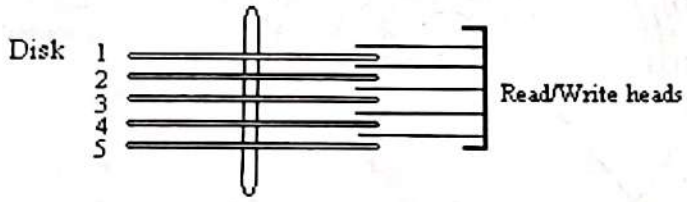


Figure 7.5: Hard disk architecture

In some hard disk systems, data are stored in the same way as they are on diskettes. A series of tracks are divided into sectors when the disk is formatted. As their name suggests, hard disks are made out of a rigid substance that is capable of storing a greater amount of data than the soft materiel used for diskettes. Hard disk drives for microcomputers can be internal unit into the computer cabinet and nonremovable or outside the computer cabinet and connected to it by a short cable. A hard-disk have the following characteristics:

- It has rigid metal platters connected to a central spindle
- The entire disk unit (disk and read/write heads) is placed in a sealed container
- Air that flows through the container is filtered to prevent contamination;
- Disks are rotated at very high speed (usually around 3,600 RPM)
- A disk drive can have four or more disk platters in a sealed unit.

Capacity: A hard disk system is actually a stack of platters, and the term cylinder is used to refer the same track across all disk sides.

For example, track 1 on every disk is in cylinder 1. Hard disks generally store 512 bytes of data in a sector, but a hard disk has more sectors per track. Characteristics of a hard disk are given below:

Number of tracks = No. of cylinders × No of heads
Number of sectors = No. of tracks × No. of sectors/track
Capacity = No of sectors × 512 bytes/sector

3.6 Optical Disk

Nonerasable Optical Disks

Optical disks have been used for music, video and audio recording for some time. Like the magnetic disk, data on optical disk are recorded in sectors within tracks. The recording technique commonly used with optical disks prohibits new data from being recorded over the old. This is because the physical makeup of the disk is altered during the writing process.

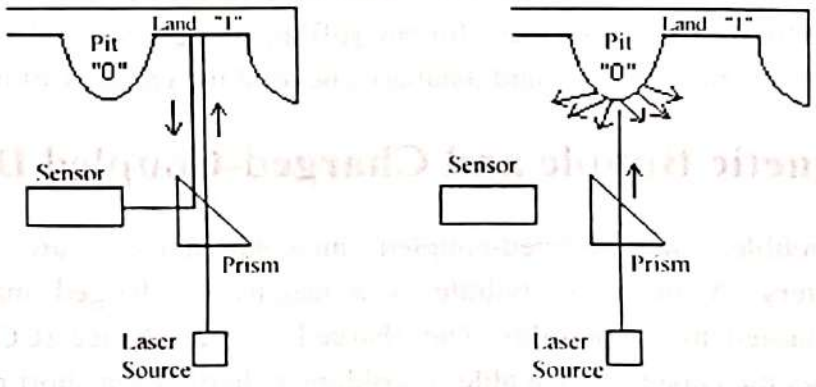


Figure 7.6: Data reading from a CD-ROM

Writing is accomplished by momentarily directing a laser beam on the surface of the disk. The laser beam causes a change on the surface approximately one micron (millionth of an inch) in diameter. Data are stored in the form of lands and pits as shown in (Figure7.6). To read from an optical disk, light is focused on a tiny area of the surface of

the disk passing below the read head. The intensity of the light reflected from this spot is monitored by a light sensor. If the light reflected from this spot is of a high intensity, it is interpreted as 1 otherwise it is interpreted as 0

Initially the CD-ROM drives read data at 150 KB per second and this speed is known as single speed. Today's CD-ROM drives read data many times faster then the first model and their speeds are identified by 16x, 32x, 48x etc. For example 48x stands for 48×150KB (or 7.2 MB) per second.

The life expectancy of data stored optically is approximately three times the shelf life of magnetically recorded information. In addition, since there are no heads flying above the surface, there is no possibility of heads colliding with the disk surface. Optical media have an enormous storage capacity, ten to thirty times that of magnetic media. For example, up to 100 gigabytes (1 gigabyte = 1 billion bytes) can be stored on a 14-inch disk. Optical disks are ideal for storing large volumes of relatively permanent data such as high-resolution graphics, photographic images, or huge volumes of text.

Erasable Optical Disks

A recording technique that combines optical and magnetic technologies is used for erasable optical disks. This technique utilizes a laser beam to heat a minute spot on the disk, altering its properties so that it can be magnetized by a magnetic field applied simultaneously. A magnetic field alone does not affect the data stored on the surface of the disk. However, applying the laser beam to the surface without a magnetic field erases the data recorded within the range of the laser beam. This device is not a replacement for magnetic media, but a replacement for microfilm, microfiche, and paper document and for the storage of massive files and databases beyond the capacity of magnetic media.

3.7 Magnetic Bubble and Charged-Coupled Devices

Magnetic bubble and charged-coupled memory devices are used with some microcomputers. A magnetic bubble is a negatively charged magnetic field in a positively charged magnetic film. The charged-coupled device (CCD) is made up of many microscopic capacitors capable of holding a charge for a short period of time. The presence of a charge represents a binary 1 and its absence represents a 0.

Bubble and CCD devices are less affected by extreme environments (heat, cold, humidity, etc.), requires less maintenance, and are much faster. Of the two, bubble memory is more desirable because of its greater storage capacity and nonvolatility. CCDs are volatile, but are faster than magnetic bubble devices.

3.8 Comparison of Primary Memory and Secondary Memory

Characteristics	Primary Memory	Secondary Memory
Location with respect to the CPU	Inside/Outside and directly accessible by CPU	Outside and indirectly accessible by CPU
Cost	Most expensive	Less expensive then primary storage
Capacity	Lower as compared to secondary memory	Several thousand times higher as compared primary memory
Average access time	In billionths of a second	In millionths of a second
Data processing directly from storage.	Yes	No, data must first be moved into primary memory
Means of storing information	Semiconductor chips	Magnetic disk, tape and optical disk

3.9 Exercise

Multiple choice questions

1. Which one is not a secondary storage?

   a. HD b. FD
   c. RAM d. CD-ROM.

2. Secondary storages are

   a. nonvolatile b. volatile
   c. nonvolatile and volatile d. none of these.

3. In-terms of speed a secondary storage as compared to a primary storage is

   a. slower b. faster
   c. similar d. none of these.

4. Size of micro-floppy is

   a. 5.25-inch b. 4.5-inch
   c. 3.5-inch d. none of these.

5. Considering speed HD as compared to FD is

   a. faster b. slower
   c. similar d. none of these.

Questions for short answers

6. Mention the use of secondary storage in a computer system.
7. Briefly describe a floppy disk.
8. What is meant by tracks and sectors of a floppy disk?

9. Give the difference between hard disk and floppy disk.
10. What the following acronym stands for: CD-ROM, WORM and CCD?
11. Distinguish between primary memory and secondary memory.

Analytical questions

12. Describe magnetic tape with advantages and disadvantages.
13. Describe floppy disk and hard disk with a comparison between them.
14. What is the CD-ROM? Describe its working principal.
15. Calculate the capacity of a 2-sided floppy disk having 80 tracks and 36 sectors per track.
16. Calculate the capacity of a hard disk having 1632 cylinders, 12 heads, 54 sectors/track.

Chapter 8

Computer Software

Lesson 1: Introduction and Classification
Lesson 2: Application Package Programs
Lesson 3: High Level Languages

Lesson 1: Introduction and Classification

1.1 Objectives

On completion of this lesson you will know:

- Concept of software
- Difference between system software and application software
- Components of system software
- Types of application software.

1.2 Software

Software of a computer system is intangible rather than physical. It is the term used for any type of program. Software consists of statements, which instruct a computer to perform the required task. Without software a computer is simply a mass of electronic components. For a computer to input, store, make decisions, arithmetically manipulate and output data in the correct sequence, it must have access to appropriate programs. Thus, the software includes all the activities associated with the successful development and operation of the computing system other than the hardware pieces. Hardware does the actual computing, and the software drives the computer. Figure 8.1 shows the communication between users and hardware. Software plays the role to connect the users and the hardware. The application software convey the user intention to system software and the system software transfers and translates the intention for the hardware.

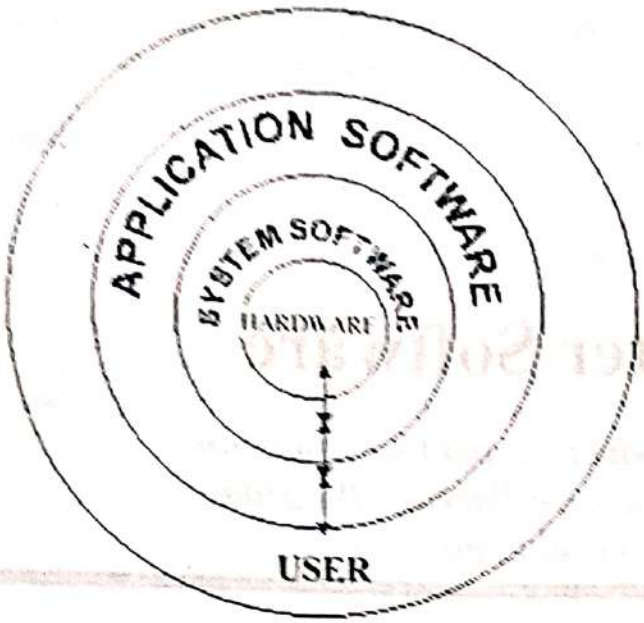


Figure 8.1: User and hardware communication

Software activities cover three major areas:

- Application programs,
- Systems programs, and
- Documentation.

An application program performs a specific, well-defined task for a particular application.

Systems programs consist of programs associated with a computing system other than application programs. Systems programs are usually available from the computer manufacturer as a part of the overall system. They aid in the creation of application programs.

Documentation includes everything put down on paper, such as a statement of the problems, flowcharting and coding, instruction sets, procedures, and the like. It plays an important role in software development.

1.3 Classification of Software

Software is generally classified as:

- System software
- Application software.

Figure 8.2 shows classification of computer software. System software consists of programs which facilitate the use of the computer by the users. These programs, perform such standard tasks as organizing and maintaining data files, translating programs written in various languages to a form acceptable to the hardware, scheduling jobs as well as aiding in other areas of computer operations. On the other hand, application software includes programs designed to perform specific user applications.

Application programs can be developed by the users themselves using suitable programming languages. There are many programming languages which allow us to write computer instructions in a text form. BASIC, FORTRAN, COBOL, PASCAL, C, Java are some well known programming languages. However, writing application programs using such languages requires expertise and experience in programming and only experienced programmers can write good quality application programs.

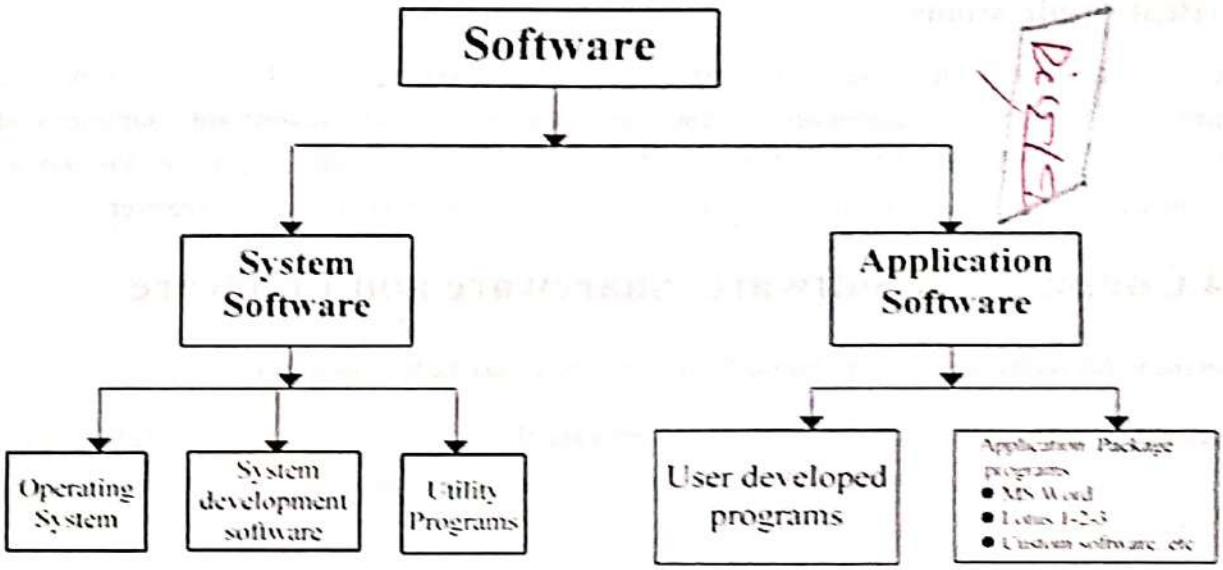


Figure 8.2 Classification of software

Many application programs are now available commercially. Such programs, developed by experienced programmers can be readily employed for user applications. These programs are popularly known as application packages or simply packages. MS Word, WordPerfect and WordStar for word-processing, Lotus 1-2-3, and MS Excel for spreadsheet analysis and dBASE, FOXPRO, and ORACLE for database management are some popular application packages.

Application program refers to all programs that enable computer users to apply the computer to the work they do. Application programs fall into two categories:

- Horizontal applications and
- Vertical applications.

Horizontal Applications

They are general purpose programs that address the needs of many people, such as writing document (word processing), working with numbers (spreadsheets) and keeping track of information (database). There are three types of horizontal applications.

Personal productivity programs: Most popular programs, examples are, word processing programs, spreadsheet programs and database programs etc.

Multimedia and graphics programs: Popular presentation programs, an example is, MS Power Point.

Internet Programs: Programs for using the Internet, for instance, e-mail programs, web browsers etc.

Vertical Applications

They are generally designed for a particular line of business or for a division in a company, for instance, garments production tracking program, restaurant management program etc. Vertical software designed for a customer of specific jobs is known as custom software. Generally vertical software is costly due to small size of market.

1.4 Commercial Software, Shareware and Freeware

Commercial software is copyrighted and one must pay before using it.

Shareware software is also copyrighted software that one can use on a "try before buy" basis. If a user likes to buy it after using it for a specified trial period, the user must pay the registration.

Freeware reefers to copyrighted software given away free, with the understanding that one can not sell it for profit.

1.5 Exercise

Multiple Choice Questions

1. Broadly, two categories of software are
    a. System software and horizontal software
    b. Vertical software and application software
    c. System software and application software
    d. none of these

2. Which one is not horizontal software?

a. MS Word b. MS Access
c. MS Excel d. none of these

Questions for short answers

3. Define the terms: software, system software, application software and package program.
4. Distinguish between hardware and software.
5. List names of at least five package programs.
6. Using a diagram, show the communication between user and hardware.
7. Differentiate between horizontal and vertical applications.
8. What is meant by custom software?
9. What do you understand by shareware and freeware?

Analytical questions

10. What do you understand by the term software? Critically discuss classification of software.
11. Write a short note on application software.
12. Discuss application programs with classification.

Lesson 2: Application Package Programs

2.1 Objectives

On completion of this lesson you will know:

- Types and characteristics of application packages
- Names of common application packages
- Features of some popular packages.

2.2 Introduction

Package programs are developed by experienced programmers and marketed by software vendors for widely used applications. The package generally achieves the goals of lower cost and reduced risk of errors. Whenever feasible, the use of an appropriate application package should be considered as an alternative to program development.

Integrated packages combine several applications into a single program package. Popular examples of integrated packages are Microsoft Word, WordPerfect, MS Office, Lotus 1-2-3, Symphony etc. The packages provide several standard applications such as word processing, spreadsheet analysis and databases. Data can be easily exchanged between the programs of the integrated packages.

2.3 Examples of Package Program

There are many different kinds of programs available in the market to solve user problems. These are known as package programs. Package programs are very popular and widely used in different disciplines. The main categories and their examples are given in Table 8.1.

2.4 Advantages of Package Programs

Advantages and benefits of using packages for popular applications are briefly described here.

Word Processing

- Document can be typed and then stored on a disk for later use.
- Easy text entry and editing.
- On-screen formatting: such as bold, italic, underlining, and highlighting etc.

- Spelling checker, thesaurus, and grammar checker to improve accuracy of writing.
- A variety of print options to customize the program for printer.
- The capability to change the font and style of texts.
- The capability to interface with graphics and other software packages.

Table 8.1: Popular Package programs

Categories	Popular Package Programs
Word Processors	Word, WordStar, WordPerfect, Latex etc.
Spreadsheets	Lotus 1-2-3, Excel, Quicken, QuattroPro, SuperCalc.
Data Management	dBASE, Access, ORACLE, FoxBASE+, Idealistic, SuperBase Professional etc.
Integrated Software	PFS: First Choice, Microsoft Works, Enable OA, WordPerfect, MS Office etc.
Desktop Publishing	PageMaker, Quark Xpress, Venture Publisher, Professional page, PFS: First Publisher, Express Publisher, Publish it! etc.
Graphics	PC Paintbrush, CorelDRAW, Harvard Graphics, Freelance Graphics etc.
Presentation	Microsoft Powerpoint, Corel Presentation, Lotus Freelance etc.
CAD	AutoCAD, Generic CADD, ORCAD, MiniCAD plus, Autosketch, TurboCAD etc.
Communications	PROCOMM PLUS, CrossTalk, PFS: First Choice (Communications), MicroPhone II, White Knight, SmartCom II, PC-Dial, PC-Talk etc.
Educational	World Atlas, The Human Brain, Spelling Bee, Fantavision, HyperStudio etc.
Recreational	Indy 500, Chuck Yeager's Advanced Flight Trainer, Monopoly, Hard Ball, Omar-sharif etc.
Personal Information Manager (PIM)	Microsoft Outlook, Lotus Oraganiser etc.

Spreadsheet Analysis

- Data entry can be automated and checked for errors.
- Easy editing
- The column-and-row format are easy to understand and use.

- Formulas perform calculations and are saved with spreadsheet.
- One can sort, copy, and move columns and rows with formulas intact.
- One can print a spreadsheet in a variety of formats.
- Built-in graph generator to create graphs from the data in a spreadsheet.

Data Management

- One can reuse the data, access data easily, edit, sort, search, and arrange the data, create report, print labels etc.
- One can arrange data easily by using a database program's search capabilities.
- Printing features enable one to generate mailing labels and reports.

Integrated Software

- Data are compatible among applications. As an example, one can transfer data from database or spreadsheet to word processor or vice versa.
- The menu system is generally the same, lessening the number of menus and options.
- Having an all-in-one package means that less time is spent for loading, opening and closing files.
- Integrated software costs less than the same number of applications purchased individually.

Desktop Publishing

- On-screen formatting and layout facility of text and graphics.
- The capability of assembling documents quickly and making changes easily.
- Import text from any word processor or type text directly in the program.
- Print on a variety of printers, from dot-matrix printers to postscript laser printers.

Graphics and CAD

- One can use the art to create over and over again without losing the original quality.
- With on-screen graphics, it is not required to go back to the drawing board each time.
- One can make small modifications to the original art until achieving the desired design.
- One can create a variety of charts easily with presentation graphic software.
- Most paint programs provide a variety of shading and filing techniques.
- The art tools in a graphics program give control over the precision of creations.

- Draw programs produce graphics in postscript printer format.
- With graphics programs one can import art directly into popular desktop publishing programs.

2.5 Popular Package Programs

Microsoft Word

Microsoft Word (MS Word) possesses a wide range of powerful and easy-to-use features. Word is popular among PC and Mac users. With Microsoft Word, one can integrate text with graphics and revise information within a document. Microsoft Word has special features such as print preview, side-by-side columns, built-in outlining, and automatic pagination. Word for Windows allows one to open multiple windows on the currently open document or more than one document at one time. Word is designed to help print reports, memos, and documents and performs all the basic word processing techniques, such as deleting, italicizing and highlighting. Word also has flexible tabs to produce great-looking tables and can mix fonts and font sizes easily. Microsoft Word has spelling checker, thesaurus, equation editor, sorting facility, drawing facility, spreadsheet facility to generate graphs and an undo command. Many users prefer Word because of its compatibility to other systems. Using word it is possible to share data between PC and Mac systems. It is also compatible with WordPerfect for Windows, PageMaker and many other windows based software.

WordPerfect

WordPerfect (WP) is another popular word processing package. A full-featured word processor with an extensive range of features, WordPerfect appeals to users who demand many sophisticated tasks from their word processor. Easy text entry, powerful thesaurus and spell-checking routines, support for a wide range of fonts and printers, capability to create customs dictionaries, on-screen formatting, print preview, graphics interfacing and the column creating facilities are the main features of this program. WordPerfect also has a master document features for assembling one large document from a series of subdocuments, and a tutorial to help the inexperienced users. One also can use WordPerfect to program, to plan, and to draft writing projects.

Microsoft Excel

Microsoft Excel is one of the most popular spreadsheet packages originally available only for the Macintosh. Microsoft Excel for other microcomputers works with Microsoft Windows. Many operations that require typing in other applications can be handled with the mouse in the Excel. It also has options for producing a variety of reports in different fonts, a feature usually limited or unavailable in many other spreadsheets packages.

Lotus 1-2-3

Another popular spreadsheet package is Lotus 1-2-3. It combines three main features: spreadsheet, database and graphics. With lotus 1-2-3, users can create simple or complex spreadsheets, use the spreadsheet as a database, or graph spreadsheet information. A wide range of people use Lotus 1-2-3. From accounts to financial planners, business owners, and managers, Lotus 1-2-3 is the spreadsheet package of choice. 1-2-3's many features include an easy-to-use spreadsheet layout, an understandable control panel, a wide range of functions, easy printing procedures, graph creation and printing and macro capability. Lotus 1-2-3 for windows is available.

Access

Access is the Microsoft's formidable desktop database platform and Access 2000 is the third in the line of 32-bit Access versions. Microsoft Access and its database engine dominate the desktop database market. Access is designed for creating multiuser applications where database files are shared on networks, and Access incorporates a sophisticated security system to prevent unauthorized persons from viewing or modifying databases.

Access has a unique database structure that can combine all related data tables and their indexes, forms, reports, and VBA (Visual Basic for Applications) code within a single .mdb database file. Like all members of Office 2000, Access 2000 offers a variety of new Internet-related features for creating HTML documents for use on intranets and the Internet.

A primary reason for Access's success is that it duplicates on the PC desktop many of the features of Client/Server relational database systems, also called SQL database.

All members of Office 2000 and Visual Basic 6.0 now share a common (VBA) version 6.0. Access 2000 also shares the new HyperText Markup Language (HTML). Millions of copies of Access are in use for applications ranging from cataloging recipes to conducting electronic commerce through the Internet.

dBASE

The dBASE is an integrated software package; which has menu oriented database processing and programming facilities. It is a data manager; a piece of software that lets the user have freedom in the conceptualization and creation of databases for all types of business applications. Since business depends on timely information dissemination, the value of a powerful programmable utility for database generation, maintenance, and

query cannot be overstated. The dBASE can be executed on a variety of microcomputers, under any one of the popular operating systems. Different version of dBASE is known as dBASE III and dBASE III +, dBASE IV etc.

Microsoft Works

Microsoft Works is an integrated program that combines: word processing, spreadsheet, database and communications. Microsoft Works has enthusiastic following in both the PC and the Mac worlds. Because of easy features and friendly user interface, the applications in Microsoft Works are easy to understand and use. Its word processor enables one to enter, edit, save, and print texts and reports. Specialized search-and-replace features, thesaurus and spelling·checker also help to ensure the accuracy of work. Using Microsoft Works spreadsheet, one can create simple or complex works, perform what-if calculations, and use formulas. The data manager gives a great deal of flexibility in generating data-entry forms, working with fields, displaying data, and generating reports. The communications module accesses the world outside a computer by transferring files to and from remote computers. Finally, the features in Microsoft Works can be used to format, merge data between applications, and exchange information with other programs and modules.

PFS: First Choice

PFS: First Choice, one of the most popular integrated packages for microcomputers combines the following applications:

- Word processing
- Database management
- Communications
- Spreadsheet
- Graphics.

It has a multitude of users from sophisticated business managers to the hobbyists. Some users include business people who can type correspondence, balance books, and manage files, and business owners and managers who can analyze and maintain information. Other users include support staff working in data entry, creating visual presentations, or producing reports. PFS: First Choice automates many business tasks traditionally done by hand. With this program, one can perform accounting tasks, such as analyzing data and creating and printing spreadsheets. The word processor can be used to type correspondence, produce manuals, print documents, create presentations, and so on. PFS: first Choice also enables one to graph data, print graphs, plot graphs, create slide shows, and perform basic data-entry tasks. With the database management part of PFS: First

Choice, one can design database forms, create mailing labels, edit and analyze data, update files, retrieve files, and create and print database reports.

PageMaker

PageMaker is a popular desktop publishing software package for microcomputers. Also called a page-composition program, PageMaker enables one to combine text and graphics on a page to compose documents. PageMaker accept text from a variety of popular word-processing programs and graphics in popular file formats. PageMaker can be used to format a page in a number of columns and formats, use different fonts, import sophisticated graphics, create on-screen graphics, create master pages, and perform many high-end functions. Although it is a high-end desktop publishing program, many people find the program easy to learn and use. Used with Microsoft Windows, PageMaker has familiar menus, pop-up dialog boxes and displays that most find easy to understand and use.

Harvard Graphics

Harvard Graphics is a popular presentation graphics program for PC users. Used primarily in business, Harvard Graphics is a feature-laden program with many enhancement options, but most users can find their way around easily. A comprehensive help system assists in learning the use of the software. Few features of the program are:

- The Chart Gallery, offering several chart types in a variety of formats.
- The Draw utility, which can add and customize art to charts.
- The ability to put charts in an automated slideshow.
- The choice of animating arts giving the illusion of real animation in presentations.
- Supports for data imported from a variety of spreadsheet programs, such as Lotus 1-2-3 and Microsoft Excel.
- Output on display, printer, plotter, or film recorder devices.
- Templates and macro creating facility.

AutoCAD

AutoCAD is one of the best CAD programs in the market. It offers the following features:

- An easy-to-use interface with pull-down menus and dialog boxes.
- Support for a mouse or digitizer.
- Capability to layer, rotate, copy, mirror, move, stretch, and scale.
- 3D wire frame modeling.
- Many different views.

The version of Auto CAD which can be used with Microsoft Windows is easy-to-use.

2.6 Exercise

Multiple Choice Questions

1. Which one is not word processing software?

   a. MS WORD
   b. WordPerfect
   c. WordStar
   d. MS Excel

2. Which one is database software?

   a. MS WORD
   b. DBSE
   c. ORACLE
   d. PC Paintbrush

3. Which one is not a graphics software

   a. Harvard Graphics
   b. Freelance Graphics
   c. PageMaker
   d. Access

Questions for short answers

4. Distinguish between single application and integrated packages.
5. List popular desktop publishing packages and database management packages.
6. List the features of Microsoft word.
7. List the features of a graphics program.
8. List the features of a CAD program.

Analytical questions

9. What is an integrated package? Discuss its features and advantages. Give one example with its capability.
10. What is a graphic package? Give at least two examples of graphic packages and discuss the characteristics of any one of them.

Lesson 3: High Level Languages

3.1 Learning Objectives

On completion of this lesson you will know:

- Generation of programming languages
- Different programming languages.

3.2 Programming Languages

A Computer does not have a brain like ours and it cannot do anything by itself. A computer cannot understand human languages, so an instruction for a task has to be told in a special computer language. One can give an instruction to a computer which is to be carried out straight away. It is also needed to store a set of instructions which are to be carried out over and over again or at a later time.

A set of instructions written in a computer language is called a program. Writing a program is called programming and the languages used are called programming languages. A program can be stored on a disk, so that it is available to be run when required.

Inside the computer, operations are done using electronic pulses. A special program translates instructions into electronic pulses that the computer can understand. In a program the instructions contain special words and symbols and the order of the words is very important. The pattern of electronic pulses for an instruction is called machine code. One could write instructions directly in machine code, but it is very difficult. In assembly language, instructions are given by short forms, called mnemonics. High level languages are much easier to understand than assembler mnemonic or machine code. Instructions of a program written in high level languages are translated by computers, in machine language with the help of the compilers before execution. Programming languages can be classified on the basis of generations. These are as follows:

- First generation (1GL): machine languages (based on 0 and 1)
- Second generation languages (2GL): assembly languages (based on special code)
- Third generation (3GL): high level languages (similar to English language, e.g., BASIC, C etc)
- Fourth generation language (4GL): nonprocedural languages.

High Level Languages

There are several high level programming languages in use. Following computer

languages are popular and have widespread applications:

- BASIC and Visual Basic (VB)
- C and C++
- FORTRAN
- PASCAL and Delphi
- COBOL
- Smalltalk
- Java and JavaScript
- HTML etc.

BASIC, QBASIC and Visual Basic: BASIC stands for Beginner's All purpose Symbolic Instruction Code. It is a comparatively easy language for beginners. Initially BASIC was the commonly used language for microcomputers. It was developed by J.G. Kemeny and Thomas Curtz of Dartmouth College of USA in 1964. Many different versions of BASIC are in use. Most microcomputers use QBASIC devised by a company called Microsoft. BASIC has several advantages over the other computer languages. they are:

- It is easy to learn and use.
- It presents several powerful additional facilities not available in many other languages.
- It is designed for interactive use rather than batch processing mode. The user can correct errors as and when they arise during program execution.

The language is easily adaptable on all computers. BASIC is easy and suitable for non-professionals.

Microsoft Visual Basic (VB) is a widely used program development package. It is by far the most successful rapid application development (RAD) tool for Microsoft Windows applications. VB enables a programmer to develop an application quickly by designing the graphical user interface. The programmer doesn't have to worry about any of the code that generates the user interface; it is handled automatically by the VB compiler, which creates an executable program capable of running on its own. Using VB, even a novice programmer can develop an impressive application. A huge variety of accessory tools enable VB programmers to quickly accomplish tasks such as building interfaces to databases and increasing the functionality of Web pages.

VB script: It is a scripting language based on VB. It was created for writing scripts (short program) that can be embedded in Web pages.

C and C++: Dennis Ritchie, a programmer at AT&T Bell Laboratories in USA designed the C language in the mid 1970s. C allows a programmer to write code with low-level access to the hardware but with high-level structured programming concept. C is a popular languages. C proved suitable for writing programs that had previously been written exclusively in assembly languages. AT&T UNIX operating system was written in C. The language is portable to microcomputers. Most PC applications and system programs are written in C.

A more recent version of C is C++, which incorporates object oriented features, but does not force programmers to adhere to the object oriented model. Thus, it provides flexibility and the high execution speed. The C++ language is in widespread use for professional program development. Software giant Microsoft Corporation uses C++ for application development.

FORTRAN: FORTRAN stands for Formula Translation. It was the first programming language developed by IBM in 1957. It has been modified many times and the later versions were given the name like FORTRAN II and FORTRAN IV, the most popular version which was developed in 1966. Latest additions are FORTRAN 77 and FORTRAN 90. FORTRAN syntax is very rigid. It is popular in engineering and scientific calculations.

Pascal and Delphi: Pascal is popular on small computers. Pascal was designed and developed by Nicholas Whirth to teach computer programming to students. It belongs to ALGOL family. A Pascal program consists of nested blocks for structured programming. Like most older programming languages, Pascal has been updated to reflect new approaches to programming. An object oriented version of Pascal provides the foundation for Delphi, a compiler created by Borland International that closely resembles Visual Basic.

COBOL: It stands for Common Business Oriented Language. It is a popular business language for data processing. It was designed in late 1960s with a view to provide a common language for business applications. COBOL is independent of any specific computer system, and thus a program written in the language can be transferred to another computer with marginal changes only.

CORBA: It stands for Common Object Request Broker Architecture, most commonly known as middleware. CORBA defines how objects can communicate with each other across a network, even if the objects are written in dissimilar programming languages. Microsoft Corporation has its own standard, called Component Object Model (COM),

which works best in Windows platforms.

Smalltalk: Developed in the early 1970s at Xerox Corporation. Smalltalk is considered as 100 percent pure object oriented programming language. Although the language is not often chosen for professional software development, more than a few corporations have chosen Smalltalk for mission-critical applications. However, Smalltalk is facing stiff competition from the newest object-oriented languages on the scene, Java.

Java: It is one of the most powerful and relatively new programming languages. It is an object oriented programming language, created by Sun Microsystems, is revolutionizing the programming applications for the World Wide Web (WWW) and corporate intranets and extranets. Java is similar to the C++ and is much simpler, secure and platform independent programming language. Java is also specifically designed for real-time, interactive, Web-based network applications. Java applications consisting of small application programs, called applets, can be executed by any computer and any operating system anywhere in a network.

The ease of creating Java applets and distributing them from network servers to client PCs and network computers is a major reason for Java's popularity. Applets can be small special-purpose application programs or small modules of larger application programs. Applets can reside at Web sites on a network server until needed by client systems, and are easy to distribute over the Internet, intranet or extranets. Applets are platform independent too – they can run on Windows, OS/2, UNIX and Macintosh systems without modifications.

Like VBScript, **JavaScript** is a scripting language designed for use on Web pages. JavaScript is not based on Java. It was created by Netscape Communications. It is simple, easy-to-learn language for writing scripts on Web pages, and it is in widespread use on the Web.

HTML: It stands for HyperText Markup Language, it is a page description language that creates hypertext or hypermedia documents. HTML inserts control codes within a document at points that create link (known as hyperlink) to other parts of the documents or to other documents anywhere on the WWW. HTML embeds control codes in the ASCII text of a document that designate titles, headings, graphics, and multimedia components, as well as hyperlinks within the document. Several programs in the top software suites automatically convert documents into HTML formats. These include Web browsers, word processing and spreadsheet programs, database managers, and presentation graphics packages. These and other specialized HTML editor programs

provide a range of features to design and create multimedia Web pages without formal HTML programming.

3.3 Some Other High Level Languages

ALGOL: It is a short name for Algorithmic Language. Originally developed in 1958, it is a more precision language than FORTRAN for scientific and numeric calculations.

Forth: It is a powerful language designed only for fourth generation computers. It is useful in home computing.

RPG: RPG Stands for Report Program Generator. It is a commercial language specially developed for reporting.

ADA: This language was designed by U.S. Defense Department for its real-time applications and is also suitable for parallel processing.

PL/1 (Programming Languages/1): This language combines the features of both COBOL and FORTRAN and can be used for scientific as well as business applications. It was introduced in the late 1960s by IBM. Its application is limited to only IBM computers.

Snobol: It was developed in 1962. It is quite different from other languages in the sense that it can be used for processing messages, patterns and strings.

LISP: LISP stands for List Processor. It is an artificial intelligence programming language.

3.4 Exercise

Multiple Choice Questions

1. Which one is not 3GL language

a Visual Basic b. Java
c. C d. Assembly language

2. C language was originally developed by

a. Babbage b. Denis Ritchie
c. Jon Von Neuman d. none of them

3. Which language is specifically designed for real-time, interactive and Web based network applications?

a. COBOL b. BASIC
c. Java d. none of these

4. HTML stands for

a. HyperText Markup Language b. HyperText Making Language
c. Hyper Touch Markup Language d. none of these

Questions for short answers

5. Define the terms: high level language, source code and object code.
6. Name the programs used for converting source code to object code.
7. List the names of six high level languages.
8. Write a note on programming language C.
9. List the features of Java.
10. List the Features of HTML.
11. List the features of VB.

Analytical questions

12. Name the levels of programming languages and discuss differences between them.
13. Write a short essay on high level programming languages.
14. Discuss the importance of Java and JavaScript for software development.
15. Discuss the importance of C and C++ in current software development market.
16. Discuss the importance of VB and VBScript for software development.
17. Discuss the importance of HTML for Web applications.

Chapter 9

System Software and Operating System

Lesson 1: System Software
Lesson 2: Introduction to Operating Systems
Lesson 3:Processing and Services
Lesson 4: Popular Operating Systems

Lesson 1: System Software

1.1 Objectives

On completion of this lesson you will know:

- The concept of system software
- Role of operating system and BIOS
- Function of language translators
- Testing, debugging and diagnostic programs.

1.2 Introduction

Systems software refers to programs that assist the users to generate, debug, test, modify application programs, and then to execute them. These programs are generally written by computer manufacturers or software developers. The system programs can be used by different users and different application programs. The programs of system software can be grouped into:

- Operating systems
- Language translators, and
- Utility programs.

The most important component of system program is the operating system. It is a collection of program modules. Collectively they manage all the hardware, provides user-interface and some other user facilities. Other system programs include language translators (i.e. assembler, compiler, interpreter), text editors and some utility programs.

An operating system is a software system. It acts as an interface between a user of a computer and the computer hardware. The operating system provides an environment in which a user may execute programs. The goals of an operating system are to:

- Make the computer system convenient for users
- Make efficient use of the computer hardware.

The operating system controls and coordinates the use of the hardware for different application programs. The basic resources of a computer system are provided by its hardware, software and data. The operating system provides the means for the proper use of these resources in the operation of the computer system.

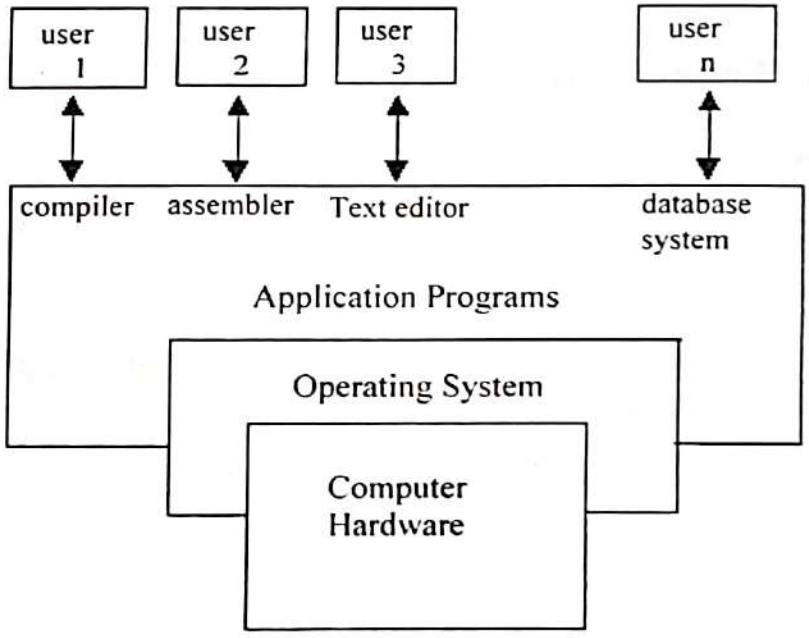


Figure: 9.1: Overview of a computer system.

An operating system is an important part of a computer system. A computer system can be viewed to have four components (Figure 9.1):

- Hardware (CPU, memory, I/O devices etc.)
- Operating system
- Applications programs (compilers, database systems, video games, business programs etc.)
- Users (people, machines or other computers).

1.3 The role of BIOS

When starts, the computer system has no idea what drives it has and can not even load its operating system. To solve this problem, a computer uses the Basic Input /Output System (BIOS) to turn it on. This small program is permanently stored in a read-only memory (ROM) chip. Since it is a small program embedded in a chip, it occupies a place between the software and hardware. For this reason it is often called firmware. BIOS performs a number of important functions. It interprets key strokes, displays characters on the screen, handles communication through the computer's ports, and tests the system while turning it on.

When a computer is turned on, the CPU hands over control to the BIOS, and the BIOS runs the power on self test or POST. The POST makes sure that all chips on the system board works well. Results of such tests are shown on the monitor at the time of turning on. Finally when the BIOS finds the operating system, in hard disk or floppy disk, it hands over control to the operating system.

1.4 Language Translators

Machine language

The machine language is written in words comprising of 1s and 0s. Computers execute commands or instructions in machine language. Programming in machine language has the following disadvantages:

- The program must be written in machine language (i.e., with 1s and 0s)
- Entering machine language program is a tedious process
- Error detection and correction is tedious and consumes much time
- Programs written in machine language for a specific machine cannot be used for another type of machine.

Assembler

Programming in assembly language is easier than programming in machine language. The programmer is required to write a source code for each instruction in the program and the assembler translates it to machine language. The programmer maintains total control of the computer operation. Assembly language offers the greatest degree of efficiency in operation and execution of programs.

Many shortcomings of machine language programming are overcome by programming in assembly language. Assembly instructions of a computer have mnemonics associated with them. The assembly program is written using these mnemonics. Assemblers are

programs that translate assembly level codes into equivalent machine codes. For programming in assembly language, it is not necessary to write programs in absolute binary, octal or hexadecimal notations.

Compiler and Interpreter

High level languages, are widely used. They have been developed to enable the users to program the machines in a language similar to the language of the users. For example BASIC, Pascal, C and Java use scientific notations and subroutines in programming formats that are similar to mathematical formulas and equations. On the other hand, COBOL contains business jargons widely used in business.

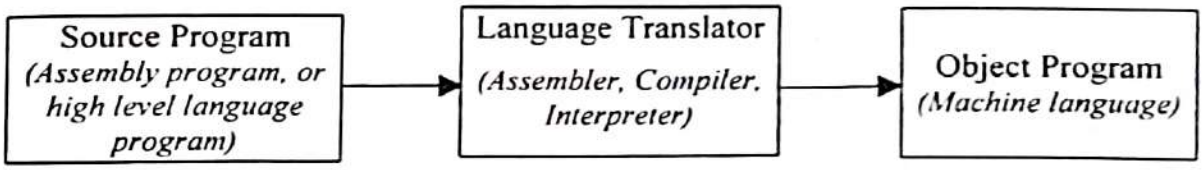


Figure 9.2: Language translators.

A Compiler is a program that translates programs written in high level language into machine language. A program written by a programmer in a high level language, is called a source program. The source program when converted into machine language by a compiler, is referred to as an object program. Unlike the assembler, the compiler generates several machine language instructions for each source statement. A compiler can be made to interface with several different computers. It is not written for any specific computer. A user can write programs that can be adapted to several computing systems. Thus, the user can formulate problems efficiently without precise knowledge of computer architecture. Compilers are complex programs compared to assemblers and they require more storage space than assemblers. An assembler or a compiler produces the object program, which is loaded into the computer memory before execution.

An Interpreter is a translator program that differs from the assembler or the compiler on several significant points. The interpreter does not prepare an object program. It translates and immediately executes each instruction of the source program. Thus an interpretive language is also an interactive language; it enables the user to load one instruction into the computer at a time and have it translated and executed. This process allows the programmer to check the results immediately. The interpreter is also a debugging tool and is useful during the program development stage. In the interpretive mode the execution time of the program is extended, but the program developmental effort and the program development time are decreased.

Comparison of Compiler and Interpreter

Compiler	Interpreter
Compiler prepares an object program from source program	Interpreter translates and immediately executes each instruction of the source program
Debugging is complex and time consuming	It is a debugging tool and is useful during program development stage
Lower execution time	Higher execution time
Requires higher program development effort and time	Requires less program development effort and time

1.5 Text Editor

While developing a program, it is often necessary to edit it. Editing involves making corrections or modifications to instructions, deletions and additions of instructions, re-sequencing instructions etc. A special system program called editor greatly simplifies the editing tasks. An editor enables the user to retain the program in the memory and modify it as required. A portion of the memory, called buffer, is reserved for editing. Lines of instruction are entered from an external I/O device and stored in the buffer for editing by the editor. The edited text can be printed by a printer or recorded on suitable I/O devices.

1.6 Testing, Debugging and Diagnostic Programs

Programs created by us are prone to error. It is not realistic to expect a newly created program to run correctly the first time. Before a program can be put into productive use it must be debugged. A bug is a defect or imperfection in a program. In the context of programming it is an error in a program that is responsible for undesirable results. Testing of a program involves validation of program design. Thus testing involves data that are intentionally used to reflect the worst-case situations.

In order to assist the programmer in locating and rectifying errors or bugs in programs, several tools are available. These tools could be hardware aids such as logic analyzers or software aids such as debug programs. A debug program is loaded into the memory with the object program that is to be debugged. If errors occur during run, the debug program is activated to identify, locate, and correct the error. A debug program generally includes several features that aid the programmer in debugging tasks. Some of the common features are described below.

Display Register Contents: This feature allows the programmer to check contents of certain registers of the CPU during the execution of a program. The contents of the registers can be printed or displayed on appropriate devices. It is also possible to modify the contents of some registers if necessary.

Display/Replace memory Contents: The programmer can access the contents of a memory location and display it. The contents can then be modified if necessary.

Memory Dump: This feature makes it possible to print the contents of a group of memory locations. Thus a programmer can view the contents of desired memory locations and determine accuracy of data.

Output Memory Content: The contents of a memory are transferred and stored in some other mass storage devices such as disks or magnetic tapes.

Breakpoints: Breakpoints are interrupts in the object program. They are used to examine the current status of certain critical CPU registers, memory locations, or I/O ports. When the execution of the object program reaches a breakpoint, its execution is temporarily halted. The debug program then allows the programmer to examine and alter the contents of the selected register, memory locations, or I/O ports to correct errors.

Disassembly: This is the reverse of the assembly process. The operation consists of reading out the contents of the program memory into assembly language.

1.7 Exercise

Multiple Choice Questions

1. An operating system is a program which acts as an interface between

   a. user and application program b. user and hardware
   c. application program and hardware d. none of these

2. BIOS means

   a. Basic Input/Output System b. Base Input/Output System
   c. Basic Input/Out Services d. none of these

3. Which one translates and immediately executes each instruction?

   a. assembler b. compiler
   c. interpreter d. none of these

4. Which register allows programmer to check the contents during the execution?

   a. status register b. Display register
   c. general purpose register d. none of these

Questions for short answers

5. What is the role of BIOS?
6. What is meant by firmware?
7. Distinguish between machine language, assembly language and high level language.
8. What is the main function of a language translator?
9. Distinguish between assembler and compiler.
10. Give the main characteristic of an interpretive language.
11. List the main features of a debug program.
12. List the main functions of a text editor.

Analytical questions

13. What is a language translator? Discuss characteristics of different types of language translators.
14. Discuss the disadvantages of machine language programming.
15. Distinguish between compiler based languages and interpreter based languages
16. What is a debug program? Discuss the major features of a debug program.

Lesson 2: Introduction to Operating Systems

2.1 Objectives

On completion of this lesson you will know:

- Tasks of operating systems
- Types of operating systems
- File management
- Utility programs.

2.2 The Tasks of an Operating System

The most important task of an operating system is to process commands and to provide coordination between different processing tasks.

Processing Commands

The operating system interprets instructions entered through the keyboard. When one uses an application program, the program's commands are interpreted by the operating system. For example, one might open a document by clicking a button on the program's toolbar. At another time the user might do so by dragging the file onto the application's icon and dropping it. The operating system interprets these commands and copies the document on the disk of the computer.

Multitasking

Modern operating systems allow to multitask, that is, to perform multiple tasks, at the same time with different programs. For example, one could download a large file from the Internet with one program, while editing a document with another. Older operating systems could run only one program at a time. Since most systems have only one microprocessor, they allow multiple programs to run by quickly switching back and forth between them. Although only one program is actually active at a time, the processor switches so fast that they all appear to be active. This process is called time-slicing or task-switching because it slices up the CPU's time as it switches between the running applications. Today's system use a form of multitasking called preemptive multitasking. In this type of multitasking access to the microprocessor is controlled so that one application can not seize it and hold it for long a period. One application can "preempt" another application to get a fair share of the processor's time.

Multithreading

Most applications process data and commands sequentially, i.e. when one task is finished

they begin another. The application therefore follows a single thread from the beginning of a session to the end and individual operations are strung like beads on a necklace. A multithreaded operating system on the other hand lets application programs start and run two or more threads simultaneously. It is similar to multitasking within a single program. For example, one thread in a word processing program can be reading the pressed keys while a second thread displays data on the screen and a third thread prints the document.

Multi-user Support

Some operating systems are designed so that many users can be connected to the system at the same time. Multi-user operating systems are also multitasking because the multiple users run their programs at the same time.

Multiprocessor Support

Some applications need huge processing power. For example, any one doing forecasting, modeling, simulation, or computer aided design needs enormous computing power. Some operating systems support multiple processors - in some cases hundreds or even more.

Miscellaneous Tasks

In addition to the processing tasks discussed, an operating system also performs the following tasks:

- The operating system continually monitors the system and if it detects something wrong, it outputs an error message.
- The operating system manages the use of memory and runs each program in its own protected space so that a problem with one program will not affect others.
- The operating system allocates peripheral devices for different tasks.
- To conserve power, power management cuts power at those times when it is not needed.
- Manages file stored on disks.

2.3 Operating System Characteristics

There are a variety of criteria, some of which can be used to compare operating systems. These are as follows:

- Character-based or graphical user interface based
- Single tasking or multitasking
- Single threading or multithreading
- Weak or strong security

- 8,16,32 and 64 bits processed at a time
- Portable or not (a portable operating system is not dependent on a specific microprocessor.)
- Proprietary or open (proprietary operating systems work only with one type of computer, open operating systems run on a variety of machines).

2.4 Types of Operating Systems

Desktop Operating Systems

Operating Systems that run on desktop computers are the least powerful and least secure operating systems. They are also the most common. Three common types are:

- Windows,
- Apple's System 8, or
- Rapidly fading DOS.

Operating Systems for Servers

Currently, computers in networks are connected to one another not directly but through powerful computers called servers that provide security to the networks and route traffic from one network to another. These complicated tasks require special operating systems. These powerful operating systems are also used for tasks (such as engineering design, order processing, and payroll) critical to a company's continued existence, called mission critical applications. Such tasks were previously run only on larger mainframe computers. These powerful 32- or 64-bit operating systems are secure, multitasking, multithreading, and open. All now possesses graphical user interfaces developed for them. All but OS/2 are portable in that they run on more than one type of microprocessor.

- **Windows NT** (NT stands for New Technology) is the fastest-growing server operating system. The upgraded version of NT is called Windows 2000.
- **OS/2** (Operating System/2) competes with Windows and is a powerful operating system but has had trouble catching on.
- **UNIX** a powerful operating system for server was originally developed at AT&T's Bell Labs in 1973.

Operating Systems for Mainframe

In the early days of computer, mainframe and minicomputers dominated the computing world, and each computer company developed its own operating system. For example, some IBM mainframe computers use an operating system called open VMS and Digital Equipment Corporation minicomputers use one called VMS. These companies developed

not just their own operating systems but also the applications that ran on top of them. These are called proprietary systems because they are all owned by a single proprietor. Because there were so many competing operating systems, it was hard to make computers talk to one another. Although costly, these operating systems are both powerful and secure and capable of handling a variety of tasks.

Handheld operating systems

Windows CE, visually similar to Windows 95, uses a small keyboard and a touch-sensitive screen.

Embedded operating systems

Generally, a microprocessor is used inside a small device such as a watch, and it needs an operating system. Such an operating system is stored in a memory chip instead of a hard drive and turns on instantly when the device is turned on.

2.5 File Management

Filenames

When files are saved, they are assigned filenames so that users can identify them later. Filenames have two parts:

- A name, and
- An extension.

One may think of them as first and last names. The name is chosen to identify the file later. For DOS, a user has to use limited number of characters for the file name. However, a user can use long, descriptive filenames in newer operating systems.

The extension is separated from the name by a period. Its purpose is to identify the file format or the application that created it. For example, a .DBF extension means the file is a database and .doc often means a word processing document.

Folders

A folder contains related document together. A well organized file cabinet uses hanging folders to group related documents. If a further breakdown is needed, manila folders are inserted into the hanging folders. It is easy then to locate and pull out the documents needed. The same is true of hard drives. To master computing, one needs to understand the concept of folders and their use for grouping together related files. A folder is also known as directory.

Drives

Most of the computer systems have more than one drive. Drives are assigned letters for identification. For example, a drive into which user can insert a floppy disk may be drive A or B. A computer looks to hard drive called C when user turns it on. Additional drives might be D and E. As the number and type of drives vary from system to system, so does the letter or name of each.

Floppy drive	A & B
Hard drive	C
CD-ROM drive	D
Zip drive	E

Figure 9.3: Drives and drive names

Trees

A tree is one of the ways to illustrate the organization of folders on a drive. In this view, all folders branch off from the drive. If a folder contains other folders, those folders are shown as a second branch from the first.

When using a tree, one can expand and collapse the entire tree or any branch. This allows a user to alternate between a summary of the drive's contents, and details of each file. On Windows, the plus sign indicates that a branch has been collapsed and it contains hidden folders. The minus sign indicates that the branch has been expanded.

Paths

A user has to follow a "path" to get files stored in folders on a disk. For example, if a file named Nasima.doc is in a folder named *Letters* on drive C, the path to that folder is *C:\Letters\Nasima.doc*. If Nasima.doc is in a folder named Memos, the path is *C:\Memos\Nasima.doc*. The drive specification in the path is C: and the folder's name is separated from it by a backslash.

Path to a file in a folder: The backslash is also used when a user has folders within folders. If the Letters folder contained a folder named Vendors, the path to it would then be *C:\Letters\Vendors\Nasima.doc*.

Path to a file in a subfolder: In most cases, a user does not need to type paths. Most new programs and operating systems let users to use a mouse to open folders to find a file.

File managers

Operating systems include a separate program that a user can use to explore drives,

folders, and files. There are a variety of these but they all allow users to display the folders on a drive arranged like a tree. Clicking any of the folders displays a list of its contents.

When browsers and search engines were first developed for the Web, people soon discovered they could locate and get to a document anywhere in the world more easily than they could locate and get to one on their own system. As a result, browser technology is now being put to use in file management. Microsoft's Active Internet Platform technology allows users to browse documents as well as the Web. In fact, a document on a local hard drive looks the same as a page anywhere on the Web.

Search engines on the Web keep track of millions of documents by searching them out and listing words they contain in an index. When a user searches a word, the search engine looks it up in the index and lists the name and address of every document in which it can be found. A user can go to the document just by clicking its name. The search engines make looking for documents so easy that personal versions are available for users own system.

2.6 Utilities

In addition to a file manager, operating systems also include utility programs that one can use to manage disks and files. The list of such utility programs is almost endless.

Disk Formatting

Disk formatting utility makes the disk ready to store data. Formatting creates a directory of the locations of each file called a file allocation table (FAT). When a user saves a file, the operating system stores it in a group of sectors, called cluster, the smallest storage unit the computer can address. To do so, it looks in the file allocation table for empty clusters and then stores the file in one of those locations. The address of the cluster where the file begins and the one where it ends are then listed in the file allocation table. When a user opens the file, the operating system uses those addresses to locate the clusters and open the file stored in them.

Fragmenting

When a user first uses a disk, data are laid down in a linear fashion. Each part of a file is stored in adjacent clusters. However, as a user begins to delete old files and add new ones, things begin to become less orderly, because fragments of files are scattered in clusters located in different areas on the disk. This file fragmentation slows the computer down because it takes longer for the drive to locate, save, and retrieve the fragmented files. To fix this problem, a user needs to regularly defragment or defrag the disk. This

procedure is often overlooked by computer users, who then wonder why their system gets slower and slower as time goes by. In many cases, the system will even begin to misbehave; programs crash or files do not open. Often these problems can be fixed just by defragging the disk.

Disk scanning

A newly manufactured disk is not always perfect. Some areas of the disk may not be usable. The manufacturer scans the disk and electronically marks bad areas as unusable so that no data are stored there. As the user uses the disk, other areas may become bad. For this reason, a utility program is used to scan the disk at regular intervals to locate and mark any bad sectors. If this is not done, the operating system may store parts of a file in them and then the user will not be able to retrieve that part of the file. In addition, scanning finds and fixes problems with files. For example, when systems crash, parts of files may be left scattered on the disk and become lost. The disk scanning program locates these lost clusters and stores them together in a separate file. In some cases, a user can open these files and recover lost work.

File deletion and recycle bin

Many of the files created by users on a computer eventually become outdated and no longer needed. To remove these files from the disk, users delete them. On many newer operating systems, this procedure does not actually delete them from the disk. Instead, it moves them to a separate area of the disk called a recycle bin or trash can. If user decides later that she/he wants to recover the file, she/he can open the recycle bin and restore it to the disk. When absolutely certain that the user will never need the files in the recycle bin, the user can make it empty. This procedure deletes the files from the disk and one can no longer recover them. However, if a user ever deletes a really important file, it can also be restored. There are programs that can recover deleted files, provided any subsequent commands did not place new files in the place on the disk where they are stored.

2.7 Exercise

Multiple Choice Questions

1. Which one is not the server operating system?

   a. NT b. UNIX
   c. OS/2 d. DOS

2. In general, which letter is considered for Hard Disk Drive?

   a. A b. B
   c. C d. none of these

3. Formatting creates a directory of the locations of each file called

   a. FAT b. BAT
   c. DAT d. none of these

4. Which utility program could improve the speed of disk?

   a. Fragmentation b. Defragmentation
   c. disk scanning d. none of these

Questions for Short Answers

5. Differentiate between multitasking and multithreading.
6. List the tasks of an operating system.
7. Write the characteristics of operating system.
8. Name at least two operating systems for server and desktop systems.
9. Differentiate between file and folder.
10. What is the necessity of disk formatting?
11. What is fragmentation?
12. What is the use of recycle bin or trash can?
13. What do you understand by proprietary software system?

Analytical Questions

14. Discuss the main tasks of an operating system.
15. Describe the operating systems for different computer systems.
16. Discuss at least three utility programs.

Lesson 3: Processing and Services

3.1 Objective

On completion of this lesson you will know:

- Different types of processing
- Services given by operating systems.

3.2 Types of Processing

Batch processing: Batch processing involves collecting transactions together over some interval of time and then processing the whole batch.

Batch systems allow automatic job sequencing by a memory resident monitor and improves the overall utilization of the computer. The computer no longer had to wait for human operation. CPU utilization is still low, because of the slow speed of the I/O devices relative to the CPU.

Batch processing is used in banks for balancing checking accounts. A bank can make a record of all checks of the day and compute account balancing in a batch at the end of the day.

On-line processing: In on-line processing the computer and terminals act interactively. If the program allows the user to enter data or look at results before continuing with the next step it is sometimes known as conversational mode. Many computers use batch and on-line processing at the same time.

Spooling: Buffering is an approach to improve system performance by overlapping the input, output, and computation of a single job. Spooling allows the CPU to overlap the input of one job with the computation and output of other jobs.

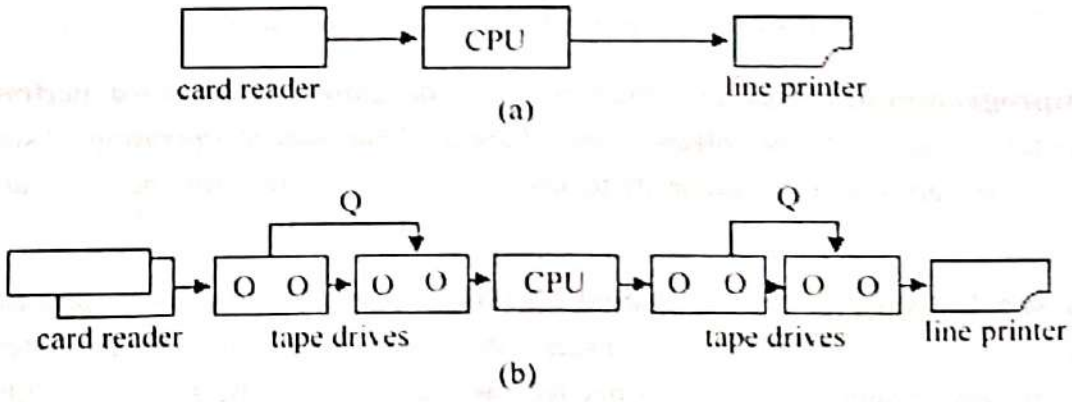


Figure 9.4: (a) On-line and (b) Off-line operation of I/O devices

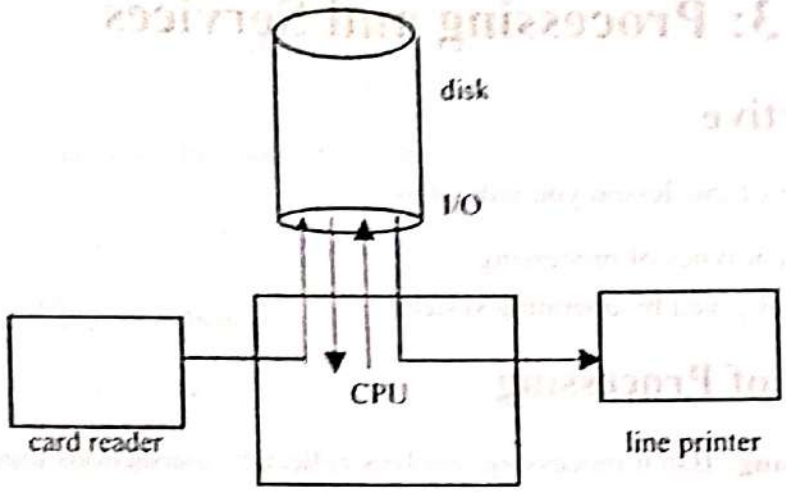


Figure 9.5: Spooling

Spooling also provides a pool of jobs which have been read and are waiting to be run. This pool of jobs supports the concept of multiprogramming. With multiprogramming, several jobs are kept in memory at one time; the CPU is switched back and forth between them in order to increase CPU utilization and to decrease the total time needed for the execution of a job.

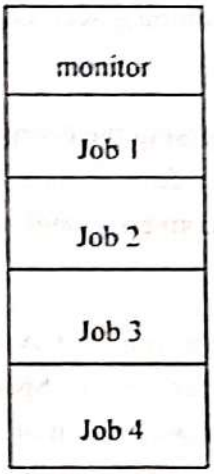
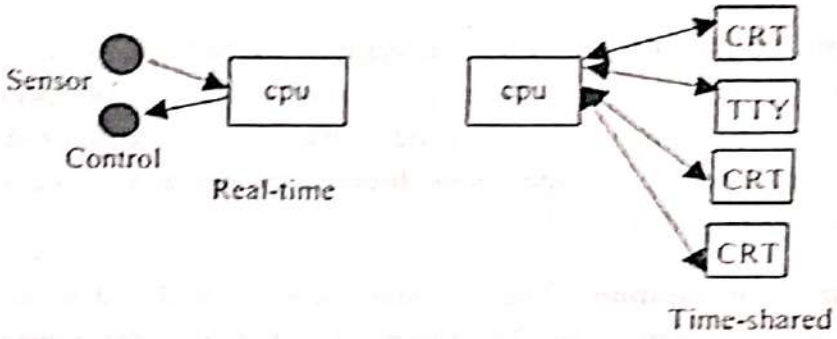


Figure 9.6: Memory layout for multiprogramming

Multiprogramming: Multiprogramming, was developed to improve performance of computer systems. It also allows time sharing. Time-shared operating systems allow many users (upto several hundred) to use a computer system interactively at the same time.

Time sharing system: A time-sharing operating system allows many users to share the computer simultaneously. Since each action or command in a time-shared system tends to be short, only a short CPU time is needed for each user. As the system switches rapidly

from one user to the next, users are given the impression that each user works on a separate computer, while actually one computer is shared among the users.



9.7: Real-time and time-shared systems

Real-time Systems: Another form of operating system is the real-time system. A real-time system is often used as a control device in a dedicated application. Transducers and sensors bring data to the computer. The computer must analyze the data and possibly adjust controls to sensor inputs. Systems which control scientific experiments, medical systems, industrial control systems, and some display systems are the examples of real-time systems.

Virtual Memory: The size and sophistication of a program is limited by the amount of main memory available. The virtual memory technique uses some rapidly accessible secondary memory to augment main memory. Portions of programs are swapped into main memory from virtual memory (also called virtual storage) as needed. This gives an illusion that more main memory is available than actually the case. Virtual memory is usually implemented by segmentation or paging or a combination of them.

Segmentation is the process of dividing a program into segments of different sizes and placing the segments in memory wherever they fit. Paging is similar to segmentation except that programs are divided into equal sized portions. As with segmentation, the operating system keeps track of page locations by constructing page tables. As pages are of fixed size, the use of paging results in less wastage of real storage space. The best memory management scheme is to combine segmentation and paging by first segmenting programs and then further subdividing each segment into pages.

3.3 Types of Services

An operating system provides an environment for the execution of programs. It provides certain services to programs and to the users of those programs. These services are provided for the convenience of the programmers, to make the programming task easier. Some common classes of services are given below.

Program execution: The system must be able to load a user program into memory and run it. The program must be able to end its execution, either normally or abnormally.

Input/output operations: A running program may be required to input data and output results. An I/O operation may involve a file or a device. For specific devices, special functions may be desired (such as, rewind a tape, clear the screen and so on). Since a user program cannot execute I/O operations directly, the operating system must provide some means for I/O operations.

File system manipulation: Programs may need to read and write files and the users create and delete files by name. The operating system provides numerious file operations.

Error detection: The operating system needs to be aware of possible errors in the system. Errors may occur in the CPU and memory (such as memory error or power failure), in I/O devices (such as parity error on tape or printer out of paper), or in the user program (such as an arithmetic overflow, an attempt to access illegal memory location or using too much CPU time). The operating system takes appropriate actions if error occurs.

Resource allocation: In some computer systems multiple users or multiple jobs run at the same time and resources should be allocated to each of them. Resources are managed by the operating system.

Accounting: It may be necessary to keep track of which users use how much and what kind of computer resources. Keeping of records may be required for the purpose of billing or for statistics.

System Calls

Operating system services are provided in many ways. Two basic methods are system calls and systems programs.

Some basic services are handled through the use of system calls. System calls provide the interface between a running program and the operating system. System calls are generally available as assembly language instructions, and are usually listed in the assembly language manuals.

System calls can be roughly grouped into three categories: process or job control, device and file manipulation, and information maintenance.

Systems Programs

Systems programs solve common problems and provide a convenient environment for program development and execution. Systems programs can be divided into several

categories. These are discussed below.

File manipulation: These programs are used to create, delete, copy, rename, print, dump, and list files, and manipulate directory operations.

Status information: These programs can be used to ask the operating system for the date, time, amount of available memory or disk space, number of users, or similar status information.

File modification: Text editors can be used to create files and modify the content of files stored on disks or tapes.

Programming language support: Compilers, assemblers, and interpreters for common programming languages (such as FORTRAN, Pascal, BASIC, C) are often provided with the operating systems.

Program loading and execution: Once a program is assembled or compiled, it must be loaded into memory for execution. The operating system may provide loaders, linkage editors for program execution.

3.4 Exercise

Multiple choice questions

1. The basic methods of services of an operating system are between
   a. system calls and resource allocation b. system calls and system programs
   c. system programs and error detection d. none of these.

2. Batch processing involves
   a. collection of processing b. single processing
   c. real-time processing d. none of these.

Questions for short answers

3. List the type of processing.
4. Distinguish between batch processing and multiprogramming.
5. What is spooling?
6. What is the function of operating system in application program execution?
7. How does an operating system handle error detection?

Analytical questions

8. Explain the nature of real-time system.
9. Discuss the common services provided by the operating system.
10. What are system calls and systems programs? Discuss the categories of system programs.

Lesson 4: Popular Operating Systems

4.1 Objectives

On completion of this lesson you will know:

- Popular operating systems
- Characteristics and scope of operating systems

4.2 Outline

From the fastest supercomputer and largest mainframe, right down to the smallest embedded microprocessor in a microwave oven or watch, every computer uses an operating system. Over the years, a variety of operating systems have been developed and many remain in use. When the IBM PC set the standard for microcomputers, DOS (Disk Operating System) became the most widely used microcomputer operating system. Today, DOS has been replaced by more powerful operating systems such as Windows, UNIX, Windows NT, Linux and OS/2 etc. These operating systems make it possible to run tasks on less expensive microcomputers that were once performed on expensive mainframes and minicomputers.

4.3 Important Operating Systems

Microsoft Disk Operating System (MS DOS)

MS-DOS, developed by Microsoft corporation, became the defacto standard for machines designed with Intel microprocessors. MS-DOS was selected by IBM for its PCs. This established the popularity of MS-DOS. Since IBM hardware and its compatibles hold a large share of the microcomputer market, application software became abundant for MS-DOS. The IBM version of MS-DOS is known as PC-DOS.

DOS is helpful to organize disks and use them efficiently. It helps to create and manage files on disks, make copies of them or remove unwanted files from the disks.

MS-DOS 5.0 comes equipped with a graphical user interface (GUI) called the DOS Shell. This interface helps the user with everyday tasks such as starting a program or formatting a floppy disk. With the GUI the desired task can be easily selected and started with the mouse, by just pointing to the item on the screen and clicking the mouse button.

The DOS shell also provides an overview of the contents of the hard disk and floppy disk, any time. Several programs can be started at once but only one program runs at a time,

the others are suspended.

Because DOS was developed for early 16-bit processors, it can not take full advantage for advanced capabilities of Intel 32-bit or higher microprocessors. However, use of DOS is declining rapidly.

Mac Operating System (Mac OS)

Mac OS introduced the graphical user interface bringing the UNIX ideas to microcomputer. The original Mac OS released in 1984, consisted of System and Finder. By late 1980s, the Mac's operating system was the most technologically advanced in personal computing, but Apple computer was unable to capitalize on its lead, and Mac OS fell behind Microsoft Windows. Mac OS is widely considered to be the easiest operating system for computer users. Mac OS released an improved version in 1998, which is up to the standard of Microsoft Windows 98.

Windows 3.XX

Microsoft Windows 3.XX is an extension to the DOS operating system and cannot run without DOS. Its is a graphical user interface whose features extend far beyond those of the DOS Shell. Although it was designed for 32-bit, full advantage of it is limited as it has to run in cooperation of DOS.

Microsoft released Windows in 1984. The goal of this graphical user interface is to make the computer friendlier to the users. With Windows, applications files are presented as symbols (icons) on the screen. Simply pressing a key or clicking the mouse activates a command. Owing to the success of Windows, Windows-compatible application software became very popular. Also Windows accessory programs are available to accomplish many tasks within Windows itself. Windows permits multitasking whereby several programs can be run at once. However, it uses cooperative multitasking, which means that the failure of any one application is likely to crash the whole system.

Microsoft Windows 95, 98 and 2000

Microsoft Windows 95 is an operating system which fully supports 32-bit processing capabilities. It also supports preemptive multitasking, which enables the operating system to regain control if an application crashes.

Windows 95, or Win 95, is an operating system, poised between Windows 3.XX (a 16-bit operating system) and Windows NT (32-bit with advanced features). Win 95 is designed to run many existing 16-bit applications designed for Windows 3.XX as well as new 32-bit applications.

Microsoft Windows 98, or Win 98 (released in 1998), is an improved version of Win 95 that offers

- Better stability
- Improved Internet connectivity
- Drivers for new peripherals.

A new version of Windows called Windows 2000 is available with a lot of additional facilities which include networking applications.

Microsoft Windows CE

Microsoft Windows CE is a simplified version of Windows, designed for palmtop computers. Windows CE includes an interactive scheduling calendar, an address book for contacts, electronic mail, and Web browsing. By means of an automatic active synchronization program, users can quickly synchronize the corresponding utilities on their computers, Windows CE includes handwriting recognition and support for voice recording as well.

Microsoft Windows NT

Microsoft Windows NT (New Technology) is a 32-bit operating system, specifically designed for client/server systems, formerly a strong hold of UNIX. It has two components to support namely, client and server.

Windows NT Workstation for desktop computers is similar to but more powerful than Win 98. It is faster than Win 98 and supports network environment. Windows NT desktops can be linked to servers running Windows NT server. Windows NT server offers

- Strong security to network resources
- Remote network administration
- Web server scope to make Web pages available
- Directory services to map files and applications available on the network.

UNIX

It is a multi-user operating system for powerful 16-bit and 32-bit machines. UNIX set the standard for multi-user systems for personal computers. Developed at AT&T's Bell Laboratories in USA, UNIX held monopoly in the multi-user OS market for quite some time before other operating systems were introduced. It was first written in the assembly language. Later in 1973, it was rewritten in C language. This gaves it portability i.e., it could be run easily on different types of machines. This is one important reason for the popularity of UNIX. There are several implementations of this system. The gratest

success of UNIX lies in client/server computing, a type of computer usage that is widely found in corporations today. UNIX also provides

- Highest level of security for applications and access control and
- Faster services.

Linux

Linux a new flavor of UNIX operating system, is the fastest growing operating system for Intel based personal computers. Two important characteristics make Linux attractive. These are:

- It is powerful
- It is free

Linux brings all the maturity and sophistication of UNIX to the PC, including multitasking, virtual memory, Internet support, multiprocessor support and graphical user interface.

Linux is distributed with General Public License. In practice, most people buy a Linux distribution, a CD-ROM containing Linux and a collection of drivers, utilities, GUI interfaces and application programs.

Linux, a derivation of UNIX, is a freeware operating system. However, Linux can not run the most popular Microsoft Office applications. Linux is gaining increasing acceptance for Web and other applications.

Xerox PARC

Xerox PARC (Palo Alto Research Center) was the first GUI (Graphical User Interface) operating system developed in 1970. Xerox PARC originated the idea of desktop icon, on screen fonts, windows and pull-down windows. Although Xerox released a GUI based computer in 1981, the company was never able to capitalize on its innovations.

4.4 Exercise

Multiple Choice Questions

1. The UNIX system, later in 1973, was rewritten in

   a. Pascal language b. C language
   c. Visual Basic d. Assembly language.

2. Microsoft first released windows in

   a. 1980 b. 1982
   c. 1984 d. 1986.

3. Which operating system provides the highest security?

   a. DOS
   b. Windows 3.XX
   c. Windows 98
   d. UNIX

Questions for Short Answers

4. Distinguish between DOS and Windows 3.XX
5. What is meant by NT? Write the features of Windows NT.
6. Differentiate between UNIX and Linux.
7. What is Windows CE?
8. Write the features of Linux.
9. Why UNIX is known as powerful operating system?
10. What is meant by PARC.

Analytical Questions

11. Discuss the changes in operating systems from DOS to Windows 98
12. Describe the features of UNIX and Linux.
13. Discuss the GUI based operating systems.

Chapter 10

Database Concepts

Lesson 1: Introduction to Database
Lesson 2:Database Software
Lesson 3:Database Structures
Lesson 4: Database Access and Development

Lesson 1: Introduction to Database

1.1 Objectives

On completion of this lesson you will know:

- Basic idea of databases
- Concept of database management system.

1.2 Basic Concepts

Data may be logically organized into characters, fields, records, files, and databases, just as writing can be organized in letters, words, sentences, paragraphs, and documents. Examples of these logical data elements are shown in Figure 10.1.

Character

The most basic data element is the character, which consists of a single alphabetic, numeric, or other symbol. The bit (or byte) is a more elementary data element, but those terms refer to the physical storage elements provided by the computer hardware. From a user's point of view, a character is the most basic element of data that can be observed and manipulated.

Field

The next higher level of data is the field, or data item. A field consists of a group of

characters. For example, the group of alphabetic characters in a person's name forms a name field, and the group of numbers in a sales amount forms a sales amount field. A data field represents an attribute (a characteristic or quality) of some entity (object, person, place, or event). For example, an employee's salary is an attribute that is a typical data field used to describe an entity that is an employee.

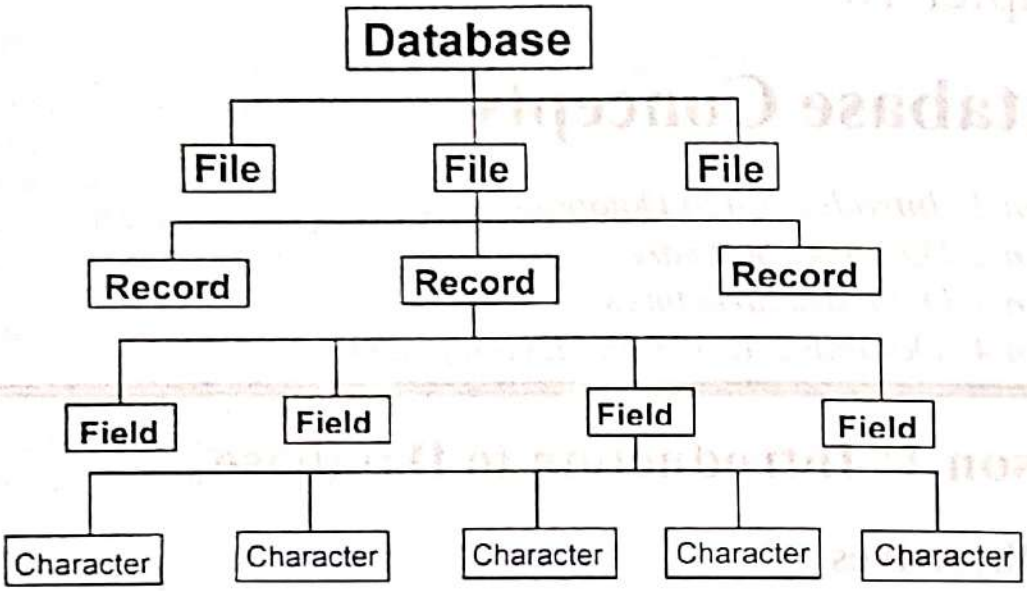


Figure 10.1: Basic database components

Record

Related fields of data are grouped to form a record. A record represents a collection of attributes that describes an entity. An example is the payroll record for a person, which consists of data fields describing attributes such as the person's name, social security number (in some countries), and rate of payment. Fixed-length records contain a fixed number of fixed-length data fields. Variable-length records contain a variable number of fields and field lengths.

File

A group of related records forms a data file, or table. Thus, an employee file contains the records of the employees of a firm. Files are frequently classified by the application for which they are primarily used, such as a payroll file or an inventory file, or the type of data they contain, such as a document file or a graphical image file. Files are also classified by their permanence, for example, a payroll master file versus a payroll weekly transaction file. A transaction file, therefore, would contain records of all transactions occurring during a period and might be used periodically to update the permanent records contained in a master file. A history file is an obsolete transaction or master file retained

for backup purposes or for long-term historical storage called archival storage or data warehouse.

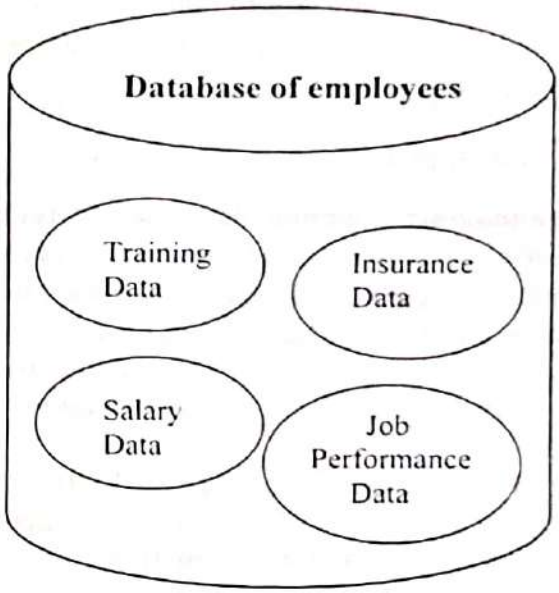


Figure 10.2: A personnel database files.

Database

A database is an integrated collection of logically related records or objects. An object consists of data values describing the attributes of an entity, plus the operations that can be performed upon the data. A database consolidates records previously stored in separate files into a common pool of data records that provides data for many applications. The data stored in a database are independent of the application programs using them and of the type of secondary storage devices on which they are stored. For example, a personnel database consolidates data formerly segregated in separate files such as payroll files, personnel action files, and employee skills files. Figure 10.2 shows the personal database of employees.

1.3 Database Management Systems

Database management systems consolidate data records and objects into databases that can be accessed by many different application programs. In addition, an important software package called a database management system (DBMS) serves as a software interface between users and databases. This helps user easily access the records in a database. Thus, database management involves the use of database management software to control how databases are created, interrogated, and maintained to provide information needed by end users.

For example, customer records and other common types of data are needed for several different applications in banking, such as check processing, automated teller systems, bank credit cards, savings accounts, and installment loan accounting. These data can be consolidated into a common customer database, rather than being kept in separate files for each of those applications (Figure 10.3).

A database management system involves three basic activities:

- Updating and maintaining common databases to reflect new transactions and other events requiring changes to an organization's records.
- Providing information needed for each end user's application programs that share the data in common databases. This sharing of data is supported by the common software interface provided by a database management system package. Thus, end users and programmers do not have to know where or how data are physically stored.
- Providing inquiry/response and reporting capabilities through a DBMS package so that end users can easily interrogate databases, generate reports, and receive quick responses to their requests for information.

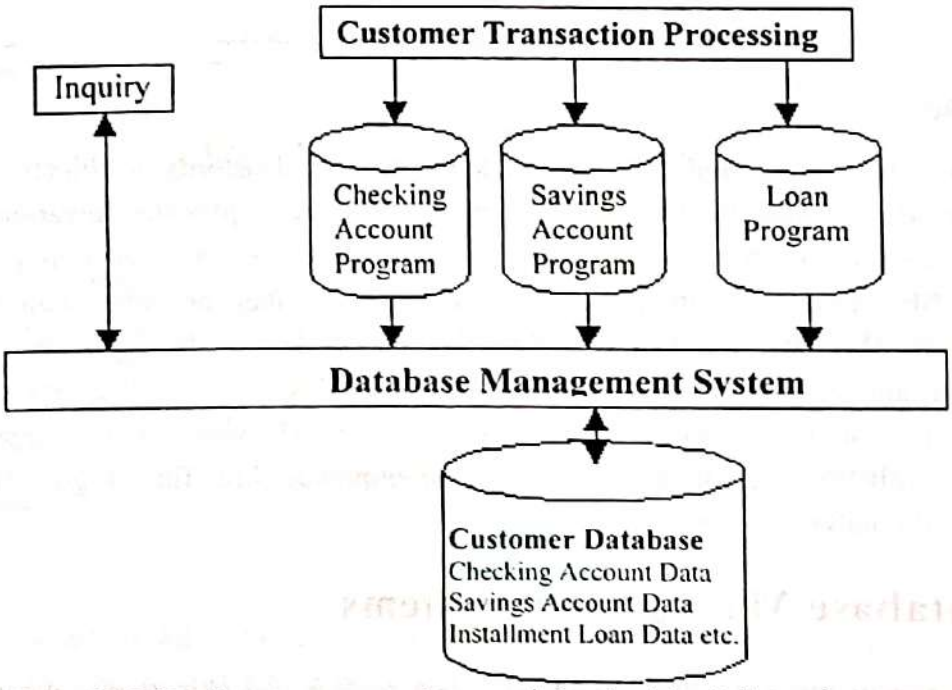


Figure 10.3: Database management approach in a banking information system.

1.4 Benefits and Limitations of Database Management

The database management approach provides end users with several important benefits. Database management reduces the duplication of data, it integrates data so that they can be accessed by multiple programs and users. Programs are not dependent on the format of the data and the type of secondary storage being used. Users are provided with

inquiry/response and reporting capabilities that allows them to easily obtain information they need without having to write programs. Programming is simplified, because programs are not dependent on either the logical format of the data or their storage location. Finally, the integrity and security of the data stored in databases can be increased, since access to data and modification of the database are controlled by DBMS software, data dictionary, and database administrator function.

The limitations of database management arise from its increased technological complexity. Thus, a database management approach can pose problems in data resource management. Developing large databases of complex data types and installing a DBMS can be difficult and expensive. More hardware capability is required, since storage requirements for the organization's data, overhead control data, and the DBMS programs are greater. Longer processing times may result from high-volume transaction processing applications. Finally, if an organization relies on centralized databases, its vulnerability to errors, fraud, and failures are increased. Yet problems of inconsistency of data can arise if a distributed database approach is used. Therefore, the security and integrity of an organization's databases are major concerns of the organization's data resource management effort.

1.5 Exercises

Multiple Choice Questions

1. A field is consists of
   a. group of characters b. group of records
   c. group of files d. none of these

2. A historical file contains
   a. all transaction data b. obsolete transaction data
   c. data for single transaction d. none of these

Questions for Short Answers

3. Differentiate between field and record.
4. What is meant by a file?
5. What is a data warehouse?
6. Differentiate between database and database management system.
7. What do you understand by customer database?

Analytical Questions

8. What is a database? Show basic components of a database.
9. What is database a management system? What are the three basic activities of a database management system?
10. Write the benefits and limitations of the database management system.

Lesson 2: Database Software

2.1 Objectives

On completion of this lesson you will know:

- Database management software
- Types of database systems
- Data resource management

2.2 Database Management Software

A database management system (DBMS) is a set of programs that controls the creation, maintenance, and use of the database and its end users. Database management packages are available for micro, midrange, and mainframe computer systems. The four major uses of a DBMS are discussed below.

Database Development

Database management packages (e.g., Microsoft Access) allow end users to easily develop the databases they need. However, large organizations with client/server or mainframe-based systems usually place control of enterprise-wide database development in the hands of database administrators (DBAs) and other database specialists. This improves the integrity and security of organizational databases. Database developers use the data definition language (DDL) in database management systems to develop and specify the data contents, relationships, and structure of each database specifications. Such information is cataloged and stored in a database of data definitions and specifications called data dictionary, which is maintained by the DBA.

The Data Dictionary: A data dictionary is a computer based catalog or directory containing meta-data, that is, data about data. A data dictionary includes a software component to manage a database of data definitions, that is, meta-data about the structure, data elements, and other characteristics of databases. For example, it contains the names and descriptions of all types of data records and their interrelationships, as well as information outlining requirements for end users' access, use of application programs, and database maintenance and security. Data dictionaries can be accessed by the database administrator to report the status of any aspect of a firm's meta-data. Some active data dictionaries automatically enforce standard data element definitions whenever end users and application programs use a DBMS to access databases. For example, an active data dictionary would not allow a data entry operator to enter a name of a customer that

exceeds the defined size of that data element.

Database Interrogation

The database interrogation capability is a major benefit of a database management system. End users can use a DBMS by asking for information from a database using a query language or a report generator. The user can easily receive an immediate response in the form of video displays or printed reports. The query language feature lets user easily obtain immediate responses to ad-hoc data requests: the users merely key in a few short inquiries. The report generator feature allows a user to quickly specify a report format for information.

SQL and QBE: SQL, or Structured Query Language, is a query language found in many database management packages. The basic form of an SQL query is:

SELECT ..... FROM ..... WHERE .....

After SELECT, list the data fields to be retrieved. After FROM, list the fields or tables from which the data must be retrieved. After WHERE, specify conditions that limit the search to only those data records which are required. For example, suppose a financial manager wants to retrieve the names, social security numbers, departments, and salaries of all employees who are financial analysts from the employee and payroll files in a human resources database. Then the user might use the SQL query to display such information.

Another query language in some database management packages is QBE, or query by example. QBE's point-and-click capabilities make it easier for end users. This method displays boxes for each of the data fields in one or more files. User then uses keyboard or mouse to fill in, click on, query, or check boxes to indicate which information is wanted.

Graphical and Natural Queries: Most end users have difficulty correctly phrasing SQL and other database language queries. So most end user database management packages offer GUI (graphical user interface) point-and-click methods, which are easier to use and are translated by the software into SQL commands. Other packages are available that use natural language query statements similar to conversational English.

Database Maintenance

In general, the databases of an organization need to be updated continually to reflect new business transactions and other events. Other miscellaneous changes must also be made to ensure accuracy of the data in the databases. This database maintenance process is accomplished by transaction processing programs and other end user application

packages, with the support of the DBMS. End users and information specialists can also employ various utilities provided by a DBMS for database maintenance.

Application Development

Application development processes can use the internal 4GL programming language and built-in software development tools provided by many DBMS packages to develop custom application programs. For example, one can use a DBMS to easily develop the data entry screens, forms, reports, or Web pages of a business application. A DBMS also makes the job of application programmers easier, since they do not have to develop detailed data-handling procedures using a conventional programming language every time they write a program. Instead, programmers can include data manipulation language (DML) statements in the programs that call on the DBMS to perform necessary data-handling activities.

2.3 Types of Databases

Six major conceptual categories of databases that may be found in computer using organizations are as follows:

- Operational Database
- Analytical Database
- Data Warehouse
- Distributed Database
- End User Database
- External Database.

Operational Database: These databases store detailed data needed to support the operations of entire organization. They are also called subject area databases (SADB), transaction databases, and production databases.

Analytical Database: These databases store data and information extracted from selected operational and external databases. They consist of summarized data and information needed by an organization's managers and other end users. Analytical databases are also called management databases or information databases. They may also be called multidimensional databases, since they frequently use a multidimensional database structure to organize data. These are the databases accessed by the on-line analytical processing (OLAP) systems, decision support systems, and executive information systems.

Data Warehouse: Data warehouse is currently a subject of widespread interest in the business sector, specially for business trend analysis. Basically, data warehouse keeps the

long term history of regular processes. This history can be utilized to generate quick report and business trend analysis, which in turn, provides support for management decision. Data warehouse offers a scope for decision support system (DSS). This is also called on-line analytical process (OLAP) and can be developed from on-line transaction process (OLTP), i.e. the regular operation. In general data warehouse offers the following benefits:

- Historical data support for long time
- Scope to generate quick report
- Business trend analysis
- Correct management, financial and marketing decisions.

More precisely, a data warehouse is a subject oriented, non-volatile and time variant integrated collection of data in support of management's decisions. In data warehouse – data are organized by splitting into multiple dimensions of categories. Time is the most common dimension. Operational data are time stamped. All data within a certain time range are derived from the operational data. Dimensions can be thought of as the key used for referencing the information. On the other hand, granularity refers to the level of detail requirement. This affects the amount of aggregation or summarization activities that has to be done to the source data. It is a continual process when operational data are brought into to refresh the warehouse data.

Distributed Database: Many organizations replicate and distribute copies or parts of databases to network servers at a variety of sites. These distributed databases can reside on network servers on the World Wide Web, on corporate intranets or extranets, or on other company networks. Replication and distribution of databases are done to improve database performance and security. Ensuring that all of the data in an organization's distributed databases are consistently and concurrently updated is a major challenge of distributed database management.

End User Database: The end user databases consist of a variety of data files developed by end users at their workstations. For example, users may have their own electronic copies of documents they download from the World Wide Web, generate with word processing packages, or receive by electronic mail. Or they may have their own data files generated using spreadsheet and DBMS packages.

External Database: Access to a wealth of information from external databases is available for a fee from commercial on-line services, and with or without charge from many sources on the Internet, especially the World Wide Web. Web sites provide an endless variety of hyper-linked pages of multimedia documents for a user to access. Data

are available in the form of statistics on economic and demographic activity from statistical data banks. Or one can view or download abstracts or complete copies of hundreds of newspapers, magazines, newsletters, research papers, and other published material and periodicals from bibliographic and full text databases.

2.4 Data Resource Management

Data form important resource and one must learn to manage data properly to ensure the success and survival of an organization. Database management is an important application of information systems technology to the management of a firm's data resources. However, other major data resource management efforts are needed to offset some of the problems that can result from the use of a database management approach. These are:

- Database administration
- Data planning, and
- Data administration.

Database administration: Database administration includes responsibility for developing and maintaining the organization's data dictionary, designing and monitoring the performance of databases, and enforcing standards for database use and security. Database administrators and analysts work with systems developers and end users to provide their expertise to major system development projects.

Data planning: This is a corporate planning and analysis function that focuses on data resource management. It includes the responsibility for developing an overall data architecture for the firm's data resources that ties in with the firm's strategic mission and plans, and the objectives and processes of its business units. Data planning is done by organizations that have made a formal commitment to long-range planning for the strategic use and management of their data resources.

Data administration: This involves the establishment and enforcement of policies and procedures for managing data as a strategic and corporate resource. This means administering the collection, storage, and dissemination of all types of data in such a way that data become a standardized resource to all end users in the organization. The focus of data administration is the control of data in support of an organization's business functions and strategic business objectives. Data administration may also include responsibility for developing policies and setting standards for corporate database design, processing, and security arrangements, and for selecting database management and data dictionary software.

2.5 Exercises

Multiple Choice Questions

1. The basic form of SQL query is

   a. SELECT -------- WHERE ------ FROM
   b. SELECT -------- FROM -------- WHERE
   c. SELECT --------- WHERE ------ FORM
   d. None of these

2. OLAP stands for

   a. on-line array processing
   b. off-line analytical processing
   c. on-line analytical processing
   d. none of these

3. Data warehouse keeps record of

   a. all data
   b. historical data
   c. backup
   d. none of these

Questions for Short Answers

4. What is the database management system?
5. What is meant by DDL?
6. Define meta-data.
7. Differentiate between SQL and QBE
8. List the types of databases.
9. What is meant by DML?
10. What is data warehouse?
11. What are the components of data resource management?

Analytical Questions

12. Describe the database software.
13. What are the types of databases? Discuss briefly.
14. What is meant by data resource management? Briefly describe the components of the data resource management approach.

Lesson 3 Database Structures

3.1 Objectives

On completion of this lesson you will know:

- Fundamental database structures
- Characteristics of database structures.

3.2 Introduction

In all information systems, data resources must be organized and structured in some logical manner so that they can be accessed easily, processed efficiently, retrieved quickly, and managed effectively. Thus data structures and access methods ranging from simple to complex have been devised to efficiently organize and access data stored by information systems. Database management system packages are designed to use a specific data structure to provide end users with quick and easy access to information stored in databases. Five fundamental database structures are:

- Hierarchical
- Network
- Relational
- Object oriented and
- Multidimensional models.

3.3 Hierarchical Structure

In the hierarchical model, all records are dependent and arranged in multilevel structures, consisting of one root record and any number of subordinate levels. Thus, all of the relationships among records are one-to-many, since each data element is related to only one element above it. The data element or record at the highest level of the hierarchy is

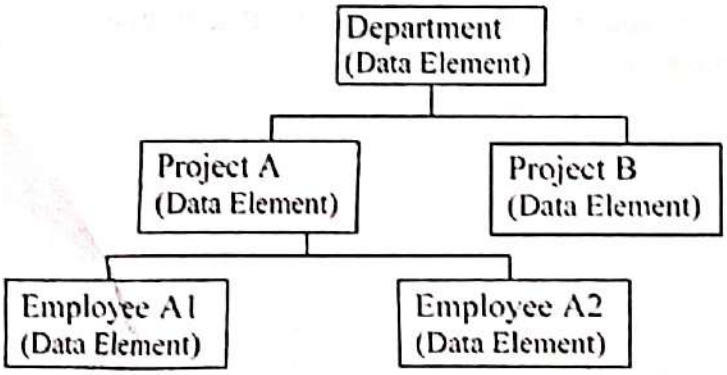


Figure 10.4: Hierarchical Structure

called the root element. Any data element can be accessed by moving progressively downward from a root and along the branches of the tree until the desired record (for instance, the employee data element) is located. Figure 10.4 shows the typical hierarchical database structure.

3.4 Network Structure

Network database structure model can represent more complex logical relationships. It allows many-to-many relationships among records; that is, the network model can access a data element by following one of several paths, because any data element or record can be related to more than one project record. Thus, one could locate all employee records for a particular department, or all project records related to a particular employee. Figure 10.5 shows a typical network database structure.

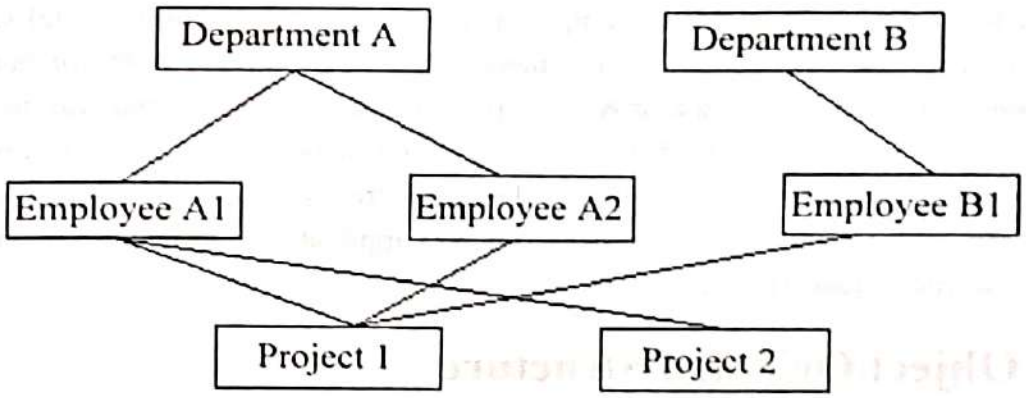
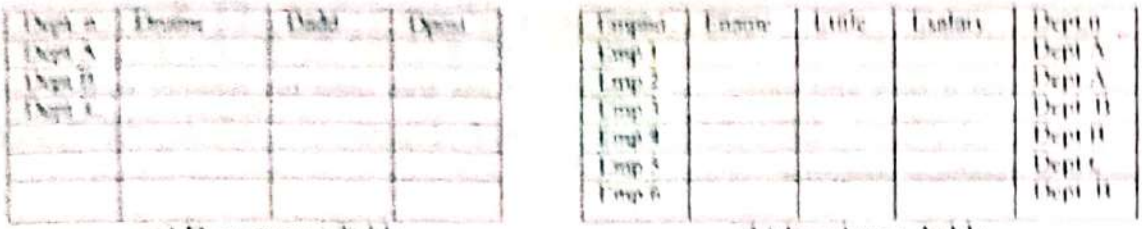


Figure 10.5: Network Structure

3.5 Relational Structure

This type of model has become the most popular and useful of the database structures. In the relational model, all data elements within the database are viewed as being stored in the form of simple tables. Figure 10.6 illustrates the relational database model with two tables representing some of the relationships among departmental and employee records. Other tables, for this organization's database might represent the data element relationships among projects, divisions, product lines, and so on. Database management system packages based on the relational model can link data elements from various tables to provide information to users. For example, a DBMS package could retrieve and display an employee's name and salary from the employee table in Figure 10.6(b), and the name of the employee's department from the department table in Figure 10.6(a), by using their common department number field (Dept. n) to link or join the two tables.



(a) Department Table (b) Employee Table

Figure 10.6. Relational Structure

3.6 Multidimensional Structure

Multidimensional structures can be visualized as cubes of data and cubes within cubes of data. Each side of the cube is considered a dimension of data. Each dimension can represent a different category, such as product type, region, sales channel, and time. Each cell within a multidimensional structure contains aggregated data related to elements along each of its dimensions. For example, a single cell may contain the total sales for a product in a region for a specific sales channel in a single month. A major benefit of a multidimensional database is that it is a compact and easy-to-understand way to visualize and manipulate data elements that have many interrelationships. So multidimensional databases have become the most popular database structure for the analytical databases that support on-line analytical processing (OLAP) applications, in which fast answers to complex business queries are expected.

3.7 Object Oriented Structure

The object-oriented database model is one of the key technologies of current software market. An object consists of data values describing the attributes of an entity, plus the operations that can be performed upon the data (Figure 10.7). This encapsulation capability allows the object-oriented model to better handle more complex types of data (graphics, pictures, voice, text) than other database structures.

The object-oriented model also supports inheritance; that is, new objects can be automatically created by replicating some or all of the characteristics of one or more parent objects. Thus, in Figure 10.7, the checking and savings account objects can both inherit the common attributes and operations of the parent bank account object. Such capabilities have made object-oriented database management systems (OODBMS) popular in computer-aided design (CAD) and in a growing number of applications. For example, object technology allows designers to develop product designs, store them as objects in an object-oriented database, and replicate and modify them to create new product designs. In addition, multimedia Web-based applications for the Internet and

coopetable Intranets and extranets have become a major application area for object technology

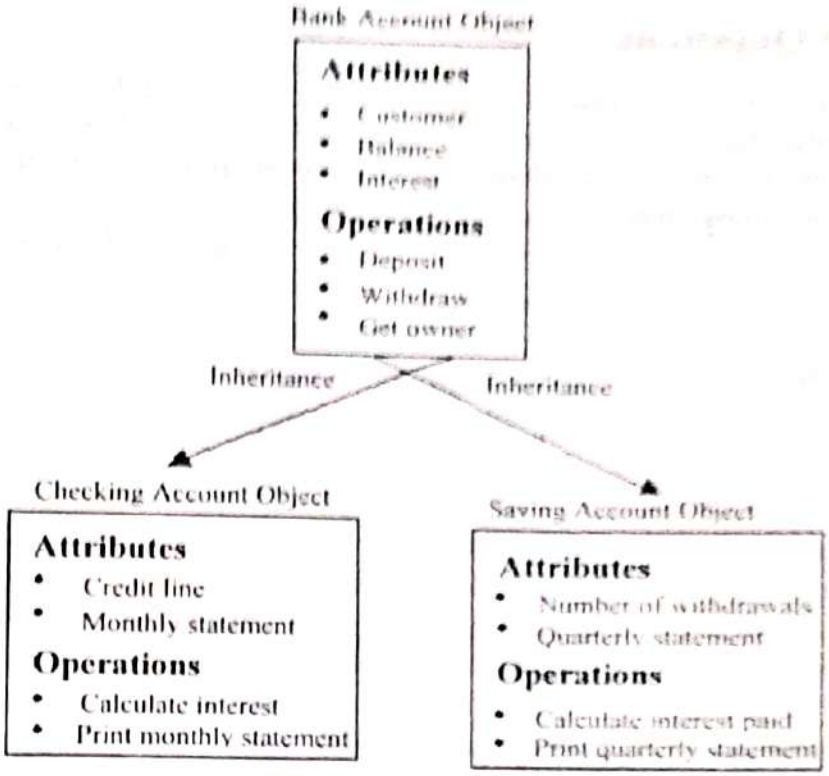


Figure 10.7: The checking and savings account objects can inherit common attributes and operations from the bank account object.

3.8 Exercise

Multiple Choice Questions

1. The Most popular database structure is

   a. multidimentional
   b. relational
   c. network
   d. none of these

2. Object oriented technology is very useful for

   a. multimedia
   b. graphics
   c. multimedia based web application
   d. none of these

Questions for Short Answers

3. List the types of five fundamentals database structures.
4. What is the relational database?

5. Differentiate between relational and multidimensional database structures.
6. What is object oriented database structure?
7. Differentiate between hierarchical and network database structure.

Analytical Questions

8. Differentiate between relational and multidimensional database structures.
9. What is the object oriented database structure? Discuss with a diagram.
10. What is the hierarchical database structure? What are its limitations as compared to a network database structure?

Lesson 4: Database Access and Development

4.1 Objectives

On completion of this lesson you will know:

- Database access
- Database development.

4.2 Database Access

In database maintenance, records or objects have to be continually added, deleted, or updated to reflect business transactions. Data must also be accessed rapidly so information can be produced in response to end user requests.

Key Fields

All data records usually contain one or more identification fields, or keys, that identify the record to be located. For example, the Social Security number (in many countries) of a person is often used as a primary key field that uniquely identifies the data records of individuals in student, and customer files and databases. Other methods also identify and link data records stored in several different database files. For example, hierarchical and network databases may use pointer fields. These are fields within a record that indicate (point to) the location of another record that is related to it in the same file, or in another file. Hierarchical and network database management systems use this method to link records so that they can retrieve information from several different database files. On the other hand, relational database management packages use primary keys to link records. Each table (file) in a relational database must contain a primary key. This field uniquely identifies each record in a file and must also be found in other related files.

URLs

The files and databases on the Internet use URLs (Uniform Resource Locators) for data access. The pages of hyper-linked text and multimedia documents on the World Wide Web and Web sites are accessed by URLs with the format shown in Figure 10.8. For example, the URL for the home page of the ACM Web site is http://www.acm.org. Thus, a Web site relies on URLs instead of traditional keys and pointers to identify and express the relationships between the interconnected pages of its hypermedia database.

- Web: http://server.address:port/homepage.html
- FTP: ftp://server.address/complete.filename

Figure 10.8: The general form of Uniform Resource Locators for major Internet services.

Sequential Access

Sequential access method uses a sequential organization, in which records are physically stored in a specified order according to a key field in each record. For example, payroll records could be placed in a payroll file in a numerical order based on employee social security numbers. Sequential access is fast and efficient when dealing with large volumes of data that need to be processed periodically. However, it requires that all new transactions be stored into the proper sequence for sequential access processing. Also, most of the database or file may have to be searched to locate, store, or modify even a small number of data records. Thus, this method is too slow to handle applications requiring immediate updating or responses.

Direct Access

In Direct Access methods, records do not have to be arranged in any particular sequence on storage media. However, the computer must keep track of the storage location of each record using a variety of direct organization methods so that data can be retrieved when needed. New transactions data do not have to be sorted, and processing requiring immediate responses or updating is easily handled. There are a number of ways to directly access records in the direct organization method. Three widely used methods to accomplish such direct access processing are:

- Key transformation
- Index
- Index Sequential Access.

Key transformation performs an arithmetic computation on a key field of record and uses the number that results from that calculation as an address to store and access that record. Thus, the process is called key transformation because an arithmetic operation is applied to a key field to transform it into the storage location address of a record.

Index is another direct access method used to store and locate records. A new data record is stored at the next available location, and its key and address are placed in an index. The computer uses this index whenever it must access a record.

Index Sequential Access Methods are used to store record in a sequential order on a magnetic disk or other direct access storage devices based on the key field of each record. In addition, each database contains an index that references one or more key fields of each data record to its storage location address. Thus, an individual record can be directly located by using its key field to search and locate its address in the database index, similar to locate key topics in a book by looking them up in its index. As a result, if a few record must be processed quickly, the index is used to directly access the record needed.

4.3 Database Development

Personal or small database development is relatively easy using PC database management packages. However, developing a large database of complex data types might be a complex task. In many companies, developing and managing large corporate databases are the primary responsibility of the database administrator and database design analysts. They work with end users and systems analysts to model business processes and the data they require. Then they determine:

- Data definitions which should be included in the database and
- Structure or relationships that should exist among the data elements.

Data Planning and Database Design

Database development may start with a top-down data planning process. Database administrators and designers work with corporate and end user management to develop an enterprise model that defines the basic business processes of the enterprise. Then they define the information needs of end users.

Next, end users must identify the key data elements that are needed to perform their specific activities. This frequently involves developing entity relationship diagrams (ERDs) that model the relationships among the many entities involved in business processes. Figure 10.9 illustrates some of the relationships in a purchasing/receiving

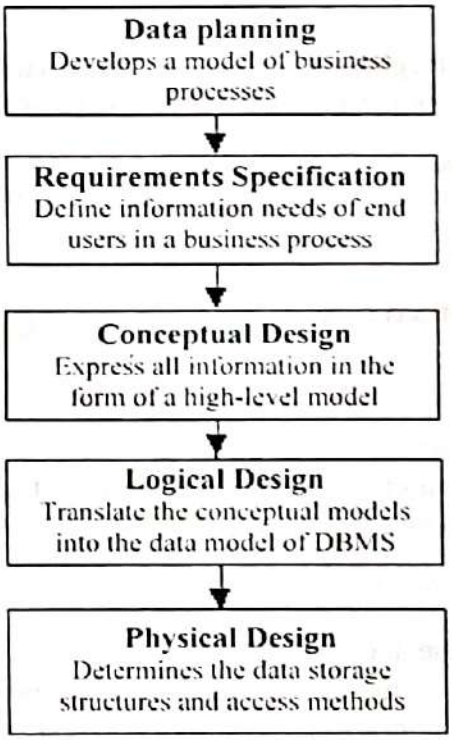


Figure 10.9: Database development process

system. End users and database designers could use ERD models to identify what supplier and product data are necessary in the purchasing/receiving process.

Each data model defines the logical relationships among the data elements needed to support a basic business process. For example:

- Can a supplier provide more than one type of product?
- Can a customer have more than one type of account?
- Can an employee have several pay rates or be assigned to several project workgroups?

Answering such questions will identify data relationships that have to be represented in a data model for a business process. These data models then serve as logical frameworks on which to base the physical design of databases and the development of application programs to support the business processes of the organization.

Data models represent logical views of the data and relationships of the database. Physical database design takes a physical view of the data (also called the internal view) that describes how data are to be physically stored and accessed on the storage devices of a computer system.

4.4 Exercise

Multiple Choice Questions

1. URL is an acronyme of

   a. Universal Register Locator b. Universal Resource Locator
   c. United Resource Locator d. none of these

2. Direct access methods are of

   a. two types b. three types
   c. four types d. none of these

Questions for Short Answers

3. What is the key field?
4. What is meant by URL?
5. What are the three direct access methods?
6. Differentiate between sequential and direct access methods.
7. Show the database development process using a flowchart

Analytical Questions

8. Briefly discuss the database access methods.
9. Describe the database development process using a flowchart.
10. Differentiate between direct and sequential access methods. Describe the different direct access methods.

Chapter 11

Software Development

Lesson 1: Introduction to Software Development
Lesson 2: System Analysis
Lesson 3: System Design
Lesson 4: System Development
Lesson 5: System Implementation

Lesson 1: Introduction to Software Development

1.1 Objectives

On completion of this lesson you will know:

- Software crisis
- Systems and personnel
- System modeling

1.2 Software Crisis

Many companies face shrinking budgets for software development. This means that the scale of investment in software projects is greatly reduced. This results in the need for shorter development times and lower manpower costs. With the increasing swing towards distributed systems, a new set of powerful visual tools is available. These tools bring with them a new approach towards system development. However, the following issues need to be resolved to overcome this crisis:

- Designing systems with open interfaces
- Ensuring reusability and extensibility of modules

- Developing modules that are tolerant to changes
- Improving productivity and decreasing cost
- Improving quality of software
- Managing time schedules
- Industrializing software development process.

Traditional methodologies can not meet the above mentioned challenges and the object oriented methodology is gaining ground to meet the challenges.

1.3 System and Analyst

System

A system is a collection of components purposefully organized to accomplish a certain task. There are natural systems, such as the cardiovascular system. However, many systems that have been planned and deliberately put into place by people are known as artificial systems. For example, a library has a system for serving a member (borrow, return and search of books).

System Analysis

System analysis is the process of studying an existing system to determine how it works and how it meets user needs. System analysis lays the groundwork for improvements to the system. The analysis involves an investigation, which in turn usually involves establishing a relationship with the client for whom the analysis is done and with the users of the system.

Client

The client is the person or organization contracting to have the work done. The users are people who contact with the system (usually employees and consumers). For instance, in a library system, the client is probably the librarian, and the users are the members of the library who have library cards.

System Analyst

System analyst fills the role of change agent. That is, the analyst must be the catalyst or persuader who overcomes the reluctance to changes within an organization in the development of the new system. The common industry phrase is user involvement, and nothing can be more important to the success of the system. Users must be involved in the process from beginning to end. The system analyst must monitor the user pulse regularly to make sure that the system being planned is one that will meet user needs.

Functions of a System Analyst

The system analyst has three principle functions which are briefly stated here.

Coordination: A system analyst must coordinate schedules and system-related tasks with a number of people: the programmers working with the system; the system users, from bottom to top management; the vendors selling equipment; and a host of others.

Oral and written communication: The system analyst may be called upon to make oral presentations to clients, users, and others involved with the system. The system analyst provides written reports on the results of the analysis and the goals and means of the design.

Planning and design: The system analyst, with the participation of members of the client organization, plans and designs the new system. This function involves all the activities from the beginning of the project until the final implementation of the system.

System design

System design is the process of developing a plan for an improved system, based on the results of the system analysis. For instance, an analysis of a library system may reveal that members stand in unacceptably long lines waiting to borrow books. A new system design might involve plans to have employees press buttons that match search items, causing a display on an overhead screen.

The system analyst normally performs both analysis and design. In some computer installations a person who is mostly a programmer may also do some system analysis and thus have the title programmer/analyst. Traditionally, most people who have become system analysts have done so by way of programming. Starting out as a programmer helps the analyst appreciate computer related problems that arise in analysis and design work. Thus, programmers often depend on system analysts for specifications from which to design programs.

1.4 System Development Cycle

In the early days of business computing, information system development was a disorganized or an ad-hoc process frequently producing discouraging results. To improve the quality of information systems and services an approach for system development life cycle (SDLC) is developed. The system development life cycle can be considered in five phases. Each phase is intended to address key issues and to produce outcomes, which form the input for the next phase. These phases are as follows:

- **Preliminary investigation** – determining the problem and scope
- **Analysis** – understanding the existing system
- **Design** – planning the new system
- **Development** – doing the work to bring the new system into working condition
- **Implementation** – converting to the new system.

These phases are summarized in Table 11.1.

Table 11.1. System development life cycle and activities

Phases	Activities
Preliminary investigation	• Problem definition • Nature of problem • Problem scope • Objectives of the problem • Feasibility test • Report of the preliminary investigation
System Analysis	• Data gathering – Written documents – Interviews – Questionnaires – Observation – Sampling • Data analysis – Charts – Tables • System requirements • Report to the management
System design	• Preliminary design • Prototyping • Detail design – Output requirement – Input requirement – Files and databases – System processing – Controls and backup
Systems Development	• Scheduling activities • Programming • Testing
Implementation	• Training • Equipment conversion • File conversion • System conversion • Auditing • Evaluation • Maintenance

1.5 Waterfall Model

The SDLC described above is a systematic step by step process. However, it is not always wise to keep going if work in a later stage turns up problems with the work performed in an earlier stage. For example, in the system analysis phase, the analyst may discover that the problem has not been formulated correctly. To overcome this problem,

the most popular waterfall model builds correction pathways into the model that enable a return to a previous phase (Figure 11.1). It is the widely used way to implement the system development life cycle.

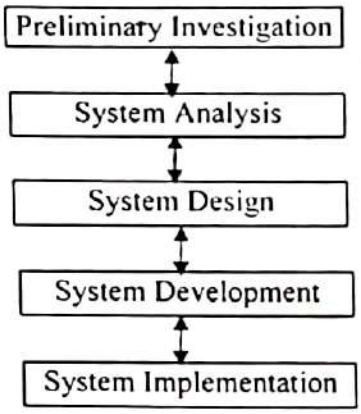


Figure 11.1: Waterfall Model for software life cycle

1.6 Exercise

Multiple choice questions

1. The person or organization contracting to have the work done is

   a. user b. programmer
   c. client d. system analyst

2. In an organization the analyst fills the role of

   a. user b. client
   c. change agent d. programmer

Questions for short answers

3. Differentiate between clients and users.
4. Briefly describe the functions of a system analyst.
5. Write down the steps of the software development life cycle.
6. What is the advantage of Waterfall model?

Analytical Questions

7. Describe the software development life cycle with a diagram.
8. What is meant by software crisis? What are the issues to be resolved to overcome this crisis?
9. What is Waterfall model? Describe with a diagram.

Lesson 2: System Analysis

2.1 Objectives

On completion of this lesson you will know:

- Preliminary investigation for analysis
- Methodology for system analysis
- Methodology for data analysis.

2.2 Preliminary Investigation

The Preliminary investigation-often called the feasibility study or system survey - is the initial investigation of the problem. It consists of the groundwork necessary to determine whether the project should be pursued or not. As a system analyst, it is required to determine what the problem is and what to do about it. The net result will be a rough plan for proceeding with the project.

Before deciding whether to proceed, the analyst must be able to describe the problem. To do this, the analyst will work with the users. One of the tools will be an organization's chart, which is a hierarchical drawing showing the organization's management by name and title. Many organizations already have such a chart. If the chart does not exist, the analyst must ask some questions and then make the chart. To work effectively within the organization, the analyst needs to understand what are the lines of authority through the formal communication channels.

Problem Definition

Initial aim of the analyst is to define the problem. Analyst and the users must come to an agreement on these points-analyst must agree on the nature of the problem and then designate a limited scope. In the process, analyst will also determine the objectives of the project.

Nature of the problem

Sometimes what appears to be the problem turns out to be, on a closer look, only a symptom. For example, suppose the analyst is examining customer complaints of late deliveries. The brief study may reveal that the problem is not in the shipping department, but in the original ordering process.

Scope of the problem

It is very difficult to establish the scope of the problem, because problems tend to expand

if there is no firm boundaries. Limitations are also necessary to stay within the eventual budget and schedule. In the beginning the analyst and user must agree on the scope of the project: what the new or revised system is supposed to do and not to do. If the scope is too narrow, it may not meet user needs.

Objectives of the problem

The analyst will soon come to understand what the user needs, that is, what the user thinks the system should be able to do. An analyst may want to express these needs as objectives. For instance, analyst must examine the objectives for the process and has to meet with the people who run the existing system or those who already know what such a system must do. In the next phase of system analysis, analyst will produce a more specific list of system requirements based on these objectives.

Feasibility Test

A feasible project can be completed successfully. Feasibility has three dimensions:

- Technical
- Operational and
- Economic feasibility.

Technically feasible project can be accomplished with existing and proven technology.

Operationally feasible project can be accomplished with the organization's available facilities.

Economically feasible is usually answered by a cost/benefit analysis. The purpose of cost estimates is to help classify what the system is going to do.

Project Proposal

At the end of preliminary investigation, the project leader has to write a project proposal that introduces:

- Nature of the existing system's problem
- Explanation of the proposed solution
- Direct and indirect benefits of the proposed system
- Details of the proposed project plan
- Conclusion with recommendation.

At this point, management is to considered one of the three choices:

- Drop the matter
- Fix the problem immediately, if it is simple or
- Authorize the analyst to go on to the next phase for a closer look.

2.3 System Analysis

2.3.1 Introduction

This phase starts when the management has decided to continue. It is to be noted that the purpose of system analysis is to understand the existing system. A related goal is to establish the system requirements. The best way to understand a system is to gather data about it; data must then be organized and analyzed. During the analysis phase the analyst will be concerned with:

- Data gathering and
- Data analysis.

It is to be noted that the system being analyzed may or may not already be a computerized system.

2.3.2 Data Gathering

There is no standard procedure for gathering data because each system is unique. Data gathering is expensive and requires a lot of work and time. But there are certain sources that are commonly used:

- Documentation
- Interviews
- Questionnaires
- Observations
- Sampling

Sometimes analysis requires all these sources, but in most cases it will be appropriate to use some and not others. All references to data-gathering techniques assume that analyst has the proper authority and the cooperation of the client organization before proceeding.

Documentation

The documentation includes procedures, manuals, reports, forms, and any other kind of material on the problem that analyst finds in the organization. The analyst may find very few documents and no trials to follow. Sometimes the opposite is true: the analyst finds so many documents that it is difficult to assess them. Thus, judgement is required, or the analyst will spend hours reading outdated reports or manuals that no one follows. However, it is better to take time to get a copy of each form used by the organization.

Interviews

This method of data gathering has advantages and disadvantages. A key advantage is that

interviews are flexible; as the interviewer can change the direction of question for a productive area of investigation. Another bonus is that the analyst can probe with open-ended questions that people would balk at answering on paper. Analyst will find that some respondents yield more information in an interview than they would if they had to commit themselves in writing. The analyst can also observe the respondent's voice inflection and body motions, which may tell him more than words alone. Finally, of course, there is the bonus of getting to know clients better and establishing a rapport with them - an important factor in promoting user involvement in the system from the beginning. Interviews have certain drawbacks. They are unquestionably time-consuming and therefore expensive. There are two types of interviews:

- Structured and
- Unstructured.

A structured interview includes only questions that have been planned and written out in advance. The interviewer sticks to those questions and asks no others. A structured interview is useful when it is desirable - or required by law - to ask identical questions. However, the unstructured interview is often more productive. An unstructured interview includes prepared questions, but the interviewer is willing to vary from the line of questioning and pursue other subjects if they seem appropriate.

Questionnaires

Questionnaires can be used to get information from large groups. Questionnaires allow people to respond anonymously - the respondents just complete forms and presumably, they respond more truthfully. Questionnaires may be used to verify information gathered from interviewing few of the users. However, owing to the large number of respondents, sometimes a trend or problem pattern emerges that would not be evident from a small number of interviews. Questionnaires do have disadvantages, however. Some people will not return questionnaires because they are wary of putting anything on paper, even anonymously. The ballot-box type ("yes" or "no"), and the qualified response type (agreement or disagreement with the question on a scale from, say, 1 to 5) are two common examples of questionnaires. In general, people prefer a questionnaire that is quick and simple. Analysts also prefer simple questionnaires because their results are easier to tabulate. The analyst should probably save open-ended questions for an interview.

Observation

As an analyst and observer, one can go into the organization and find who interrelates with whom. In particular, the analyst may observe how data flows: from desk to desk, fax

to fax, or computer to computer and note how data come into and leave the organization. Initially, analyst can make arrangements with a group supervisor and make everyone aware of the purpose of the visit. It is essential to visit on more than one occasion so that the people under observation become used to his presence. One form of observation is participant observation; in this form the analyst temporarily joins the activities of the group. This practice is not especially popular but can be useful in studying a complicated organization.

Sampling

Analyst may need to collect data about quantities, costs, time periods, and other factors relevant to the system. For example: the number of phone orders that can be taken by an order entry clerk in an hour. This type of question may be best answered through a procedure called sampling. Instead of observing all the clerks filling orders for an hour, the analyst can pick a sample of 3 or 4 clerks. Or, in a case involving a high volume of paper output, such as customer bills, the analyst could collect a random sample of a few dozen bills.

2.3.3 Data Analysis

There are a variety of tools – charts and diagrams – for analyzing data, but not all of them are appropriate for every system. Analyst should become familiar with the techniques, then use the tools that suit at the time. In general there are two typical tools:

- Data flow diagrams (DFD)
- Decision table.

The reasons for data analysis are related to the basic functions of the system analysis phase: to show how the current system works and to determine the system requirements. In addition, data analysis materials will serve as the basis for documentation of the system.

Data Flow Diagram

A data flow diagram is a sort of road map that graphically shows the flow of data through a system. It is a valuable tool for depicting present procedures and data flow. Although data flow diagrams can be used in the design process, they are particularly useful for facilitating communication between analyst and the users during the analysis phase. Analyst would probably make notes about what goes on where. But that is only the data – gathering function, now the analyst must somehow analyze the findings. Analyst could come back on another day with pages of narrative for the manager to review or, instead, show an easy-to-follow picture. Most users would prefer the picture.

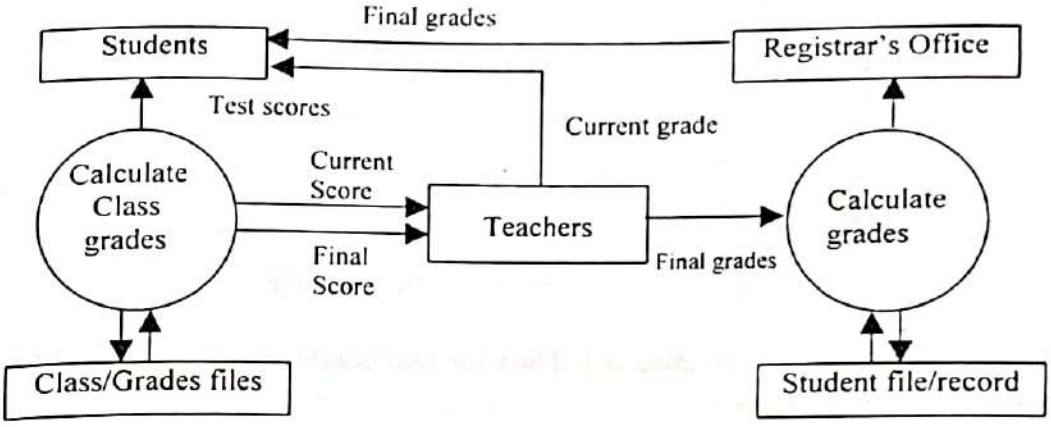


Figure 11.2: A data flow diagram shows how data moves through the existing system.

There are variety of notations for data flow diagrams. The notations used here have been chosen because they are easy to draw and read. The elements of a data flow diagram are processes, files, sources and sinks, and vectors.

Processes, represented by circles, are the actions taken on the data – comparing, checking, stamping, authorizing, filling, and so forth. A file is a repository of data – a disk file, a set of papers in a file cabinet, or even mail in an in-basket or blank envelopes in a supply bin. In a DFD, a file is represented by an open-ended box.

A source is a data origin outside the system under study. An example is a payment sent to a department store by a customer; the customer is a source of data. A source is represented by a square.

A sink is a destination for data going outside the system; an example is the bank that receives money deposits from the accounts receivable organization. A sink is also represented by a square.

Vectors are simply arrows, lines with directional notations showing the flow of data. A vector must come from or go to a process circle, or bubble.

Decision Table

A decision table, also called a decision logic table, is a standard table of the logical decisions that is made regarding potential conditions in a given system. Decision tables are useful in cases that involve a series of interrelated decisions; and helps to ensure that no alternatives are overlooked. Programmers can code portions of a program right from a decision table. Figure 11.3 shows the format of a decision table.

Heading	Rules				
	1	2	3	4	5
Condition stub (if....)					
⋮					
Action Stub (then...)					

Figure 11.3: The format of a decision table.

A condition can be true (T) or false (F). Thus for two conditions there are four possible combinations (as an examples):

	1	2	3	4
Purchaser is a member	T	T	F	F
Purchase exceeds Tk. 1000	T	F	T	F

The actions for this example could be:

- 15% discount
- 10% discount
- 5% discount
- No discount.

As an explanation, in this case rule 2 says that the purchaser is a member and the purchase dose not exceed Tk. 1000. The decision table illustrating the discount policy is shown in Figure 11.4.

		1	2	3	4
Conditions	Purchaser is a member Purchase exceeds Tk. 1000				
Actions	15% discount	✓			
	10% discount		✓		
	5% discount			✓	
	No discount				✓

Figure 11.4: Example of a purchase decision table

2.4 System Requirements

The purposes of gathering and analyzing data are to:

- Understand the system and,
- Establish the system requirements.

The description of the system is quite broad in the preliminary investigation phase, but at

this stage the analyst is ready to list precise system requirements. The analyst needs to determine and document specific user needs. A system that a bank teller uses, for instance, needs to be able to retrieve a customer record and display it on a screen within few seconds.

If appropriate, analyst may be able to open the user's eyes to system capabilities they may not have considered. Users who are used to leafing through computer-produced printed reports, for example, might find screen output easier and more timely. On the other hand, as an analyst, one may have to reign in unrealistic user expectations. A user may have seen or heard about computer systems that are fancier or more sophisticated than what circumstances (or budget) allow.

The importance of accurate requirements cannot be overemphasized, because the design of the new system will be based on the system requirements. Furthermore, the analyst and management must come to clear agreement on the system requirements, since a misunderstanding can result in a poor evaluation of the new system and even cause a delay in project completion.

2.5 Report to Management

Once analyst has finished the system analysis phase, he/she has to present a report to management. This comprehensive report, part of the continuing process of documentation, summarizes the problems the analyst found in the current system, describes the requirements for the new systems, includes a cost analysis, and makes recommendations on what course to take next. If the project is significant, analyst may also make a formal presentation, including visual displays. If implementing the new system presents significant problems, this might be a good point for management to stop the project, because the investment is still small relative to the amount that will be invested when programming begins. If management decides to pursue the project, analyst move on to the next phase.

2.6 Exercise

Multiple choice questions

1. What are the two very common tools for data analysis?

    a. Data flow diagram and decision table
    b. Database package & Decision table
    c. ERD and database package
    d. None of these

Questions for short answers

2. What is meant by technical feasibility?
3. What are the components to be included in a project proposal?
4. What is DFD?
5. What is meant by ERD model?
6. What is the purposes of decision table?
7. Differentiate between structured and unstructured interviews.
8. What is meant by sampling? Write down the necessity of sampling for data gathering.

Analytical questions

9. Discuss briefly the components of preliminary investigation.
10. What is meant by data gathering? Describe the sources that are commonly used for data gathering.
11. Describe the data analysis methodologies in system analysis.

Lesson 3: System Design

3.1 Objectives

On completion of this lesson you will know:

- Preliminary design procedure
- Prototyping and its advantages
- Procedure for detail design of a system.

3.2 Introduction

System design activity is divided into two sub-phases:

- Preliminary design – the analyst establishes new system concept
- Detail design – the analyst determines exact design specifications.

The reason this phase is divided into two parts is that an analyst wants to make sure that management approves the overall plan before spending time and money on the details of the new system.

3.3 Preliminary Design

The initial task of preliminary design is to review the system requirements and then consider some of the major aspects of a system. These are:

- Should the system be centralized or distributed?
- Should the system be on-line?
- Can the system be run on the users' personal computers?
- How will input data be captured?
- What kind of reports will be needed?

The questions can go on and on. A key question that should be answered early on is whether packaged software should be purchased, as opposed to having programmers write custom software. That is, instead of designing, developing and implementing a new system from scratch, analyst may be able to obtain an existing system that meets requirements. This may be tricky because clients often think that their problems are unique. However, if the new system falls into one of several major categories, such as accounting or inventory control, then many software vendors offer packaged solutions. A packaged solution should meet at least 75% of client requirements. For the remaining 25%, the client can adjust ways of doing business to match the package software or, more expensively, customize, or alter, the package software to meet the client's special needs.

Finally, some systems analysts, especially those who work independently as consultants, specialize in certain types of problems and become expert in acquiring and implementing certain commercial software package to solve those problems.

However, if the analyst proceeds with an in-house design, then, together with key personnel from the user organization, he/she has to determine an overall plan. In fact, it is common to offer alternative plans, called candidates. Each candidate meets the user's requirements but with variations in features and costs. The chosen candidate is usually the one that best meets the user's current needs and is flexible enough to meet future needs. The selected plan is expanded and described so that it can be understood by both the user and the analyst.

At this stage it is wise to make a formal presentation of the selected plan, or possible alternatives. The point is that analyst does not want to commit time and energy to - nor does the user wants to pay for – a detailed design until the analyst and the user agree on the basic design. Such presentations often include a drawing of the system from a user's perspective.

3.4 Prototyping

The idea of building a prototype - a sort of modeling of the system - has taken a sharp upward turn in popularity recently. Considered from a systems viewpoint, a prototype is a limited working system - or subject of a system - that is developed quickly, sometimes in just a few days. A prototype is a working model that can be tinkered with and fine-tuned. The idea is that users can get a clear picture of what the system might be like before it is fully developed. If they are not satisfied, they can revise their requirements before a lot has been invested in developing the new system.

Prototyping Tools

The prototype approach exploits advances in computer technology and uses powerful high-level software packages to allow analysts to build systems quickly in response to user needs. In particular, the fourth-generation languages could be the right choice for prototyping. One of their key advantages is that they can be used to produce something quickly. The systems produced can then be refined and modified as they are used, in a continual process, until the fit between user and system is acceptable.

Reasons for Prototyping

Most of the organizations use prototyping on a limited basis. For example, an organization may make a prototype to demonstrate a particular screen output or an

especially complex or questionable part of a design. That is, prototyping does not necessarily have the scope of the final system. Some organizations develop throwaway prototypes that they use only to get a grip on the requirements; then they begin again and go through the systems development life cycle formally. Other organizations start with a prototype and keep massaging it until it becomes the final and accepted version. In either case, a prototype forces users to get actively involved.

Results of Prototyping

A prototype of a whole system will initially include:

- Minimum input data
- No validation checks
- Incomplete files
- Limited security checks
- Sketchy reports, and
- Minimum documentation.

However, actual software uses real data to produce real output. It is to be noted that prototyping is an iterative process; the system is changed again and again based on the lessons learned by creating the prototype.

3.5 CASE Tools

CASE (Computer-Aided Software Engineering) tools goes beyond the concept of prototyping and has become a significant factor in the development of systems. CASE tools provide an automated means of designing and changing systems. In fact, integrated CASE tools can automate most of the systems development life cycle.

CASE software is available for personal computers. The system analyst can produce designs right on the computer screen. Thus, a key ingredient of a package of CASE tools is a graphics interface. What is more, that screen is usually part of a personal computer. Other important CASE ingredients are data store - often called a data dictionary or even an encyclopedia, and the ability to generate a program automatically, right from the design.

CASE tools have several advantages. Foremost is the ability to note inconsistencies in the system design. Such tools can also make global changes related to a single change; for example, if someone changes a name in one place, the CASE tool automatically changes it throughout the design specifications. CASE tools provide consistency, speed, increased productivity, and cost savings.

3.6 Detail Design

Once the users have accepted the design proposal, the system analyst must develop detailed design specifications, or a detail design. This is a time-consuming part of the project, but it is relatively straightforward. In this phase every fact of the system is considered in detail. Here is a list of some detail design activities:

- Designing output forms and screens,
- Planning input data forms and procedures,
- Drawing system flowcharts,
- Planning file access methods and record formats,
- Planning database interfaces,
- Planning data communications interfaces,
- Designing system security controls, and
- Considering human factors.

This list is comprehensive, nor will all activities listed be used for all systems. Some analysts choose to plan the overall logic at this stage, preparing program structure charts, pseudo-code etc. Normally, in the detail design phase, parts of the system are considered in this order:

- Output requirements
- Input requirements
- Files and databases
- Systems processing
- Systems controls and backup.

Output Requirements

First of all, analyst must know exactly the client's demand that is the system's output. The analyst must also consider the medium of the output - paper, computer screen, microfilm, and so on. In addition, he/she must determine the type of reports needed (summary, exception, and so on) and the contents of the output – what data are needed for the reports. The form of the printed outputs is also a consideration; they may be custom-printed if they go outside the organizations' to customers or stockholders. Analyst may wish to determine the report format by using a printer spacing chart, which shows the position of headings, the spacing between columns, and the location of date and page numbers. Analyst may also use screen reports, mock-ups on paper of how the screen will respond to user queries.

Input Requirements

Once desired output is determined, analyst must consider what kind of input is required to produce it. First the input medium must be considered. Next the analyst consider

content again - what fields are needed, the order in which they come, and so on. This in turn may involve designing forms that will organize data before it is entered. Analyst needs to plan some kind of input validation process, a check that data are reasonable as well as accurate – he/she would not expect a six-figure salary, for example, for someone who works in the mail room. Finally, it is necessary to consider input volume, particularly the volume at peak periods and whether the system can handle it. A mail-order house, for instance, may have to be ready for higher sales of expensive toys in the holiday season than at other times of the year.

Files and Database

It is necessary to consider how the files in a computer system will be organized: sequentially, directly, or by some other method. Analyst also needs to decide how the files should be accessed. They might be organized as indexed files but be accessed directly or sequentially. Also it is needed to determine the format of records making up the data files. If the system has one or more databases, or accesses databases used in other systems, then the analyst will have to coordinate with the database administrator, the person responsible for controlling and updating databases.

Systems processing

Now it is necessary to show the flow of data in the new system. One method is to use standard ANSI flowchart to illustrate what will be done and what files will be used. Note that a systems flowchart is not the same as the logic flowchart used in programming. The systems flowchart describes only the big picture. On the other hand, a logic flowchart represents the flow of logic within a single program.

Systems Controls and Backup

To make sure data are input, processed, and output correctly and to prevent fraud and tampering with the computer system, analyst will need to institute appropriate controls. In a batch system, in which data for the system are processed in groups, begin with the source documents, such as time cards or sales orders. Each document should be serially numbered so that the system can keep track of it. Documents are time-stamped when received and then grouped in batches. Each batch is labeled with the number of documents per batch; these counts are balanced against totals of the processed data. The input is controlled to make sure data are accurately converted from source documents to machine-processable form. Data input to online systems is backed up by system journals, files that record every transaction processed at each terminal, such as an account withdrawal through a bank teller. Processing controls include the data validation procedures.

It is also important to plan for the backup of system files: copies of transaction and master files should be made on a regular basis. These file copies are stored temporarily in case the originals are inadvertently lost or damaged. Often the backup copies are stored off site for security.

3.7 Exercises

Multiple choice questions

1. Prototyping provides scope for

   a. detail development b. quick development
   c. slow development d. none of these

Questions for short answers

2. What major aspects are to be considered for prototyping design?
3. What are the advantages of prototyping?
4. What is meant by CASE?
5. What are the activities involved in detail design?
6. What are the components of detail design?
7. What do you understand by systems controls and backup?

Analytical questions

8. Briefly describe the preliminary design and its necessity.
9. What is prototyping? Describe the methodology, advantages and disadvantages of prototyping.
10. What are the components for detail design? Briefly describe the components.

Lesson 4: System Development

4.1 Objectives

On completion of this lesson you will know:

- Scheduling for system development
- Program development life cycle
- Testing methodology.

4.2 Components of System Development Process

Components of a system development are as follows:

- A schedule to monitor the principal activities in system development
- Programming and
- Testing the system.

The description of each of the above procedures are given below.

4.3 Scheduling the Activities

Scheduling the activities of system development is an important component for success. A Gantt chart is a simple device, easy to understand and implement. However, organizations that want more control use project management software, which offers additional features, such as allocating people and resources to each task, monitoring schedules, and producing status reports. In particular, project management software tracks the impact of schedule slides on future dependent activities. For example, if a program depends on the design of certain files, then a delay in file design may delay the development of the related program. Although project management is most carefully needed in the development phase, most projects of any size are monitored from the initial phases.

A Gantt chart is shown in Figure 11.5 as an example. In the example the chart shows the work to be accomplished over a given period. It does not, however, show the number of work hours required. As a supervisor, it would be common practice for analyst to ask others on the development team to produce individual Gantt charts of their own activities.

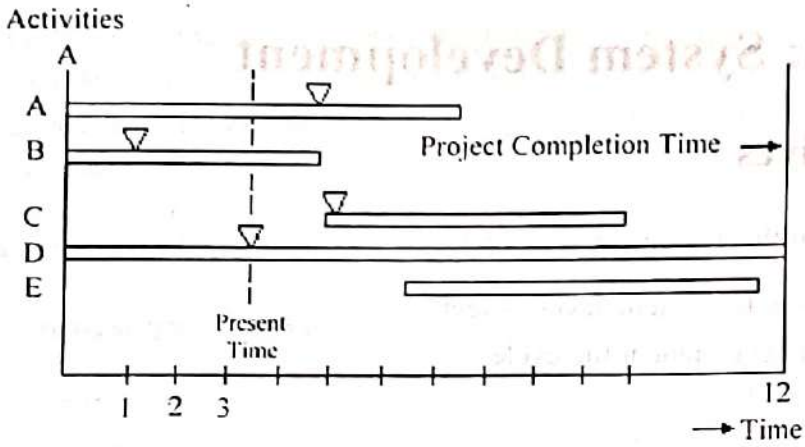


Figure 11.5: Gantt Chart shows the scheduled tasks and milestones of a project (∇ denotes current status).

4.4 Programming

Until this point there has been no programming, that is, unless some prototyping was done. Usually, before programming begins, analyst needs to prepare detail design specifications. Program development tools must be considered. Some of this work may already have been done as part of the design phase, but usually programmers participate in refining the design at this point. Design specifications can be developed through detailed logic flowcharts and pseudocode, among other tools. There are five main steps in the program development process. These are:

- Defining the problem
- Planning the solution
- Coding the program
- Testing the program
- Documenting the program.

Defining the problem

As a programmer, one has to meet with the users from the client organization to analyze the problem, or with a systems analyst who outlines the project. Specifically, the task of defining the problem consists identification of input (given data) and the requirement i.e. output. Eventually, a written agreement is to be produced that among other things, specifies the kind of input, processing and output required. This is not a simple process. It is closely related to the process of system analysis, in turn, software development process, as discussed earlier.

Planning the Solution

Most usual manner of planning the solution to a problem are to draw a flowchart and to

write pseudocode, or possibly both. Essentially, a flowchart is a pictorial representation of step-by-step solution to a problem. It consists of arrows representing the direction the program takes and boxes and other symbols representing actions. It is a map of what program is going to do and how it is going to do it.

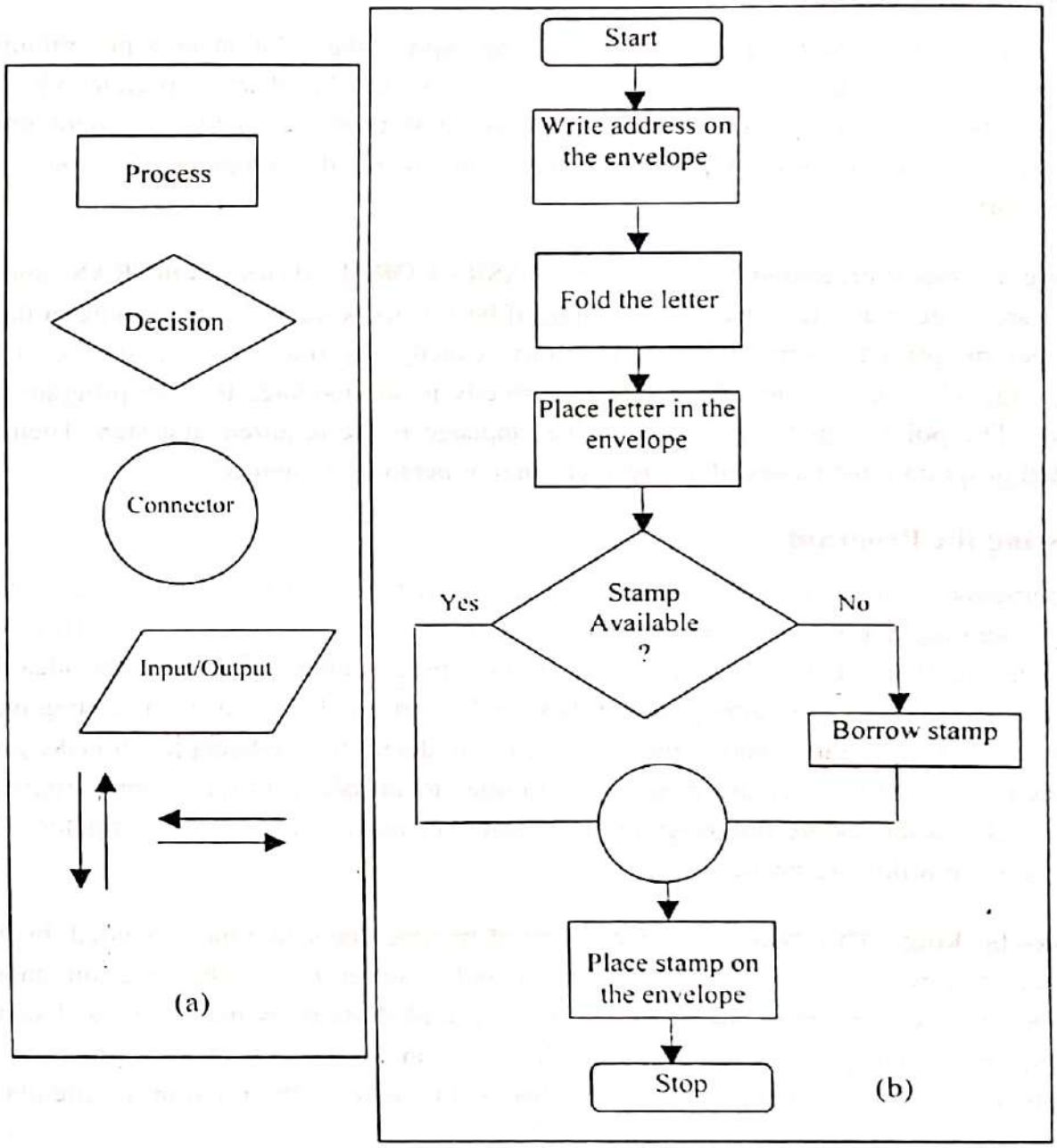


Figure 11.6: Flowchart, (a) symbols, (b) a flowchart for a specific job.

On the other hand, pseudocode is a nonstandard description that gives state of solution with more precision than one can in plain language but with less precision than is required when using a formal programming language.

Pseudocode permits to focus on the program logic without having to be concerned about the precise syntax of a particular programming language. However, pseudocode is not executable on the computer.

Coding the Program

Next step is to code the program - that is, to express the solution in a programming language. Programmer will translate the logic from the flowchart or pseudocode - or some other tool - to a programming language. It is to be noted that a programming language is a set of rules that provides a way of instructing the computer what operations to perform.

There are many programming languages, BASIC, COBOL, Pascal, FORTRAN, and C, C++ are some examples. One can find himself/herself working with one or more of these. To get program to work, one has to follow exactly the rules and the syntax of the language. Of course, using the language correctly is no guarantee that the program will work. The point is that correct use of the language is the required first step. Then the coded program must be keyed, using a terminal or personal computer.

Testing the Program

Experts insist that a well-designed program can be written correctly the first time. In fact, they assert that there are mathematical ways to prove that a program is correct. However, the imperfections of the world still exist, so most programmers get used to the idea that their newly written programs probably have a few errors. This is a bit discouraging at first, since programmers tend to be precise, careful, detail-oriented people who take pride in their work. Still, there are many opportunities to introduce mistakes into programs. Eventually, after coding the program, a programmer must test it on the computer. This involves the following phases.

Desk-checking: This phase, similar to proof-reading, is sometimes avoided by the programmer who is looking for a short-cut and is eager to run the program on the computer once it is written. However, with careful desk-checking one may discover several errors and possibly save time in the long run. In desk-checking a programmer simply sits down and mentally traces, or checks, the logic of the program to attempt to ensure that it is error-free and workable.

Many organizations take this phase a step further with a walkthrough, a process in which a group of programmers involve to review and offer suggestions.

Translating: A translator is a program that,

- Checks the syntax of the program
- Translates the program into a form the computer can understand.

A by-product of the process is that the translator tells improper use of the programming language in some way. These types of mistakes are called syntax errors. The translator produces descriptive error messages. For instance, if in someone mistakenly write N=(2*(1+J) – which has two closing parentheses instead of one – he/she will probably get a message that says, "UNMATCHED PARENTHESES". Programs are most commonly translated by a compiler. A compiler translates entire program at one time. As shown in Figure 11.7, the translation involves original program, called a source module, which is transformed by a compiler into an object module. Prewritten programs from a system library may be added during the link/load phase, which results in a load module. The load module can then be executed by the computer.

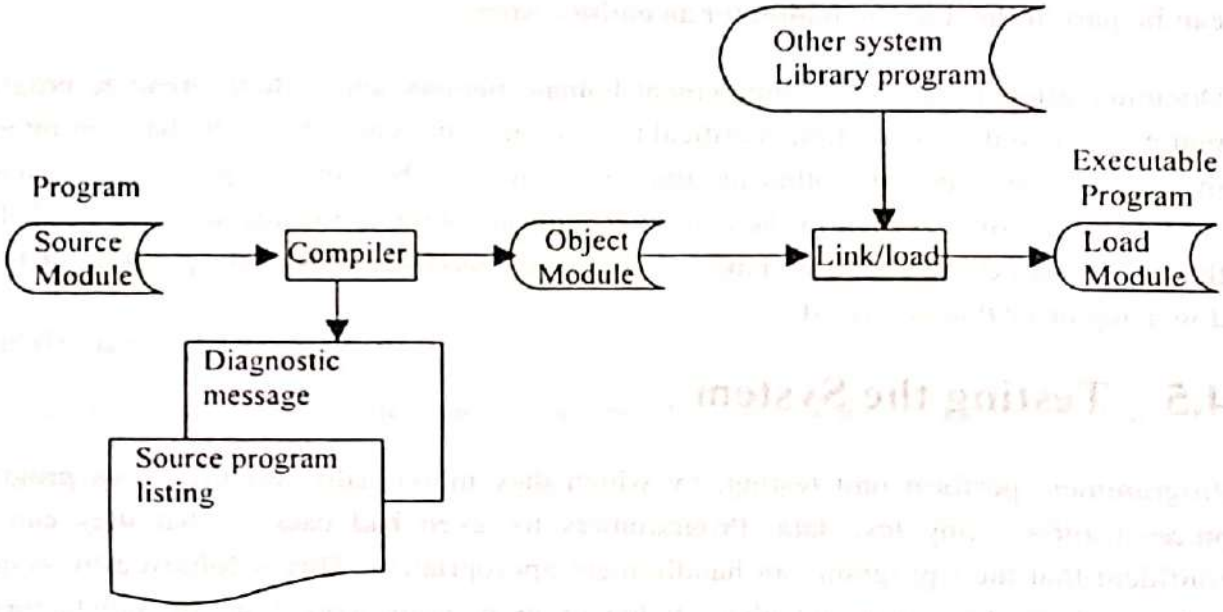


Figure 11.7: Preparing a program for execution.

Figure 11.7 shows that an original program, the source module, is translated by the compiler into an object module, which represents the program in a form the machine can understand. The compiler may produce diagnostic messages, indicating syntax-errors. A listing of the source program may also be output from the compiler. After the program successfully compiles, the object module is linked in the Link/load phase with system library programs as needed, and the result is a load module, or executable program

Debugging: Debugging means detecting, locating, and correcting bugs (mistakes), usually by running the program. These bugs are logic errors, such as telling a computer to

repeat an operation but not telling it how to step repeating. In this phase programmer runs the program using test data. The programmer must plan the test data carefully to make sure he/she tests every part of the program.

Program Documentation

Documenting is a necessary process, although, many programmers do not find interest to do that. Documentation is a written detailed description of the programming cycle and specific facts about the program. Typical program documentation materials include the origin and of the problem, a brief narrative description of the program, logic tools such as flowcharts and pseudocode, data record descriptions, program listings, and testing results.

Comments in the program itself are also considered an essential part of documentation. Many programmers document as they code. In a broader sense, program documentation can be part of the documentation for an entire system.

Documentation is needed to supplement human memory and to help organize program planning. Also documentation is critical to communicate with others who have an interest in the program, especially other programmers who may be part of a programming team. And since turnover is high in the computer industry, written documentation is needed so that those who come after can make any necessary modifications in the program or track down any error that is missed.

4.5 Testing the System

Programmers perform unit testing, by which they individually test their own program pieces (units), using test data. Programmers try even bad data so that they can be confident that their program can handle them appropriately. This is followed by system testing, which determines whether all the program units work together satisfactorily. During this process the development team uses test data to test every part of the programs. Finally, volume testing uses real data in large amounts. Volume testing sometimes reveals errors that do not show up with test data, especially errors in storage or memory usage. In particular, volume testing of online systems will reveal problems that are likely to occur only under heavy usage. Similar to every phase of the project, documentation is an ongoing activity. Documentation describes the program logic and detailed data formats.

In addition, the results of this phase are to be documented. The resulting report, usually referred to as the detail design specifications, is an outgrowth of the preliminary design document. The report is probably large and detailed. A presentation often accompanies

the completion of this stage.

4.6 Exercises

Multiple Choice Questions

1. Gantt chart is used to
    a. scheduling the design activities
    b. scheduling the system development activities
    c. scheduling program development activities
    d. none of these.

Questions for Short Answers

2. What are the components in system development process?
3. What is meant by Gantt chart?
4. Write down the steps involved in program development life cycle.
5. What is a flow-chart?
6. What do you understand by pseudocode?
7. Differentiate between pseudocode and source code.
8. Differentiate between program testing and the system testing.
9. What is program documentation?

Analytical Questions

10. What is meant by Gantt chart? Describe the advantages of Gantt chart using a diagram.
11. Briefly describe the steps involved in program development life cycle.
12. What do you understand by testing of a system? Differentiate between program testing and the system testing.

Lesson 5: System Implementation

5.1 Objectives

On completion of this lesson you will know:

- Components to be considered in implementation phase
- Problems encountered throughout the process.

5.2 Introduction

Implementation is the final phase of software development. However, a good deal of effort is still required to achieve the success in the software project development. The activities at this phase are:

- Training
- Equipment conversion
- File conversion
- System conversion
- Auditing
- Evaluation
- Maintenance.

5.3 Implementation Phases

Training

Users of a system require well documented training. An important training tool is the user's manual, a document prepared to aid users not familiar with the computer system. The user's manual can be an outgrowth of the other documentation. Some organizations employ technical writers to create the user's manual while the system is being developed. But documentation for the user is just the beginning. Any teacher knows that students learn best by doing. Besides, users are as likely to read a thick manual as they are to read a dictionary. The message is clear: users must receive hands-on-training to learn to use the system. The trainer must prepare exercises that simulate the tasks users will be required to do. The trainer must consider class space, equipment, data, and the users' schedules.

Equipment Conversion

Equipment considerations vary from almost none to installing a mainframe computer and

all its peripheral equipment. If some one start implementing a small – or medium – sized system on established equipment in a major information systems department, then perhaps his/her equipment considerations will involve no more than negotiating scheduled run time and disk space. If one is to purchase moderate amount of equipment, such as terminals or personal computers, then he will be concerned primarily with delivery schedules and compatibility.

The planning for such a purchase, of course, must begin long before the implementation phase. A major equipment purchase will need site preparation advice from vendors and other equipment experts. The analyst will need to know the exact dimensions and weight of the new equipment to fit it through doors and locate it in various parts of a building. The systems should be protected from damage by dirt, water, and fire. The analyst may need to consider electrical capacity and wiring hookups as well as new flooring - probably flooring that is raised to hide cabling and ease access for repairs to large computers and related equipment. Finally, most medium to large machines need air conditioning and humidity control, and the analyst may be the one who does the actual installation.

File Conversion

It would be very tricky if the existing files are handled manually. The data must be prepared in such a way that they are accessible to computer systems. All the contents of the file drawers in the personnel department, for instance, must now be keyed, or possibly scanned, to be stored on disk. Some scheme must be used to input data files and keep them updated. The analyst may need to employ temporary help. The big headache during this process is keeping all file records up-to-date when some are still manual and some have been keyed in preparation for the new system.

If the analyst is modifying an existing computer system and he/she may have files already in computer-accessible form, may need a program to convert the old files to the format needed for the new system. This is a much speedier process than having to key in data from scratch. Nevertheless, it is not unusual for file conversion to take a long time.

System Conversion

This stage is considered to convert an old system and begin using the new one. There are four ways of handling the conversion.

Direct conversion: Direct Conversion means the user simply stops using the old system and starts using the new one – a somewhat risky method, since there is no other system to

fall back on if anything goes wrong. This procedure is the best followed only if the old system is in unusable condition.

Phased conversion: Phased conversion is one in which the organization eases into the new system one step at a time so that all the users are using some of the system.
Pilot conversion: In contrast, in a pilot conversion the entire system is used by some of the users and is extended to all users once it has proved successful. This is the most useful when a company has several branch offices or separate divisions.

Parallel conversion: In Parallel conversion - the most prolonged and expensive method – the old and new systems are operated simultaneously for some time, until users are satisfied that the new system performs to their standards.

System conversion is often a time of stress and confusion for all concerned. During this time users are often doing double duty, trying to perform their regular jobs and simultaneously cope with a new computer system. Problems seem to appear in all areas, from input to output. Clearly, this is a period when patience is needed.

Auditing

Security violations, whether deliberate or unintentional, can be difficult to detect. Data begins from some source, perhaps a written source document or a transaction, for which there must be a record log. Eventually, the data – no matter how they were originally keyed – form a part of the system on some media, probably disk. Once the data are on disk, it is possible for an unauthorized person to alter them in some illicit way. To guard against this situation, the system analyst designs an audit trail to trace output back to the source data. In real-time systems security violations can be particularly elusive unless all transactions are recorded on disk for later references by auditors. Modern auditors no longer shuffle mountains of paper; instead, they have computer programs of their own to monitor application programs and data.

Evaluation

For sometimes the systems analyst and someone from the client organization evaluate the system against preset criteria directly related to the requirements that were determined during the systems analysis phase. Some organizations prefer to bring in an independent evaluating team on the assumption that independent members will be free from bias and expectations. However, evaluation explores how well the system is meeting the original requirements, budgets, schedules, and so forth. Out of such evaluation will come adjustments that will improve the system.

Maintenance

In many cases, maintenance is considered as a separate phase, one that begins only when the initial system effort is implemented and complete. However, maintenance is an ongoing activity, one that lasts the lifetime of the system. Monitoring and necessary adjustments continue so that the computer produces the expected results. Maintenance tasks also include making revisions and additions to the computer system.

Those who are interested to become programmers may envision themselves writing programs for new systems, but they are more likely, at least at first, to be maintaining programs in existing systems. This type of work has advantages and disadvantages. It can be an advantage to learn from programs written by others but, unless strict programming procedures have been in place since the program's inception, a distinct disadvantage to unravel layered changes added by many programmers. Some maintenance programmers find satisfaction in having a deep understanding of a complicated system, so that everyone knows they can fix any problem. This is contrasted with participating in the development of a new system, which has unknown perils at every turn and is scrutinized regularly for schedule and budget overruns.

5.4 Problems Encountered in SDLC

One may have the impression that, by simply following a recipe, a system can be developed. In fact, inexperienced analysts sometimes have the impression that there is a formula for developing systems. Each system is unique, however, no one formula can fit every project. It would be correct to say that there are guidelines.

Historically, even analysts who followed the guidelines were not always successful in developing systems. System analysts have been embarrassed to find that they were not always good at estimating time, so schedules constantly slipped (budget overruns are one of the obvious results of schedules). Some observers, in fact, think that system analysis is so ambiguous that analysts do not even know when they are finished. Sometimes it seems that the definition of project completion is the point at which analysts have run out of time on the schedule.

Another frequent problem has been imperfect communication between analysts and users. Poor communication results in poorly defined specifications, which, results in a supposedly complete system that does not do what the user expects.

5.5 Exercise

Questions for Short Answers

1. List the activities involved in the software implementation phase
2. What is meant by file conversion?
3. What are the ways of system conversion?
4. What is meant by auditing?
5. What are the components involved in system evaluation?
6. Describe briefly the maintenance of a system.

Analytical Questions

7. Discuss the equipment conversion, file conversion and system conversion methodologies.
8. Briefly describe the implementation steps in software development cycle.

Chapter 12

Computer Networks and the Internet

Lesson 1: Introduction to Computer Networks
Lesson 2: Local Area Networks
Lesson 3: Wide Area Networks
Lesson 4: Introduction to the Internet
Lesson 5: Internet Services and Protocols
Lesson 6: Intranet and Extranet

Lesson 1: Introduction to Computer Networks

1.1 Objectives

On completion of this lesson you will know:

- Types of computer networks and their characteristics
- Use of computer networks
- Some network terminologies
- Function of network software.

1.2 Introduction

A computer network is a system of interconnected computers. The computers of a network communicate with one another and share applications, data, voice and video, and hardware components. There are three main types of computer networks, they are:

- Local Area Network (LAN)
- Wide Area Network (WAN) and
- Metropolitan Area Network (MAN).

A LAN provides high speed communication between computers situated in a limited

area, typically within a building or a small area like a college/university campus. LANs are usually owned and managed by a single organization.

A WAN spreads over a large geographical area and may connect computers within a country or cover many countries. WANs may be provided by telephone and telegraph organizations, they may be private to a single organization or they may be operated by a group of organizations. Traditionally, WANs have been implemented using circuit switching and packet switching technologies. A WAN can connect a number of LANs. The Internet is a popular example of a WAN.

MANs cover an area of a typical city. They offer a simple and fast way to link different sites of an organization for exchange of information. MANs use technologies similar to LAN technologies.

Use of Computer Networks

Computer networks provide many benefits. Major uses of computer networks are:

- Simultaneous access to programs and data.
- Sharing peripheral devices like printers, scanners etc.
- Personal communications using e-mail
- Making backup of information
- Aiding communication by teleconferencing and video-conferencing
- Protecting information by account name and password.

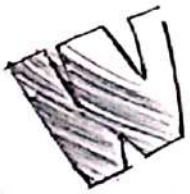
1.2 Some Network Terminologies

Network Topology: Topology of a network describes the way the computers and the nodes of the network are interconnected. There are a number of possible topologies:

- Bus topology
- Star topology
- Tree topology
- Ring topology
- Mesh topology.

Links: The physical connections that connects the nodes are known as links. A link may be through a pair of wires, a coaxial cable, an optical fibre or through a satellite. The data transmission rates of the links are expressed by kbps (Kilo bits per second), Mbps (Mega bits per second), Gbps (Giga bits per second) or Tbps (Tera bits per second). Examples are 64 kbps, 2 Mbps and 2.4 Gbps etc.

Media: The most commonly used media for computer networks are twisted pair, co-axial cable and optical fibre. Wireless techniques include broad-cast radio waves, microwaves



and infrared.

Network Protocols: Computers of a network must obey some rules when communicating with each other. This set of rules forms the protocol for a computer network. The structural set of modules that implements the communication functions is called protocol architecture. An example of common protocol architecture is the TCP/IP protocol suite.

1.4 Network Software

Software controls the operations of computer networks. The software that manages the resources of the network is often called the network operating system. Servers in LANs rely on network operating systems, such as Novell Netware, IBM OS/2 Warp Server, or Microsoft Windows NT Server. A server network operating system must meet stringent set of requirements than a client operating system such as Windows 98 or Mac Operating System. A variety of communication software packages are available for microcomputers, especially for Internet Web browsing, like Microsoft Explorer, Nestcape Navigator, Microsoft Outlook etc. Several functions are commonly provided by communication software packages, examples are: access control, transmission control, network management, error control, security management etc.

Access control is responsible for establishing the connection between terminals and computers in a network. Access control activities include connecting links through modems, establishing communication parameters such as speed, mode and direction, automatic telephone dialing and redialing, logging on and off with appropriate account numbers.

Transmission control function allows computers and terminals to send and receive commands, messages, data and programs. Data and programs are usually transmitted in the form of files and thus this activity is also called file transfer.

Error control functions involve detection and correction of errors. Communication software and processors detect errors in transmission by different methods, including parity checking and cyclic redundancy checks (CRC). Most error correction methods involve retransmission, in which a signal is sent back to the computer to retransmit the previous message which has been received with error.

Network management function manages communication in a computer network. It also monitors network activity and the use of network resources by end users. Security management is required to protect a computer network from unauthorized access.

Network operating systems, or other security programs, restrict access to the data files and other computing resources in networks. Data transmissions can also be protected by encryption and authentication techniques.

1.5 Exercise

Multiple Choice Questions

1. A network that connects computers in a building is

   a. WAN
   b. MAN
   c. LAN
   d. Campus Area Network

2. Which is not a network operating system?

   a. Novell Netware
   b. Windows NT Server
   c. IBM OS/2 Warp Server
   d. MS Word

3. Which is not an Internet Web browser?

   a. Microsoft Explorer
   b. MS Excell
   c. Netscape Navigator
   d. MS Outlook

Questions for short answers

4. List the possible topologies for computer networks.
5. List some common uses of computer networks.
6. Name any three server operating systems and any two client operating systems.

Analytical questions

7. Name different types of computer networks and briefly explain their characteristics.
8. Explain the terms: network topology, links and network protocols.
9. Explain the major functions of network software.

Lesson 2: Local Area Network

2.1 Objectives

On completion of this lesson you will know:

- General characteristics of LANs
- LAN topologies and media
- Network interface cards
- LAN standards
- Clients, servers and peers.

2.2 Introduction

Local area networks provide interconnection of a variety of data communicating devices within a small area. The key ingredients that determine the characteristics of a LAN are:

- Topology
- Transmission media
- Interface card and protocol.

Major features of LANs are given below:

- Multiple computers are connected together
- Computers are spread over a small geographical area
- Channels are of relatively high capacity (Mbps, Gbps)
- Channels are relatively error free (generally 1 error in 1,000,000,000 bits)

2.3 LAN Topology

Three basic topologies for LANs are: bus topology, star topology and ring topology (Figure 12.1). In bus network all computers are connected by a single cable with a terminator at each end. The bus is the simplest and the most widely used topology for LANs. In star topology all computers are connected to a central switch, called hub. A hub is the central controlling device. Two main types of hubs are: active hub and passive hub. In a ring topology computers are connected in a closed loop. Data travels around the ring through the nodes which act as repeaters. A mesh network uses direct links to connect some or all the computers of the LAN. A variation of bus topology is the tree topology which may have several branches connecting computers.

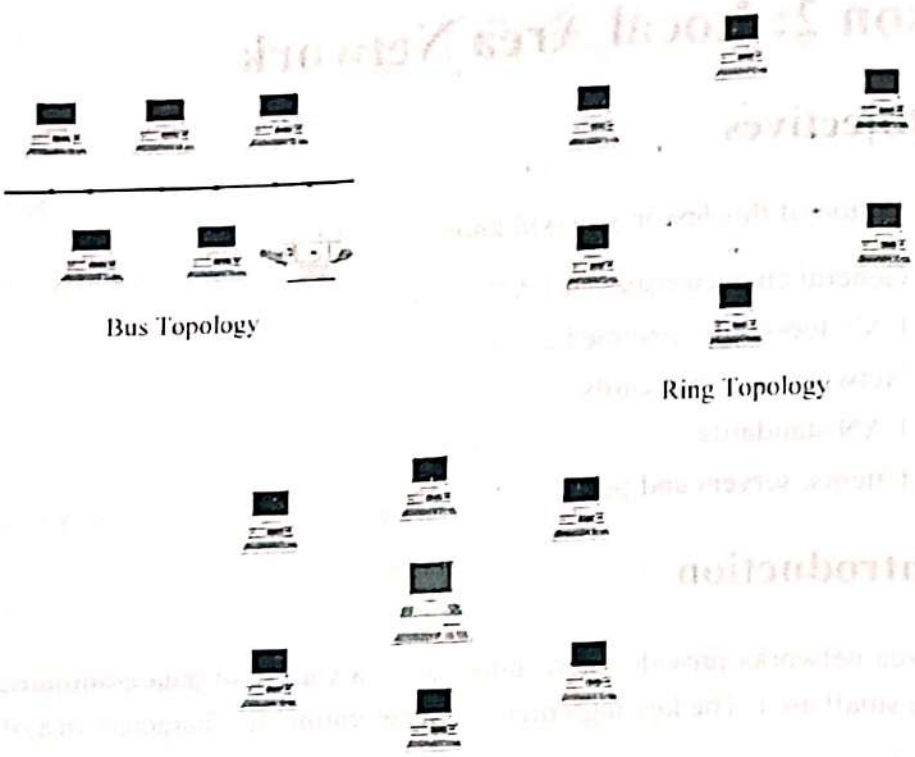


Figure. 12.1: Bus, star and ring topologies.

2.4 Transmission Media

Three main transmission media for LANs are: metallic cable (usually copper), optical fiber and wireless. The metallic cable includes twisted pair and coaxial cable. Optical fibers include single-mode(or mono-mode) and multi-mode fibers. Wireless media include microwave and infrared. Names of media with examples of maximum transmission speeds are given below.

Media	Maximum Speed
Twisted pair (UTP and STP)	2 Mbps-100 Mbps
Coaxial Cable (baseband/broadband)	264 Mbps-550Mbps
Microwave (satellite/terrestrial)	100Mbps
Wireless Radio (for LAN)	3.5 Mbps
Infrared (for LAN)	4 Mbps
Fiber optic cable	3 Gbps

Twisted pair: Two types of twisted pair cables are UTP (unshielded twisted pair) and STP (shielded twisted pair). Twisted pairs having different characteristics have evolved over the years; the most modern type is the category 5 UTP which is used for high speed

LANs. Twisted pair wiring is extensively used in home and office telephone systems and wide and local area networks.

Coaxial cable: It consists of one copper conductor within and insulated from another conductor of longer diameter (usually a copper braid). They are available in many styles. The most commonly used types in computer networking are thick-net and thin-net. Coaxial cables allow much higher speed data transmission compared to twisted pair cables.

Fiber Optic Cable: This cable consists of one or more hair-thin filaments of glass (or plastic) fiber wrapped in a protective jacket. Two types of mostly used fiber optic cables are: multi-mode and single-mode cables. The multi-mode has a core diameter of 62.5 microns (1 micron =1/1,000,000 meter) and a cladding (glass surrounding the core) diameter of 125 microns. The single-mode has a core diameter of 8.3 micron and a cladding diameter of 125 microns. Currently available fiber optic cables can carry data at transmission rates of several gigabits or more.

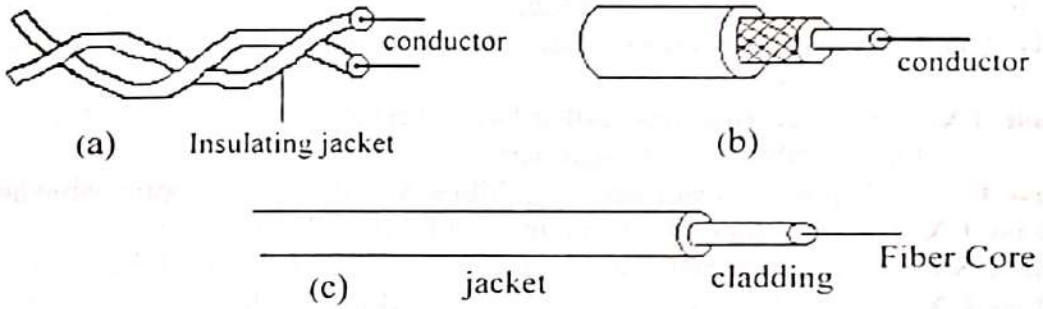


Figure 12. 2: (a) Twisted pair, (b) Coaxial cable and (c) Fiber optic cable.

Wireless: Wireless can be divided into wireless radio and wireless infrared. Wireless radio is extremely useful in situations where running cable is prohibitively expensive or impossible. Infrared solutions require a clear line of sight and are greatly affected by weather such as rain or fog. They are used for indoor LANs.

2.5 Network Interface Cards and Protocols

A network interface card (NIC), or a LAN adapter, plugs into a slot on the motherboard to connect it to the LAN. The NIC serves a number of purposes:

- It makes the physical connection between the computer and the network.
- It converts parallel data of the computer's bus into serial data for the LAN.
- It boosts the strength of a signal so that it can flow through the transmission link.

Only one computer is allowed to transmit in a LAN. Signals would interfere if there are

more than one transmissions at a time. For this reason, a LAN needs a medium access control (MAC) scheme to prevent overlapping of signals. The NIC determines the access method used by a network. The most common medium access control methods are: Ethernet, Token Ring, FDDI, Arcnet etc.

2.6 LAN Standards

Several standards for LANs are commercially available. Most of the standards were developed by a committee known as IEEE 802, sponsored by the Institute of Electronic and Electrical Engineers (IEEE) of USA. All the standards have subsequently been adopted as international standards by ISO (the International Organization for Standardization). Some of the commercially available LANs are listed below:

10 Base5: Bus Topology, Data rate - 10 Mbps, Medium - coaxial cable
10 Base2: Bus Topology, Data rate - 10 Mbps, Medium - coaxial cable
10 Base-T: Star Topology, Data rate - 10 Mbps, Medium - UTP
10 Base FP: Star Topology, Data rate - 10 Mbps, Medium - Fiber optic cable
FDDI: Ring Topology, Data rate - 100 Mbps, Medium - Fiber optic cable
100 VG-Any LAN: Star Topology, Data rate - 100 Mbps, Medium - Twisted pair and Fiber optic cable.
100 Base TX: Star wired Bus (also called Fast Ethernet), Data rate-100 Mbps, medium - Optical fiber and Twisted pair
100 Base F: Star Topology, Data rate - 100 Mbps, Medium - Fiber optic cable and UTP
1000 Base LX: Gigabit Ethernet, Data rate – 1000 Mbps, Medium – Fiber optic cable
1000 Base SX: Gigabit Ethernet, Data rate – 1000 Mbps, Medium – Fiber optic cable
1000 Base CX: Gigabit Ethernet, Data rate – 1000 Mbps, Medium – Twisted pair cable
1000 Base TX: Gigabit Ethernet, Data rate – 1000 Mbps, Medium – Twisted pair cable
ATM LAN: Star Topology, Data rate - Scalable upto several hundred Mbps, Medium - Fiber optic cable

2.7 Client, Server and Peer

Computers of a LAN can be of three types: clients, peers and servers. A Client computer uses, but does not provide, network resources. On the other hand, servers provide network resources. Servers run network operating systems, such as, Windows NT Server, or Novell NetWare. Clients run client operating systems, such as, MS-DOS, or OS/2. Peers run peer network operating systems, such as, Windows 98, or Mac Operating System. On the basis of the roles played by the computers attached to them, networks can be divided into following three types.

Server based LAN: This network, also called client-server network, contains server and client computers.

Peer network: This network, also called peer-to-peer network, does not contain a server and the independent computers share their resources through the network.

Hybrid Network: It is a client-server network that also has peers sharing resources through the network.

2.8 Exercise

Multiple choice questions

1. Which one is the most readily available and widely supported medium?

   a. Optical fiber b. Infrared
   c. Microwave d. UTP

2. Which one represent the fastest LAN?

   a. 100 Base SX b. 10 Base T
   c. 1000 Base TX d. 100 Base F

3. Which can concurrently provide and request services?

   a. Server b. Client
   c. Peer d. None

Questions for short answers

4. Give the major features of LANs.
5. Name the common transmission media of LANs.
6. Explain the terms: 10 Base 5, 10 Base T, FDDI, 100 Base F.
7. Distinguish between a client and server in connection with LANs.
8. Name the common transmission media and mention their maximum data rates.

Analytical Questions

9. With the help of illustrations briefly explain the following network topologies: a) star topology, b) bus topology and c) ring topology.
10. Name the common transmission media and briefly explain their characteristics.
11. What is a network interface card explain its characteristics.
12. What are client, server and peer? Briefly explain a) server based network, b) peer network and c) hybrid network.

Lesson 3: Wide Area Network

3.1 Objectives

On completion of this lesson you will know:

- General characteristics of a wide area network (WAN)
- Bandwidth and switching characteristics of WANs
- Modems and communications through telephone lines.

3.2 General characteristics

A wide area network connects computers located across a wide geographical area. For example an organization may have its offices scattered over different cities and each office needs to share data with other offices. In this case computers and LANs of different sites may be connected by public communication facility (like telephone network) or private communication network to realize a WAN. A WAN for a particular organisation is called enterprise-wide network. The Internet is a popular example of WAN as it connects thousands of computers and LANs around the world.

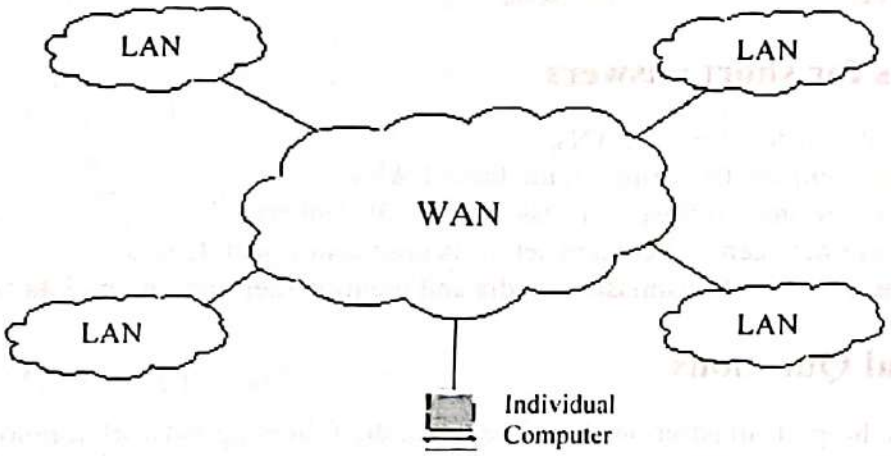


Figure 12.3: A wide area network.

Intermediate devices like gateways and routers and transmission facilities like telephone and data networks, fiber optic links, satellites are generally used for realizing WANs. The major features of WANs are listed below.

- Multiple computers are connected together
- Computers are spread over a wide geographic area
- A WAN usually interconnects multiple LANs

- Communication links between computers are provided by telephone networks, public data networks, satellites etc.
- Links are of low capacity (that is low data rate)
- Bit error rate is higher (1 in 100,000) compared to that for a LAN.

3.3 Bandwidth

Speed and capacity of computers and communication networks can be classified by bandwidth. The speed and capacity of data networks are generally measured in bits per seconds (bps). Low speed local communication channels are typically used for transmission rates from 300 to 56000 bps, but can now handle up to several Mbps for ADSL connections. They are usually UTP lines commonly used for voice communications, but are also used for data communications by PCs, video terminals and fax machines. STP lines can be used for speed up to 100 Mbps for short distances. High speed broadband digital channels allow transmission rates from 256,000 bps to several billion bps; typically they are microwave, satellite transmission and fiber optic cables.

3.4 Switching

Switching techniques used for WAN are circuit switching, message switching and packet switching. Normal telephone service relies on circuit switching. In circuit switching a switch establishes and maintains a link between a sender and receiver until the communication session is completed. In message switching, a complete message is transmitted in a block from one switching node to another.

In packet switching a message is subdivided into groups called packets. In X.25 packet switching, packets are 128 characters long, while they are of variable length in frame relay packet switching. Frame Relay is considerably faster than X.25 and is used by many large companies for their wide area networks. ATM (Asynchronies transfer mode) is a fast emerging high speed cell switching network technology. An ATM system, using BISDN (Broadband Integreted Services Digital Network), organises voice, video, and other data into fixed cells of 53 bytes and send them to their next destination. ATM networks are becoming popular for the fastest high-capacity multimedia services for voice, video and other high speed applications. Currently available network technology with data rates for WAN are listed below.

Network Technologies	Typical Data Rate
DDN : Digital Data Network	2.4 kbps-2Mbps
PSN : Packet Switching Network	2.4 kbps-64kbps
Frame Relay Network	56/64 kbps-2Mbps
ISDN : Integrated Services Digital Network	64/128 kbps-2Mbps
ATM : Asychronous Transfer Mode	25/155 Mbps-2.4 Gbps

3.5 Communication over Telephone Lines

Although digital telephone lines are gaining popularity, most of the telephones are analog in nature. Attaching a computer to an analog phone line requires a modem. A telephone converts the sound into an electric signal that flows through the phone line. The telephone at the receiving end converts the electric signal back to sound. Both the sound and telephone signals are analog in nature. A computer works with 0's and 1's and a modem is required to translate the digital signals into analog signals so that they can travel through telephone lines. "Modem" is the short for modulator and demodulator. In the modulation phase, the modem turns the computer's digital signals into analog signals and in the demodulation phase, it does the reverse conversion. Figure 12.4 shows how two computers communicate through telephone lines using the modems.

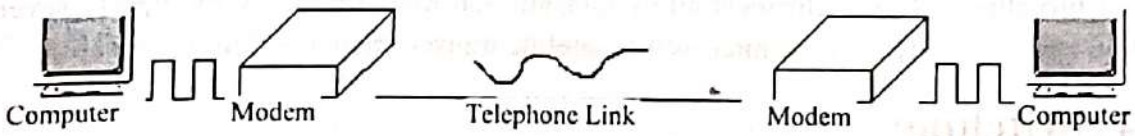


Figure 12.4: Modems connecting computers through telephone lines.

A modem can be a simple circuit board that plugs into one of the expansion slots of a PC or it can be an external device that plugs into a serial port of a PC. Choosing a modem is not so simple. The International Telecommunication Union (ITU) has developed a number of standards for modems as listed below.

Standard	Speed
V.22 bis*	2,400 bps
V.32	9,600 bps
V.32 bis	14,400 bps
V.32 terbo*	19,200 bps
V.34	28,800 bps
V.42	57,600 bps
V.42 bis	57,600 bps

*bis means second and turbo beans third in French.

While buying a modem the main factors to be considered are:

- Transmission speed
- Data compression
- Error correction
- Internal versus external

A company called Microcom developed popular standards for early modems, called

Microcom Network Protocol, or MNP in short. Many manufactures adopted the MNP standard and its successors. The common ones are MNP2, MNP3, MNP4, MNP5 etc. MNP5 and V.42 bis modems incorporate both data compression error correction protocols. Most modems for personal computers can also emulate a fax machine and can exchange faxes with other fax modems or fax machines.

Information superhighway

Open, high-speed, digital networks with fiber optic and satellite links and the widespread use of the Internet have made the concept of an information superhighway feasible and captured the interest of both business and government. In this concept, local, regional, nationwide, and global networks could be integrated into a vast network of networks, with interactive multimedia capabilities. The information superhighway system would connect individuals, households, business, news and entertainment providers, government agencies, libraries, universities, and many other institutions, and would support interactive voice, data, video, and multimedia communications.

The information superhighway could provide a national information infrastructure and economic network with an economic impact equivalent to the international railway and highway systems combined. The information superhighway is needed to create an information infrastructure that would dramatically increase business efficiency and competitiveness by improving communications, electronic commerce and collaboration, and information gathering. In any event, the information superhighway promises to be a catalyst for the development and improvement of the Internet and changes in electronic communications and services in the future.

3.6. Exercise

Multiple choice questions

1. Which of the technologies provide the highest speed?

   a. Frame Relay b. ISDN
   c. DDN d. ATM

2. Which of the following improves the reliability of a modem?

   a. Demodulation b. Encryption
   c. Compression d. Error detection

Questions for short answers

3. Describe the general characteristics of WANs.
4. Explain speed and capacity of data networks.
5. List the currently available network technologies for WAN and mention their data speeds.

Analytical questions

6. Briefly Discuss the switching technologies for WANs.
7. What is a modem? Explain how two distant computers can communicate through telephone line using modems.

Lesson 4: Introduction to the Internet

4.1 Objectives

On Completion of this lesson you will know:

- Evolution of the Internet
- Common terminologies associated with the Internet
- Working principle of the Internet

3.2 Evolution of the Internet

The Internet spanning over the globe is the largest and most important network of the world. It is a shared network of government and other agencies and organisations from all over the world. It is evolving into the information superhighway of the future. Thousands of business, educational and research networks now connect millions of computers and users to each other. It has become the key platform for rapidly expanding information and entertainment services, and business applications which includes electronic commerce and enterprise collaboration.

The Internet evolved from ARPANET– a research network established by United States Defense Department in 1969. The ARPANET was created with a view to enable academic and government researchers to communicate by e-mail and to share data and computing resources. Each message sent through the Internet has a unique address and any Internet server of the network can forward it to its destination.

The Internet is not owned by a single authority and it has no headquarter or governing body. The Internet Society in Reston, USA is one of several volunteer groups of individuals and corporate members who promote use of the Internet and development of its new communication standards.

The Internet is expanding very fast. At present (in 2001) the Internet is doubling in size each year, growing to over 150 million host computers and over 800 million users. The monthly growth rate of the Internet (in 1999) was about 10% and some network experts expect that the Internet will interconnect millions of networks in the future.

4.3 Some Terminologies

Internetwork: Many LANs are connected to each other through wide area networks. These interconnected networks are generally called internetworks.The Internet is the

largest internetwork. Such networks enable end users to communicate with the workstations on other LANs, or to access the computing resources and databases at other locations or organizations. These interconnected networks rely on internetwork processors, such as switches, routers, hubs, or gateways, to make internetworking connections (Figure 12.5).

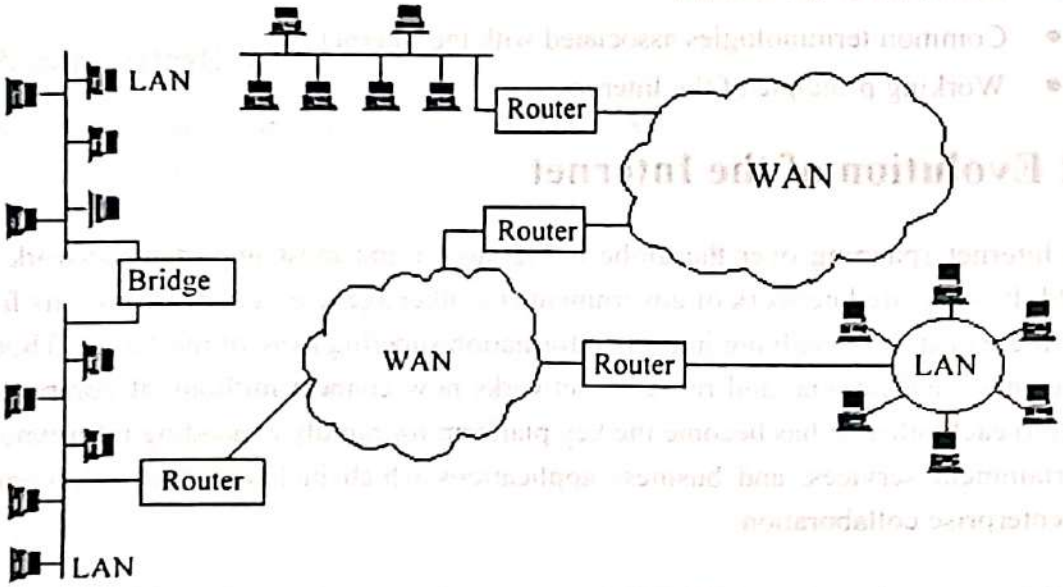


Figure 12.5: An internetwork.

Repeaters: Repeaters also known as relays connect segments of a network, and since signals get weaker and less distinct with distance, these small devices refresh and enhance them before sending them along. Their job is to extend the network as far as possible, perhaps to another building.

Bridges: Bridges are also known as data link relays, which connect segments of local area networks.

Routers: Routers control movement of messages through networks. The router keeps the addresses of other nodes in a table. If the address of a message sent though the router is on the network, the message is forwarded. If the address is not on the network, the router forwards the message to another network by sending it to a router.

Gateways: Gateways are used to connect different types of networks. Since two networks may not share a common protocol these devices translate each network's protocol so the other network can understand the data.

Hubs: A hub is a switching processor. Hubs provide automatic switching among

connections called ports for shared access to a network's resources. Workstations, servers, printers, and other network resources are connected to ports provided by the hub.

Firewall: The metal between an automobile's passenger area and engine compartment is called a firewall because it is designed to prevent an engine fire from spreading into the passenger compartment. A firewall is to prevent unauthorized users on the Internet from reaching the files and programs on the local area network. When a computer connected to the outside Internet is also connected to an inside local area network, a firewall is installed between the Internet and the local network. If an organization wants to provide interactive services for outsiders, it puts a file server outside the firewall. E-mail, mailing lists, and news services are store-and-forward service, where the outsiders do not have interactive access to computers inside the firewall. One of the reasons firewalls are necessary is to prevent attacks that can be launched against a network connected to the Internet.

4.4 How the Internet Works

Every computer connected to the Internet uses the same set of rules and procedures to control timing and data formats. The set of commands and timing specifications used by the Internet is called TCP/IP (Transmission Control Protocol/Internet Protocol) suite. The TCP/IP suite includes the specifications that identify individual computers and exchange of data between computers. TCP/IP consisting of five layers of protocols.

Application or Process Layer
Host-to-Host Layer (TCP)
Internet Protocol (IP)
Network Interface
Physical Layer

Figure 12.6: Five layers of TCP/IP protocol suite

On the Internet there are two sets of basic rules, the Internet Protocol (IP) and the Transmission Control Protocol (TCP).

Internet Protocol (IP): The first step in getting data to the right place is to get the address right and in a form that can be read by everyone that handles the message along the line. On the Internet the IP governs the way data are addressed and routed along the Internet. To get a message from one user to another, the message is put into an Internet/Protocol "envelope" and addressed to the recipient. Routers then move it through the system to the user with that address.

Transmission Control Protocol (TCP): If a huge graphic file is sent to someone else on the network, it could tie up other's work as they need to wait for the transfer to be finished. To eliminate this problem, another protocol, called TCP, breaks message up into packets, each containing up to 1500 characters. It numbers each of the units, puts each into an IP packet (envelope), and sends them. Each packet may travel a different route and take a different time to arrive at its destination. Some may even get lost or damaged in transmit. At the recipient's end TCP extracts the data from each envelope, checks each for accuracy, and reassembles them into their original order. If it finds that any data are lost or damaged, it requests that those packets be retransmitted.

Most computers are not connected directly to the Internet; instead, they are connected to smaller networks that are connected through routers or gateways to the Internet. That is why the Internet is generally described as "the network of networks" (Figure 12.7)

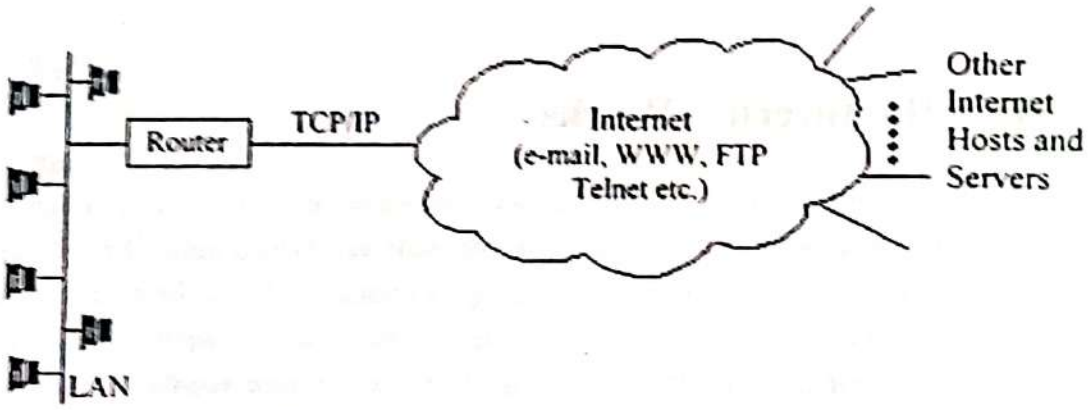


Figure 12.7 A typical Internet connection.

The core of the Internet is the backbone connections that connects the local networks together and the router controls the way for each piece of data to find its destination. In the literature the Internet backbone is portrayed as a big cloud. The network backbone is a large network with many connections to other networks. In USA, Canada and EEC most network-backbones are operated by private telecommunication companies and network service companies. In many countries, including Bangladesh, backbones are also provided by the government telecommunication company.

4.5 Exercise

Multiple choice questions:

1. The Internet evolved from

   a. ARPANET b. BERNET
   c. USENET d. ERNET

2. Which is not an internetworking device?

a. Router
b. Bridge
c. TCP
d. Gateway

Questions for short answers:

1. What is the ARPANET?
2. Distinguish between: repeater, bridge and router.
3. Briefly explain the use of IP and TCP.

Analytical questions:

1. What is the Internet? Briefly describe the development of the Internet.
2. Briefly explain the terms: repeater, hub, bridge, router, gateway and firewall.
3. With the help of an illustration explain working principal of the Internet.
4. Briefly explain backbone network in connection with the Internet.

Lesson 5: Internet Services and Protocols

5.1 Objectives

On completion of this lesson you will know:

- Common services available through the Internet
- Internet address and Domain Name System
- Electronic mail and World Wide Web
- Internet access and ISP.

5.2 Internet Services

Numerous application services have been developed for the Internet using TCP/IP protocol suite. The most popular Internet applications are e-mail, browsing the sites on the World Wide Web and participating special Internet newsgroups. Figure 12.8 illustrates the layering structure of the TCP/TP protocol suite to include a few of the application services included in the protocol suite. The popular TCP/IP protocols are described below.

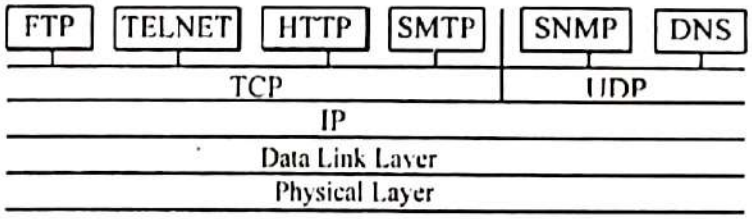


Figure 12.8: Some applications of the TCP/IP suite.

HTTP (HyperText Transport Protocol): The WWW organises Internet related resources for multimedia information, education, entertainment, electronic commerce etc. The HTTP conveys information between a Web browser and a Web server.

SMTP (Simple Mail Transfer Protocol): E-mail is the process of exchanging messages and files among millions of Internet users. SMTP is used for e-mails in the Internet.

FTP (File Transfer Protocol): FTP is an Internet tool for transferring data files, programs, reports, articles, magazines, books, pictures, sounds and other types of files from thousands of sources. The process of transferring a file from a network computer to the user's computer is called downloading and the reverse process is known as uploading.

TELNET: It provides remote terminal access to Internet hosts. By this software tool one

can log on to and use thousands of Internet computers around the world.

POP (Post Office Protocol): It enables PC users to access mail from a mail server.

SNMP (Simple Network Management Protocol): It provides the exchange of information for the management of a network itself.

OTHER APPLICATIONS: Internet Relay Chat for real-time text conversations with Internet users, long distance phone calls at nominal costs, desktop video-conferencing, listening to radio programs, watching television, enjoying films and games, exploring virtual world etc.

5.3 Internet Address

The computer organizing a transaction must identify its intended destination with a unique address. Every computer on the Internet has a four-part numeric address, cal'ed the Internet Protocol address or IP address. The IP address contains routing information that identifies destination location. Each of the four parts is a number between 0 and 255, so an IP address looks like this: 108.64.125.116.

Computers have no trouble working with such numbers, but humans are not so skilled. Therefore, computers connected to the Internet use an address system called a Domain Name System (DNS), where an address uses words instead of numbers.

DNS addresses have two parts: an user name, followed by a domain that identifies the type of institution that uses the address, such as .com for commercial or .edu for schools, colleges, and universities. Microsoft's DNS address is microsoft.com. Within the United States, the last three letters of the domain usually tell the types of institution. The following table lists the most common types.

Domains	Type of organization
.com	(Business) Commercial
.edu	Educational
.gov	US Government
.mil	US Military
.net	Gateway or host
.org	Other organisations

Large institutions and big corporations may divide their domain address into smaller sub-domains. A business with many branches might have a sub-domain for each office. Sub-domains may be broken into even smaller sub-domains, like: du.bangla.net

Outside of the United States, domains usually identify the country in which the system is located, such as .in for India, .jp for Japan, .ca for Canada or .fr for France. A geographic domain address may include a sub-domain. An example could be: www.telemagazine.ac.uk

DNS and IP addresses identify individual computers. A single computer might have many users, each of whom must have an account. Some of the largest domains may have huge number of different user names. When a message is sent to a person that person's user name in the address identifies the recipient. The standard format is: the user name first, "at" symbol @, and finally the DNS address. Therefore, Abdur Rahim's e-mail address might be: arahim@titas.com.

5.4 Electronic-mail

The most common use of the Internet is for the e-mail. Anyone with an e-mail account can send messages to other users of the Internet. Most e-mail programs also permit users to attach data files and program files to messages. E-mail is not a live connection between the sender of a message and its recipient. There is always a delay between the sending time of a message and its arrival time at the distention. Sometimes that delay is just a few seconds, but it is always there.

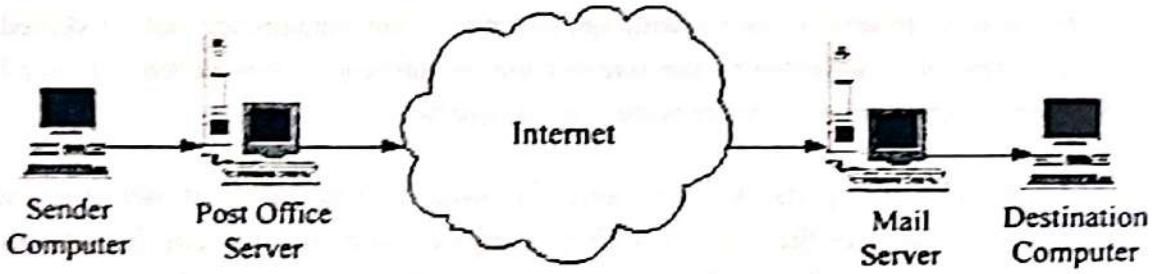


Figure 12.9: Path of an e-mail message.

When a message is sent the computer passes it to a post office server, which identifies the destination address and passes it through the Internet to a mail server, where the message is stored in a mailbox until the recipient looks for it. Figure 12.9 shows the path of an e-mail message from origin to destination. The sender computer creates the message and sends it to a local post office server. The post office server passes the message through the Internet to the recipient's mail server which passes it to the recipient's computer. E-mail programs generally combine the following features.

- A text editor for writing and editing messages.
- Arrangements for storing names and e-mail addresses.
- Ability to send messages to a post office server.
- Ability to get messages from a mail server.
- A manager that organizes and stores the sent and received messages.

There is an e-mail program included in Windows 98 and many others are available separately. One of the most popular e-mail programs is Eudora, which exists in versions for both Windows and Macintosh operating systems. Besides one-to-one, Internet e-mail is also used for one-to-many, in which the same message goes to a list of names.

5.5 The World Wide Web

The World Wide Web (the Web or WWW) was created in 1989 at the European Particle Physics Laboratory in Geneva, Switzerland. Originally it was created as a method for incorporating footnotes, figures, and cross-references into online hypertext documents in which a reader can click on a word or phrase in a document, and immediately jump to another file. The second file may be located on the same computer or anywhere else on the Internet. The WWW organizes widely scattered resources into a seamless whole.

The Web browsers have changed the way people use the Internet. Web pages are now used to distribute news, interactive educational services, product information and catalogs, highway traffic reports, live audio and video, among many other things. Interactive Web pages permit users to consult databases, order products and information, and submit payment with a credit card or other account number.

The latest Web browsers, including Netscape Navigator and Microsoft's Internet Explorer, can open file viewers and other application programs automatically when they receive graphic images, audio, video, and other files. Users also can import live, interactive data (such as financial information that changes frequently) and executable programs from the World Wide Web.

The organization of the World Wide Web is built on a set of rules called HyperText Transfer Protocol (HTTP) and a page description language called HyperText Markup Language (HTML). HTTP uses Internet addresses in a special format called a Uniform Resource Locator, or URL. The format of URLs is: type://address/path

Here type specifies the type of server in which the file is located. The address part is the address of the server, and finally the path is the location within the file structure of the server. As an example the URL for the IEEE membership renewal is:
http://www.ieee.org/renewal

A document that uses HTTP is known as a Web page. A Web page contains information about a particular subject with links to related Web pages. Many Web sites contain a top-level home page that has pointers to additional pages with more information. For example, a university might have a home page with links to a campus guide, Telnet access to the library, and separate pages for individual departments and offices.

Thousands of new Web pages appear every day, but they are not always easy to find. The World Wide Web does not provide a basic structure for locating resources. To fill this gap, search tools are available online. One of them is Yahoo!, whose URL is http://www.yahoo.com. Yahoo! is an extensive directory with a menu that lists thousands of web sites, organized by topic, and a search engine that looks for specified words in titles and addresses of the Web sites in the directories. Other popular search tools include InfoSeek, Lycos, Magellan, Excite, to name a few.

5.6 Internet Access

The rapid growth in the use of the Internet by both individuals and businesses resulted in many organizations selling Internet access to both individuals and business. These organizations are popularly known as ISPs (Internet Service providers) or local hosts. Two popular types of Internet access are:

- Dial-up telephone Access
- Leased line access.

Most individuals use dial-up to access the facilities of an Internet Service Provider. However the businesses use leased lines to connect local area networks to the Internet. Figure 12.10 illustrates the two most common methods of Internet access. The leased (or direct) line is expensive and the line is dedicated for the exclusive use of the organization.

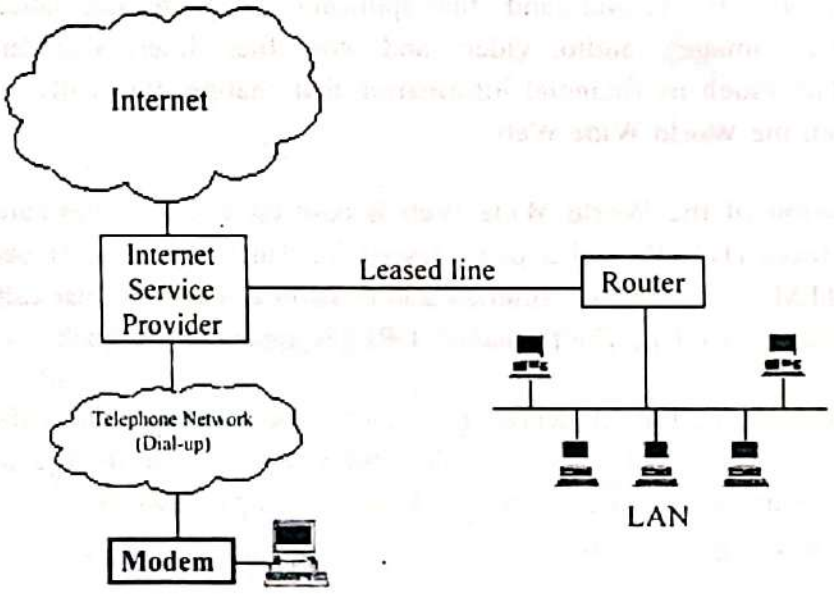


Figure 12.10: Common Internet access method.

So it is possible to use the line for any length of time. On the other hand, for the dial-up connection, one has to ring up the number of the service provider's computer and get a

connection for accessing the Internet. The user is charged for the duration of the call. Normally the data transfer speed through the dial-up line is low.

It is normally convenient to connect a LAN to the Internet through a leased line than to have an individual dial-up connection. Because a LAN can provide support for numerous computers and there is always a high probability that one or more users require access to the Internet at any particular point of time. In addition, a leased line provides immediate access from the Internet to any application residing on the LAN, such as Web Server, an FTP Server or an e-mail system. Thus a leased line connection facilitates bidirectional access to and from the Internet.

5.7 Exercise

Multiple choice questions

1. Which is not an Internet service protocol

   a. FTP b. HTTP
   c. SMTP d. RFC

2. An IP address has

   a. 2 numeric parts b. 3 numeric parts
   c. 4 numeric parts c. 5 numeric parts

3. The protocol for WWW is

   a. TELNET b. FTP
   c. SMTP d. HTTP

Questions for short answers

4. List common Internet services.
5. Distinguish between DNS address and IP address.
6. What is an Web browser?
7. Briefly explain electronic mail service.

Analytical questions

8. Using an illustration show the layers of TCP/IP suite and some of its application protocols.
9. What is an Internet protocol address? Distinguish between IP address and DNS address.
10. Without mentioning country domains list six top level domains and mention their significance.
11. Explain at least two types of Internet access methods.
12. Write short notes on a) WWW and b) E-mail

Lesson 6: Intranet and Extranet

6.1 Objectives

On completion of this lesson you will know:

- Intranet and intranet services
- Intranet technology
- Webmaster and extranet.

6.2 Introduction to intranet

An intranet uses Internet technology to deliver an organization's internal information. This includes integration of e-mail, FTP, Mail Server and Web server with the internal applications. The user interface for intranet is provided by Web browsers. Web servers are the most visible part of the intranet since this is where the organization's Web pages would be hosted and accessed by the clients.

The objective of an intranet is to organize an individual's desktop with minimal cost, time and effort to be more productive, cost-efficient, timely and competitive. With an intranet, access to all information, applications and data can be made available through the same browser. Intranets connect people together with Internet technology, using Web servers, Web browsers, and data warehouses in a single view. All clients and servers of intranets must support the TCP/IP protocol. An intranet does not have to be connected to the Internet. However, moving mail and other information across the Internet to clients and partners might be needed and an Internet connection is desirable. The Internet is always starved of bandwidth, but intranets, even on the slowest LANs with bandwidth of 10 Mbps, have no bandwidth issues. An application designed for the intranet may be choked on the Internet, while those for the Internet may run in a flash on an intranet. Since the Internet is an open environment, security is a major issue. Intranets, on the other hand, are secured and confined to an organization

Intranets provide a lot of choice and flexibility. Existing applications do not have to be rewritten to the new client-server environments. New versions of word processors, spreadsheets and database programs have built-in Internet capabilities. A Web page can be produced from an existing document by simply saving it as an HTML document. However, policies for using the technology within an organization are extremely important.

Security in the intranet environment is provided through Secure Socket Layer (SSL), Secure Electronic Transactions (SET), and Secure MIME (S/MIME) to provide

confidentiality, data integrity, authentication and digital signatures. Access to internal information systems from outside can be regulated through the installation of firewalls. Finance, Sales and Marketing, Manufacturing, Research and Development, Personnel and Customer Support departments of an organization may be benifitted by intranets

6.3 Intranet services

An intranet provides Internet services within an organization. An intranet client is a universal browser using TCP/IP protocol. The servers in an organization, may support any services on any operating system. As long as they support TCP/IP protocol stack, they are part of the intranet. Figure 12.11 shows the organisation of an intranet.

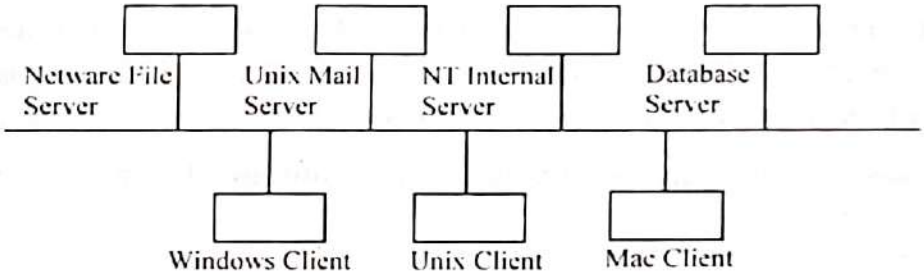


Figure 12.11: A multiclient, multiserver intranet.

The requirements or services for an intranet may include the following:

- Mail Services
- File Transfers
- Web Services
- Audio Services
- Video Services.

The easiest way to start an intranet is by providing e-mail services from a standards-based server, i.e. SMTP mail server with client using Netscape Navigator or the Microsoft Internet Explorer browser to retrieve mail, using POP3 protocol, from the mail server. The simplest way for setting up a server is by installing a Linux system which has built-in free SMTP mail server software. No expensive gateways are required. This is the easiest way to get started and demonstrate the usefulness of an intranet. The essential components of an intranet include the following:

- A network
- TCP/IP on servers and clients
- Hardware for hosting intranet services
- Software: Mail server as the minimum, Web server for hosting Web pages
- Mail servers
- Browsers.

The optional items include:

- HTML editors
- Productivity tools which are Web-aware
- E-mail Remote User Agents
- Proxy servers
- CGI
- Java
- ActiveX.

The essential components make an intranet operational, but the optional items make it truly useful by delivering all, or most of the intended intranet services. The Web services, database linkages, applications requiring users to fill forms, graphics and beautification of sites are enabled by the tools mentioned above. Applications on an intranet help in cost saving by reductions in printing and distribution costs. The largest benefit is the increased access to information. An intranet achieves the following in an organization:

- Reduction of costs of printing, paper, software distribution, mailing, order processing
- Reduced telephone expenses
- Easier and faster access to technical and marketing information
- Easier and faster access to remote locations
- Increased access to competitive information
- Latest and uptodate, research base
- Easier access to customers and partners
- Collaborative group working
- Increased accuracy and timeliness of information
- Just-in-time information.

It is the business needs that should drive an intranet. All or some of the above benefits will make a business case for an intranet. Requirements must, therefore, be studied carefully, and the intranet tailored towards realising them to provide the organization with the much-needed tool for operational efficiency, better turn-around time, better knowledge base, more productivity, and competitiveness to survive in the market place.

6.4 Intranet implementation

The intranet should be implemented in phases. If a network is already in place, the steps essentially revolve around selecting servers and software, the operating system and special tools for content creation. The existing databases, and software applications may be integrated into the intranet. This calls for a proper plan to be prepared. The important steps to implementation follows.

Planning: This step includes scope of services and facilities; mail, Web sites, home pages department-wise, personnel, administration of the site, layout of pages, connecting with existing databases and applications, access to intranet and connecting with the Internet.

Subnets: If the number of servers and clients is large, there may be many subnets. IP addressing plan should be worked out in detail.

Servers and Software: Server hardware platforms and the software that will be used on them for hosting mail and Web services need to be finalised. For intranet users, browsers and mail clients have to be standardised. A server platform, may host both mail and Web services. As the intranet load grows, this could be split into multiple-servers. Web server, mail server, print server, Domain Name Server (DNS), workflow applications etc could be implemented on different hardware systems.

Operating system: An operating system may be chosen from the following popular server operating systems: Unix, Linux, Windows NT, NetWare, Mac OS, OS/2, AS/400. The choice of operating system will determine the software requirements of the Web server, mail server etc. While Unix was the operating system on RISC systems, non-Unix platforms are equally aggressive solutions now. The OS should be installed on the server, complete with the TCP/IP suite of protocols. Web server and mail server have to be configured as intranet software. DNS services should preferably be implemented from the very beginning. IP addresses and subnet masks should be allocated. It is better to use TCP/IP addresses duly registered with InterNIC so that connection to the Internet at a future date avoids address conflicts.

Proxy Servers: If the intranet requirements include the users connected on it to share a single Internet dial-up connection on the network, a proxy server is essential. It is the proxy server that dials into the Internet connection and all the nodes access the Internet through it.

Content Creation Software: HTML editors, Web-enabled office productivity tools are essential on client machines which are equipped with browsers and e-mail software.

Training: The users on mail and Web applications may require training for discussion group, mailing list server, converting existing documents and reports for the intranet, and making them available to all. Skills to create new contents using the office-suites, productivity tools, HTML editors etc are also required.

Existing Databases: Set up a group of programmers to study the existing databases, and to Web-enable them for intranet.

Web Publishing: The Web is like a magazine, and content creation is like Web publishing. The magazine is on the Web server, and is used by the employees of an organization who are on the intranet. The contents should be catchy, easy to access, and uptodate. Web design and maintenance are the major issues on an intranet.

6.5 Webmaster

Intranets are managed by managers who have been designated as webmasters. In the Internet world, the webmaster is responsible for the setting up and maintenance of Web sites. This term and the role borrowed from the Internet, is also used in the intranet. A webmaster is responsible for the creation and maintenance of intranet servers and services, and is responsible for updating the content of the Web sites, mail directories, integrating with databases, connection with the Internet etc.

A webmaster must know the organization's activities very well, besides being familiar with the Internet and being thorough with programming languages and network technology. The webmaster should be able to visualise the present and future goals of Web sites. Emerging technologies and business models must fascinate the webmaster. Since the webmaster is required to have many types of skills, and since no one person may have them, the greatest ability that one is looking for is that of a desire to learn and acquire new skills. The webmaster must clearly understand the organization's problems and challenges, develop a cross-organizational plan for the intranet, translate it into a technical plan to support the plan, and deploy support tools to provide the desired intranet services, with an eye to the future for full intranet services.

As the intranet grows, the webmaster may cease to be just one person. In larger organizations, several people collectively may act as the webmaster or as an intranet team. The webmaster can enable intranet applications to be ready for other sites of an organization through the Internet gateway. This role converts an intranet to an extranet.

6.6 Extranet

AN extranet is a network that links some of the intranet resources of an organization with other organizations and individuals. For example, extranets enable customers, suppliers, subcontractors, consultants, and others to access selected intranet Web sites and other databases. Organization can establish private extranets among themselves, or use the Internet as part of the network connections between them. Many organizations use virtual private networks (VPNs) to establish secure intranets and extranets. A VPN is a secure network that uses the Internet as its main backbone, but relies on the firewalls and other security features. For example, a VPN would enable a company to use the Internet to

establish secure intranets between its distant branch offices and manufacturing plants, and secure extranets between itself and its customers and suppliers.

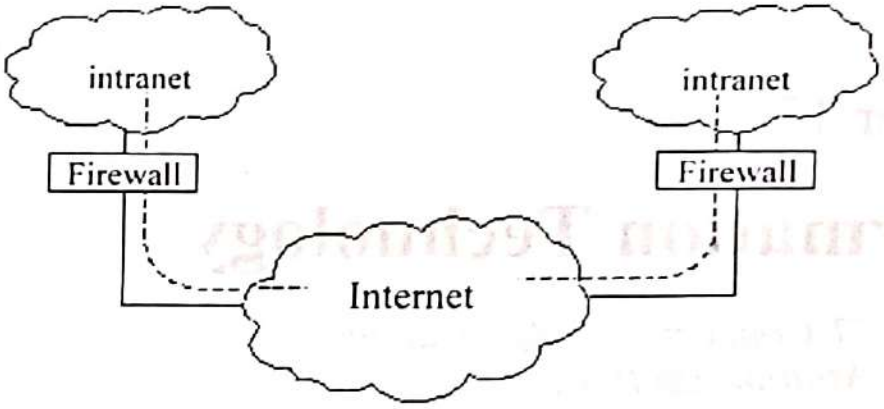


Figure 12.12 : An extranet realised by connecting two intranets.

6.7 Exercise

Multiple choice questions

1. An intranet uses

   a. LAN technology
   b. MAN technology
   c. Internet technology
   d. None of them

2. Clients and Servers of an intranet must support

   a. OSI model
   b. TCP/IP
   c. HTTP only
   d. TELNET

Questions for short answers

3. List common services available through an intranet
4. Comment on the security of an intranet system
5. List essential components of an intranet
6. List the advantages of an intranet

Analytical questions

7. What is an intranet? Distinguish between intranet and extranet
8. Using an illustration describe a multiserver and multiclient intranet system
9. Explain the important steps for implementation of an intranet
10. Briefly explain the responsibilities of the webmaster

Chapter 13

Information Technology

Lesson 1: IT Concepts and Applications
Lesson 2: Multimedia Basics
Lesson 3: Multimedia System Development
Lesson 4: Electronic Commerce

Lesson 1: IT Concepts and Applications

1.1 Objectives

On completion of this lesson you will know:

- Concepts of information technology
- Characteristics of information systems
- Routine applications of computers
- Use of computers in management
- Scientific, industrial and medical applications of computers

1.2 Information Technology

Information Technology (IT) refers to the creation, gathering, processing, storage, and delivery of information and processes. IT is currently a subject of widespread interest. Major activities of information technology are:

- Information technology can process raw data into meaningful and useful information
- It can recycle processed information and use it as data in another processing step
- It can package information in a new form which is easier to understand, and is more attractive or more useful.

The most important component of IT is information processing. Computer hardware and

software are used for information processing. There are five steps for processing information, collectively called information processing cycle (Figure 13.1). These are:

- Input
- Processing
- Output
- Storage and retrieval
- Distribution and communication

The term IT is gradually being replaced by ICT, meaning "Information and Communication Technology", and the term "Communication" indicates that information revolution needs an effective electronic communication infrastructure.

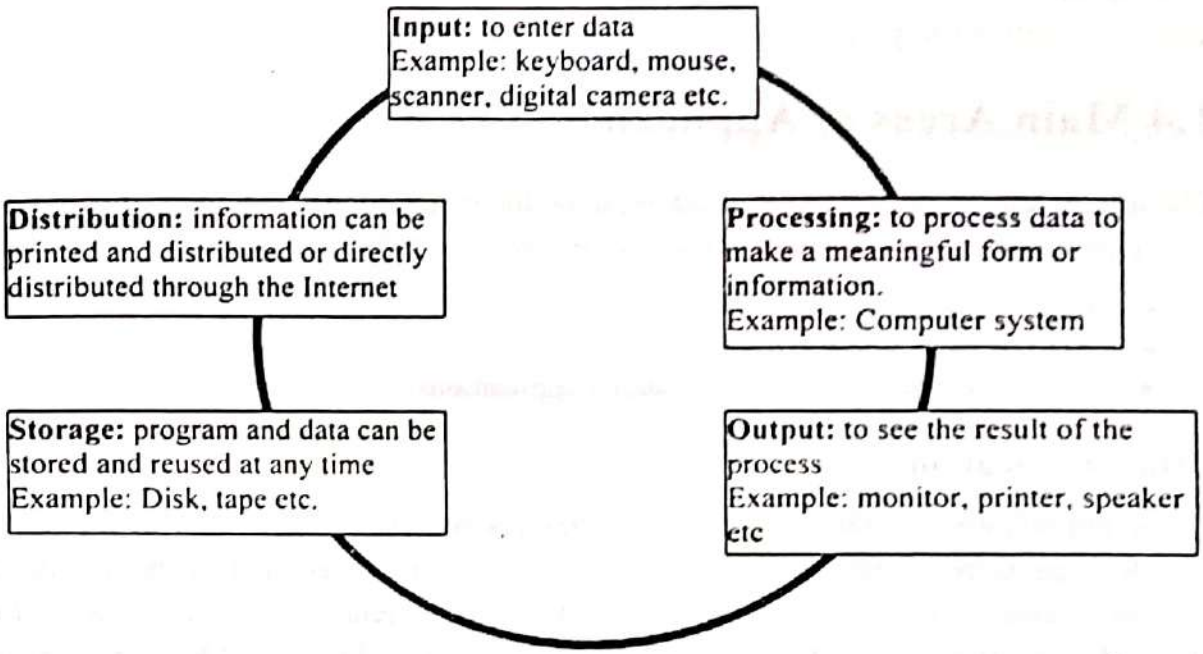


Figure 13.1: Information processing cycle and equipment

1.3 Information System

An information system comprises of a set of people, procedures and equipment. It is used for the management and better control of an organization. The function of an information system is to collect store, process and present data to manage the business of an organization. Processing of data is an integral part of an information system. Information systems may be classified as:

- Deterministic
- Probabilistic, and
- Adaptive

Deterministic or mechanistic: The output from this system is uniquely determined, (example: evaluating formula).

Probabilistic or stochastic: A degree of uncertainty exists so that the output is specified in terms of averages or distributions (example: random number simulations).

Adaptive or cybernetic: A system adapts itself to changes in environment (example: a stock control problem where the re-order strategy depends on changes in demand).

In real life few systems are truly deterministic. However deterministic models are often created for systems which are stochastic or even adaptive. Modeling is used for designing a computer system to represent a real system; the computer system may be studied for analysis, control and processing of various activities of the real system.

1.4 Main Areas of Application

The aim of this section is to focus attention on the major application areas of computers. Computer applications can be divided into following broad areas:

- Office automation
- Management applications
- Scientific, engineering and research applications.

Office automation

These applications include the use of computers for clerical activities in public and private organizations. Office automation involves the use of computers, in conjunction with other electronic equipment, to automate the basic secretarial and clerical tasks of the office. The routine nature of office work makes office procedures suitable for automation. Reduced equipment costs now makes it cost effective. Word processing is the basic job in an office.

Some office automation systems go far beyond the function of word processing on networks. PCs or workstations are connected to a network which provides access to a wide variety functions including:

- Word processing
- Spreadsheet analysis
- Drawing diagrams
- Database applications
- Electronic mail facilities
- Access to other networks for information exchange
- Electronic commerce.

There may be a number of file servers and print servers connected to a network. Popular example is a client-server arrangement. In most client-server arrangements an application program (the client) runs on a PC or workstation and communicates through the network with the server program which runs on another computer. The server machines are usually more powerful and are popularly dedicated to provide access to:

- A printer
- Electronic mail
- Database
- Other systems.

Management Applications

A commercial organization often needs information on which to base decisions concerning the current operations and future plans. This information should be timely and accurate. Examples of the use of computers for management and control are given below.

Stock control: A computer can process data quickly and thereby make available information on stock levels. A computer can be used to link widely separated warehouses, thus treating them as one vast stock control system. Customer orders can be served from an alternative warehouse if an item is not available in the local one. Thus the computer enables stock holdings to be kept to a minimum and thereby reduces cash requirement for the business.

Production control: It is an extremely complex area especially in factory automation. Production should respond quickly to changes in demand and other circumstances. This requires accurate up-to-date information which should also be timely. On-line systems help to get data quickly enough to influence current events. A computer can make machine loading, materials control, machine utilization etc. A computer is more efficient because of its ability to make complex calculations and shift data quickly and accurately.

Labour control: Some of the information used in the control of labour can be obtained from the payroll and personnel records. An analysis of labour hours for idle time, sickness and absence, can aid forecasting future requirements. Results of analysis help to highlight these problems accurately and actual performance can be measured quickly against planned performance. Reports for the shop-floor manager aid control of routine operations and can help better utilization of manpower.

Network analysis: Network analysis is used for planning and control of complex projects. Examples are the building of a factory, the installation of new plant and the manufacture of an aircraft or ship. A model is used to show each stage in the project and

its dependence on other stages. When the necessary data have been supplied, the program produces a plan for the management. The plan can be quickly modified by the program in the light of actual progress, and revised plans produced at a short notice. The program also indicates the interdependent activities that take longest time to complete and could hold up the project.

Linear programming: This is used to find the optimum solution which maximizes profits or minimizes costs. A considerable amount of data are supplied to construct a computer model. The computer has the storage, capacity and the speed to calculate and execute every feasible solution and thus to find the best one.

Financial modeling: A model can be constructed for a company's finances comprising of its resources, income and expenditure. This can be used to simulate the effects of different policies. An example is the forecast of the results of different investment policies. It is possible by a series of computer runs, with variations of the basic assumptions to obtain a forecast of the likely effects of alternative policies. The usefulness of the computer lies in its high speed, which enables it to execute instructions quickly.

Scientific, Engineering and Research Applications

These applications include the use of computer for complex mathematical calculations, analysis of data of various nature, design, analysis and control of physical systems etc. Here the main emphasis of data processing is research and development for scientific, engineering and industrial applications. Few areas are described below.

Weather-forecasting: Weather-forecasting demands vast computational powers. This is an area which requires the super-computers.

Mathematical and statistical analysis: These analyses require large calculations and solutions of mathematical and statistical problems. The applications include research in physics, chemistry, geology, archaeology, medicine, astronomy, social science etc. Some commercial problems have mathematical bias, and require mathematical analysis to determine the optimum use of resources.

Design work: Computers can be used as a design tool in engineering and other disciplines. CAD (Computer Aided Design) is used in electronic, electrical, mechanical, aeronautical, and civil engineering, architecture and many other disciplines.

Medical applications: Computer can be used as an aid to medical research by analyzing data. The computer acts as a large bank of data about known medical conditions. Once a

computer systems has been set up by medical experts an ordinary doctor can be taken through a question and answer session by the computer until a correct diagnosis is made.

Computers can be used to hold details of patient data. Microcomputer systems are now used for this purpose in increasing numbers. Computerized health records can be used by medical officers, health visitors, and researchers.

Education: Computers are used extensively as a part of study in computer science itself. They are used as a way of aiding the understanding of a wide variety of other subjects. The computer can provide instructions and ask questions. This kind of activity is called Computer-Aided Learning (CAL) or Computer-Aided Instructions (CAI). Computers are also used for a number of other applications in education; examples are marking of multiple choice examination papers and processing examination results.

Manufacturing: Some examples of use of computers in manufacturing are stock and production control, engineering design etc. The industrial design, manufacturing and testing processes are becoming increasingly computerized. The terms CAD, CAM (Computer-Aided Manufacture) and CADMAT (Computer-Aided Design Manufacture and Testing) are familiar words in industries now.

Robots: The word "robot" comes from a Czech word whose meaning is "to labour". For many years the term "robot" was associated with science fiction rather than science fact. That association has changed now. The basic difference between industrial robots and other automated machines is that a robot can be programmed to carry out a complex task and be reprogrammed to carry out another complex task. A complex task is a series of actions involving mechanical manipulations.

1.5 Other Applications

Computers are used for a variety of applications and there are some applications that do not fall into either of the applications mentioned above. Some of these applications are discussed below.

Expert Systems

An expert system is a program package that can perform the function of a human expert. Some of the initial expert systems were used for medical diagnosis. A medical consultant, assisted by staff, takes part in a lengthy exercise in which both the knowledge and the decision-making procedures are transferred to the computer. As a result the computer can ask questions and draw conclusions from the answers and a junior doctor, aided by the computer, could be as expert as a consultant.

There are many applications for expert systems, examples are company law, investment, finance and personnel management etc. Expert systems normally have the following features:

- An organized base of knowledge in the form of a database.
- A user interface to support diagnostic discussions with the user.
- A facility to hold details of the consultation.
- An inference engine, that is a software, which can use the knowledge base and current consultation to formulate further questions or draw conclusions from actions.
- A knowledge acquisition system, to update the knowledge base.

Although the basic components of an expert system are bound to vary from one discipline to another the basic structure is the same. The user of the expert system sits before a PC or/and takes part in a question and answer session in which data about the problem are typed in. At various stages during the session, or at the end, the system makes an assessment of the problem and recommends actions. Expert system packages are not confined to large computers; several are available now for personal computers.

Intelligent Knowledge-Based Systems (IKBS)

These systems are intended to provide an expert consultative service to management and are also called expert systems. A typical system is an interactive one and contains a large database. The system is programmed with a decision-making strategy developed by careful consultation with experts in the field. The user takes part in a question and answer session in which data about the problem are supplied through the keyboard. At the end of the session the system makes an assessment of the problem and suggests actions.

Application Packages

Use of an application package is an alternative to developing programs. Many small organizations cannot afford to employ a programmer, and individual users may not be capable of writing programs. As an alternative; many firms and individuals purchase programs. An application package generally include a full description of the operation of the program, user instructions and documents. There are many large organizations which find it more economic to purchase application packages rather than writing their own applications. Some advantages connected with the use of application packages are:

- No need to employ a programmer for program development
- The program is ready immediately
- The developer of the application package employs skilled programmers
- It is possible to get help or share experience with other users of the package.

There are, however, some disadvantages in the use of application packages as compared

with program development. The most important are:

- The programs may be too general and not readily matched to the needs of the user.
- It may be difficult to adapt the program to the changing needs and growth of the user's application.

1.6 Exercise

Multiple Choice Questions

1. How many steps are there in the information processing?

   a. two c. three
   b. four d. five

2. CAL stands for

   a. Computer Aided Learning c. Computer Aided Lesson
   b. Computer Assisted Lesson d. None of these

Questions for short answers

3. What is Information Technology?
4. List the steps for information processing.
5. What are the types of information systems?
6. Mention the common areas of applications of computers.
7. List the operations incorporating office automation.
8. Explain the terms: CAD, CAL, IKBS, robot, E-mail and application package.
9. Give the advantages of using application packages.
10. Name the components of computer application in management.

Analytical questions

11. What is meant by information technology? Discuss the cycle of information processing.
12. What do you understand by office automation? Discuss the common jobs for office automation.
13. Discuss, with examples, the importance of computer for management applications.
14. Discuss scientific, engineering and research applications of computers.
15. What is an expert system? Explain its characteristics.

Lesson 2: Multimedia Basics

2.1 Objectives

On completion of this lesson you will know:

- Multimedia systems
- Multimedia hardware
- Multimedia software
- Compression and decompression

2.2 Introduction

The combination of text, graphics, animation, video and sound for presentation to the consumer is known as multimedia. A TV show (or movie or advertisement) is an example of multimedia. "Interactivity" characteristics of multimedia makes it more attractive and exciting. In interactive multimedia, users can choose their own path through the presentation. Most Web pages include graphics along with the text, and many also offer animations, videos, and sounds. The Web's navigation method, called hypertext, enables users to browse as they please. By blending multimedia with the Web, hypermedia becomes possible; in many Web pages, one can click parts of a graphic to access a different page. In hypermedia, media other than text becomes the vehicle for navigating to new material.

2.3 Multimedia Hardware and Software

Hardware

A multimedia system is a computer with a sound card, a CD-ROM drive and speakers. For professional multimedia production, someone may need additional equipment, such as:

- Digital camera
- Pen-based graphics tablet
- Stereo microphones
- Video capture card to display full-motion video
- 3D video accelerator for playing games
- 3D sound card capable of reproducing sounds.

Software

A multimedia presentation may involve some or all of the following:

- Bit-mapped graphics
- Vector graphics
- Edited photographs
- Rendered three-dimensional images
- Edited videos, and
- Synthesized sound.

The following sections briefly introduce some of the software used to create multimedia productions.

Paint Programs

These programs are used to create bit-mapped graphics, which are composed of tiny dots, each corresponding to one pixel on the computer's display. Although paint programs enable artists to create pictures easily, the resulting bit-mapped image is difficult to edit. One must zoom the picture so that the individual pixels can be edited, and enlargement may produce an unattractive distortions called the jaggies. Professional paint programs such as Fractal Design Painter enable users to create beautiful painterly effects, even if they are not good at art.

The more colors shown on-screen, the more memory is required to store them. A file that stores 8 bits of color information per pixel can display a color depth of only 256 colors. For realistic color photographs, a color depth of 16.7 million colors is needed (24 bits per pixel). Few standard formats for paint programs are as follows:

- Graphics Interchange Format (GIF) is a 256 color file format. It is often used for Web page design.
- Joint Photographic Expects Group (JPEG) for up to 16.7 million colors and is also a Web standard.
- Portable Network Graphics (PNG) is a new alternative to GIF.
- Windows Bitmap (BMP) is a standard bit-mapped graphics format developed for Microsoft Windows.

Image Editors

Image editors are sophisticated versions of paint programs designed for editing and transforming – but not creating complex bit-mapped images. These programs make use of automated image processing algorithms to add a variety of special effects to photographic images. They also enable skilled users to combine photographs in a way that leaves few traces behind. Leading image editors such as Adobe PhotoShop have been used by professional design studios for years.

Drawing Programs

Drawing programs are used to create vector graphics, in which each on-screen object is

stored as a complex mathematical description. In practice, every object in a vector graphic can be independently edited and resized without introducing edge distortions. To compose an image with a drawing program a user can create independent lines and shapes; colors and textures can then be added to these shapes. Professional drawing programs, such as Macro-media Freehand and Adobe Illustrator, save files by outputting instructions in the PostScript language.

Animation Programs

Like a movie, an animation consists of a series of still images displayed at a frame rate high enough to create the illusion of movement. Animators create each of the still images separately. In computer animation, the computer provides tools for creating the animation as well as running it. It is easy to create a simple animation using the GIF file. Also stored in the file is a brief script that tells the application to play the images in a certain sequence and to display each for a set period of time. Because Web browsers can read GIF files and play the animations, GIF animations are common on the Web.

3D Rendering Programs

A 3D rendering program adds three-dimensional effects to computer graphic objects. The results are strikingly realistic and can be rotated in any direction to achieve just the result the artist is looking for. The rendering technique and ray tracing add amazing realism to a simulated three-dimensional object by adding variations in color intensity that would be produced by light falling on the object from multiple directions.

Video Editors

Video editors enable users to modify digitized videos, such as videos imported from a video camera by means of a video capture board. With a video editor, a user can cut segments, resequence the video, add transitions, compress the file, and determine the frame rate. A video editor also enables to save a video file to some or all of the following formats.

- **Moving Pictures Expert Group ((MPEG):** A family of video file formats and compression standards for full-motion video.
- **Quick Time:** A video file format developed by apple computer. The latest version, Quick time-3, plays full-screen, broadcast-quality video as well as CD-Quality audio and is widely used in Multimedia CD-ROM productions.
- **Video for Windows:** The native video file format for Microsoft Windows (often called AVI, because these files use the .avi file name extension). This format is not adequate for full screen, broadcast-quality video.

Because a huge amount of data must be stored to create realistic looking video on a

computer, video file formats use compression/decompression techniques. For the best playback, special video adapters are required.

Audio software

For capturing and processing sound for multimedia presentations a variety of programs are available, including sound mixers, compression software, bass enhancers, synthesized stereo, and even on-screen music composition programs. Digital audio waveforms (recorded live sound or music) can be saved in a variety of file formats. These are as follows.

- **MP3:** A sound file format with almost unbelievable characteristics. Using MP3 (also known as MPEG3), user can compress CD-quality digital audio by a factor of 12.1 with no perceptible loss in sound quality.
- **Sun/Next (AU) Sounds:** A low fidelity monaural format now widely used owing to the small size of AU files. It is often used to distribute sample sounds over the Internet.
- **WAV Sounds:** The default Microsoft Windows sound file format WAV sounds can be saved to a variety of quality levels, including mono to CD-quality stereo. WAV sounds usually are not compressed, so they tend to take up a lot of disk space.
- **Streaming audio** formats enable Internet-accessed sounds to start playing almost immediately after the user clicks the audio link. The most popular streaming audio format, Real Audio, can deliver voice-quality audio over modem connections.
- **Musical Instrument Digital Interface (MIDI)** files, do not contain waveforms. Instead, they contain a text-based description designed to derive synthesizers, electronic devices that simulate musical sounds. Sound cards have built-in synthesizers.

Authoring Tools

To create multimedia presentations, user has to use authoring tools. These tools enable one to specify which multimedia objects to use (such as text, pictures, videos, or animations), how to display them, and how to enable the user to interact with the presentation. To take full advantage of an authoring tool's capabilities, it is often necessary to learn a scripting language. A leading authoring package is Macromedia Director.

2.4 Compression and Decompression

Multimedia files require huge amounts of storage space. For example, an average-sized hard disk could store only a dozen audio CDs, and there would not be much room left for system software or applications. To reduce the size of multimedia files, most software uses comparison/decompression algorithms.

There are two approaches for compression, called lossless compression and lossy compression. In lossless compression, the original file is compressed so that it can be completely restored, without flaw, when decompression occurs. In lossy compression, the original file is processed so that some information is permanently removed. However, users probably can not notice the loss. Lossy compression techniques eliminate information that is not perceived when people see pictures or hear sounds.

2.5 Exercise

Multiple Choice Questions

1. GIF stands for

   a. General Interchange Format
   b. Geographic Interchange Format
   c. Graphics Interchange Format
   d. None of these

2. The native video file format for Microsoft Windows is called

   a. MP3
   b. JPEG
   c. AVI
   d. none of these

Questions for Short Answers

3. What do you understand by multimedia?
4. What do you understand by hypertext?
5. What are the functions of 3D rendering programs?
6. Give the hardware requirement for multimedia production.
7. Define GIF, JEPG and BMP.
8. What are the functions of image and video editors?

Analytical Questions

9. Discuss the hardware requirement for multimedia production.
10. Describe the major software requirement for multimedia production.
11. What are the functions of video editors and authoring tools?
12. Discuss compression and decompression in connection with multimedia production.

Lesson 3: Multimedia System Development

3.1 Objectives

On completion of this lesson you will know:

- Steps involved in multimedia system development
- Applications of multimedia systems.

3.2 Multimedia System Development

Multimedia system development process involves a number of steps like (Figure 13.2) traditional software system development and is known as Multimedia System Development Life Cycle (MSDLC). The steps are as follows:

- Defining the system
- System Design
- Tools selection
- Authoring/rendering
- Testing

Defining the system to be developed

To develop a multimedia presentation or program, it is valuable to understand precisely who the user is. To compose a multimedia program the developer should consider the following:

- Level of understanding of the audience
- Expectation of the users
- Goal of presentation (training /entertainment)
- Allocated time
- Type of presentation required (text, graphics, sound, video, animation or combination)
- Interactive/non-interactive presentation
- Impact of the presentation to the audience (instant or long-term).

Having all these details firmly in hand will help the multimedia developer to create a successful product.

System Design

Most common way to start design is by composing an outline of the sequences and blocks of information that will appear on the screen. This determine the amount of information - text, graphics, clickable objects which will be presented on each screen. It

is also the time to establish a navigation methodology for the user. The developer has to decide whether there will be a navigation bar with arrows leading from scene to scene, or there will be text or graphic objects that the users should click to jump around the entire program.

The storyboard can be used for a great deal of animation or many different scenes. Used by film directors for productions ranging from 30-second television commercials to feature-length motion pictures, the storyboard consists of sketches of the scenes and actions. Mapping out a storyboard helps the author to recognize gaps in logic. Some multimedia authoring programs provide facilities for drawing and organizing the frames of a storyboard.

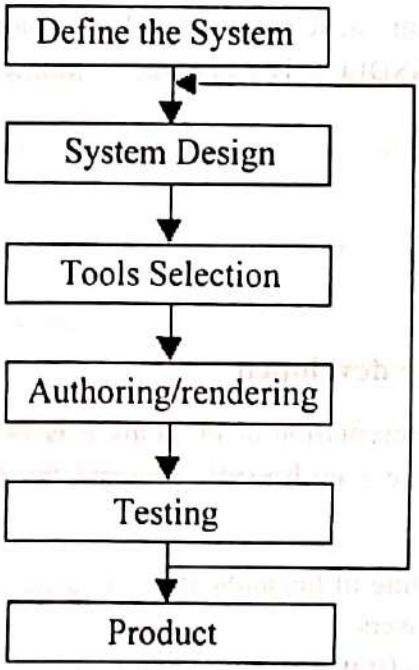


Figure 13.2: Life cycle for multimedia product development

Tools Selection

Multimedia product requires many type of software tools, for instance, creating text often requires a word processor; working with digital images requires graphics software; using video requires a video-capture program and editing software; sound often requires its own editing software. All of this software is used to generate the content. When the content is ready, it needs to be assembled in a process called multimedia authoring. This process requires still another type of software, which can understand different types of media, combine them, control the sequences in which they appear, and create navigational tools for the user interface.

The tools used for multimedia depends largely on the variety of media to be included. For a simple text-and-graphics slide show type of presentation, a presentation software program should be enough. One could enliven the offering by adding preexisting graphics from the hundreds of commercially available CD-ROM collections of clip art, clip sound, and clip video. These clip collections store small files in the proper file formats for use in a wide range of multimedia programs.

A user can use sophisticated sound editing program, to splice together segments or add sound effects electronically, for example, echoes to the sounds. Users can do the same kind of electronic editing for video.

The most popular programs for combining the elements of a multimedia presentation is Macromedia Director. In Director, the multimedia author assembles each element – text, graphics, sound, and video into separate "tracks". The program helps the author to synchronize all the elements. For instance, a crash sound effect is heard when the two animated objects collide. The director generates files that contains the entire multimedia presentation ready for distribution on disk or CD-ROM, or to be played over an intervactive television or through the Internet.

Authoring/Rendering

After creation of all the contents, it is required to put them all together. For a complex product created with the use of a sophisticated tool such as Director, the multimedia authoring generally is performed by a skilled multimedia developer or programmar

Testing

Multimedia product should be tested by the users. By going through this testing, the developer can locate flaws ahead of time and repair them before unleashing the finished product. During the test the developer should consider the following points:

- Is it difficult to read (in case of descriptive text)?
- Is it user friendly?
- Is it interactive and easy to navigate?

The final audience and the developer very often have different points of view. The developer must learn to regard problems the user detects with the program as constructive criticism. The reason testing is so valuable is that it is too easy to lose sight of the audience once the heavy duty authoring starts.

3.3 Applications

Use of multimedia is common in many computer applications, because graphics, sounds,

animations and video can often do a more effective job of involving the user than text alone. Other applications are:

- Multimedia is a prerequisite of all kinds of games
- Computer-based training
- Dictionaries (tells how to pronounce difficult word)
- Encyclopedias (important sound and video clips)
- Business presentations
- Web applications
- Advertisement, etc.

3.4 Exercise

Multiple Choice Questions

1. Storyboard can be used for

   a. single scene
   b. two scenes
   c. more then two scenes
   d. none of these

2. A popular program for combining the elements of multimedia presentation is

   a. Macromedia Freehand
   b. Adobe Illustrator
   c. Macromedia Director
   d. none of these

Questions for short answers

3. Name the steps for multimedia system development life cycle
4. What is a storyboard?
5. What is meant by authoring
6. What are the steps to be considered for testing a multimedia product?

Analytical Questions

7. With the help of a flow-diagram discuss the steps for multimedia system development life cycle.
8. Describe the steps involved in multimedia system development life cycle.
9. Discuss the applications of multimedia.

Lesson 4: Electronic Commerce

4.1 Objectives

On completion of this lesson you will know:

- Concepts of e-commerce
- Benefits and limitations of e-commerce
- E-commerce technology and applications
- Electronic payment systems and security methods.

4.2 Introduction

Electronic commerce, or e-commerce, is currently a topic of wide-spread interest in the business sector. E-commerce is defined as the use of telecommunications or wide area networking to carry out business activities and financial transactions. However, in the age of the Internet, intranet and extranet, e-commerce is more than just buying and selling products online. It also includes on-line process of developing, marketing, delivering and servicing products, and services purchased by global Internet communities. Extranet access of inventory databases by customers and suppliers, intranet access of customer records by sales and customer services, and involvement in product development via Internet newsgroups and e-mail exchanges are also included in e-commerce.

4.3 Benefits of E-Commerce

The benefits of e-commerce can be summarized as follows:

Benefits	Reasons
Reduced transaction costs	Elimination of paperwork and labour saving
Improved cash flow	Faster processing and exchange of information
Reduced inventory levels	Shorter order cycle; reduced ordering costs
Higher information quality	Increased accuracy and accessibility of information
Increased operational efficiency	Improved internal operations owing to time and cost reduction and better information management.
Better customer service	Shorter lead times, more timely information about transaction status
Increased ability to compete	Increased ability to reach new markets, and to provide better service at low cost
Improved trading partner relationships	Enhanced trust through increased sharing of information, increased ability to participate in just-in-time program

4.4 E-commerce Technologies

E-commerce architecture and activities are dependent on the following:

- The Internet, intranet and extranet are the network infrastructures or founda of e-commerce.
- Security of information, marketing, transaction process and payment serv must be provided to the customer.
- Trading and business partners rely on the Internet and extranets to exchange accomplish secure transactions using electronic data interchange (EDI).
- Company employees depend on a variety of Internet resources to commun and collaborate in support of e-commerce activities.
- Internet service professionals and end users can use a variety of software too develop and manage contents and operations of the Web sites.

EC Technology Services	Examples
Application services	• Business-to-consumer • Business-to-business • Internal business processes
Brokerage and data management	• Interactive order processing • Secure payment systems • Clearing houses for virtual malls
Interface services	• Interactive catalogs • Directory support functions • Software agents
Secure messaging	• Secure HyperText Transfer Protocol • Encrypted E-mail and EDI • Remote programming interface
Middleware services	• Hypermedia document management • Web/legacy systems interface
Network infrastructure	• Internet, intranet, and extranet • Client/server and other networks

Figure13.3: E-commerce technology services.

4.5 Categories of E-commerce

The Internet, intranet, and extranet provide vital electronic commerce links betwe components of a business, customers, suppliers, and business partners. Three categories of electronic commerce applications are:

- Business-to-consumer (B2C)
- Business-to-business (B2B)
- Internal business processes.

Business-to-Consumer Commerce: In business-to-consumer (B2C) commerce, bu must develop attractive electronic marketplaces to sell products and servi consumers. For example, companies may offer multimedia Web sites that provide

storefronts and virtual shopping malls, interactive order processing, and secure electronic payment systems.

Business-to-Business Applications: Business-to-business (B2B) category of electronic commerce involves both electronic business market places and direct market links between businesses. For example, many companies offer the business community a variety of marketing and product information on the World Wide Web. Others also rely on electronic data interchange (EDI) via the Internet or extranets for direct computer-to-computer exchanges of business documents with their customers and suppliers.

Internal Business Process: All business functions and many business processes are influenced by electronic commerce activities. For example, many inter-networked enterprises are customer-to-customer-driven and market-driven. They continually monitor and evaluate online information of their customers, suppliers, and competitors from their Web sites and discussion groups. This information is available via intranets to all business functions, and is used to shape a company's product development, marketing programs, customer services, and competitive strategies.

4.6 Electronic Payment Systems and Security

Electronic Funds Transfer

Electronic funds transfer (EFT) systems constitute a major form of electronic commerce system in banking and retailing industries. EFT systems use a variety of information technologies to capture and process money and credit transfers between banks and businesses and their customers. For example, banking networks support teller terminals at bank offices and automated teller machines (ATMs). Banks may also support pay-by-phone services allowing bank customers to use their telephones as terminals to electronically pay bills. Major forms of EFT systems are outlined below.

Banking and financial payments
• Large-scale or wholesale payments (e.g., bank-to-bank transfer)
• Small-scale or retail payments (e.g., automated teller machines and cash dispensers)
• Home banking (e.g., bill payment)
Retailing payments
• Credit cards (e.g., VISA, MasterCard etc.)
• Private credit/debit cards (e.g., JC Penney Card)
• Charge cards (e.g., American Express)
Online electronic commerce payments
• Token-based payment systems Electronic cash (e.g., CyberCoin) Electronic Checks (e.g., NetCheque) Smart cards or debit cards (e.g., Mondex Electronic Currency Card)
• Credit card-based systems Encrypted credit cards (e.g., SSL, CyberCash, or SET encryption) Third-party authorization numbers (e.g., First Virtual)

Figure 13.4: Types of electronic fund transfer systems with examples.

Secure Electronic Payments on the Internet

One of the most visible and contentious topics in the Internet commerce today is the security of Internet transactions. While the technical difficulties may be soluble, a less obvious problem is how to define standards to let all consumers and merchants do business with each other while satisfying security requirements of the financial institutions involved.

When a user makes an online purchase on the Internet, his credit card information is vulnerable to interception by network sniffers–software that easily recognizes credit card number formats. Several basic security measures are used to solve security problems:

- Encrypting (code and scrambling) the data passing between the customer and merchant
- Encrypting the data passing between the customer and the company authorizing the credit card transaction
- Taking sensitive information offline.

Many companies use the Secure Socket Layer (SSL) security method developed by Netscape Communicator that automatically encrypts data passing between user Web browser and a merchant's server. However, sensitive information is still vulnerable to misuse once it is decrypted (decoded and unscrambled) and stored on a merchant's server. So a digital wallet approach, such as the Cyber Cash payment system, was developed. In this method one can add security software add-on modules to the Web browser. This enables user's browser to encrypt credit card data in such a way that only the bank that authorizes credit card transactions for the merchant gets to see it.

Another security mechanism is Secure HyperText Transfer Protocol (SHTTP). To ensure a secure conversation between a Web client and a server, SHTTP works by negotiating the type of encryption scheme used between client and server. SHTTP and SSL can both be used for improved security.

The Secure Electronic Transaction, or SET, is another standard for electronic payment security. In this method, e-commerce software encrypts a digital envelope of digital certificates specifying the payment details for each transaction. SET has been agreed to by VISA, Master Card, IBM, Microsoft, Netscape, and most other industry players. Therefore, SET is expected to become the dominant standard for secure electronic payments on the Internet.

Other electronic payment systems include micro-payment systems like DigitalCash and CyberCoin. These technologies create digital currency or digital cash, sometimes called e-cash, for making payments that are too small for credit card transactions. Encryption and authentication techniques are used to generate strings of data that can be handled like currency for making cash payments. For example, Web sites like ESPNET Sports Zone, Discovery Online, and Rocket Science Games let users chat with superstars, download video segments, or play video games by using digital cash micro-payment systems.

4.7 Drawbacks of E-commerce

E-commerce activities include several pitfalls, some are:

- Access, security and privacy problems
- Skill shortage
- Small cash transaction problem
- Stock shortage
- High delivery cost for long distance
- Slow download time owing to poor information infrastructure in many countries
- Very high delivery time for far distances

4.8 Exercise

Multiple Choice Questions

1. E-commerce is business through:
   a. LAN
   b. Physical stores
   c. Internet
   d. None of these

2. Transaction through the Internet as compared to traditional business is
   a. Faster
   b. Similar
   c. Slower
   d. None of these

3. Encryption means
   a. Encoding
   b. Encoding and Scrambling
   c. Scrambling
   d. None of these

Questions for short answers

4. What is e-commerce?
5. What are the benefits of e-commerce?
6. What are the drawbacks of e-commerce?
7. Name at least two security schemes for e-commerce.
8. What is meant by encryption?
9. Define B2B and B2C.

Analytical questions

10. Describe the benefits and drawbacks of e-commerce.
11. Discuss the applications of e-commerce and name at least two large e-eommerce companies.
12. Explain the security in electronic payment systems.
13. Discuss technology infrastructure for e-commerce.

Chapter 14

Computers and Society

Lesson 1: Impact of Computers on Society
Lesson 2: Access, Security and Privacy
Lesson 3: General Maintenance
Lesson 4: Selection of a Microcomputer

Lesson 1: Impact of Computers on Society

1.1 Learning Objectives

On completion of this lesson you will know:

- Interaction between computers and society
- Major causes of dependency on computers
- Impact of computers in the society
- Illegal and controversial computer-related issues
- Challenges for the future computer applications.

1.2 Computers and Society

Storage, processing and dissemination of information play a very important role in modern society. Computers telecommunications, data and computer communications contribute tremendously for these activities. Computers and electronic communications are getting closer and closer every day. Efficiency of production of wealth depends heavily on various kinds of automation, and computers are used to automate methods for processing information. Apart from information processing, computers play important role in many other areas of industry. Computers, electronic communications and other technologies associated with automation come under the general term of information

technology. Information technology influences our lives and its impact on individuals, organizations and society is increasing day by day.

Major technological innovations generally affect the lives of individuals. New inventions lead people and society to view world through new eyes and to accept new values. Apart from benefits, many technological advances have harmful effects in society. This lesson explores the values of technological advances of computers and information technology on society.

1.3 Causes of Dependence on Computers

Present society is heavily dependent on computers and their applications. Among others, the two major causes for this dependence are:

- Ability to calculate and compute
- Handling and processing of information.

Computers can perform calculations at a speed beyond human ability and with accuracy and consistency which people can not match. Some of the knowledge we process today would be unknown without the fast computational ability of computers. The rapid social, economic cultural, and technological changes cannot be imagined without the fast computational speed of present day computers.

Ability to record and store information, process that information at high speed, and reproduce the information in a variety of ways makes the computers indispensable for modern society. Information is generated in such a volume that it is impossible to handle information by conventional methods.

Present world is so much dependent on computers that if they could be swept away overnight, the airlines communications and financial operations of most part of the world would collapse. There are many more examples like these. It is not possible for average persons to appreciate the direct and indirect impact of computers in society. But it is true that computers have penetrated human society far more deeply than the overage persons can realize.

1.4 Impact of Computers

Commerce, industry, education, administration, transportation, communication etc. and many other organizations, professions and individuals depend on computers and their applications directly or indirectly. Impact of computers on few important areas of human activities is discussed briefly.

Commerce: Banking, insurance, credit companies, airlines, shopping centers and many business concerns need computers for their daily operations. They form the majority users of computers. Financial world cannot survive without assistance from computers. Commerce needs quick communication and businesses depend much on computer communications, computer bulletin boards, e-mail, e-commerce and the Internet. Impact of such items can only increase in the future. Financial success of a nation cannot be imagined without the successful use of computers in business activities.

Industry: Many industries such as the steel, chemical and oil companies rely on computers for the actual control of plant processes. This trend can only increase, in the future. Microprocessors are used in harsh environments: in extreme cold and in dirty and noisy atmosphere. Many of the tasks involved in assembly work as in the motor-car, washing machine and television industries are undertaken by processes controlled by microcomputers.

Many sophisticated industrial control and manufacturing processes cannot be imagined without microcomputer boards, embedded microprocessors and robots. Computer aided designing (CAD), computer aided manufacturing (CAM) and computer aided design manufacture and testing (CADMAT) are popular for many industries.

Education: In future much of the traditional teaching will be performed with the help of computers. This form of teaching is known as computer-assisted instruction (CAI) and computer-assisted learning (CAL). Technically this is possible and indeed exists in some countries. In the institutions of higher education the computer has become an indispensable tool for undergraduates and postgraduates for arts, science, business, technology etc. Many of the current projects undertaken by postgraduates would be impractical without computers.

Interactive multimedia, electronic libraries and the information superhighway play a major role in providing a range of interactive courses and services. Thus the educational institutions provide virtual learning environment, besides a physical campus. Such institutions are termed as virtual institutions of learning.

Administration: Many of the administrative tasks are profaned by computers. Word-processing, spreadsheet analysis, database management, pay roll, automatic order processing are some examples. Interpersonal and inter-office messaging, which is popularly known as electronic mail, is essential for quick and effective way of communication for administration. Videoconferencing is a common place for official and administrative activities in some nations. Automatic offices of the future will be dependent on computers, electronic communications and computer networking.

Transportation: Sea and air transportations of today relay on computers and computer communications. Many road transport companies in advanced countries use cellular and satellite communications to prepare optimized routes for the vehicles and for sending instructions to the drivers and operators. Computer controlled traffic systems provide efficient management of traffic in some major cities of the world. Without computers air traffic control would be virtually impossible at the busy airports. Ships in the congested shipping lanes (such as the English channel) are regulated by computer communication through satellites.

Networks and Communications: Information networks provide services to end users through communication links between microcomputers. The major components of computer networks are the central computer, database, the communication links and microcomputers. The end users desiring services can access the database in the central computer. The databases contains data and screens of information that are presented to the users. A great variety of wide area networks and databases already exists for end users. The Internet which is the network of networks wrap around the world and it is the biggest source of information and the most popular means of communication for education, research, business, entertainment and many other social and cultural activities. The future telephone will be incorporated into the microcomputer and one will not only hear but also see the person on other end of the line. The users will be able to pass data and information back and forth as if they were sitting at the same table.

Other Professions: Law, medicine, libraries, armed forces and many other professions and working environments make use of computers and therefore depend upon computers. A good percentage of children at school works directly or indirectly with computers. This emphasizes computer literacy and computer education in the society.

Individuals: Even if we are not directly concerned with computing in our everyday work, as individuals we are affected by them in one way or other. Air passengers, hospital patients, the shoppers in large store are some of the people for whom computers process information. Tax, electricity, telephone or gas bills are processed by computers and salaries are calculated by computers. Indeed, it is difficult to find anyone who in some way or another does not depend on computers.

1.5 Controversies

Society depends very much upon the computational and information processing skills of computers. The benefits of using computers are obvious and have been discussed above. However computer revolution has generated intense controversies. Some of these heated controversies are listed below:

- Privacy and misuse of personal information
- Monitoring data on job activities
- Electronic fund transfer and cashless society
- Effect of automation on job
- Computer crimes.

Privacy: A great deal of data about individuals are stored in computers, and the trend to store such data is increasing. Some of the data are of personal or private nature and there is a natural concern that they should not be misused. Some countries have passed or are about to pass the data protection act and create data protection authority to stop the misuse of personal data and to cheek violation of privacy.

Computer Monitoring: In computer monitoring, computer continuously gather and assimilate data on job activities to measure performance of the employees. Computer monitoring has proved itself a useful tool for increasing productivity. However, workers usually complain that being constantly observed and analyzed by a computer automatically add unnecessary stress to them.

Casheless Society: In electronic fund transfer (EFT) money is transferred electronically from bank to bank and from account to account by computers. Use of automatic teller machines (ATM) and payroll transfer systems are increasing. Some banks offer home banking services through their computers. Money may be eliminated in future resulting in a casheless society! It has some advantages, but the critical issue is the misuse of personal information as the system needs every thing from a person's life style to his or her location to be monitored.

Jobs and Unemployment Issues: Computers usually affect the work of staff within the organization. As a result, some jobs may be created and some may be lost. This creates a demand for training and retraining. Loss of jobs owing to computerization gives rise to alarm. However, only a very small proportion of the unemployment is directly caused by new technology. In certain applications jobs likely to be lost are: some office and factory jobs. Whether job losses result in permanent unemployment is another matter. It depends on the process of reemployment and retraining of the affected persons.

Computer Crimes: There are many types of computer crimes. However, three aspects of computer crimes are often reported and deserve attention because of their social importance. They are:

- Hacking and Computer fraud
- Computer viruses
- Copyright piracy.

The term "hacker" identifies a programmer who works in an skillful but illegal way. The name has become associated with individuals who make a hobby of making unauthorized access to computer systems. This anti-social behavior often leads to loss and inconvenience to the individuals. Legislation has been passed in many countries to make hacking illegal. The technique used by hackers to bypass system security are similar to those used by individuals concerned in computer fraud. Computer systems should be made as safe and secure as possible against hacking, fraud and viruses.

A computer virus is a program which attaches itself to another program. It replicates itself and causes undesirable effect on the programs it becomes attached to. Computer viruses usually infect systems by disks which have been infected. When a virus infected program is run the virus, modifies its host, and also replicate itself. Some viruses are merely annoying. Others cause data to be corrupted or deleted. There are many virus detection and protection packages in use.

Software piracy is the use of unauthorized copies of software. It is like any other breach of copyright and the individuals responsible can face stiff penalties. Many countries, including Bangladesh, passed regulation to prevent software copy right piracy.

1.6 Future Challenge

The outlook for innovative and beneficial computer applications is very bright. The area of applications that are socially, economically and technologically feasible is expanding steadily. Use and expansion of computer networks and on-line database will tremendously influence business, communications, office automation, manufacturing, retail and financial services, publishing, transportation, health care, government and administration, education, entertainment etc. Artificial intelligence will eventually revolutionize computing in the future generation.

Computers can improve the quality of life and contribute to improve our economy. The challenge is to harness the power of computers and direct it to the benefit of society. The poor countries find it difficult to invest money for expanding computer education and computerization. However, careful steps are essential to use computers to improve productivity and socio-economic condition of our society.

1.7 Exercise

Multiple Choice Questions

1. Major two causes of computer dependence are

   a. ability to calculate and compute
   b. intelligence and thinking ability
   c. thinking and processing ability
   d. none of these

2. Hacker is a programmer who works in an skillful but

a. legal way
b. efficient way
c. illegal way
d. none of these

Questions for short answers

3. What are the main components of information technology?
4. Briefly discuss the influence of computers on a) education, b) communication, c) commerce, and d) industry.
5. Briefly discuss the influence of computer networks on society.
6. Explain the term cashless society and virtual institution of learning.
7. List some heated controversies of wide-spread use of computers.
8. What is a computer crime? List common computer crime.
9. Write short notes on a) computer virus, b) hacking and c) copyright piracy.

Analytical questions

10. Discuss the future influence of information technology in society.
11. Discuss critically: Computers are both asset and a danger to the society.
12. What developments are likely to occur in computer technology in the foreseeable future? How will such developments influence Bangladesh society?
13. Discuss how the nature of jobs should change by the influence of information technology.
14. Discuss critically how computers are changing activities in the field of a) education, b) commerce, c) industry, d) communication, and e) transportation.
15. Write a short essay on computer crimes.
16. Explain virtual learning environment and virtual institution of learning.

Lesson 2: Access, Security and Privacy Issues

2.1 Objectives

On completion of this lesson you will know:

- Access control of information systems
- Security of information systems
- Privacy of information

2.2 Introduction

With the introduction of computers, the need for automated tools for protecting files and other information stored in computers became important. This is especially important for a shared system, such as a time-sharing system or a distributed system. In such systems data can be accessed over a public telephone or data network. The authority of a person or organization to some specific data is defined as access. The generic name for the collection of tools designed to protect data and thwart hackers is computer security. On the other hand, the right of persons to control distribution of information is termed as privacy.

In order to assess the access, security and privacy needs of an organization effectively and to evaluate and choose various tools and policies, the responsible manager needs systematic way of defining requirements and characterizing approaches to satisfy the requirements. This is difficult enough in a centralized data-processing environment. With the use of local area and wide area networks the problems are compounded. A list of persons who cause security problems and their objectives are given in Table 14.1.

Table 14.1: Persons who cause security problems and their objectives

Adversaries	Objectives
Sales Representative	Claiming to represent all of the country, not just a city
Businessman	Discovering a competitor's strategic marketing plan
Hacker	Testing out someone's security system; steal data etc.
Accountant	Embezzling money from the company
Con-man	Stealing credit card number for sale
Ex-employee	Getting revenge for being fired
Student	Having fun with other's security, for instance, e-mail
Stockbroker	Denying a promise made to a customer by e-mail.

2.3 Access Control

Various means have been devised to give access to authorized people. These means fall

into four broad categories:

- ID card or key
- Password or ID number
- Sign document
- Biometrics.

The user may have a key, badge, token, or plastic card to give him or her physical access to the computer room or a locked-up terminal. A card with a magnetized strip, for example, can give the user access to the bank account via a remote cash machine. Taking a step further, some employees begin each business day by donning an active badge, a clip-on identification card with an embedded computer chip. The active badge, which is becoming increasingly common, presents a challenging problem: balancing an employee's privacy against a corporation's desire for efficiency and control.

Password and ID number: Standard items are system password or identification number for bank cash machine. Cipher locks on doors require that a user knows the correct combination of numbers.

Sign document: Sign documents provide identification of a person. Though a signature is difficult to copy, forgery is not impossible. For this reason, signatures lend themselves to human interaction better than machine interaction.

Biometrics: Some security systems use biometric devices which identify a person through finger print, voice intonation or other biological characteristics. Even newer is the concept of identification by the retina of the eye, which has a pattern that is harder to duplicate.

Some systems use a combination of the preceding categories. For example, access to an automated teller machine requires both something like a plastic card and a personal identification number (PIN).

2.4 Security

Security is a system of safeguards designed to protect a computer system and data from deliberate or accidental damage or access by unauthorized persons. Network security measures are needed to protect data during their transmission. Basic security issues of different information systems are described in this section.

Software Security

Software security has been an industry concern for years. Initially, there were many

questions, for instance, who owns custom-made software? Is the owner the person who wrote the program or the company for which the author wrote the program? What is to prevent a programmer from taking copies of programs from one job to another? The answers to these questions are well established. If the author of the software – the programmer-is an employee of an organization, the software belongs to the organization, not the programmer. The programmer may not take the software along to the next job. If the programmer is a consultant, however, the ownership of the software produced should be spelled out specifically in the contract.

Data Security

There are five critical planning areas for data security:

- Development and implementation of security safeguards, such as passwords
- Determination of appropriate policies and standards to protect data against unauthorized disclosure, alteration or destruction
- Inclusion of security precautions at the development stage of new automated systems, rather than after the systems are in use
- Review of state laws related to security
- Maintenance of historical records associated with computer abuse.

There are several data protection techniques; these will not individually (or even collectively) guarantee security, but at least they make a good start.

Internal Controls: Internal controls are controls that are planned as part of the computer system. One example is a transaction log. This is a file of all accesses or attempted accesses to certain data.

Passwords: A password is a secret word or number, or a combination of the two, that must be typed on the keyboard to gain access to a computer system. Good data protection systems change passwords often and also compartmentalize information by passwords, so that only authorized persons can have accesses to certain data. Cracking passwords is the most prevalent method of illicit entry to computer systems.

Secured Waste: Discarded printouts, printer ribbons, and the like can be sources of information to unauthorized persons. This kind of waste can be made secure by the use of shredders or locked trash barrels.

Auditor Checks: Most companies have auditors and in the course of their duties, auditors frequently review computer programs and data. From a data security standpoint, auditors might also check to see who has accessed data during periods when that data is not usually used. They are also on the lookout for unusual numbers of corrected data

entries, usually a trouble sign.

Separation of Employee Functions: By limiting employee duties so that doubling up on job functions is not permitted, a computer organization can restrict the amount of unauthorized access. That is, in an installation where the computers - mainframes or minis - are behind locked doors, only operators have physical access to them. Unfortunately, separation of functions is not practical in a small shop; usually one or more employees perform multiple functions. And, of course, separation of functions does not apply in a personal computer environment.

Built-in Software Protection: Software can be built into operating systems in ways that restrict access to the computer system. One form of software protection system matches a user number against a number assigned to the data being accessed. If a person does not get access, it is recorded that he or she tried to tap into some area to which they were not authorized. Another form of software protection is a user profile: information is stored about each user, including the files to which the user has legitimate access. The profile also includes each person's job function, budget number, skills, areas of knowledge, access privileges, supervisor, and loss-causing potential. These profiles are available for checking by managers if there is any problem.

Network Security

Many people have access to the network system, often from remote locations. However, to begin with, network operating systems provide basic security features, such as user identification and authentication, probably by password. Sophisticated network systems can permit network supervisors to assign varying access rights to individual users. All users, for example, could access word processing software, but only certain users could access payroll files. Some network software can limit how many times users can call up a particular file and generate an audit trial of who looked at which files. One fundamental approach to network security is to dedicate one computer, called a firewall, whose sole purpose is to talk to the outside word. A firewall will provide an organization with greatly increased security because only one network computer is accessible to people outside the network, and that one computer accepts only appropriate access. Among its other chores, a firewall computer can call back all remote - access terminals.

In addition to monitoring access to the network, organizations must be concerned about unauthorized people intercepting data in transit, possibly thieves or industrial spies. Data being sent over communication lines may be protected by scrambling the messages - that is, putting them in code that can be broken only by the person receiving the message. The process of scrambling messages is called encryption. The American National Standards

Institute has endorsed a process called the Data Encryption Standard (DES), a standardized public key by which senders and receivers can scramble and unscramble messages. Although the DES code is well known, companies still use it because the method makes it quite expensive to intercept coded messages. Encryption software is available for personal computers. A typical package, for example, offers a variety of security features: file encryption, keyboard lock, and password protection.

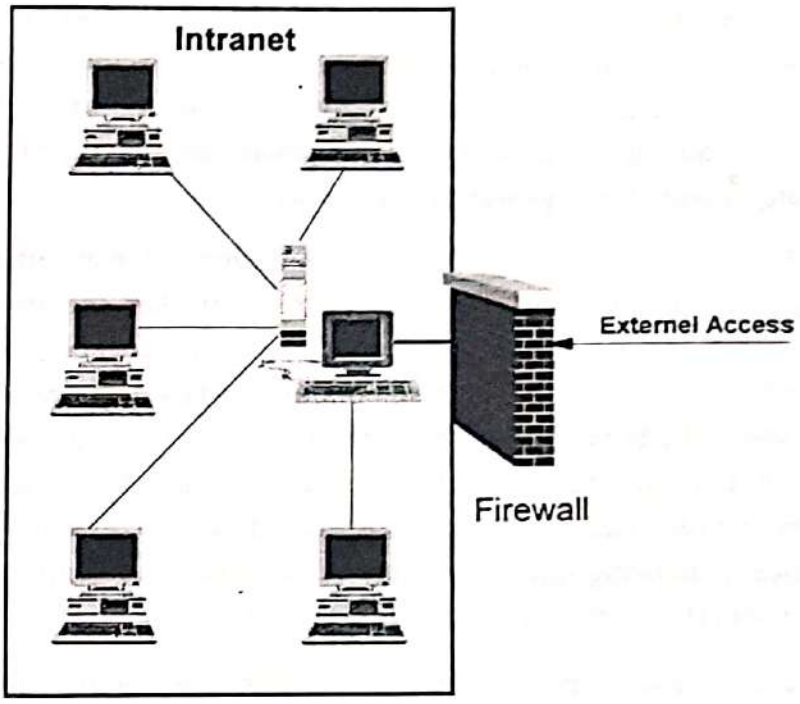


Figure 14.1: A firewall protecting an intranet

Personal Computer Security

In addition to theft, personal computer users need to be concerned about the computer's environment. Personal computers in business are not coddled the way bigger computers are. They are designed, in fact, to withstand the wear and tear of the office environment, including temperatures set for the comfort of people. Most manufacturers discourage eating and smoking near computers and recommend some specific cleaning techniques, such as vacuuming the keyboard. The response to these recommendations are directly related to the awareness level of the users. Most personal computer data are stored on diskettes, which are vulnerable to sunlight, heaters, cigarettes, scratching, magnets, theft, and dirty fingers. The data are vulnerable as well. Hard disks used with personal computers are subject to special problems too. If a computer with a hard disk is used by more than one person, files on the hard disk may be available for anyone to browse through.

Several precautions can be taken to protect disk data. One is to use a surge protector, a device that prevents electrical problems from affecting data files. The computer is plugged into the surge protector, which is plugged into the outlet. Diskettes should be under lock and key. The most critical precaution, however, is to back up files.

2.5 Privacy

Privacy can be defined as the right of persons to control the distribution of information about themselves. By definition, invasion of privacy would include collection or distribution of information in such a way that the person to whom it pertains finds it objectionable. Although it is assumed that individuals have the right to privacy, that right is not explicitly stated in any government document.

In developed countries, business is highly computerized and many citizens use electronic means to pay for purchases, the buyer's every step is tracked and analyzed. When some one purchase an item over the phone with his credit card, chances are good that his/her purchase is added to a record of the types of things. Marketers can amass quite a bit of information about the buyer simply from tracking the purchasing history and combining it with other public information, such as address, occupation and age. A company can learn a lot about purchaser also by matching records and combining facts from different databases, such as banking records and payroll records. All this information can then be used to target specific marketing information to the buyer.

Many people view such comprehensive data gathering as an invasion of privacy. They cite the potential for the information to be used inappropriately or for purposes unrelated to that for which the information was originally disclosed. For instance, it may be possible to correlate a medical database with an insurance database to learn the names of policy holders who had a positive result on a particular medical test. The fear is that the insurance company could then cancel the policy based on the test.

Buying and selling data and matching services are a big business. Although most countries have laws to prohibit the release and cross-matching of information in government agency database, there are no corresponding laws to cover data kept in private sector databases.

Another important concern is e-mail privacy. Millions of employees routinely use electronic mail to send messages to colleagues in their companies and to others outside the firms. Recent court decisions (in USA) ruled that this mail is not private and that company personnel may review messages to be certain that they pertain to company business. Some individual believes that e-mail should have the same privacy protection as postal mail.

2.6 Relevant Issues

Disaster, Losses and Recovery Plan

Initial considerations of a user might be of the hardware, that is, the computer and its related equipment. But loss of hardware is not a major problem in itself; the loss will be covered by insurance, and hardware can be replaced. The true problem with hardware loss is the diminished processing ability that exists while managers find a substitute facility and return the installation to its former state. The ability to continue processing data is critical. Some information industries, such as banking, could literally go out of business in a matter of days if their computer operations were suspended. Loss of software should not be a problem if the organization has heeded industry warnings and used common sense - to make backup copies of program files. A more important problem is the loss of data. Imagine trying to reassemble lost or destroyed master files of customer records, accounts receivable, or design data for a new airplane. The costs would be staggering. The disaster recovery plan should include the following items:

- **Priorities:** A list of priorities identifies the programs that must be up and running first. A Bank, for example, would give greater weight to account inquiries than to employee vacation planning.
- **Personnel requirements:** The plan should comprise procedures for notifying employees of changes in locations and procedures.
- **Equipment requirements:** A list of needed equipment and where it can be obtained will speed recovery efforts.
- **Facilities:** Most organizations cannot afford consortiums, so the recovery plan should include a list of alternative computing facilities.
- **Capture and distribution:** This part of the plan outlines how input and output data will be handled in a different environment.

Computer installations actually perform emergency drills. At some unexpected moment a notice is given that "disaster has struck," and the computer professionals must run the critical systems at some other site.

Worms and Viruses

A worm is a program that transfers itself from computer to computer over a network and plants itself as a separate file on the target computer's disks. A virus, as its name suggests, is contagious. That is, a virus, a set of illicit instructions, passes itself on to other programs with which it comes in contact. A virus may be dealt with by means of a vaccine, or anti-virus program, a computer program that stops the spread of and often eradicates the virus.

Transmitting a Virus: Although many viruses are transmitted over bulletin boards, the most common method is by passing diskettes from computer to computer. More

insidious viruses attach to the operating system. Viruses attached to the operating system itself have greater potential for mischief.

Damage from Viruses: In addition to simply displaying a message, many viruses do significant damage, often including destruction of files. Most viruses remain dormant until triggered by some activity. For example, a virus called Jerusalem B activates itself every Friday the 13th and proceeds to erase any file user may try to load from the disk. Another virus includes instructions to add 1 to a counter each time the virus is copied to another disk. When the counter reaches 4, the virus erases all data files. But this is not the end of the destruction, of course; the three copied disks have also been infected.

Prevention: Viruses are most commonly passed via diskettes. In addition to that, viruses use many other means to propagate - bulletin boards, local area networks, and electronic mail. If a personal computer has a disk drive, a modem, or a network connector, it is vulnerable. Furthermore, viruses are rampant on some college campuses, a source of considerable annoyance to students. Use the following common sense approaches to new files:

- Never install a program unless the diskette comes in a sealed package.
- Be especially wary of software that arrives unexpectedly from companies with whom one has not done business.
- Use virus-scanning software to check any file, no matter what the source, before loading it onto hard disk.
- If diskette was used in another computer, scan it to see if it caught a virus.

Although there have been isolated instances of viruses in commercial software, viruses tend to show up on free software acquired from friends or through electronic bulletin board systems. Anti-virus software can be installed to scan hard disk every time while booting the computer or, if possible, at regularly scheduled intervals.

Back Up Files

Although organizations recognize the value of data and have procedures in place for backing up data files on a regular basis, personal computer users are not as devoted to this activity. In fact, one wonders why, with continuous admonishments and readily available procedures, some people still leave their precious files unprotected.

Users could use software incorrectly or simply input data incorrectly; it may be some time before the resulting erroneous data are detected, and user need to go back to the time when the data files were still acceptable. Sometimes the software itself can harm data. A hard disk could physically malfunction making files inaccessible. Although none of these are too likely, they certainly do happen. It is even less likely that user lose his hard disk files to fire or flood, but this is also possible. It is most likely that user will accidentally delete some files himself/herself. Finally, there is always the possibility of

user files being infected with a virus. Experts estimate that average users experience a significant disk loss every year.

Some people simply make another copy of their hard drive files on diskette. This is not too laborious. If a user is at all vulnerable to viruses, he/she should back up all the files on a regular basis. A better way is to back up all files on a tape. One can also use software that will automatically back up all files at a certain time of day, or on command. Sophisticated users place their files on a mirror hard disk, which simply makes a second copy of everything the user puts on the original disk; this approach is expensive. It is better to keep backed up files in a cool, dry place off site.

2.7 Exercises

Multiple Choice Questions

1. Which part of human body is used for access control?

   a. hair c. nose
   b. eye retina d. finger

2. Which is transmitted through the Internet?

   a. virus c. worms
   b. bugs d. none of these

3. Backup of database keeps

   a. similar copy c. concise history of data
   b. names of files d. none of these

Questions for Short Answers

4. Define access, security and privacy of information systems.
5. What are the categories of access control?
6. What are the critical planning areas for data security?
7. What are the items one should include in the disaster recovery plan?
8. Briefly describe the methods to prevent viruses and worms of information systems.

Analytical Questions

9. What is meant by access control? Explain the methodology for access control in information systems?
10. What is meant by security of information systems? Discuss the methods of network security.
11. Describe the disaster, losses and recover plans of an information system.
12. What is meant by virus and worm? Discuss the necessity to make backup for data security.

Lesson 3: General Maintenance

3.1 Learning Objectives

On completion of this lesson you will know:

- Different factors which affect computers
- Diagnosis of common problems
- Precaution against viruses
- Preventive maintenance

3.2 Factors Affecting Computers

Common physical factors affecting computers are heat, dust, water, corrosive agents, power line problems and magnetism. Each of them may cause the breakdown of the computer system.

Heat

It is one of the most destructive factors that damages computers. An electronic device dissipates about 50% of its power as heat energy. Heat should be removed as quickly as it is produced. Heat can be checked by installing fan and running the computer in a safe temperature range. Ambient temperature range for computers should be between 60^0F and 85 ^{0}F. Another source of heat is rays of sun. A computer should be installed in a cool place away from sun rays.

Dust

Dust acts as a thermal insulation on the board. It prevents heat to radiate from inside the computer. A blower with compressed air can be used to remove dust. A dust free cloth wetted with water should not be used to clean the circuit board. Moreover, dust can clog air intake area to power supply or hard disk drive and space between the floppy disk drive head and the disk. To avoid these problems a dust cover should be used when the machine is not used. Smoking should always be avoided near computers. The printers should be periodically cleaned by blowing out paper dusts.

Water

Water or liquid should not be allowed near computers and drinking should be avoided near computers. In case of accidental spilling of water or liquid on the computer or keyboard, the computer should be disassembled, cleaned, dried and then reassembled.

Corrosive Liquids

Salt in sweat and cleaning fluids are corrosive. Oxidation of circuit contacts is another problem. When a connector becomes oxidized, it does not function properly. Tea and coffee contain tannic acids, carbonated liquids have carbonic acid. Drinking tea or coffee should be avoided near computers.

Powerline Problems

Good and clean power supply is the most important environmental factor for a computer. Common power line problems are:

- Brownouts
- Blackouts
- Transients and
- Noise.

Brownouts are sags for lower voltages, blackouts are the total loss of line voltage, transients are large potentially damaging spikes of voltages or currents and noise is any unwanted electrical disturbance in the power line. There are two kinds of approaches to remove powerline problems. They are:

- Condition the power being supplied and
- Providing an auxiliary or backup power source.

Voltage stabilizers are normally used to condition electrical power against brownouts, transients and noise. The uninterrupted power supply (UPS) stores energy when the line voltage is present and then delivers power to the computer when a blackout occurs. The UPS should be used when it is important to save data and files against total loss of power.

Magnetism

Magnetism causes permanent loess of data on hard or floppy disks. Magnetic field is present around phones, electric motors, stereo speakers, cathode ray tubes (CRT) etc. Stray elector-magnetism like radiated electro-magnetic interference (EMI), power line noise (due to overvoltage, undervoltage, transient spike and surge) and electrostatic discharges can cause problems for computer.

Computer Virus

A computer virus is a piece of software which infects data or disks and has multiplication capability of itself. They affect computers by

- Deleting files
- Corrupting screen displays
- Corrupting data
- Slowing down operation of a system,
- Displaying unwanted messages etc.

A sophisticated virus can spread undetected for a long time. A boot virus attacks boot sector of a disk. Viruses affect files with extensions EXE, COM, SYS, DRV, OVR, OVI, BIN. Some common viruses are Jerusalem, Stone (Marijuana), Dhaka virus, Vienna, CIH virus, April First etc.

There are many antivirus programs which detect clean and prevent viruses. Some antivirus programs are Norton Anti-virus (NAV), Central Point Anti-Virus (CPAV), Toolkit, Macaffe etc. For virus prevention, antivirus programs are used to detect and disinfect viruses. Floppy disks should not be used unless they are checked or reformatted. Backups of data should be maintained against virus attacks.

Preventive Maintenance

To avoid servicing cost one should perform the preventive or routine maintenance of computers. But in case of complicated failures of computers the experienced engineers or the suppliers should be contacted.

Common factors affecting computers are heat, dust, water, power-line problems, magnetism, corrosive liquids etc. Temperature of the computer room should be between 65 ^{0}F and 85 ^{0}F. The room should be dust free and drinking near computers should be avoided. A computer should be protected against static electricity and magnetism. Stabilizers or uninteruptable power supplies should be used against power-line problems. Antivirus programs should be used to scan, detect, clean and prevent computer viruses.

3.3 Exercise

Multiple Choice Questions

1. Which one should be used to remove dust?

   a. wetted cloths c. dry soft cloth
   b. blower d. tissue paper

2. Which one may oxidize the connection in the circuit board?

   a. water c. air
   b. sugar d. dust

Questions for short answers

3. List the physical factors that affect the performance of a computer.
4. Explain the terms blackout, brownout, transient and EMI.
5. Distinguish between a stabilizer and a UPS.
6. Give advantages of preventive maintenance.

Analytical questions

7. Explain how dust affects the performance of computers. What precautions should be taken against dust?
8. What is a computer Virus? How does it affect computer systems? Discuss virus protection measures.
9. Why clean supply of power is important for computers? Explain the power line problems that affect computes.
10. What measures should be adopted to prevent computers against power line problems?

Lesson 4: Selection of Microcomputers

4.1 Learning Objectives

On completion of this lesson you will know the criteria for selection of:

- Microprocessors for microcomputers
- Primary and secondary memories
- Monitors and cards for monitors
- Operating system and application software.

4.2 Components of Microcomputer

Selection of a microcomputer means selection of different components of a microcomputer. Most important components are processor, memory (cache, hard disk and floppy disk), monitor, operating system and application software.

Choosing a Processor

The microprocessor is the central component of a microcomputer and it determines the main features. It is also a major factor in determining the computer's price as the other components of the computer must be compatible to it.

In general, the higher the processor's clock speed, the faster the computer. However, clock speed comparisons make sense only when user looking at one brand and model of processors. For example, a Pentium III running at 750 MHz is faster than one with 650 MHz.

For selecting the PC, it is important to consider the role of the computer, both initially and in the longer term. It is always better to buy high-performance, bare-bone systems having upgrade potential.

Main Memory

Memory chips have become inexpensive in recent years and tendency of using larger programs are increasing. New versions of programs seem to grow exorbitantly requiring huge disk space and large main memory. Thus more main memory for both RAM and cache, is the better option. When buying a computer it is important to look out for slots on the motherboard for extra RAM, otherwise adding memory later becomes relatively expensive.

Program manufactures usually give the minimum requirements for the programs. Below the specified limit, a program won't run. For satisfactory operation, considerably more memory is almost always necessary. However, determining the actual amount requires some experience with the particular program. It is better to check with the dealer, consult computer journals, or ask an experienced user.

In almost every PC, there is some facility to expand the main memory. PC motherboard manufactures usually equip their products with special sockets and sell standard memory modules for memory expansion.

Hard Disk Drive

As the data transfer rates have been standardized, the essential selection criterion should be the smallest possible average access time. IDE and EIDE disks were once hot sellers and are still quite popular. However, there are better alternatives now.

The special properties of the SCSI interface make it the model of choice now because of speed and higher disk capacity. Single-user computers may be equipped with an IDE disk. This disk combines high speed and advanced drive technology at a reasonable price. Attention should be paid to the average access time. Access times greater than 20 ms are no longer attractive. The controller and the hard disk must match.

The size for the hard disk depends on the software and the type of work to be done. The size should be about 40 GB plus. Sufficient space should be available for normal office applications at this capacity. Programs that use graphic interfaces often require large capacity.

Floppy Disk Drive

The 3.5-inch floppy disks are smaller, convenient and they have higher storage capacity than 5.25-inch floppy disks. A computer with a single drive, should have a 3.5-inch with 1.44MB capacity. Difference can be observed between the 3.5-inch drives of various manufactures. The mechanism for ejecting the floppy disks is poorly designed in some disk drives. Inserting and ejecting a floppy disk should be tried before buying a disk drive. When the floppy disk drive is opened, the disk should come out just far enough to be removed easily. The loudness of the drive should also be checked. Some drives make noise particularly when formatting a floppy disk.

CD-ROM and DVD drive

A CD-ROM drive's speed rating refers to how fast the drive can transfer data. For instance, a 24X drive transfer data at 3,600kbps or 3.6MB per second. The faster CD-

ROM drives are especially useful for playing audio and video files. A revised format for CD-ROM disks, called DVD (Digital Video Disk) has increased the capacity of CD-ROM size disk to 4.7GB or more.

Selecting Monitor

The video card and monitor form a unit and they must match. This applies not only to the type (VGA card for VGA monitor) but also for the technical characteristics.

Recommendations for monitors are similar to those for video cards. Basic equipment today should be at least a VGA monitor with a resolution of 1024 × 768, and a refresh rate greater than 60 Hz, in non-interlaced mode. Higher resolution are important for bigger monitors.

Currently, systems offer a special bus design that directly connects the video circuits with the microprocessors, increasing performance considerably. AGP or Accelerated Graphics Port is one of the best system, which transfers video data much more quickly than the standard PCI interface. To display 1600X1200 resolution with a color palette of 16.7 million colors, for instance, the video card should be equipped with 8MB of Video RAM (or VRAM).

The monitor's dot pitch is another important issue to consider. One should not select monitor with a dot pitch larger than 0.28mm and the smaller is the better.

Monitor quality can be imagined by displaying dark letters against a light background. The display should not flicker, a non-interlaced monitor refreshes the entire screen with each pass of the cathode gun without flicker. Attention should be paid to the lowercase letters e, s, and a. There should not be a ring around the individual letters. It is good to have a demonstration of the monitor. Several hours of work at an unsuitable monitor can bring on headache for the user.

The interaction between the video card and the screen or monitor is important. SVGA cards and their corresponding monitors have become inexpensive now and the older graphics standards should not be recommended. A SVGA card can operate both with a monochrome or a colour screen.

Any video card should have a refresh rate of more than 60 Hz in graphics mode. The advanced graphics applications require refresh rates of 70 Hz.

Keyboard and Mouse

Most computers come with standard keyboard and mouse. However, some one worried

about repetitive stress injury, can consider upgrading to an ergonomic keyboard, such as the Microsoft natural Keyboard. On the other hand, any mouse that supports Microsoft IntelliMouse standard includes a wheel, which enables to scroll through document with ease.

Sound card and speaker

Sound cards and speakers are most important components to take full advantage of multimedia capabilities. One should select a standard sound card that offers wavetable synthesis and which uses stored samples as well as PCI interfaces. Higher quality branded speakers with amplifier and volume control should be selected for higher sound quality.

Network cards

In order to connect the computer to a network, a network interface card (NIC) is essential. Some common NIC cards are 10Mbps (10 baseT) or 100Mbps (100 baseT) cards.

Modem

To log on to the campus network or to the Internet by means of a telephone connection, one has to select a modem with maximum baud rate. Todays standard is 56kbps V.90 protocol.

4.3 Application Software

Current computer magazines should be consulted to gather information about a given program before buying it. There are special interest magazines dealing only with freeware and shareware programs. Dealers and the forums of on-line services, if available, should be consulted for information. Application areas and the extent of use of the program should be considered when selecting software. Sometimes public domain or shareware products suffice for private use. For businesses, a commercial product is generally preferred, so that when problems do occur, it can be removed quickly. It is, of course, a great advantage if a knowledgeable dealer is there to help immediately. However, the software manufacturer is ultimately responsible for supporting its own product.

Some publishing houses produce market overviews of software programs. PC Magazine, PC World, and Info World and other local computer magazines are good source of such information. In Some magazines short descriptions of the software and hardware, the vendor's addresses, and prices, all classified by type of applications are provided. Also some books are available for choosing the right program for an application.

4.4 Selection Chart

In order to check the main features of a microcomputer or for selection of a microcomputer simple chart can be designed. Filling the blanks in the chart (Figure 14.2) with specifications provided for the computer can help to compare and select a microcomputer.

	Vendor 1	Vendor 2	Vendor 3	Vendor4
Manufacturer				
Model				
Case				
Microprocessor				
Clock speed				
BIOS				
RAM				
Cache memory				
Number of slots				
Floppy Disk Drive				
Hard Disk Drive				
Display Adapter				
Monitor				
Mouse				
Keyboard				
CD-ROM Drive				
Modem				
Network Card				
Other adapters				
Operating System				
Application Software				

Figure 14.2: Microcomputer selection chart.

Once software is selected several dealers should be asked about it. There are considerable price differences in extreme cases.

4.5 Exercise

Multiple Choice Questions

1. If the processors are of the same brand and model, what is the basis to compare processors?

   a. number of transistors in the processor
   b. size of the processor
   c. clock rate
   d. none of these

2. What are main components to be considered while selecting the main memory?

   a. RAM and ROM
   b. Cache and ROM
   c. RAM and cache
   d. None of these

3. AGP stands for

   a. Audio Graphics Port
   b. Accelerated Graphics Port
   c. Automatic Graphics Program
   d. None of these

4. Which one is flicker-free monitor?

   a. Interlaced
   b. LCD
   c. Non-interlaced
   d. None of these

5. DVD stands for

   a. Digital Video Disk
   b. Digital Video Device
   c. Digital Video Drive
   d. None of these

Questions for short answers

6. Which one is the most widely used processor in the current market?
7. How can you increase memory of your computer?
8. Differentiate between interlaced and non-interlaced monitor.
9. Differentiate between CD-ROM and DVD

Analytical questions

10. What are the considerations for selecting a hard disk?
11. What are the considerations for selecting an application software?
12. Discuss the criteria for selecting a monitor and a printer.
13. Discuss the importance of microcomputer selection chart. Prepare a selection chart for purchasing a computer for office use.

Appendix 1

ASCII Code

Symbol	ASCII	Symbol	ASCII	Symbol	ASCII
(space)	0100000	?	0111111	^	1011110
!	0100001	@	1000000	_	1011111
"	0100010	A	1000001	a	1100001
#	0100011	B	1000010	b	1100010
$	0100100	C	1000011	c	1100011
%	0100101	D	1000100	d	1100100
&	0100110	E	1000101	e	1100101
–	0100111	F	1000110	f	1100110
(	0101000	G	1000111	g	1100111
)	0101001	H	1001000	h	1101000
*	0101010	I	1001001	i	1101001
+	0101011	J	1001010	j	1101010
'	0101100	K	1001011	k	1101011
-	0101101	L	1001100	l	1101100
.	0101110	M	1001101	m	1101101
/	0101111	N	1001110	n	1101110
0	0110000	O	1001111	o	1101111
1	0110001	P	1010000	p	1110000
2	0110010	Q	1010001	q	1110001
3	0110011	R	1010010	r	1110010
4	0110100	S	1010011	s	1110011
5	0110101	T	1010100	t	1110100
6	0110110	U	1010101	u	1110101
7	0110111	V	1010110	v	1110110
8	0111000	W	1010111	w	1110111
9	0111001	X	1011000	x	1111000
:	0111010	Y	1011001	y	1111001
;	0111111	Z	1011010	z	1111010
<	0111100	[	1011011	{	1111011
=	0111101	/	1011100	}	1111101
>	0111110	]	1011101		

Appendix 2

Decimal, Binary, Octal and Hexadecimal Numbers

Decimal	Binary	Octal	Hexadecimal	Decimal	Binary	Octal	Hexadecimal
0	0	0	00	42	101010	52	2A
1	1	1	01	43	101011	53	2B
2	10	2	02	44	101100	54	2C
3	11	3	03	45	101101	55	2D
4	100	4	04	46	101110	56	2E
5	101	5	05	47	101111	57	2F
6	110	6	06	48	110000	60	30
7	111	7	07	49	110001	61	31
8	1000	10	08	50	110010	62	32
9	1001	11	09	51	110011	63	33
10	1010	12	0A	52	110100	64	34
11	1011	13	0B	53	110101	65	35
12	1100	14	0C	54	110110	66	36
13	1101	15	0D	55	110111	67	37
14	1110	16	0E	56	111000	70	38
15	1111	17	0F	57	111001	71	39
16	10000	20	10	58	111010	72	3A
17	10001	21	11	59	111011	73	3B
18	10010	22	12	60	111100	74	3C
19	10011	23	13	61	111101	75	3D
20	10100	24	14	62	111110	76	3E
21	10101	25	15	63	111111	77	3F
22	10110	26	16	64	1000000	100	40
23	10111	27	17	65	1000001	101	41
24	11000	30	18	66	1000010	102	42
25	11001	31	19	67	1000011	103	43
26	11010	32	1A	68	1000100	104	44
27	11011	33	1B	69	1000101	105	45
28	11100	34	1C	70	1000110	106	46
29	11101	35	1D	71	1000111	107	47
30	11110	36	1E	72	1001000	110	48
31	11111	37	1F	73	1001001	111	49
32	100000	40	20	74	1001010	112	4A
33	100001	41	21	75	1001011	113	4B
34	100010	42	22	76	1001100	114	4C
35	100011	43	23	77	1001101	115	4D
36	100100	44	24	78	1001110	116	4E
37	100101	45	25	79	1001111	117	4F
38	100110	46	26	80	1010000	120	50
39	100111	47	27	81	1010001	121	51
40	101000	50	28	82	1010010	122	52
41	101001	51	29	83	1010011	123	53

Appendix 3

Abbreviations and Acronyms

3GL	Third Generation Language
4GL	Fourth Generation Language

A

ACD	Automatic Call Distributor
ACK	Acknowledgement
ADT	Abstract Data Type
ADSL	Asymmetric Digital Subscriber Line
AGP	Advanced Graphics Port
AI	Artificial Intelligence
ALGOL	ALGOrithmic Language
ALU	Arithmetic Logic Unit
AM	Amplitude Modulation
AMD	Advanced Micro Devices
AMP	Asymmetric Multi Processing
ANSI	American National Standards Institute
APDU	Application Protocol Data Unit
API	Application Program Interface
APT	Automatically Programmed Tools
ARPANET	Advanced Research Project Agency NETwork
ASCII	American Standard Code for Information Interchange
AT	Advanced Technology
AT&T	American Telephone and Telegraph
ATM	Automatic Teller Machine, Asynchronous Transmission Mode

B

BASIC	Beginner's All-purpose Symbolic Instruction Code
BBS	Bulletin Board System
BCD	Binary Coded Decimal
BCS	Bangladesh Computer Society Bangladesh Computer Samity
BERNET	Bangladesh Education and Research NETwork
BIND	Berkely Internet Name Domain
BIOS	Basic Input Output System
BIP	Business Information Processing
BISDN	Broadband Integrated Services Digital Network

Bit	Binary digit
BOOTP	BOOTstrap Protocol
bps	Bits per second

C

CAD	Computer Aided Design
CADMAT	Computer Aided Design Manufacture and Testing
CAI	Computer Aided Information
CAL	Computer Aided Learning
CAM	Computer Aided Manufacturing
CAS	Content Addressable Storage
CASE	Computer Aided Software Engineering
CATV	Community Antenna Television
CBT	Computer Based Training
CCD	Charge Coupled Device
CCITT	Consultative Committee of International Telegraphy and Telephony (Now called ITU-T)
CD	Compact Disk
CD-ROM	Compact Disk-Read Only Memory
CGA	Colour Graphics Adapter
CISC	Complex Instruction Set Computer
CMMU	Cache Memory Management Unit
CMOS	Complementary Metal Oxide Semiconductor
COBOL	Common Business Oriented Language
CODASYL	COmmittee on DAta SYstems Language
COMDEX	COMputer Dealers EXposition
CP/M	Control Program for Microcomputers
cpi	Characters Per Inch
CPU	Central Processing Unit
CR	Carriage Return
CRC	Cyclic Redundancy Check
CREN	Consortium for Research and Education Network
CRT	Cathode Ray Tube
CSMA/CD	Carrior Sense Multiple Access with Collision Detection
CSPDN	Circuit Switched Public Data network
CSR	Connected Speech Recognition
CU	Contorl Unit
CWR	Connected Word Recognition

D

DASD	Direct Access Storage Devices
DAT	Digital Audio Tape
DAV	Digital Audio Video

DBA	DataBase Administration
DBMS	DataBase Management System
DCA	Document Content Architecture
DCE	Data Circuit-termination Equipment
DCL	Data Control Language
DDE	Direct Data Entry, Dynamic Data Exchange
DDL	Data Description Language, Dynamic Data Library
DDS	Digital Data Service
DEC	Digital Equipment Cerporation
DFD	Data Flow Diagram
DIP	Dual In-Line Package
DLC	Data Link Control
DMA	Direct Memory Access
DML	Data Manipulation Language
DMS	Digital Metropolitan Service
DNS	Domain Name System
DOS	Disk Operating System
DP	Data Processing
dpi	dots per inch
DQDB	Distributed Que Dual Bus
DR	Data Register
DRAM	Dynamic Random Access Memory
DSL	Data Sub-Language
DSP	Digital Signal Processor
DSS	Decision Support System
DTE	Data Terminal Equipment
DTP	DeskTop Publishing

E

E-mail	Electronic mail
EAR	Entity-Attribute-Relationship
EBCDIC	Extended Binary Coded Decimal Interchange Code
ECMA	Electronic Data Interchange For Administration, Commerce and Transport
EDP	Electronic Data Processing
EDVAC	Electronic Discrate Variable Automatic Computer
EEPROM	Electrically Erasable Programmable Read Only Memory
EFT	Electronic Funds Transfer
EFTPOS	Electronic Funds Transfer at POint of Sale
EGA	Enhanced Graphics Adapter
EIA	Electronic Industries Association
EISA	Enhanced Industry Standard Architecture
EMI	Electro-Magnetic Interference
EMS	Expanded Memory Specification

ENIAC	Electronic Numerical Integrator and Calculator
EOF	End-of-File
EPROM	Erasable and Programmable Read Only Memory
ESDI	Enhanced Small Device Interface
EU	End-User
EWP	Electronic White Page

F

F. Keys	Function Keys
FAQ	Frequently Asked Questions
FAST	Federation Against Software Theft
FAT	File Allocation Table
Fax	Facsimile
FCC	Federal Communications Commission
FCS	Frame Check Sequence
FDC	Floppy Disk Controller
FDD	Floppy Disk Drive, Fixed Disk Drive
FDDI	Fibre Distributed Data Interface
FDM	Frequency Division Multiplexing
FEA	Finite Eliment Analysis
FEP	Front-End Processor
FF	Form Feed, Flip-Flop
FGL	Fourth Generation Language
FIFO	First in First Out
FM	Frequency Modulation
FORTRAN	FORmula TRANslation
FRAM	Ferromagnetic Random Access Memory
FSK	Frequency Shift Keying
FTAM	File Transfer Access and Management
FTP	File Transfer Protocol

G

GB	Giga Byte
GHz	Giga Hertz
GIGO	Garbage In Garbage Out
GIS	Geographic Information System
GUI	Graphical User Interface
GW-BASIC	Gee Whiz! Beginner's All-purpose Symbolic Instruction Code

H

H-Zone	Hypheration-Zone
HCI	Human Computer Interface

HD	High Density
HDC	Hard Disk Controller
HDD	Hard Disk Drive
HDLC	High-level Data Link Control
HGC	Hercules Graphics Card
HIPO CHART	Hierarchical Input-Process-Output Chart
HMA	Higher Memory Block
HMI	Horizontal Motion Index
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HWD	Height/Width/Depth
Hz	Hertz

I

I/O	Input/Output
IAB	Internet Architecture Board
IAR	Instruction Address Register
IBG	Inter Block Gap
IBM	International Business Machines
IC	Integrated Circuit
ICMP	Internet Control Message Protocol
IDE	Integrated Drive Electronics
IDN	Integrated Digital Network
IEE	Institution of Electrical Engineers
IEEE	Institute o Electrical and Electronics Engineers
IEN	Internet Experiment Note
IETF	Internet Engineering Task Force
IIL	Integrated Injection Logic
IP	Internet Protocol, Information Processing
IPR	Intellectual Property Right
IPX	Internetwork Packet Exchange
IR	Instruction Register
IRC	Internet Relay Chart
IRG	Inter Record Gap
IRQ	Interrupt ReQuest
ISA	Industry Standard Architecture
ISAM	Index Sequential Access Mode
ISDN	Integrated Services Digital Network
ISO	International Standards Organisation
ISOC	Internet SOCiety
IT	Information Technology
ITU-T	International Telecommunication Union-Telecommunication
IWR	Isolated Word Recognition

J

JCL	Job Control Language
JTM	Job Transfer and Manipulation

K

KB	KiloByte
Kb	Kilobit
Kbps	Kilobits per second
KPO	Key Punch Operator

L

LAN	Local Area Network
LAP	Link Access Protocol
LAPB	Link Access Protocol Balanced
LCD	Liquid Crystal Display
LED	Light Emiting Diode
LF	Line Feed
LFU	Least Frequency Used
LHN	Long Haul Network
LIFO	Last In First Out
LIM-EMS	Lotus Intel Microsoft-Expanded Memory Specification
LIS	Land Information System
LLC	Logical Link Control
LRC	Longitudinal Redundancy Check
LSI	Large Scale Integration

M

MAC	Medium Access Control
MAN	Metropolitan Area Network
MAR	Memory Address Register
Mb	Megabit
MB	Megabyte
Mbps	Million bits per second
MBR	Memory Buffer Register
MCA	Micro Channel Architecture
MDA	Monochrome Display Adapter
MDR	Memory Data Register
MFD	Mini Floppy Disk
MFLOPS	Million Floating Point Operations Per Second
MHS	Message Handling Service
MHz	MegaHertz

MIB	Management Information Base
MICR	Magnetic Ink Character Reader
MIMD	Multiple Instruction Multiple Data
MIPS	Million Instruction Per Second
MIS	Management Information System
MMU	Memory Management Unit
MOS	Metal Oxide Semiconductor
MOTIS	Message Oriented Text Interchange System
MPU	MicroProcessor Unit
MS	MicroSoft, Message Store
MS-DOS	MicroSoft-Disk Operating System
MSI	Medium Scale Integration
MTA	Message Transfer Agent
MTBF	Mean Time Between Failure
MUX	Multiplexer

N

NAK	Negative Acknowledgement
NETBIOS	NETwork Basic Input/Output System
NI	Norton Index
NIC	Network Information Centre, Network Interface Card
NL	Natural Language
NLQ	Near Letter Quality
NMI	NonMaskable Interrupt
NOS	Network Operating System
NPDU	Network Protocol Data Unit
NPQ	Near Photographic Quality
NRZ	Non-return to Zero
ns	Nanosecond
NSAP	Network Service Access Point

O

OCR	Optical Character Reader/Recognition
ODD	Optical Disk Drive
OEM	Original Equipment Manufacturer
OHP	Over Head Panel
OLE	Object Linking and Embedding
OMR	Optical Mark Reader
OOD	Object Oriented Design
OOP	Object Oriented Programming
OR	Operation Research
OS/2	Operating System/2
OSF	Open Software Foundation
OSI	Open System Interconnection

P

PABX	Private Automatic Branch Exchange
PAL	Phase Alternate Line, Programmable Array Logic
PAB	Private Branch Exchange
PC	Personal Computer, Program Counter
PCB	Printed Circuit Board
PCI	Peripheral Component Interconnect, Protocol Control Information
PCM	Pulse Code Modulation
PDL	Page Description Language
PDN	Public Data Network
PDU	Protocol Data Unit
PERT	Progress Evaluation and Review Technique
PIN	Personal Identification Number
PIO	Parallel Input/Output
PL/1	Programming Language 1
PLA	Programmable Logic Array
PLL	Phase Locked Loop
PM	Phase Modulation
PMBX	Private Manual Branch Exchange
POP	Post Office Protocol
POS	Point of Sale
POST	Power On Self Test
ppm	Page Per Minute
PPP	Point-to-Point Protocol
PPU	Protocol Data Unit
PROLOG	PRogramming LOGic
PROM	Programmable Read Only Memory
PSDN	Packet Switched Data Network
PSK	Phase Shift Keying
PSPDN	Packet Switched Public Data Network
PTT	Post Telegraph and Telephone

Q

QAM	Quadrature Amplitude Modulation
QBE	Query By Example

R

RAID	Redundant Array of Inexpensive Disks
RAM	Random Access Memory
RDBMS	Relational DataBase Management System
RFI	Radio Frequency Interference
RGB	Red-Green-Blue

RISC	Reduced Instruction Set Computer
RLL	Run-Length-Limited
RMM	Read Mostly Memory
ROM	Read Only Memory
RPG	Report Programme Generator
RPROM	ReProgrammable Read Only Memory
RSI	Repititive Strain Injury
RTS	Request To Send

S

SAA	Systems Application Architecture
SAP	Service Access Point
SCR	Sequence Control Register
SCSI	Small Computer System Interface
SDLC	Synchronous Data Link Control
SIMD	Single Instruction Multiple Data
SIMM	Single In-Line Memory Module
SIPP	Single In-Line Pin Profile
SISD	Single Instruction Single Data
SLIP	Serial Line Internet Protocol
SMP	Symmetric MultiProcessing
SMTP	Simple Mail Transfer Protocol
SNA	Systems Network Architecture
SNMP	Simple Network Management Protocol
SONET	Synchronous Opticla NETwork
SPARC	Scalable Processor ARChitecture
SPC	Stored Programme Control
SPEC	Systems Performance Evaluation Cooperative
SQL	Structured Query Language
SRAM	Static Random Access Memory
SSD	Single Sided Disk
STP	Shielded Twisted Pair
SVGA	Super Video Graphics Array
SYSOP	System OPerator

T

TBU	Tape Back-up Unit
TCP	Transmission Control Protocol
TCP/IP	Transmission Control Protocol/Internet Protocol
TDM	Time Division Multiplexing
TE	Terminal Equipment
TEI	Terminal Endpoint Identifier
TIFF	Tagged Image File Format

TMOT	Thermo-Magnetic-Optic Technology
TPDU	Transport Protocol Data Unit
tpi	Tracks per inch
TSR	Terminate and Stay Resident
TTL	Transistor-Transistor Logic
TTY	Tele TYpewriter

U

UART	Universal Asynchronous Receiver Transmitter
UC	Upward Compatibility
UDP	User Datagram Protocol
UIMS	User Interface Management System
ULSI	Ultra Large Scale Integration
UMB	Upper Memory Block
UNIVAC	UNIVersal Automatic Computer
UPC	Universal Product Code
UPS	Uninterruptable Power Supply
URL	Universal Resource Locator
USART	Universal Synchronous Asynchronous Receiver-Transmitter
UTP	Unishielded Twisted Pair
UUCP	Unix-to-Unix Copy

V

VAN	Value Added Network
VC	Virtual Circuit
VDE	Voice Data Entry
VDT	Visual Display Terminal
VDU	Visual Display Unit
VESA	Video Electronics Standard Association
VGA	Video Graphics Array
VLIW	Very-Long-Instruction Word
VLSI	Very Large Scale Integration
VM	Virtual Memory
VRAM	Video Random Access Memory
VRC	Vertical Redundancy Check
VSAT	Very Small Aperture Terminal
VT	Virtual Terminal

W

W3	World Wide Web
WAIS	Wide Area Information Server
WAN	Wide Area Network
WORM	Write-Once-Read-Many

WP	Word Processing
WW	Word Wrap
WWW	World Wide Web
WYSIWYG	What You See Is What You Get

X

XGA	Extended Graphics Array
XMS	Extended Memory Specification
XOR	Exclusive OR
XT	Extended Technology

Y

YP	Yellow Pages

Z

ZWS	Zero Wait State

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Computer Fundamentals: Introduction of the paper "Computer Fundamentals" at the bachelor's programs of many disciplines of universities necessitated a comprehensive text for the course. The book covers the basic concepts of hardware, software and computer applications normally included in a 3-credit course on computer fundamentals. Sallent features of the book are:

- Computer organization, hardware and software
- Up-to-date information on microcomputers
- Concepts, functions and types of operating systems
- Overview of package programs and programming languages
- Database and database concepts
- Computer networks, Internet and Internet services
- Information and Communication Technology
- Maintenance and selection of computers

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Systech Publications Ltd.
Banglabazar Book & Computer Complex
38/3 Banglabazar, Dhaka-1100, Bangladesh
Phone: 88-02-7112406, 01711622565, 01714184844-5-6